Electronic Devices and Circuits

For Bachelor's of Engineering in Electronics and Communication, Electrical, and Computer Engineering

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Heritage Publishers & Distributors Pvt. Ltd.
Bhotahity, Kathmandu

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Chapter 1

Diodes

Introduction

The subject of this book is "**Electronic Devices and Circuits**". In the earlier course "**Basic Electronics Engineering**" we studied fundamental aspects of Electronics and Communication Engineering. In that course we studied basics of diode, bipolar junction transistors, field effect transistors, operational amplifiers, communication systems, and different applications of Electronics. This course is in fact a continuation of the aforementioned course with the emphasis on modeling of the semiconductor devices, single stage amplifiers configurations, output stage circuits, oscillators and waveform shaping circuits, voltage regulators etc. This is a second level course on Electronics. Thus it assumes readers have some degree of familiarity with the basic concepts of electronics.

Diode

Diode is a two terminal non-linear device. It can be of two types viz. vacuum diode and semiconductor diode. A vacuum diode works on the principle of thermionic emission of electrons. On the other hand, the semiconductor diode is a solid state device which works on the principle of **chemical bonding and crystal theory.**

Due to **small size, fast response and low cost** semiconductor diodes are the device of choice nowadays. Vacuum diodes are getting extinct day by day. This is because vacuum diodes are slow, bulky and expensive. They require high power to operate and it takes certain heating time to start and operate. Their life time is relatively small. Also they require frequent replacement. On the other hand, semiconductor diodes have relatively long lifetime and require very less power as compared to vacuum diode.

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At present, except for few high power applications almost all two terminal non-linear functions are carried out by semiconductor diodes. So, in the near future, it is certain that the use of vacuum diode will be completely replaced by semiconductor diode.

In this chapter our concern is to study semiconductor diodes. Now onwards when we say diode we will mean semiconductor diode. The important analog functions that can be carried out by the diodes are **clipping, clamping, rectification, switching** etc.

Ideal Diode

The ideal diode is a fictitious circuit element that is capable of conducting unidirectional current without any voltage drop within itself. Obviously one cannot find the ideal diode in real. It is only considered as a reference so that we can compare performance of the real diode with it.

Let v be the voltage across the diode and i be the current flowing through it as shown in the figure below:

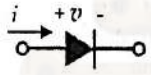


Figure: Ideal diode

Then the above diode is said to be ideal diode if:

$$i > 0 \text{ for } v \geq 0$$
$$i = 0 \text{ for } v < 0$$

The *i-v* characteristic for ideal diode is given below:

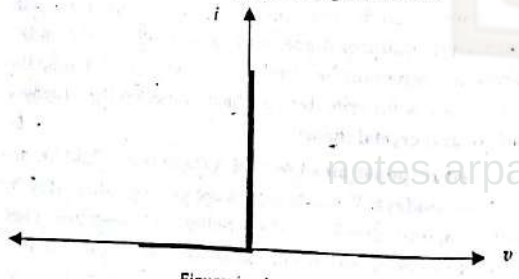


Figure: *i-v* characteristics for ideal diode

In other words, ideal diode acts as a short circuit when voltage across the diode is positive and as an open circuit if the voltage across it is negative.

This is depicted in the figure below:

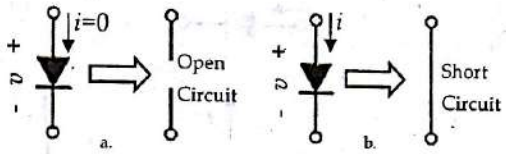
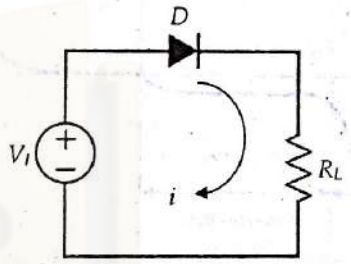


Figure: Equivalent circuit of ideal diode when
a. $v < 0$ b. $v \geq 0$

Let us consider an ideal diode connected to the source of voltage V_I as shown in the figure below:



Then

if $V_I \geq 0$ the current through the circuit will be given as:

$$i = \frac{V_I}{R_L}$$

If $V_I < 0$ the current through the circuit will be given as:

$$i = 0$$

Terminal Characteristics of the Junction Diode:

Terminal characteristics mean the voltage and current relationship across the diode when it is in operation. Here we are going to consider the *i-v* characteristics of the real Si junction diode not the ideal one. The current voltage characteristic of the junction diode is given by the figure as shown below:

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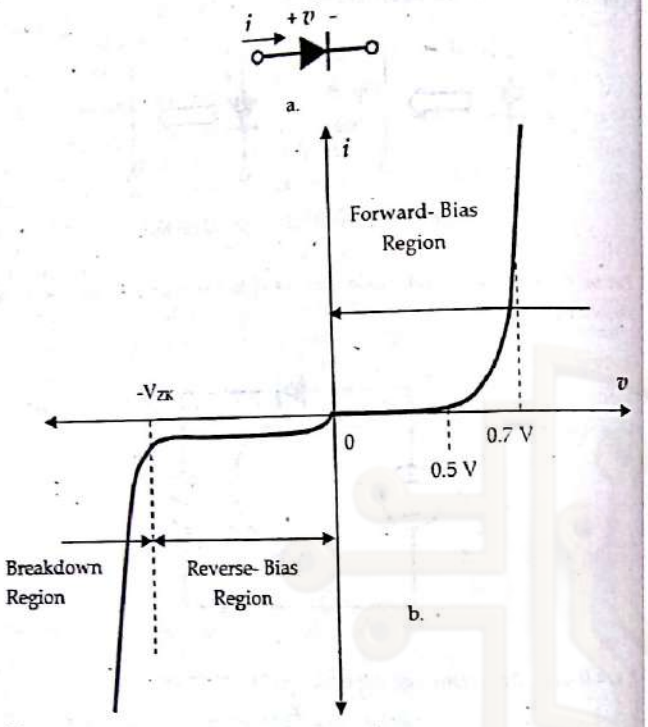

Figure:

a. Junction diode with terminal currents and voltages labeled
b. i - v characteristics of a Si based junction diode

Forward Bias Region

The diode is said to be in the forward bias region of operation when the terminal voltage of the diode v is positive. i.e. $v > 0$. From the characteristic curve we can see that the diode conducts negligibly small current for $v < 0.5$ voltage. This voltage is known as **cut-in voltage**. For the voltage between 0.5 and 0.7 the diode conducts few current. As soon as the voltage $v = 0.7$ V the diode starts to conduct appreciable amount of

current. This voltage is known as **knee voltage** and is sometimes designated as V_K.

The mathematical relation between the current i and voltage v of the diode in this region is given as:

$$i = I_S\,(e^{\frac{v}{\eta V_T}} - 1) \quad \text{.....(i)}$$

Where,

- i =diode current
- I_s =Reverse saturation current
- v =diode voltage
- η =1 to 2, is emission coefficient
- $V_T = \frac{KT}{q}$, thermal voltage
- K =Boltzmann's constant = 1.38×10^{-23}J/K
- T =absolute temperature
- q =electronic charge=1.6×10^{-19}C

Equation (i) is also known as **diode equation.**

The alternative logarithmic form of the above equation is:

$$v = \eta V_T \ln\frac{i}{I_S}$$

If $e^{\frac{V_D}{\eta V_T}} >> 1$, then the equation (i) may be written as:

$$i = I_S\, e^{\frac{v}{\eta V_T}} \quad \text{.....(ii)}$$

Let I_1 be the current flowing through the diode when voltage across its terminal is V_1 and I_2 be the current flowing through the diode when voltage across its terminal is V_2.

Then using equation (ii), we get,

$$I_1 = I_S e^{\frac{V_1}{\eta V_T}} \quad \text{... (iii)}$$

$$I_2 = I_S e^{\frac{V_2}{\eta V_T}} \quad \text{... (iv)}$$

Dividing equation (iv) by equation (iii), we get,

$$\frac{I_2}{I_1} = e^{\frac{V_2 - V_1}{\eta V_T}}$$

Taking ln on both sides, we get,

$$ln\left(\frac{I_2}{I_1}\right) = \frac{V_2 - V_1}{\eta V_T}$$

$$or, V_2 = V_1 + \eta V_T\, ln\left(\frac{I_2}{I_1}\right) = V_1 + 2.3\eta V_T\, log\left(\frac{I_2}{I_1}\right)$$

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Reverse Bias Region

The diode is said to be in the reverse bias region of operation when the terminal voltage of the diode v is negative. i.e. $v < 0$

The diode equation applies equally in the reverse bias region. In this case the voltage v is negative. As a result the exponential term becomes negligibly small as compared to 1. Thus,

$$i \cong -I_S \qquad \text{.....(v)}$$

Thus the current in the reverse bias is approximately equal to the saturation current. This is the main reason for naming I_S as **reverse saturation current**.

Breakdown Region

The diode is said to be in the breakdown region of operation when the reverse voltage exceeds certain threshold value called the break down voltage. This voltage is designated as V_{ZK}. It is written so because the voltage is at the knee of the characteristics curve and Z is for Zener breakdown which will be explained later in this chapter.

As seen from the graph, in breakdown region, the reverse current increases sharply with the corresponding change in voltage being very small. Thus it appears that even when change in current is large voltage remains constant. So this region of operation can be utilized in the voltage regulation.

The breakdown will not damage the diode permanently unless the power dissipated is within the maximum capacity that the diode can withstand. The maximum power that the diode can handle is usually written in the data sheet of the diode by the manufacturers.

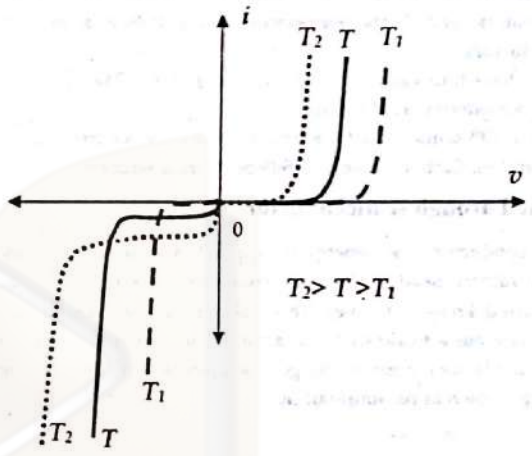
Temperature Effects on I-V characteristics of Diode

The figure shows the effect of temperature on i-v characteristics of the junction diode. In figure i-v characteristics of junction diode is given for three different temperatures T, T_1 and T_2 respectively with $T_2 < T < T_1$.

We can observe the following:

i. As the temperature is increased the curve in the forward direction shifts towards the origin. This means the forward voltage drop decrease as the temperature is increased. The diode will start to conduct at even lower value of forward voltage. It is known that the forward voltage decreases by 2 mV for every 1°C rise in temperature

ii. The reverse current increases with the increase in temperature. From the rule of thumb, the saturation current I_s doubles for every 10°C rise in temperature.



Physical Operation of Diodes

Having studied terminal characteristics of the junction diodes we are now about to study the physical phenomenon which gives rise to the above terminal characteristics.

Basic Semiconductor Concepts

Semiconductors are those materials whose electrical conductivity lies in between that of conductors and insulators. Their resistivity lies in the range from $10^{-4}\Omega$m to 0.5Ωm.

Following are the characteristics of semiconductors:

i. The resistivity of semiconductors is less than that of insulators and greater than that of conductors.

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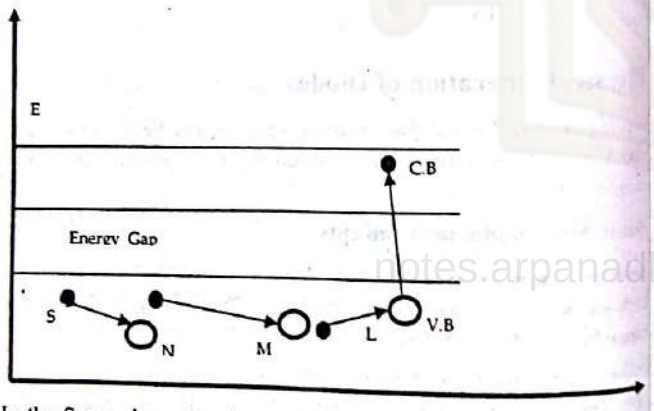
ii. They have negative temperature coefficient of resistance. This means resistance decrease with increase in temperature and vice versa.
iii. When impurity is added, there is a significant change in the concentration of charge carriers.
iv. Semiconductors are highly temperature dependent materials. Even a slight change in temperature can cause an appreciable change in carrier concentration. At low temperature, their behavior is similar to insulators and high temperature their behavior is similar to conductors.
v. They have four valance electrons in their atoms. They lie in-group IV of the modern periodic table.
vi. Out of 10^{10} semiconductor atoms, there is only one free electron.
Examples: Carbon, Selenium, Silicon, Germanium etc

Current Through semiconductor

In a semiconductor, when energy is supplied to an atom, electrons moves from the valance band to the conduction band leaving a vacancy in the valance band known as holes. This hole physically does not have any charge but it has a tendency to attract electrons. So, it acts like a positive charge but it is not a positive charge. The process of fusion of the hole and electron is known as **recombination.**



In the figure above the electron from position L moves to conduction band leaving a hole. To occupy the vacancy, electron from position M moves to position L leaving behind another vacancy at position M. Again, electron from position N occupies vacancy at M leaving a vacant position which in turn attracts the electron from position S. This process goes on and on.

Here we see that, there is a flow of holes in the valance band, which constitutes **hole current.** It is similar to flow of positive charge. In the conduction band, free electrons are available for conduction. So, in semiconductors the total current is the sum of hole current and electron current.

$\therefore \quad I = I_e + I_h$

Where;

I =total current
I_e =electron current
I_h =hole current

Types of Semiconductors

a. **Intrinsic semiconductor:**

Pure semiconductors are known as intrinsic semiconductors.

Example: Pure Silicon, Germanium etc.

Bonding in intrinsic semiconductor occurs by the formation of covalent bonds, four electrons of one semiconductor atom pairs with four electrons with one each from four other semiconductor atoms.

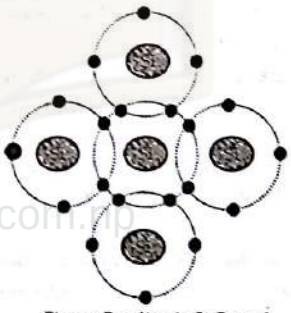


Figure: Bonding in Si Crystal

Or

it can also be depicted as:

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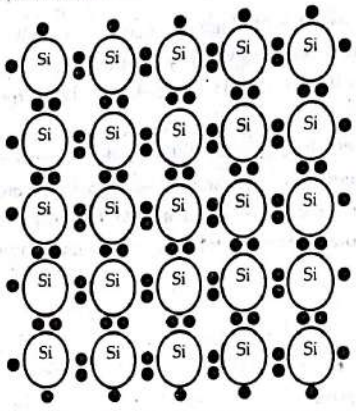

Figure: Bonding in Si Crystal (Alternative View)

At low temperature all covalent bonds are intact and there are no free electrons. However at room temperature some of the covalent bonds are broken and free electrons are formed. These free electrons leave their parent atom. As a result there is a vacancy in the parent atom. This vacancy is known as **hole**. It acts as a positive charge. The positive charge has magnitude equal to that of electron. This positive charge or hole may attracts free electron from another atom. It fills up the hole that existed in the previously ionized atom but will create hole in the next atom. This process may repeat itself with the result that we have positively charged hole moving within the silicon crystal.

Thus holes and free electrons are formed as a result of thermal ionization in the intrinsic semiconductors. These holes and electrons move randomly within the lattice of the Silicon crystal. And in the process some free electron may fill up hole. This process is called as **recombination**. The process of recombination results in the disappearance of both the holes and electrons.

The thermal ionization results in the formation of equal number of holes and electrons. And recombination results in the disappearance of holes and electron. At equilibrium, the recombination rate is equal to the thermal ionization rate or thermal generation rate. Thus, at equilibrium the concentration of electrons n and concentration of holes p are equal.

Thus,

$$n_i = n = p$$

Where,

n_i is called as intrinsic concentration of semiconductor.

The concentration is usually expressed as number of electrons or holes within $1cm^3$ of the semiconductor crystal.

From the study of semiconductor physics the intrinsic concentration of the semiconductor can be found from the following relation:

$$n_i^2 = BT^3 e^{-E_g/K.T}$$

Where,

B	= Material Specific parameter
	$=5.4*10^{31}$ for Silicon
T	=Temperature in absolute scale
E_g	=Energy gap
K	=Boltzmann's Constant

Diffusion and Drift

The electrons and holes move in the semiconductor by two possible mechanisms known as diffusion and drift.

Diffusion:

When the concentration of holes and electrons is uniform within the silicon crystal then their random motion will not result in the net flow of charge i.e. current.

But when the concentration of charge carrier (hole or electron) is different, then in such case the charge carrier will diffuse from the region of high concentration to the region of the low concentration. This process is called diffusion which results in the net flow of the charge carriers. The current because of diffusion is known as **diffusion current.**

The figure shows the intrinsic silicon bar having non uniform concentration of hole along the x-axis. The concentration of hole decreases as we move along the positive x- axis.

Due to such concentration profile hole diffusion current exists and the current density can be expressed as:

$$J_P = -q.D_p \frac{dp}{dx} \ldots\ldots\ldots(i)$$

Where,

q	=Electronic Charge= 1.6×10^{-19} C
D_P	=Diffusivity or diffusion constant for holes.

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dp/dx =Gradient or the Slope of the concentration profile

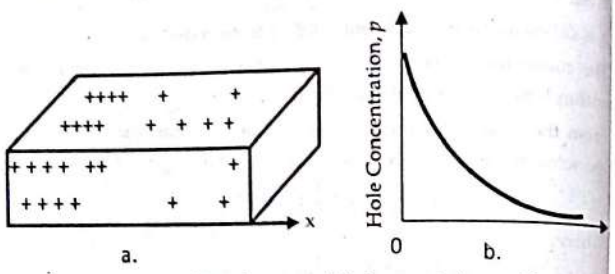

Figure: a. Intrinsic Silicon bar b. Hole concentration profile

Note that dp/dx is negative. Thus above expression suggests us there is the positive current flowing in the x-direction, as it should be because it is holes flowing in the x-direction.

Similarly, the diffusion current density resulting from the electron concentration gradient is given as:

$$J_n = q.D_n\frac{dn}{dx}.....(ii)$$

Where,

q =Electronic Charge= 1.6×10^{-19} C

D_n =Diffusivity or diffusion constant for electrons

dn/dx =Gradient or the Slope of the concentration profile

Note that dn/dx is negative. Thus the above expression suggests us that there is negative current flowing in the x-direction., as it should because it is electrons flowing in the x-direction.

Drift:

Drift is another mechanism of current flow in the semiconductors.

When an electric field is applied across a Silicon bar then the electron or holes will be accelerated by the applied electric field. The holes or electron will acquire a velocity component called **drift velocity.**

Let E be the Electric field applied and μ_P be the mobility of holes. Then the holes will drift in the direction of the applied field with the the drift velocity given by:

$$v_{drift} = \mu_p E$$

Let us consider a .Silicon crystal having hole density p and electron density n subjected to the electric field E. Thus the holes will drift in the direction of the applied field with the velocity $\mu_p E$. Thus in one second

the total charge flowing through area A perpendicular to the direction of the flow of holes is given as

Total Charge = $qp\mu_p E.A$

The charge in each hole has magnitude q. There are p such holes in unit cm of the crystal. The velocity in 1 second will be the distance in cm. Thus the expression $qp\mu_P EA$ will give total charge flowing per second through area A.

Therefore the hole current density because of drift is given as:

$$J_{P-drift} = qp\mu_P E$$

The free electrons will drift in the direction opposite to that of electric field E with the velocity$\mu_n E$. Thus in 1 second the total charge flowing through the area A perpendicular to the direction of the flow of electrons is given as:

Total Charge = $-qn\mu_n E.A$

Therefore the electron current density due to drift is given as:

$$J_{n-drift} = qn\mu_n E$$

The total drift current density is given as:

$$J_{drift} = J_{p-drift} + J_{n-drift}$$
$$= q(p\mu_P + n\mu_n)E$$

The ratio of electric field to current density gives the resistivity (ρ) of the silicon crystal. It is expressed as:

$$\rho = \frac{1}{q(p\mu_P + n\mu_n)}$$

Einstein Relationship:

The relationship between the carrier diffusivity and mobility exists and is given as:

$$\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = V_T$$

This relation is known as Einstein relationship.

b. Extrinsic semiconductor:

Impure semiconductors are known as extrinsic semiconductors. The process of mixing impurity to intrinsic semiconductor is known as doping which results to the formation of extrinsic semiconductor. Generally for 10^8 semiconductor atoms 1 impurity atom is added.

There are two types of extrinsic semiconductors viz. P-type and N-type semiconductors.

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i. **P-type semiconductor**

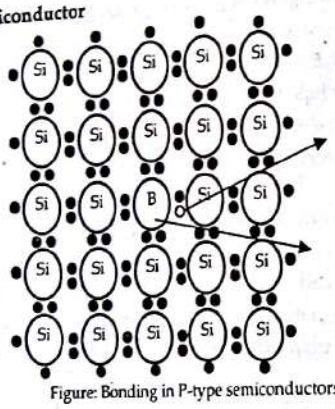

Figure: Bonding in P-type semiconductors

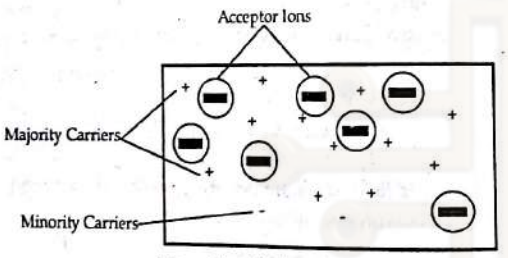

Figure: P-type Semiconductor

When intrinsic semiconductor is doped with trivalent impurities like boron, gallium, indium etc., p-type extrinsic semiconductor is formed. In p-type extrinsic semiconductor, four electrons from pure semiconductor atoms have to pair with three electrons from the impurity atom. Thus, three-three electrons from each atoms part take in paring and one electron gets unpaired. So, there remains a vacancy which is a hole. Hence, in p-type materials there is large number of such holes. It is therefore the material is called P-type (or positive type, as holes are positive).

Sometimes due to breakage of covalent bond in p-type material few thermal electrons are generated along with holes. These electrons are responsible for few leakages current and their number is very less than the number of holes. Hence, holes are present in excess and are called majority charge carriers and free electrons are present in a minimum and are called minority charge carriers.

Let p_{p0} and n_{p0} be the concentration of the majority holes and minority electrons in the p-type semiconductor under equilibrium. Let N_A be the number of the acceptor atoms or number of trivalent impurity atoms.

Then,

The concentration of holes is approximately equal to the number of the acceptor atoms.

i.e. $p_{p0} = N_A$

From the semiconductor physics,

$$p_{p0}\, n_{p0} = n_i^2$$

Thus the concentration of the minority electrons in the p-type semiconductor is given as:

$$n_{p0} = n_i^2 / p_{p0}$$

$$\therefore n_{p0} = n_i^2 / N_A$$

ii. **N-type semiconductor**

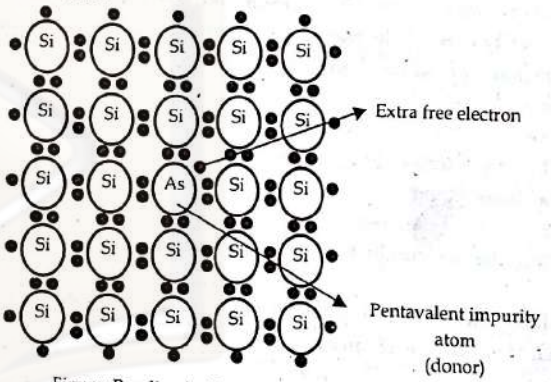

Figure: Bonding in N-type semiconductors

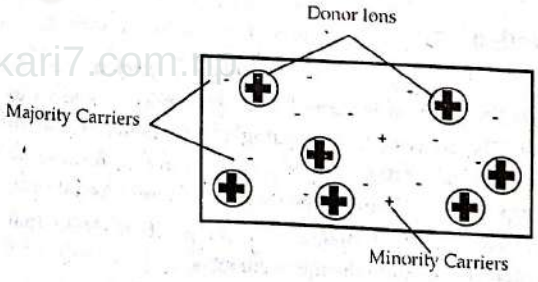

Figure: N-type Semiconductor

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When intrinsic semiconductor is doped with pentavalent impurities like arsenic, antimony etc the N-type extrinsic semiconductor is formed. In N-type extrinsic semiconductor four electrons from pure semiconductor atoms have to pair with five electrons from the impurity atom. Thus, four-four electrons from each atoms part take in pairing and one electron gets surplus. So, there is large number of such surplus electrons. It is therefore the material is called N-type (or negative type).

Sometimes due to the breakage of covalent bond in N-type material few thermal holes are generated along with electrons. These holes are responsible for few leakages current and their number is very less than the no of electrons. Hence, free electrons are present in excess and are called majority charge carriers and holes are present in a minimum and are called minority charge carriers.

Let n_{n0} and p_{n0} be the concentration of the majority electrons and minority holes in the n-type semiconductor under equilibrium. Let N_D be the number of the donor atoms or number of pentavalant impurity atoms.

Then,

The concentration of electrons is approximately equal to the number of the donor atoms.

i.e. $n_{n0} = N_D$

From the semiconductor physics,

$$n_{n0}\, p_{n0} = n_i^2$$

Thus the concentration of the minority holes in the n-type semiconductor is given as:

$$p_{n0} = n_i^2 / n_{n0}$$

$$\therefore p_{n0} = n_i^2 / N_D$$

The PN junction

When p-type semiconductor is suitably connected to n-type semiconductor, PN junction is formed and the device is known as PN junction diode. The process of connecting p-type material and n-type material is done by fabrication. A PN junction can't be formed by just connecting p-type material with n-type material. It should be fabricated.

Figure shows the simplified diagram of pn junction. Note that the terminals are taken by forming the metal contacts.

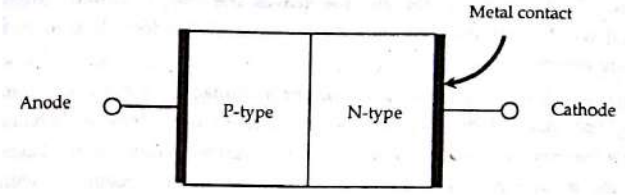

Figure: Simplified form of PN junction diode

PN junction under open circuit condition

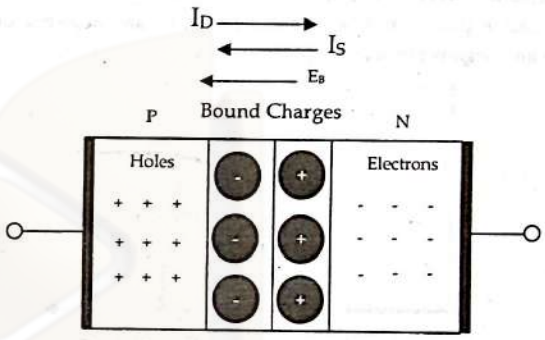

Figure: PN junction under open circuit condition

Diffusion Current I_D

The concentration of holes is more in p type material than in n type material, so the holes from p type material diffuse across junction from p side to n side. Similarly, the concentration of electrons is more in n type material than in p type material. Thus, the electrons from n type material diffuse across junction from n side to p side. These two current components add together to form the diffusion current I_D whose conventional direction is from p side to n side as shown in the figure.

Formation of Depletion layer:

In p type material holes are bound to the negative acceptor atoms or ions and in n type material free electrons are bound to the positive donor atoms or ions. When hole leaves the corresponding parent atom, it will become uncovered negative ion since the neutralizing holes are no more

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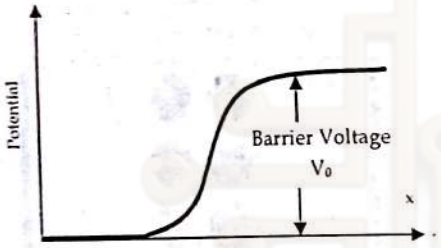
present. Similarly when free electron leaves the corresponding parent atom, it will become positive ion since the neutralizing free electrons are no more present.

When p and n type materials are brought in contact, free electrons from the n type material diffuse into the p region to recombine with holes leaving behind the positive ions near the junction. Similarly, the holes from the p type material diffuse into the n region to recombine with electrons leaving behind negative ions near the junction.

These positive ions and negative ions form a carrier depleted layer known as the depletion layer. It is called so because the layer is composed of positive and negative ions where positive ions are depleted of free electrons and negative ions are depleted of holes.



After the depletion layer is formed normal flow of holes and electrons gets stopped. The depletion layer acts as a barrier.

Junction Built-in potential

The potential difference across the depletion region is called as the barrier potential or built-in potential and it is denoted by V_o. The expression for V_o is given by:

$$V_o = \frac{kT}{q} ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Where,

- K =1.38×10^{-23}J/K= Boltzmann's Constant
- T =Temperature in absolute scale
- q =charge of one electron=1.6 × 10^{-19}C
- N_A =Acceptor concentration
- N_D =Donor concentration
- n_i =Intrinsic Concentration

Due to the barrier potential, there exists an electric field E_B which acts as a barrier for flow of majority charge carriers.

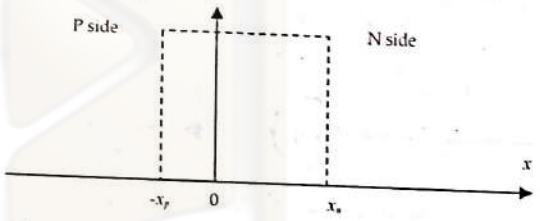
Drift Current I_S and equilibrium

There are minority charge carriers both in p side and n side. When the minority electrons in the p side moves near the depletion layer or junction, it is swept away by the barrier potential towards the *n* side. Also when minority holes in the n side moves near the junction, it is swept away by the barrier potential towards the *p* side. In other words minority carriers in either side are drifted by the field created by the depletion layer. Because of the flow of these minority carriers a current is set up known as **drift current**. It is denoted as I_S. The direction of I_S will be as shown in the figure.

When pn junction is open circuited, the diffusion current and drift current are in equilibrium i.e. they cancel each other. Thus,

$$I_D = I_S$$

Width of the depletion layer



The depletion layer or depletion region takes up both p side and n side. Equal amount of charge exists on the either side. Let x_p and x_n be the width of the depletion region in p side and n side respectively.

From the condition of charge equality:

We may write,

$$qx_pAN_A = qx_nAN_D$$

(Because no. of acceptor atoms is equal to number of holes and no. of donor atoms is equal to the number of free electrons in equilibrium.)

This may be reduced to following form:

$$\frac{x_n}{x_p} = \frac{N_A}{N_D}$$

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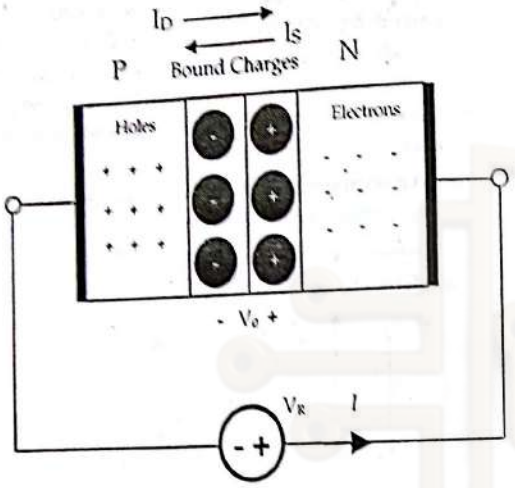
From the knowledge of semiconductor physics the expression for the width of the depletion region under open circuited condition can be expressed as:

$$W_{dep} = x_n + x_p = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0}$$

PN junction under reverse bias condition



Let us consider a pn junction reversed biased with the source of voltage V_R The negative terminal of the source is connected to the p side and the positive terminal of the source is connected to the n side. The reverse current I flow through the circuit. Let V_0 be the potential in the depletion layer initially.

After the pn junction is reversed biased, holes from the p side are attracted by the negative terminal of the source and free electrons from the n side are attracted by the positive terminal of the source. Thus for the reverse current I to flow, the free electrons leave the n side and holes leave the p side. As a result, the uncovered bound positive and negative ions are deposited near the junction. The width of the depletion layer is increased. Due to the widening of the depletion layer the junction barrier potential is increased from V_0 by the magnitude of V_R which is the

magnitude of external voltage. After that an equilibrium point is reached when,

$$I = I_S - I_D$$

It is to be noted that in reverse bias condition the current within the diode is mainly due to the drift current. After the steady state is reached the diffusion current becomes extremely small.

Thus, in the reverse bias condition of the pn junction, the current within diode is mainly because of drift current.

Width of the depletion region

From the semiconductor physics the width of the depletion region is calculated as

$$W_{dep} = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_0 + V_R)}$$

Depletion Capacitance or Junction Capacitance

Under reverse bias condition the depletion region of the pn junction can be considered as a parallel plate capacitor having equal charge on the either side.

The charge can be found by either calculating charge on n side or on p side. Let the charge in the junction be denoted as q_j.

We can find the charge q_j by calculating charge present on n side.

The charge present on n side is given as

$$q_n = qN_D x_n A$$

Thus

$$q_j = q_n = qN_D x_n A \ldots\ldots(i)$$

Also we know,

$$\frac{x_n}{x_p} = \frac{N_A}{N_D}$$

$$\therefore x_p = \frac{N_D}{N_A}.x_n$$

$$\therefore W_{dep} = x_n + x_p$$

$$= x_n\left(1+\frac{N_D}{N_A}\right)$$

$$or, W_{dep} = x_n\left(\frac{N_A+N_D}{N_A}\right)$$

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$$\therefore x_n = \left(\frac{N_A}{N_A + N_D}\right) W_{dep}$$

Substituting the value of x_n in the equation (i) we get

$$q_j = q.\frac{N_A N_D}{N_A + N_D} A.W_{dep}$$

The junction capacitance is given as:

$$C_j = \frac{\epsilon_S A}{W_{dep}}$$

Substituting the value of W_{dep}, we may write above equation as:

$$C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

Where,

$$C_{j0} = A\sqrt{\left(\frac{\epsilon_S q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}$$

The more general formula for the junction capacitance is given as:

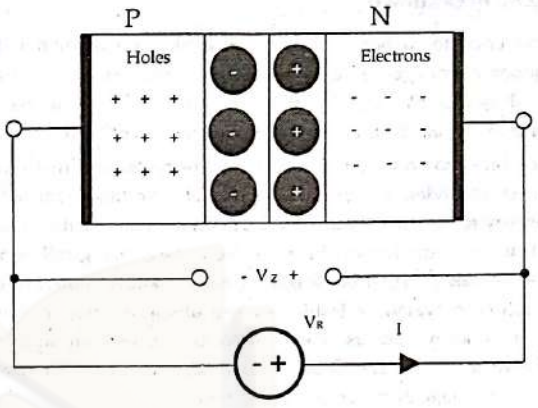
$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_0}\right)^m}$$

Here m is a constant called **grading coefficient**. Its value ranges from 1/3 to ½.

PN junction in the breakdown region

Let us consider a pn junction reversed biased with the reverse voltage V_R. The negative terminal of the source is connected to the p side and the positive terminal of the source is connected to the n side. The reverse current I flow through the circuit. Let V_Z be the reverse voltage that appears across the pn junction.

After the pn junction is reversed biased the holes from the p side are attracted by the negative terminal of the source and free electrons from the n side are attracted by the positive terminal of the source. Thus for the reverse current I to flow, the free electrons leave the n side and holes leave the p side. As a result, the uncovered bound positive and negative ions are deposited near the junction. The width of the depletion layer is increased. It keeps on increasing with the increase in the reverse bias voltage V_R or equivalently reverse current I. Due to the widening of the depletion layer the junction barrier potential V_Z and electric field across the junction also increases.



The saturation current I_S which flows due to the drift of the minority carrier is not able to maintain the reverse current I within the diode when $I >> I_S$. When the sufficient high voltage V_Z develops across the junction, a new mechanism is developed inside the diode to supply charge carriers needed to support the current I. This new mechanism is known as breakdown phenomenon. The topic below discusses breakdown phenomenon in pn junction.

Breakdown Phenomenon in pn junction

There are two possible mechanism of breakdown of pn junction. They are as follows:

Zener Breakdown:

If the pn junction breaks down with a breakdown voltage $V_Z < 5V$, the breakdown is usually by the Zener effect. Zener breakdown occurs when the electric field in the depletion layer increases to the point where it can break the covalent bonds and generate the electron hole pair. These electron hole pair will constitute reverse current which will support the external current I. Once the zener effect starts a large number of such electron hole pair will be generated with negligible increase in the junction voltage V_Z. The reverse current in the breakdown will be

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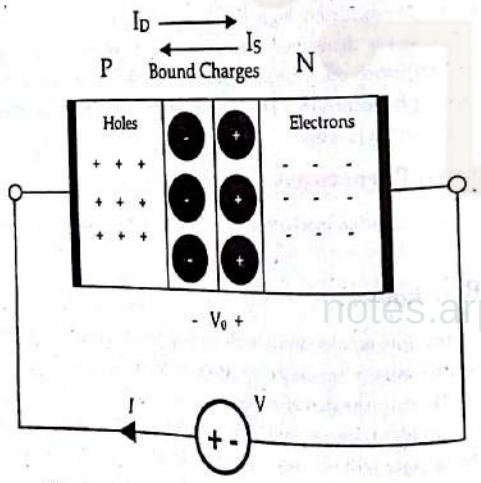
determined by the external circuit while the reverse voltage between the diode terminals will remain close to the rated breakdown voltage.

Avalanche breakdown

Avalanche breakdown occurs when pn junction breakdown with the reverse junction voltage V_Z being greater than approximately 7 Volts. If V_Z is in between 5V and 7V then the breakdown may be Zener breakdown or Avalanche breakdown or combination of both.

Avalanche breakdown occurs when the minority carriers that cross the depletion layer under the influence of the electric field gain sufficient kinetic energy to be able to break covalent bond in atoms with which they collide. Thus more carriers are liberated from the ionizing collisions. The carriers so liberated further break other covalent bonds in their way. This process occurs in avalanche fashion that results in the flow of the large number of minority carriers. These minority carriers can support any value of reverse current set up by the external circuit with the negligible change in the voltage drop across the junction.

PN junction under forward bias condition



Let us consider a pn junction excited with the voltage source V in forward bias configuration. Current I flow through the circuit because of the

source voltage. After the pn junction is forward biased holes are repelled by positive terminal of the source and free electrons are repelled by the negative terminal of the source. As a result large number of holes and electrons move towards the junction.

In order to setup the required current I, large number of holes in the p side and large number of free electrons on the n side move towards the junction. In this process recombination takes place near the junction. Also the free electrons from n side neutralize the uncovered bound positive ions and the holes from the p side neutralize the uncovered bound negative ions thereby reducing the charge stored on the depletion layer. In other words width of the depletion region is reduced to a much extent.

In forward bias condition the current flowing through the junction is mainly because of the diffusion current.

At equilibrium,

$$I = I_D - I_S$$

Derivation of the diode equation:

The figure below shows the minority carrier concentration profile in pn junction.

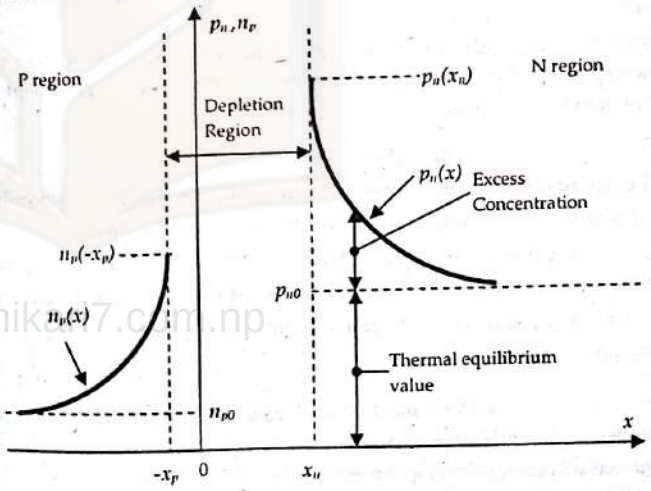


Figure: Minority Carrier distribution in the forward biased pn junction. It is assumed that the *p* region is heavily doped as compared to *n* region.

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Normally we will use

p_n to denote the minority hole concentration in n type material.

n_p to denote the minority electron concentration in p type material.

p_{n0} to denote the minority hole concentration in n type material at equilibrium.

n_{p0} to denote the minority electron concentration in p type material at equilibrium.

$p_n(x)$ to denote concentration of minority holes in n type material at a distance x from reference axis.

$n_p(x)$ to denote concentration of minority electrons in p type material at a distance x from reference axis.

Because of the source voltage V, holes are injected in the n region from p side. Similarly electrons are injected into the p region from n side.

The holes injected into the n region will cause the concentration of minority holes, p_n, in the n region to increase from the thermal equilibrium value p_{n0}. The excess concentration (p_n-p_{n0}) will be highest at the edge of the depletion region and will decrease exponentially as we move away from the junction.

Similar is the case with the concentration of minority electrons, n_p, in the p region.

From the semiconductor physics, using the law of junction, we can write the expression for the minority hole concentration at the edge of depletion region in the n-side as:

$$p_n(x_n) = p_{n0}e^{\frac{V}{V_T}} \qquad \ldots\ldots \text{(i)}$$

The distribution of the excess hole concentration in the n region is an exponentially decaying function which can be expressed as:

$$p_n(x) = p_{n0} + [p_n(x_n) - p_{n0}]e^{-(x-x_n)/L_p} \qquad \ldots\ldots\ldots \text{(ii)}$$

Where, L_P is a constant called diffusion length of holes in Silicon.

Substituting equation (i) into equation (ii)

We get,

$$p_n(x) = p_{n0} + [p_{n0}e^{\frac{V}{V_T}} - p_{n0}]e^{-(x-x_n)/L_p} \qquad \ldots\ldots\ldots\ldots \text{(iii)}$$

We know, the diffusion of holes in the n region will give rise to the hole current whose density is given by:

$$J_P = -q.D_p\frac{dp}{dx} \qquad \ldots\ldots\ldots\ldots \text{(iv)}$$

From equation (iii) and (iv), we get

$$J_P = -qD_p\frac{d}{dx}\left\{p_{n0} + [p_{n0}e^{\frac{V}{V_T}} - p_{n0}]e^{-(x-x_n)/L_p}\right\}$$

$$= q\frac{D_p}{L_p}p_{n0}\left[e^{\frac{V}{V_T}} - 1\right]e^{-(x-x_n)/L_p}$$

Under steady state $x=x_n$ therefore above relation becomes:

$$J_p = q\frac{D_p}{L_p}p_{n0}\left[e^{\frac{V}{V_T}} - 1\right] \qquad \ldots\ldots\ldots\ldots \text{(v)}$$

Similarly the electron current density at steady state can be obtained as:

$$J_n = q\frac{D_n}{L_n}n_{p0}\left[e^{\frac{V}{V_T}} - 1\right] \qquad \ldots\ldots\ldots\ldots \text{(vi)}$$

The total current density is obtained by adding equation (v) and (vi).

$$J = J_p + J_n$$

$$= q\frac{D_p}{L_p}p_{n0}\left[e^{\frac{V}{V_T}} - 1\right] + q\frac{D_n}{L_n}n_{p0}\left[e^{\frac{V}{V_T}} - 1\right]$$

$$= q\left[e^{\frac{V}{V_T}} - 1\right]\left(\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)$$

We know,

$$p_{n0} = n_i^2/N_D \text{ and } n_{p0} = n_i^2/N_A$$

$$\therefore J = q\left[e^{\frac{V}{V_T}} - 1\right]\left(\frac{D_p}{L_p}\frac{n_i^2}{N_D} + \frac{D_n}{L_n}\frac{n_i^2}{N_A}\right)$$

$$J = qn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)\left[e^{\frac{V}{V_T}} - 1\right]$$

So the current I through the diode is given as

$$I = A.J$$

$$I = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)\left[e^{\frac{V}{V_T}} - 1\right]$$

$$I = I_S\left(e^{\frac{V}{V_T}} - 1\right) \qquad \ldots\ldots\ldots\ldots \text{(vii)}$$

Where,

$I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$ is the saturation current.

Hence we have derived the **diode equation**. In order to include other non ideal effects we introduce a **fix-up parameter** in exponential term of equation (vii) which is known as **emission coefficient** and denoted as η.

Thus,

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$$I = I_S \cdot \left(e^{\frac{V}{\eta V_T}} - 1\right)$$

Diffusion Capacitance:

During the forward bias operation of pn junction diode, there is the storage of excess minority charge carriers in p region and n region. This charge storage phenomenon will give rise to diffusion capacitance. In forward bias condition diffusion capacitance is given by:

$$C_d = \left(\frac{\tau_T}{V_T}\right) I$$

Where,

τ = Mean Transit time of charge carriers.

V_T = Thermal Voltage

I =Diode Current

Junction Capacitance:

The depletion layer capacitance or the junction capacitance in forward bias condition is calculated by the following relation:

$$C_j = 2.C_{j0}$$

Where, C_{j0} is the junction capacitance at zero bias.

Modeling the Semiconductor Diode

The main objective of modeling the non-linear semiconductor device (like diode) is to achieve the linearity so that we can apply various circuit theories to obtain the parameters of the device in a convenient way with simple calculations.

Definitions

i. Modeling

The process of representing the semiconductor diode (or any other semiconductor device) by the basic components like resistor, inductor, capacitor and sources like a voltage source, current source etc without the loss of its functional behavior is known as modeling.

ii.Model

Circuitry representation of the diode (or any other semiconductor device) with the equivalent elements is known as the model of diode (or device model).

In our day to day life we have to frequently deal with two types of signals, DC signals and AC signal. In DC signal we generally require high voltage of the order of volts. So, DC signals are known as Large Signals. In AC signal we require voltage of order mV so, AC signals are known as small signal.

DC or Large Signal Model of the Diode

In DC or large signal model we deal with the application of large signals to the diode circuit.

1.Ideal Diode and Ideal diode model

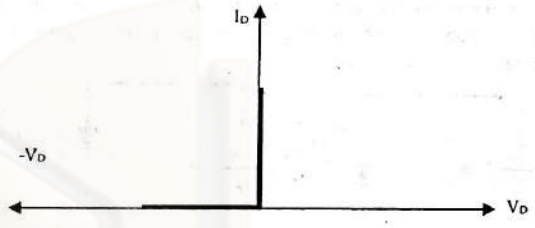

Figure: I-V characteristics for ideal diode

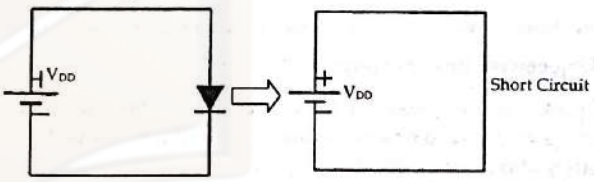

Figure: Forward Biased Ideal Diode and its equivalent circuit

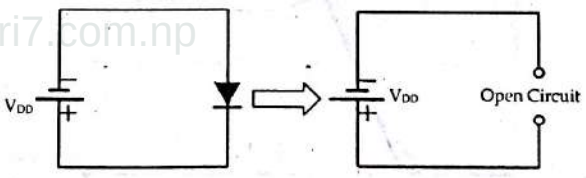

Figure: Reversed biased Ideal Diode and equivalent model modeled by an open circuit. Also, the voltage drop across the ideal diode is assumed to be zero.

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2.Constant Voltage drop model:

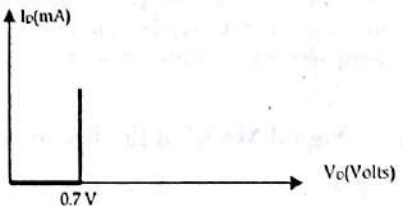

Figure: Characteristics plot for constant voltage drop model.

In this model, a diode is replaced by the constant diode voltage drop. In this model, the diode resistance is assumed to be zero. Since the diode is replaced by a constant voltage drop, this model is called constant voltage drop model.

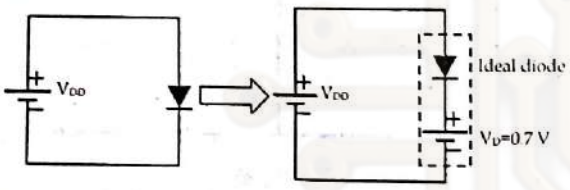

Figure: Forward biased diode and its equivalent circuit using constant voltage drop model

3.Piecewise linear model

In piecewise linear model, a diode is replaced by the ideal diode with its voltage drop V_D and internal resistance r_D. So this model is also known as **battery plus resistance model.**

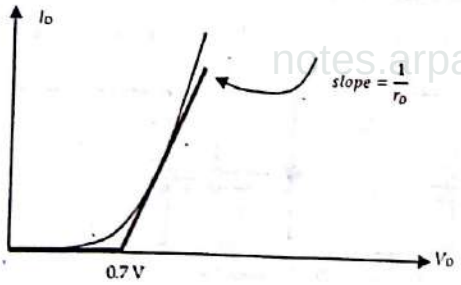

Figure: Exponential Curve approximated by two straight lines.

In this model, the exponential curve is approximated by two pieces of straight lines. So, this model is known as the piecewise linear model.

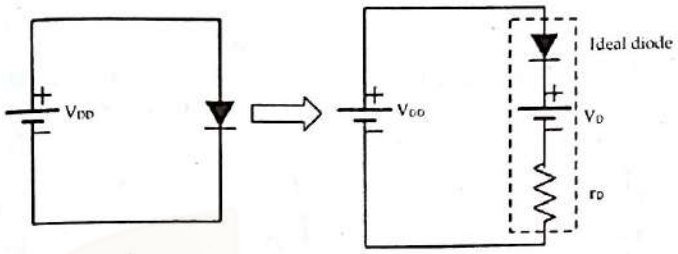

Figure: Piecewise equivalent circuit of the diode

Where,

r_D =Forward resistance

V_D =Diode Voltage drop

S· ·ll Signal analysis of diode:

In t small signal model, we study the applications of small signal or ac signals to the diode circuit. Generally the signals with the amplitude around 10mV or less are known as small signals.

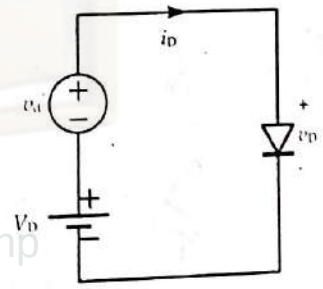

Figure: Circuit diagram for small signal analysis of diode

Figure shows the circuit diagram for the small signal analysis of diode. A signal source v_d having triangular waveform is superimposed on the dc voltage source V_D. The combination of signal source and dc voltage source provides a total voltage v_D across the diode.

$$\therefore v_D = V_D + v_d \quad \text{.....(i)}$$

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Let i_D be the total current flowing through the circuit. This total current, i_D is sum of the current contributed by dc source, I_D, and the current contributed by the signal source or ac source, i_d.

$$\therefore i_D = I_D + i_d \quad \ldots\ldots\ldots(ii)$$

It is to be noted that here we have used dc voltage even in small signal analysis. This is done to ensure the diode to be forward biased throughout the operation.

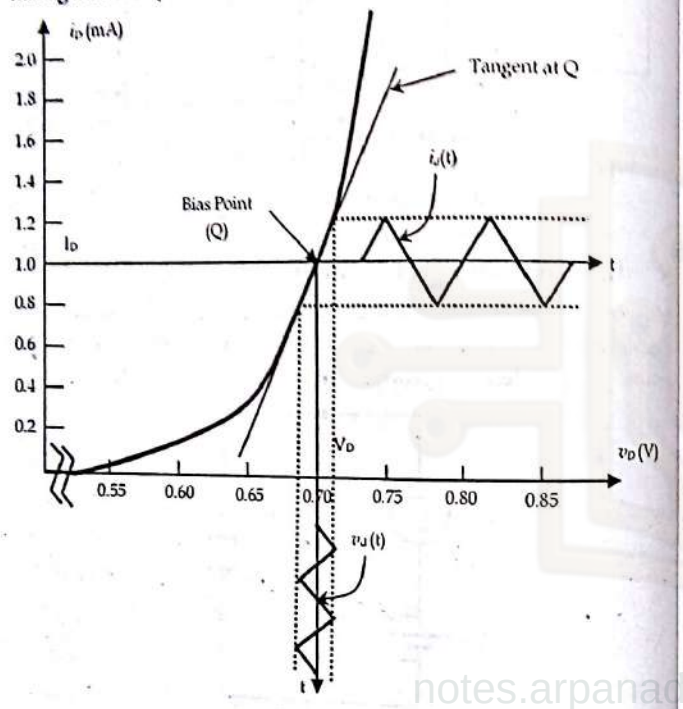

Figure: Graphical analysis of small signal model.

Figure shows the graph for small signal operation of the diode. In the figure current due to dc source is 1 mA and dc voltage is assumed to be 0.7 V. These are typical values for Si based diodes and these voltage and current appear in the circuit even when signal source is zero. The purpose is to forward bias the diode throughout the operation. The DC current and voltage in the circuit defines a Q point. The voltage signal $v_d(t)$ and the current signal $i_d(t)$ swings across the bias point Q as shown in the figure.

From the diode equation, we may write,

$$i_D = I_S e^{\frac{v_D}{\eta V_T}}$$

$$\text{or, } i_D = I_S e^{\frac{V_D + v_d}{\eta V_T}} \quad \text{from equation (i)}$$

$$\text{or, } i_D = I_S e^{\frac{V_D}{\eta V_T}} e^{\frac{v_d}{\eta V_T}}$$

Since $I_D = I_S e^{\frac{V_D}{\eta V_T}}$

$$i_D = I_D e^{\frac{v_d}{\eta V_T}}$$

$$\text{or, } I_D + i_d = I_D e^{\frac{v_d}{\eta V_T}}$$

Expanding the exponential function on the right hand side of the above equation, we get,

$$I_D + i_d = I_D\left(1 + \frac{\frac{v_d}{\eta V_T}}{1!} + \frac{\left(\frac{v_d}{\eta V_T}\right)^2}{2!} + \frac{\left(\frac{v_d}{\eta V_T}\right)^3}{3!} + \cdots\right)$$

Since this is small signal analysis and the signal voltage v_d is very small. So in the above expression we can neglect the higher powers of v_d.

If we do so,

$$\text{or, } I_D + i_d = I_D\left(1 + \frac{v_d}{\eta V_T}\right)$$

$$\text{or, } I_D + i_d = I_D + I_D\frac{v_d}{\eta V_T}$$

$$\text{or, } i_d = I_D\frac{v_d}{\eta V_T}$$

$$\text{or, } \frac{v_d}{i_d} = \frac{\eta V_T}{I_D}$$

$$\therefore r_d = \frac{\eta V_T}{I_D}$$

This is the required expression for the dynamic resistance or ac resistance of the diode.

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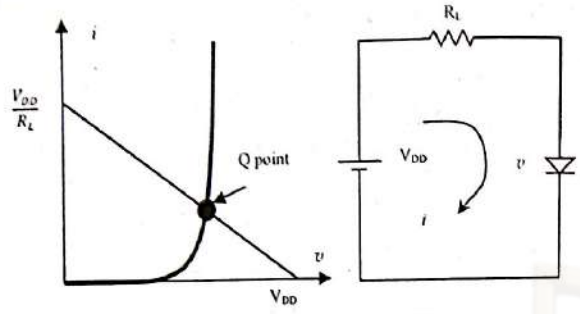
Diode Load Line



Applying KVL for the above circuit,

$V_{DD}=i R_L + v$ .....(i)

When $I = 0$ then

$v = V_{DD}$ and

When $v = 0$ then

$i = \frac{V_{DD}}{R_L}$

The equation (i) can be written as

$i = -v/R_L + V_{DD}/R_L$

This equation is in the form,

$y = c - mx$

This is the first degree equation of the straight line having a negative slope. This line has the end points (V_{DD}, 0) and ($0, \frac{V_{DD}}{R_L}$).Since v and V_{DD} are constant for the particular circuit and the different values of i can be obtained for different values of load resistance R_L. So, the straight line is also known as load line or diode load line.

The point of intersection of the load line and forward characteristics curve of the diode is known as Q-point or quiescent point or operating point. The diode starts functioning beyond this point.

Analysis of diode circuits

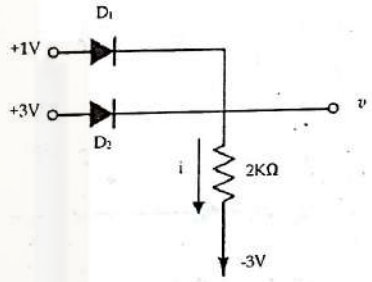
The circuit containing diode can be analyzed based on the models that we learned in previous section of this chapter. If the signals applied to the diode are large signals then we utilize large signal models whereas if the signals applied to the diode are small signal or ac signals then we utilize the small signal model of the diode.

In the following examples we will try analyze few diode circuits. But it is to be noted that for rapid circuit analysis in large signal condition the best diode model is constant voltage drop model. In some conditions we utilize ideal diode model and piecewise linear model.

Examples:

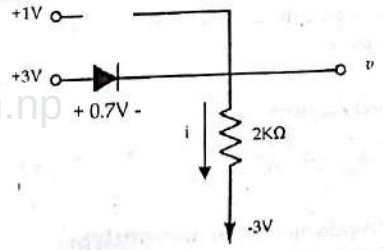
Ex.1.

Find the voltage v and i of the the following circuit. Use constant voltage drop model of diode.



Solution:

The diode D_1 will be reversed biased and the diode D_1 will be forward biased. Since we are using Constant voltage drop model the drop across the diode will be 0.7 V assuming Si based diode.



Therefore the voltage v is calculated as:

v = 3V - 0.7 V

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$= 2.3\ V$

Similarly the current i can be calculated as:

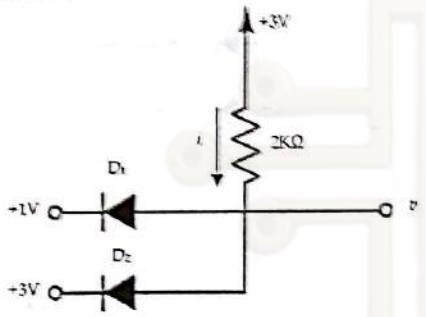
$$i = \frac{v - (-3)}{2K} = \frac{2.3 + 3}{2} = \frac{5.3}{2} = 2.65\ mA$$

Note:

- In fact the above circuit is diode OR gate. Here it is evident that when one of the inputs is high, the output is high. The output will be low only when both inputs are low.

Ex.2.

Find the voltage v and i of the the following circuit. Use constant voltage drop model of diode.



Solution:

In this case the diode D_2 is reversed biased and diode D_1 is forward biased. Using constant voltage drop model of diode, we can say that when forward biased the drop in diode D_1 is 0.7 V. Thus the voltage v will be calculated as:

$$v = 1 + 0.7 = 1.7\ V$$

Thus the current i is given as:

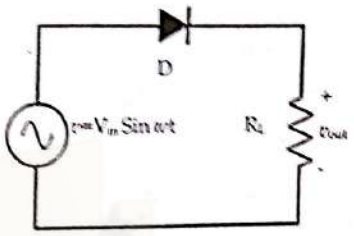
$$i = \frac{3 - v}{2K} = \frac{3 - 1.7}{2} = \frac{1.3}{2} = 0.65mA$$

Note:

In fact the above circuit is of the diode AND gate. Here it is verified that when one of the two inputs is low the output becomes low. Similarly we can easily observe that the output of the AND gate will be high only when both the inputs are high.

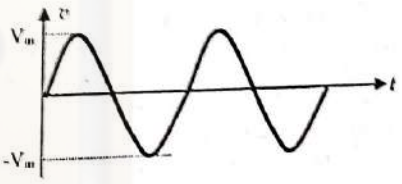
Ex.3.

Find the output waveform v_{out} of the following circuit assuming diode to be ideal



Solution:

The input waveform is:



Since we are using the ideal diode model, the diode will act as a short circuit when the input is positive and will act as open circuit when the input is negative. Thus the output waveform will be:

Operation in the reverse breakdown region - Zener diodes

We briefly studied about the operation of junction diode in the breakdown region in the earlier section of this chapter. As mentioned

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earlier, in breakdown region, the diode conducts large amount of current with the corresponding change in voltage being very small. Thus utilizing breakdown region of operation diode can be used in voltage regulation.

But it should be noted that ordinary diodes are not designed to operate in breakdown region. Thus we cannot use ordinary diode in voltage regulation. There is a special diode so called Zener diode which is designed to operate in the breakdown region. Zener diode can be used in voltage regulator circuits so as to regulate the voltage.

Zener Diodes:

Zener diode is a special purpose diode which is specially designed to operate in the breakdown region. Unlike other ordinary PN junction diodes, the Zener diode can operate in the breakdown region without any damage to the junction. For this Zener diode is heavily doped. In the forward biased condition, its characteristics are same as that of ordinary PN junction diodes. It is always connected in reverse bias mode in the circuit. Due to the very steep I-V characteristics in the breakdown region (ideally a straight vertical line) the Zener voltage remains constant over the large change in current. This property of Zener diode is utilized in the design of voltage regulator circuits so as to get stable output voltage. The operation of zener diode as a voltage regulator will be studied in the following section. The voltage at which Zener Breakdown occurs is known as reverse breakdown voltage or in specific Zener Breakdown voltage. It is designated as V_Z.The symbol of the Zener diode is:

Figure :Symbol of Zener Diode

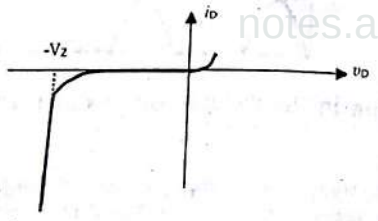

Figure: Characteristics Curve of Zener Diode

Zener diode as a Voltage regulator

In many cases it is desired to have a stable dc output. For this voltage regulators are used. Due to its property of having the constant voltage in breakdown region, the zener diode can be used as a voltage regulator. The circuit for the voltage regulator using the zener diode is as shown below.

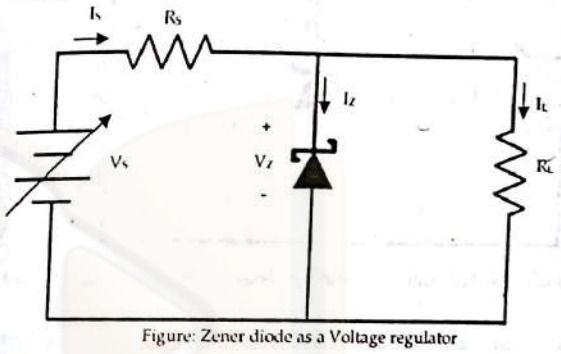

Figure: Zener diode as a Voltage regulator

In the figure, the source V_S is a variable source, R_S is source resistance and the output is taken across the load resistor.

Circuit operation

When the input voltage is less than V_Z, the Zener diode does not conduct so all the current flows through R_L and output is taken across it. When the input voltage exceeds V_Z,the zener diode comes into operation and current I_Z flows through the zener diode maintaining the output voltage constant. This is because zener diode is so doped that whenever I_Z increases, the zener resistance R_Z decreases but V_Z remains constant. Similarly, when I_Z decreases the zener resistance R_Z increases but V_Z remains constant. Since the load resistor is connected in parallel with zener diode, the voltage across it also remains constant. Thus, when input voltage is low, all the current is transferred to load and when the input voltage is high, the excess current is bypassed by the zener diode. Hence, output across R_L is stable and constant. In this way, zener diode acts as a voltage regulator.

Applying KCL to above circuit,

$$I_S = I_L + I_Z \quad \text{.....(i)}$$

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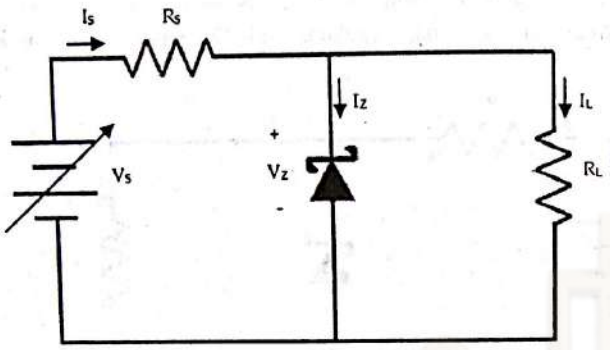
Applying KVL to above circuit,

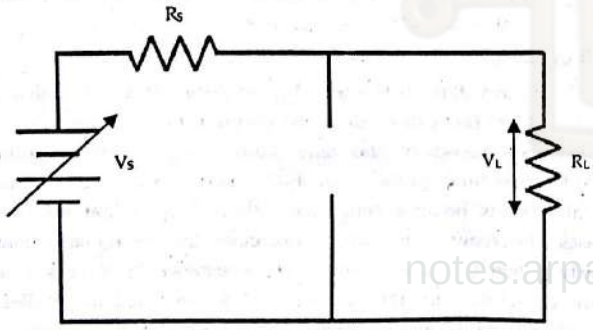
$$V_S = I_S R_S + V_Z \quad \text{.....(ii)}$$

Steps for solving circuit containing Zener diode



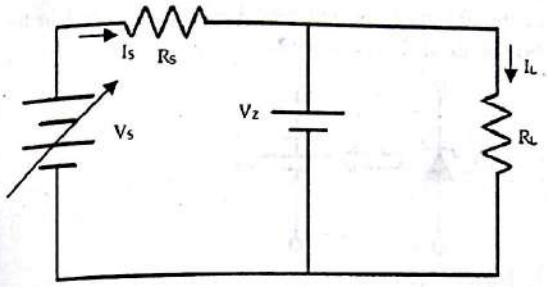
In order to solve above circuit containing zener diode following steps are followed:

i. Open circuit the zener diode and find the voltage V_L across load resistor.



ii. If $V_L > V_Z$ then zener diode conducts current. And it will act as a voltage regulator. Otherwise it will not.

iii. Replace zener diode by a voltage source equal to V_Z (Assuming point ii is satisfied)



iv. Current through load resistor is calculated as: $I_L = \frac{V_Z}{R_L}$

v. Current I_S is calculated as: $I_S = \frac{V_S - V_Z}{R_S}$

vi. Then the zener current is calculated as: $I_Z = I_S - I_L$

vii. The maximum current that can flow through the zener diode is determined by maximum power rating of the zener diode. It is given by : $I_{Z\,max} = \frac{P_{Z\,max}}{V_Z}$

Zener Diode Modeling

Zener diode is always connected in the circuit in reverse bias mode because of its capability to handle significantly large current in the breakdown region. So zener diodes are modelled based on their operation at breakdown region. The two zener diode models are ideal zener diode model and practical zener diode model.

Ideal Zener diode model:

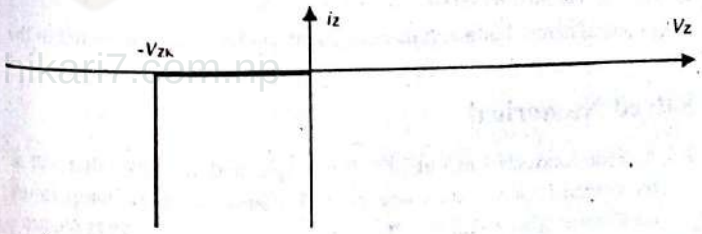


Figure: Characteristics Curve of ideal Zener Diode

Figure shows the i-v characteristics of the ideal zener diode. In the breakdown region, ideal zener diode exhibits a constant voltage.

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The ideal zener diode and its equivalent model when connected in the circuit in reverse bias mode are as shown below:

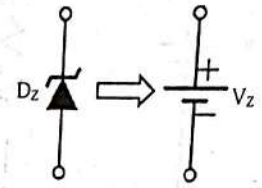


Figure : Ideal zener diode and its equivalent circuit when connected in the circuit in reverse bias mode

Practical Zener diode model:

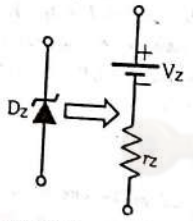


Figure: Practical zener diode and its equivalent model when connected in the circuit in reverse bias mode

The characteristics curve of practical zener diode was presented in the earlier section. Practical zener diode exhibits certain slope in the breakdown region of the characteristic curve. To model this one has to include the incremental resistance that it offers.

The practical zener diode and its equivalent model when connected in the circuit in reverse bias mode is shown in the figure.

Solved Numerical

1. A diode conducts 1 mA at 20°C. If it is operated at 100°C, what will be its current? Given data are η = 1.6 and negative temperature coefficient value = -2.2 mV/°C. **2071 Chaitra**

Solution

Let I_{D1} be the current through the diode at 20°C and I_{D2} be the current through the diode at 100°C. Let V_{D1} and V_{D2} be the corresponding voltage across the diode at 20°C and 100°C respectively.

By question,

$I_{D1} = 1mA$

$\eta = 1.6$

Negative temperature coefficient = -2.2 mV/°C

$I_{D2} = ?$

We know, for a diode,

$$V_{D2} - V_{D1} = 2.3\,\eta V_T \log\left(\frac{I_{D2}}{I_{D1}}\right) \ldots\ldots\ldots(i)$$

In this question the value of negative temperature coefficient is given. So we will not consider the effects of temperature on the saturation current. In this case, it is assumed that the temperature coefficient solely represents the effect of temperature on diode characteristics.

We may write,

$V_{D2} = V_{D1}$ + temperature coefficient × change in temperature

or, $V_{D2} = V_{D1} - 2.2\times10^{-3}\times(100-20)$

or, $V_{D2} = V_{D1} - 0.176V$

$V_{D2} - V_{D1} = -0.176V$

Assuming V_T = 25 mV = 0.025V, from equation (i),

$$\text{or, } -0.176 = 2.3\times1.6\times0.025\times\log\left(\frac{I_{D2}}{1}\right)$$

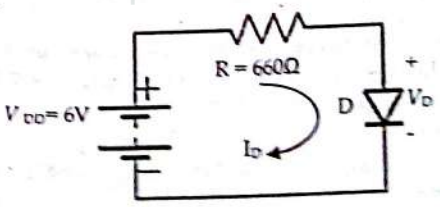
or, $\log(I_{D2}) = -1.19$

or, $I_{D2} = 10^{-1.19}$

or, $I_{D2} = 1.23\times10^{-2}$

$\therefore I_{D2} = 1.23\times10^{-2}$ mA

2. In the given circuit the diode used has its η= 1.74 and it conducts 1 mA at forward bias voltage of 0.7 V. Find the current flow in the circuit. **2070 Ashad**



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Solution

Given: $\eta = 1.74$ and $R = 660\Omega = 0.66\ k\Omega$

Let us assume $V_T = 25\ mV = 0.025\ V$,

$$2.3\eta V_T = 2.3 \times 1.74 \times 0.025$$

$$\therefore 2.3\eta V_T \approx 0.1V \quad ...(i)$$

Question does not specify us to use any specific model. Therefore, we shall use exponential diode model. In order to solve above circuit we need to solve exponential relation (from diode equation) and the linear equation (obtained by applying KVL in the circuit).

One of the simplest methods to find solution of non linear and linear equation is iterative analysis. We shall therefore use iterative procedure to solve above circuit.

The question says current through the diode is 1mA when voltage across it is 0.7 V. We shall use these values as a reference to start the iteration.

First Iteration:

From the circuit,

$$I_D = \frac{V_{DD} - V_D}{R} \quad .....(i)$$

$$\therefore I_D = \frac{6 - 0.7}{0.66} = 8.03\ mA$$

From the diode equation,

$$V_{D2} = V_{D1} + 2.3\,\eta V_T \log\left(\frac{I_{D2}}{I_{D1}}\right)$$

Substituting $I_{D2} = 8.03$ mA, $V_{D1} = 0.7$ V and $I_{D1} = 1$ mA, we get,

$$\text{or, } V_{D2} = 0.7 + 0.1 \log\left(\frac{8.03}{1}\right) \quad .....(ii)$$

$$\therefore V_{D2} = 0.7904$$

Thus the result of first iteration is $V_D = 0.7904$ V and $I_D = 8.03$ mA

Second Iteration:

Substituting $V_D = 0.7904$ in equation (i), we get $I_D = 7.8933$ mA

Again substituting $V_{D1} = 0.7904$, $I_{D1} = 8.03$ mA, $I_{D2} = 7.8933$ mA in equation (ii), we get,

$V_{D2} = 0.78965$

Thus the result of this iteration is $V_D = 0.789654$V and $I_D = 7.8933$ mA

Third Iteration:

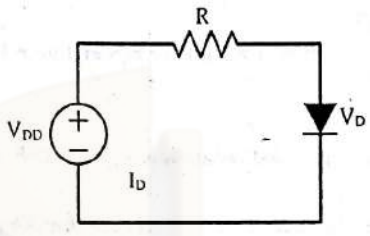
Following similar procedure, the result of this iteration is

$V_D = 0.78966$ and $I_D = 7.8945$ mA

Since the results of iterations second and third are not so much different we stop iteration here.

Hence, for the given circuit, $v_D = 0.78966$ V and $I_D = 7.8945$ mA

3. Determine the current I_D and the diode voltage V_D for the circuit as shown with $V_{DD} = 5V$ and $R = 1k\Omega$. Assume that the diode has a current of 1 mA at a voltage of 0.7 V and its voltage drop changes by 0.1 V for every decade change in current.



Solution:

This question is similar to question number 3 with slight change in language to give the details and values of source and resistance.

Thus it is evident that we should use iterative procedure to analyze above circuit.

The question says that current through the diode is 1 mA when voltage across it is 0.7 V. This is a reference for performing iterative analysis. It also says that voltage drop changes by 0.1 V for every decade change in current. This is an indirect statement for

$$2.3\eta V_T \approx 0.1V$$

First Iteration:

From the circuit, using KVL,

$$I_D = \frac{V_{DD} - V_D}{R} \quad ....(i)$$

$$I_D = \frac{5 - 0.7}{1} = 4.3\ mA$$

From diode equation we may write,

$$V_{D2} = V_{D1} + 2.3\,\eta V_T \log\left(\frac{I_{D2}}{I_{D1}}\right)$$

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$$V_{D2} = V_{D1} + 0.1 \log\left(\frac{I_{D2}}{I_{D1}}\right) \quad \text{.....(ii)}$$

Substituting $V_{D1} = 0.7$ V, $I_{D1} = 1$ mA and $I_{D2} = 4.3$ mA in equation (ii), we get,

$V_{D2} = 0.7633$ V

Thus the result of first iteration is $V_D = 0.7633$ V and $I_D = 4.3$ mA

Second Iteration:

Substituting $V_D = 0.7633$ in equation (i), we get,

$I_D = 4.2367$ mA

Substituting $V_{D1} = 0.7633$ V, $I_{D1} = 4.3$ mA and $I_{D2} = 4.2367$ in equation (ii), we get,

$V_{D2} = 0.76265$ V

Thus the result of second iteration is $V_D = 0.76265$ V and $I_D = 4.2367$ mA.

Since the values obtained in this iteration are not so different from that obtained in first iteration, therefore, we stop our iteration here.

Thus, the current through the diode and voltage across it are as follows:

$I_D = 4.2367$ mA and $V_D = 0.76265$ V

4. Find the value of the dynamic resistance if voltage in the diode is 650 mV and I_{RS} is 10 pA. ($= 10 \times 10^{-12}$A). Given $\eta = 2$ and $V_T = 25$ mV.
2070 Chaitra

Solution

$v_D = 650$ mV $= 0.650$ V

$I_{RS} = 10$ pA $= 10 \times 10^{-12}$ A

$\eta = 2$

$V_T = 25$ mV $= 0.025$ V

Therefore current through the diode is given as:

$$I_D = I_{RS} e^{\frac{v_D}{\eta V_T}}$$

$$\text{or, } I_D = 10 \times 10^{-12} e^{\frac{0.650}{2 \times 0.025}}$$

$$\therefore I_D = 4.42 \times 10^{-6} \text{ A} = 4.42 \ \mu\text{A}$$

The dynamic resistance of diode r_d is given as:

$$r_d = \frac{\eta V_T}{I_D} = \frac{2 \times 0.025}{4.42 \times 10^{-6}} = 11312.2\ \Omega = 11.3\ \text{k}\Omega$$

5. A diode conducts 1 mA at 20°C. If it is operated at 100°C, what will be its current? Given data are $\eta = 1.8$ and negative temperature coefficient value = -1.8 mV/°C. **2068 Chaitra**

Solution:

Let I_{D1} be the current through the diode at 20°C and I_{D2} be the current through the diode at 100°C. Let V_{D1} and V_{D2} be the corresponding voltage across the diode at 20°C and 100°C respectively.

By question,

$I_{D1} = 1$mA

$\eta = 1.8$

Negative temperature coefficient = -1.8 mV/°C

$I_{D2} = ?$

We know, for a diode,

$$V_{D2} - V_{D1} = 2.3\, \eta V_T \log\left(\frac{I_{D2}}{I_{D1}}\right) \text{.....(i)}$$

In this question the value of negative temperature coefficient is given. So we will not consider the effects of temperature on the saturation current. In this case, it is assumed that the temperature coefficient solely represents the effect of temperature on diode characteristics.

We may write,

$V_{D2} = V_{D1}$ + temperature coefficient × change in temperature

or, $V_{D2} = V_{D1} - 1.8 \times 10^{-3} \times (100-20)$

or, $V_{D2} = V_{D1} - 0.144$V

or, $V_{D2} - V_{D1} = -0.144$V

Assuming $V_T = 25$ mV $= 0.025$V, from equation (i),

$$\text{or, } -0.144 = 2.3 \times 1.8 \times 0.025 \times \log\left(\frac{I_{D2}}{1}\right)$$

$$\text{or, } \log(I_{D2}) = -1.39$$

$$\text{or, } I_{D2} = 10^{-1.39}$$

$$\text{or, } I_{D2} = 4.07 \times 10^{-2}$$

$$\therefore I_{D2} = 4.07 \times 10^{-2} \text{ mA}$$

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6. Prove that $V_{D2} - V_{D1} = 2.3\eta V_T \log(\frac{I_{D2}}{I_{D1}})$

Solution

Let I_{D1} be the current through the diode when voltage drop across it is V_{D1} and I_{D2} be the current through the diode when voltage drop across it is V_{D2}

Assuming constant temperature, we may write diode equation as:

$$I_{D1} = I_S e^{\frac{V_{D1}}{\eta V_T}} \quad \text{.....(i)}$$

$$I_{D2} = I_S e^{\frac{V_{D2}}{\eta V_T}} \quad \text{.....(ii)}$$

Dividing equation (ii) by equation (i)

$$\frac{I_{D2}}{I_{D1}} = e^{\frac{V_{D2} - V_{D1}}{\eta V_T}}$$

Taking ln on both side of above equation we get,

$$\ln(\frac{I_{D2}}{I_{D1}}) = \frac{V_{D2} - V_{D1}}{\eta V_T}$$

$$\text{or, } V_{D2} = V_{D1} + \eta V_T \ln(\frac{I_{D2}}{I_{D1}})$$

$$\therefore V_{D2} - V_{D1} = 2.3\,\eta V_T \log(\frac{I_{D2}}{I_{D1}})$$

7. Prove that diode voltage changes by $2.3\eta V_T$ when diode current changes by 10 times.

Solution

We know,

$$V_{D2} - V_{D1} = 2.3\,\eta V_T \log(\frac{I_{D2}}{I_{D1}})$$

If $I_{D2} = 10\, I_{D1}$, then

From above equation, we get,

$V_{D2} - V_{D1} = 2.3\,\eta V_T \log(10))$

$\therefore \Delta V_D = 2.3\,\eta V_T$

Thus, if the current through diode changes by 10 times, the diode voltage changes by $2.3\,\eta V_T$.

8. In the diode circuit η=1, V_{D1}=0.7 V, I_{D1}=1mA. Calculate V_{D2} when
   i. I_{D2} = 0.1mA ii. I_{D2} = 10mA

Solution

We know,

$$V_{D2} = V_{D1} + \eta V_T \ln(\frac{I_{D2}}{I_{D1}})$$

For V_T=0.025V,

$$V_{D2} = V_{D1} + 1 \times 0.025 \times \ln(\frac{I_{D2}}{I_{D1}})$$

$$\therefore V_{D2} = 0.7 + 1 \times 0.025 \times \ln(\frac{I_{D2}}{1})$$

If I_{D2} = 0.1 mA, then from above equation,

V_{D2} = 0.643 V

If I_{D2} = 10 mA, then from above equation,

V_{D2} = 0.758 V

9. A si diode has the saturation current of 0.1pA at 20°C.Find its current when it is forward biased by 0.55 volt. Find the current in the same diode when the temperature rises to 100°C.

Solution

20°C,

=0.1 pA

$\therefore I_{S1}$=0.1*10^{-12}A

V_D = 0.55 V

$$V_T = \frac{T}{q} = \frac{1.38 \times (273+20)}{1.6\times10^{-19}} = 0.02527\ V$$

We know, η=1 then,

$$I_{D1} = I_{S1} e^{\frac{V_D}{\eta V_T}}$$

$$\text{or, } I_{D1} = 0.1 \times 10^{-12} e^{\frac{0.55}{1\times0.02527}} = 2.83\times10^{-4}\,A = 0.283\ mA$$

At 100°C,

$I_{S2} = 2^8 \times I_{S1}$,

Since, for 10°C rise in temperature saturation current doubles.

$\therefore$ for 20°C to 100°C i.e. 80°C change in temperature, saturation current doubles 8 times.

$$V_T = \frac{KT}{q} = \frac{1.38\times10^{-23}\times373}{1.6\times10^{-19}} = 0.032\ V$$

$$\therefore I_{D2} = I_{S2} e^{\frac{V_D}{\eta V_T}}$$

$$I_{D2} = 2^8 I_{S1} e^{\frac{0.55}{1\times0.032}}$$

$$= 2^8 \times 0.1 \times 10^{-12} \times e^{\frac{0.55}{1\times0.032}} = 7.45\times10^{-4}\,A = 0.745\ mA$$

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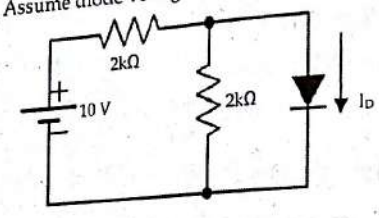
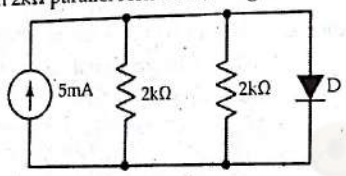
10. Find the current through the diode using constant voltage drop model. Assume diode voltage drop to be 0.7V



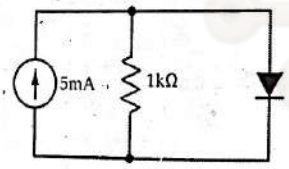
Solution

Using the source conversion technique,

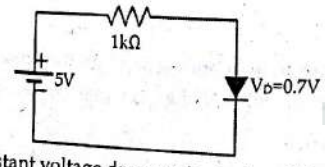
Converting 10V source with 2kΩ series resistance to 5mA current source with 2kΩ parallel resistance, we get



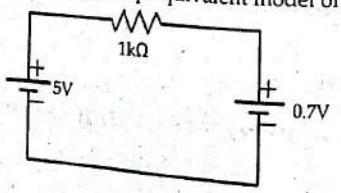
The above circuit is equivalent to:



Again converting 5mA current source with 1kΩ parallel resistance into 5V voltage source with 1 kΩ series resistance



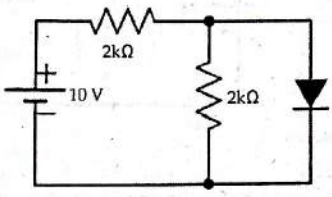
Using Constant voltage drop equivalent model of diode:



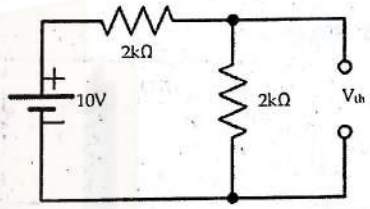
$$I_D = \frac{5-0.7}{1} = 4.3 \text{ mA}$$

OR

Using Thevenin's Theorem

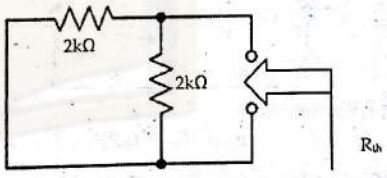


Finding V_{oc} or V_{TH} :



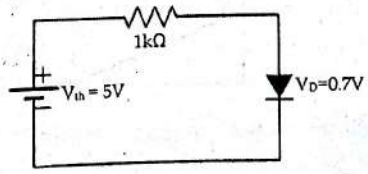
$$\therefore V_{th} = 10 \times \frac{2}{4} = 5\ V$$

Finding R_{th}:



$$\therefore R_{th} = \frac{2 \times 2}{2+2} = 1\text{kΩ}$$

Therefore we can draw following circuit,



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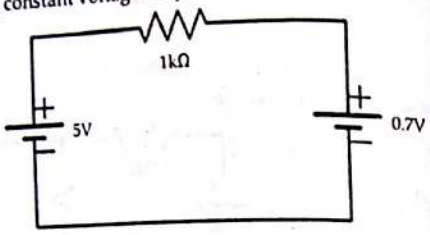
Incorporating constant voltage drop model:



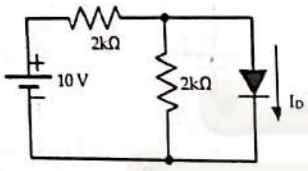
∴ Diode current, $I_D = \frac{5-0.7}{1}$

∴ $I_D = 4.3mA$

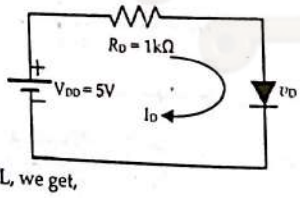
11. Draw the load line for the circuit given in question no 10.

Solution

The given circuit is:



Using repetitive source conversion



Applying KVL, we get,

$$v_D = V_{DD} - I_D R_D$$

When $I_D=0$, $v_D = V_{DD} = 5V$

When $v_D=0$, $I_D = \frac{V_{DD}}{R_D} = \frac{5}{1} = 5\ mA$

So the load line is:

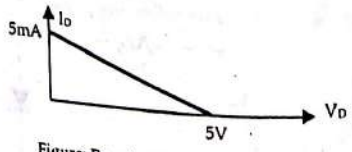
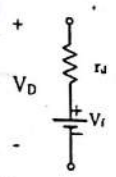


Figure: Required load line

12. Find the value of small signal resistance, V_D and V_f of the given figure:



($I_S=10^{-12}A, \eta=1.6, I_D=1mA$)

Solution

$$r_d = \frac{\eta V_T}{I_D} = \frac{1.6\times0.025}{10^{-3}} = 40\Omega$$

We know,

$I_D = I_S\, e^{\frac{V_D}{\eta V_T}}$

∴ $V_D = \ln\left(\frac{I_D}{I_S}\right) \times \eta V_T$

∴ $V_D = \ln\left(\frac{10^{-3}}{10^{-12}}\right) \times 1.6 \times 0.025 = 0.82V$

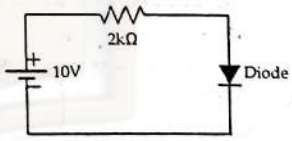
Again,

$$V_D = I_D \times r_D + V_f$$

or, $V_f = V_D - I_D \times r_D$

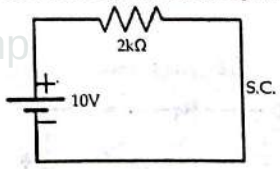
or, $V_f = 0.82 - 10^{-3} \times 40 = 0.78V$

13. Use the ideal diode model and piecewise linear model to find the current through the diode (for piecewise linear model $r_d=0.5\Omega$)



Solution

Using ideal diode model the equivalent the equivalent circuit is:



∴ $I_D = \frac{10}{2} = 5mA$

Using piecewise linear model, the equivalent circuit is:

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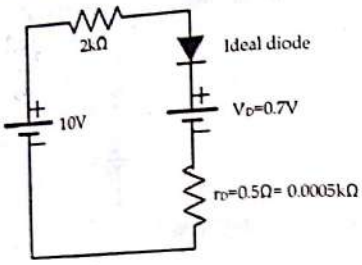
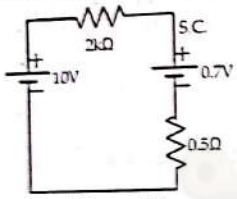
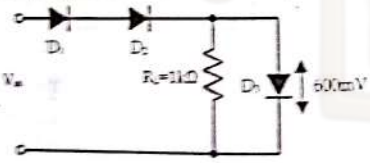


The equivalent of this circuit is:



$$\therefore I_D = \frac{10-0.7}{2+0.0005} = \frac{9.3}{2.0005} = 4.64\ mA$$

14. The circuit shown below uses three identical diodes D_1, D_2, and D_3.For each diode assume reverse saturation current is 10^{-13}A at 300K and η=1.Calculate V_{in} to get 600mV across diode D_3



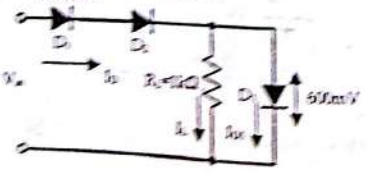
Solution

T=300K

η=1

$I_S = 10^{-13}A$

$V_T = \frac{kT}{q} = \frac{1.38\times10^{-23}\times300}{1.6\times10^{-19}} = 0.025877V$



$$I_{D3} = I_S e^{\frac{V_{D3}}{\eta V_T}}$$

$$or, I_{D3} = 10^{-13} e^{\frac{600\times10^{-3}}{1\times0.02587}}$$

$$or, I_{D3} = 1.18\times10^{-3}A$$

Again,

$$I_L = \frac{600mV}{1K\Omega} = 0.6\ mA$$

$$\therefore I_D = I_{D3} + I_L = 1.18+0.6 = 1.78mA$$

For D_1,

$$I_D = I_S e^{\frac{V_{D1}}{\eta V_T}} = 10^{-13} e^{\frac{V_{D1}}{1\times0.02587}} = 1.78mA$$

$$\therefore V_{D1} = \ln\left(\frac{1.78\times10^{-3}}{10^{-13}}\right)\times0.02587 = 0.6V$$

The diodes D_1 and D_2 are identical and current through them is also same.

$\therefore V_{D2}$=0.6V

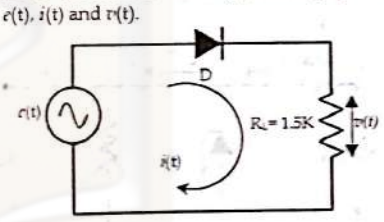
Also,

V_{D3}=600mV=0.6V

Applying KVL for the above circuit, we get,

$V_{in} = V_{D1} + V_{D2} + V_{D3}$ =0.6+0.6+0.6 =1.8V

15. For the circuit shown below find the peak value of current $i(t)$ and voltage $v(t)$ across the resistor when $e(t)=20\sin(\omega t)$. Also draw the waveform of $e(t)$, $i(t)$ and $v(t)$.



Solution

Peak voltage across resistor = 20-0.7 = 19.3V

$\therefore$ Peak current through resistor = $\frac{19.3V}{1.5K} = 12.87mA$

$\therefore$ $V_R(t)$=19.3 sin ωt

$\therefore$ i(t)=12.87 sin ωt

The waveforms for $e(t)$,V(t) and $i(t)$ are shown in the figure below.

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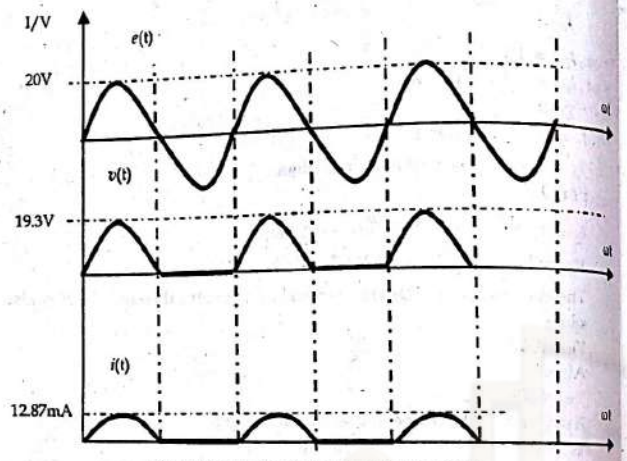
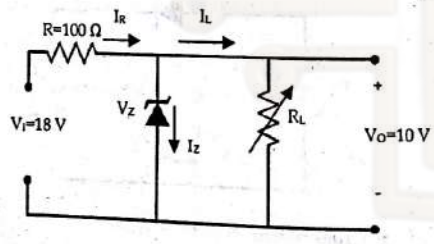


Figure: Output waveforms for e(t), V(t) and i(t)

16. Find the Zener current in the given circuit when R_L=1.2 KΩ. Assume V_Z = 10V.



Solution

If we open zener diode across its terminals and find voltage drop across load resistor R_L then we will get.

V_L (with zener open) > V_Z .

Thus, it is safe to assume that zener diode is operating in break down region.

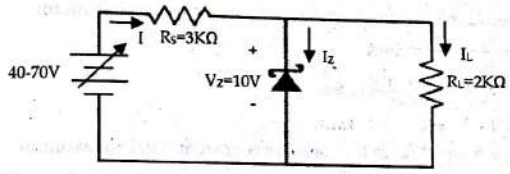
$V_O = V_Z = 10\,V$ *(Because zener diode is parallel to load resistor)*

$I_L = \frac{V_O}{R_L} = \frac{10V}{1.2\,K\Omega} = 8.3333\,mA$

$\therefore I_R = \frac{V_I - V_Z}{R} = \frac{(18-10)V}{100\,\Omega} = 80\,mA$

$\therefore I_Z = I_R - I_L = 80 - 8.3333 = 71.67\,mA$

17. Calculate the total current I, load current I_L and the Zener current I_Z in the circuit shown below. How will these values be affected if the source voltage be increased to 70V?



Solution

If we open zener diode across its terminals and find voltage drop across load resistor R_L for input voltage range then we will get,

V_L (with zener open) > V_Z

Thus, it is safe to assume that zener diode is operating in break down region.

When V_{in}= 40V,

$I_L = \frac{V_Z}{R_L} = \frac{10V}{2K\Omega} = 5mA$

$\therefore I = \frac{V_{in} - V_Z}{R_S} = \frac{(40-10)V}{3K\Omega} = 10mA$

$I_Z = I - I_L = 10 - 5 = 5mA$

When V_{in}= 70V,

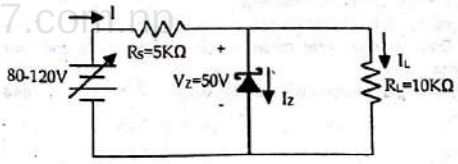
$I_L = \frac{V_Z}{R_L} = \frac{10V}{2K\Omega} = 5mA$

$\therefore I = \frac{V_{in} - V_Z}{R_S} = \frac{(70-10)V}{3K\Omega} = 20mA$

$I_Z = I - I_L = 20 - 5 = 15mA$

Thus, when input voltage varies from *40-70* V the zener current varies from *5 mA* to *15 mA*

18. Find the maximum and minimum current through the Zener diode from the figure shown below.



Solution

If we open zener diode across its terminals and find voltage drop across load resistor R_L for input voltage range then we will get.

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V_L (with zener open) > V_Z

Thus, it is safe to assume that zener diode is operating in break down region.

When V_{in}=80V, in this case zener current will be **minimum**.

$$I_L = \frac{V_Z}{R_L} = \frac{50V}{10K} = 5mA$$

$$\therefore I = \frac{V_{in}-V_Z}{R_S} = \frac{(80-50)V}{5K\Omega} = 6mA$$

$$I_Z = I - I_L = 6 - 5 = 1mA$$

When V_{in}= 120V, in this case zener current will be **maximum**.

$$I_L = \frac{V_Z}{R_L} = \frac{50V}{10K\Omega} = 5mA$$

$$\therefore I = \frac{V_{in}-V_Z}{R_S} = \frac{(120-50)V}{5K\Omega} = 14mA$$

$$I_Z = I - I_L = 14 - 5 = 9mA$$

Hence, $I_{Z\,max} = 9\,mA$ and $I_{Z\,min} = 1\,mA$

EXERCISE

1. Draw graphs of IV characteristics of ordinary pn junction diode and zener diode. Draw ac equivalent model for pn junction and derive its dynamic resistance. 2071 Chaitra
2. Describe with the help of load line and IV characteristics of the diode, a simple circuit that uses pn junction diode in forward biased state. 2071 Shrawan
3. What is pn junction diode? Explain the large signal models of pn junction diodes. 2070 Chaitra
4. What is the significance of operating point in the diode? Draw the dc load line with the help of diode circuit and its characteristics curve. 2070 Ashad
5. Explain the small signal model of pn junction diode and derive the expression for ac or dynamic resistance 2069 Chaitra
6. Explain the large signal models of pn junction diodes. 2068 Chaitra
7. Define and explain the reverse breakdown effects in diodes. 2068 Chaitra
8. Define modeling. Explain the small signal model of a semiconductor diode with necessary figure and derivation. 2068 Shrawan
9. Draw graphs of IV characteristics of ordinary pn junction diode and zener diode. Draw ac equivalent model for pn junction diode and derive its ac resistance. 2068 Baisakh
10. Define and explain reverse breakdown effect. 2068 Baisakh

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Chapter 2

The Bipolar Junction Transistors

Introduction

After having studied the junction diode which is a two terminal non linear device we are now about to study the three terminal non linear semiconductor devices. Three terminal semiconductor devices are far more useful than two terminal ones because they can be used for multitude of operation.

Transistors are basically of two types namely Bipolar Junction Transistors and Field Effect Transistors. In Bipolar Junction Transistor both holes and electrons take part in the conduction process and hence the name bipolar. In Field Effect Transistor either only holes or only electrons are responsible for the conduction process. So, Field Effect Transistors are also called **unipolar** transistors. It should be noted that when we say transistor only we automatically mean bipolar junction transistor. Field effect transistors are of two types namely **Junction Field Effect Transistor** (JFET) and **Metal Oxide Semiconductor Field Effect Transistor** (MOSFET). JFET and MOSFET will be studied in the next chapter.

In transistor and FETs, the basic principle is to apply current or voltage between two terminals to control the flow of current through the third terminal. Thus, transistor and FETs, act as controlled sources.

The Bipolar Junction Transistor (BJT)

When a third doped material is suitably connected to a PN junction diode, the resulting device is known as transistor. Or the semiconductor device which is formed by sandwiching either p-type material by n-type materials or n-type material by p-type materials is called transistor. Thus, transistors are of two types viz. npn transistor and pnp transistor.

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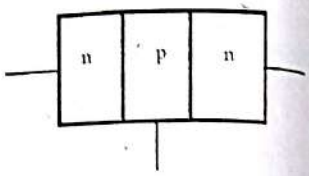
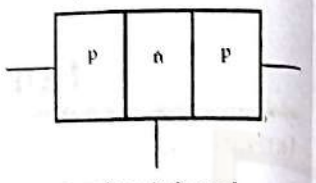
i. **npn transistor:** When a third doped material is suitably connected to a PN junction diode or crystal diode in such a way that p-type material is sandwiched by n-type materials then npn transistor is formed.

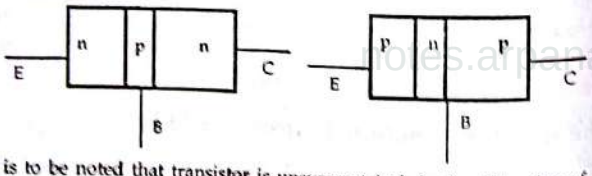


ii. **pnp transistor:** When a third doped material is suitably connected to a PN junction diode or crystal diode in such a way that n-type material is sandwiched by p-type materials then pnp transistor is formed.



In a transistor there are three basic parts. They are:

1. **Emitter (E):** The heavily doped region of a transistor which supplies charge carriers is known as emitter. For npn transistor the emitter is of n-type and for pnp transistor emitter is of p-type. Thus, for npn transistor majority charge carriers are electrons and for pnp transistor majority charge carriers are holes.
2. **Base (B):** The lightly doped region or the middle region of the transistor is known as base. For npn transistor, the base is of p-type and for pnp transistor the base is of n-type.
3. **Collector (C):** The moderately doped region of the transistor is called as collector. It lies on the other side of the emitter. For npn transistor collector is of n-type and for pnp transistor the collector is of p-type. The function of collector is to collect the charge carriers supplied by the emitter.



It is to be noted that transistor is unsymmetrical device. The size of the collector is greater than the size of the emitter and the size of the base is least. Transistor works in the following modes:

i. **Active Mode:** When the emitter base junction is forward biased and the collector base junction is reversed biased the transistor is said to be in the active mode. In this mode the transistor works as an amplifier.

ii. **Saturation Mode:** When the emitter base junction and the collector base junction are both forward biased the transistor is said to be in saturation mode.

iii. **Cutoff Mode:** When emitter base junction and collector base junction are both reversed biased the transistor is said to be in cutoff mode. The saturation and cutoff modes of transistor are used to make the transistor work as a switch.

iv. **Inverse active mode:** When emitter base junction is reversed biased and the collector base junction is forward biased the transistor is said to be in inverse active mode. Though there are very few applications of this mode of operation, it is very important. In this mode the transistor operate with the role of emitter and collector interchanged. One important use of this mode can be found in the TTL circuits where the advantage of this transistor action is taken to remove the excess stored base charge during saturation to cut- off transition of another transistor which ultimately increases switching speed.

Operation of the npn transistor in the active mode

The figure shows the working of npn transistor. When emitter base junction and collector base junction are biased for active mode i.e. when the emitter base junction is forward biased and the collector base junction is reversed biased, the npn transistor starts to operate. Two external batteries are used to establish the required bias.

Due to the forward bias in the emitter base junction EBJ, electrons in emitter are repelled by the negative terminal of the battery and injected towards the EBJ. Likewise, holes in base are repelled by the positive terminal of the battery and are injected towards the EBJ. These injected holes and electrons give rise to the emitter current. The conventional direction of this emitter current, i_E is out of the emitter lead E as shown in the figure. Since the emitter is heavily doped as compared to the base, the emitter current in the npn transistor will mostly be dominated by the electron component. Thus, in npn transistor, electrons are known as majority carriers and holes are known as minority carriers.

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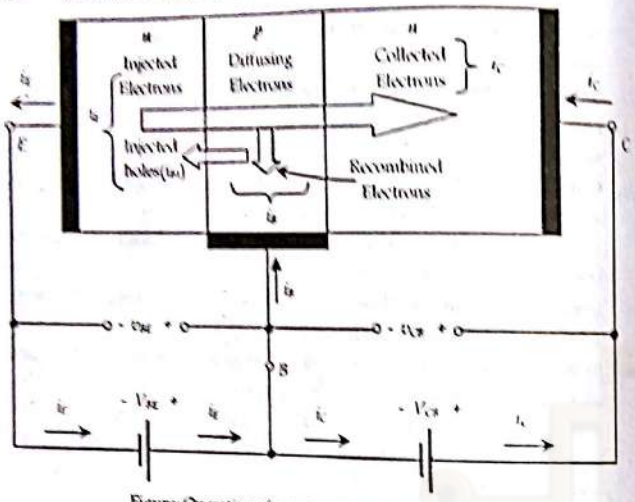

Figure: Operation of npn transistor in the active mode.

Large numbers of electrons diffuse into the base region crossing the EBJ. These diffusing current recombine with some of the holes present in the base. It gives rise to the recombination current i_{B2}. The current due to the holes injected into EBJ from the base, i_{B1}, and recombination current i_{B2} due to the recombination of diffusing electrons in the base add up to form the base current i_B which flows into the base through the Base lead B.

The holes in the base can only recombine with few electrons injected from emitter because of small size of the base and relatively low doping level. Thus large number of electrons moves towards the collector base junction CBJ. Since the collector base junction is reversed biased the electrons reaching near the CBJ are swept away by the reverse bias voltage into the collector. In other words, electrons near the CBJ are repelled by the negative terminal of the battery and attracted by the positive terminal of the battery. Thus large numbers of the electrons get collected in the collector. These current gives rise to the collector current, i_C, whose conventional direction is into the collector through lead C.

Applying KCL we get,

$$i_E = i_B + i_C$$

Thus, the emitter current is the sum of the base current and the collector current.

Note:

For the sake of simplicity, the depletion region and the bound charge carriers are not shown in the figure. Also here in the discussion we have ignored the effects of the minority carriers drift current.

Operation of pnp transistor in active mode

We can easily explain the working of pnp transistor with the similar approach that we made for npn transistor. For pnp transistor conduction is done mainly by holes. So holes are the majority charge carriers.

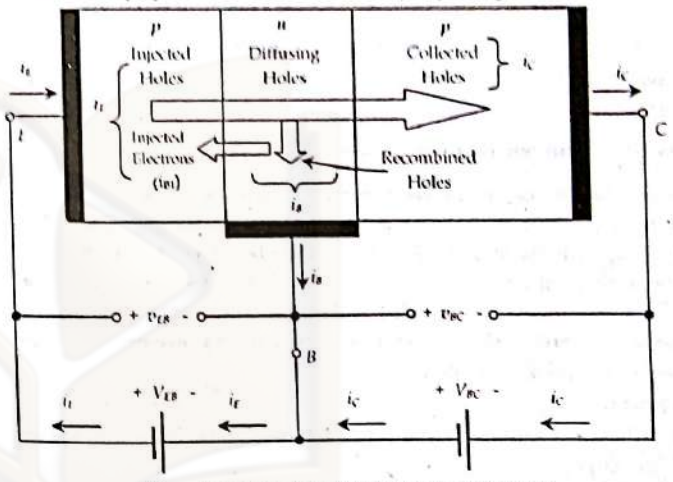

Figure: Operation of pnp transistor in the active mode.

Note:

However, the conduction inside the pnp transistor is done mainly by holes but in the external circuit the conduction is still by the electrons.

The direction for the emitter current, base current and the collector current in the conventional direction is as shown in the figure.

Applying KCL, we get

$$i_E = i_B + i_C$$

Symbols for pnp and npn transistor

While making transistor circuits it is a tedious job to draw the actual structure of the transistor. So, we use the symbols to represent the pnp and npn transistor.

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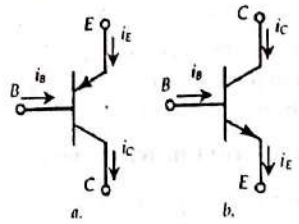

Figure:

a. Symbol of pnp transistor(pnp= pointing in point)
b. Symbol of npn transistor(npn=not pointing in)

Note that the arrow in the symbol suggests the direction of the emitter current. The transistor currents are labeled as per convention.

Symbol Conventions for voltage and current

Let us consider an npn transistor connected as shown in the figure. Here in the input side we have an ac signal source v_{bs} as well as a dc bias voltage V_{BE}. In the output side however there is just a dc bias battery V_{CC}. The currents set up in the circuit will consist of both the dc components as well as ac components. Similar is the case with terminal voltages of transistor. Here I wish to show reader the symbol convention generally used in analyzing electronic circuits.

Normally,

- DC current and voltages are denoted as a capital letter with capital subscript.
- AC currents and voltages are denoted as small letters with small letter subscript.
- The total current or voltage is written in small letter with capital subscript.

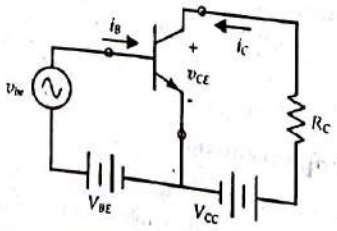

Figure: npn transistor circuit containing ac source as well as dc source.

In the above circuit,

v_{bs} represents ac source

V_{BE} and V_{CC} represent dc sources.

i_b represents base current due to source v_{bs}

I_B represents base current due to source V_{BE}

The total base current is represented as i_B and is given as:

$$i_B = i_b + I_B$$

Also the total base emitter voltage is represented as v_{BE} and is given by:

$$v_{BE} = v_{bs} + V_{BE}$$

Similarly,

Collector current due to base current i_b is represented as i_c

and collector current due to base current I_B is represented as I_C.

The total collector current is written as i_C and is given by:

$$i_C = i_c + I_C$$

The emitter current will also contain dc component I_E and ac component i_e. Therefore, the total emitter current is written as i_E and is given by;

$$i_E = i_e + I_E$$

Finally the collector to emitter voltage of the transistor also consists of dc term V_{CE} and ac component v_{ce}. Therefore, the total collector emitter voltage of the transistor is written as v_{CE} and is given as:

$$v_{CE} = V_{CE} + v_{ce}$$

Sometimes we may need to perform the dc analysis only to find the effects of dc source. In that case we ignore the effects of ac source.

If we ignore the ac source in above figure then we will obtain following dc equivalent circuit. Observe that only dc currents and voltages appear in the circuit.

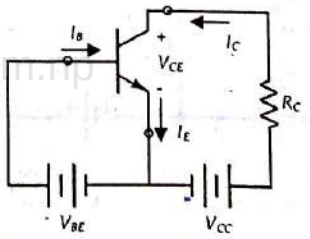

Figure: DC equivalent circuit of the previous circuit

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NPN configurations in active mode:

Transistor is a three terminal nonlinear device and it has three terminals to connect to the external circuit. To perform a specific function a device should have input and output. To give input to the device we need at least two terminals and to take output from the device we need at least two terminals. So, in total we need four terminals for a circuit to perform a specific function. But the problem is that transistor has only three terminals. This problem is solved by making one terminal common to both the input and output circuit. This is how the transistor can be operated in the active mode. In active mode transistor can be configured in three ways. Accordingly, there are three types of transistor configuration. They are

1. Common Base Configuration
2. Common Emitter Configuration
3. Common Collector Configuration

Note:

In the discussion that follows we will consider the application of only dc source to the transistor so as to simplify the discussion and to explore basics. Also the terminal for signal source is shown but we will not consider its effects. We shall postpone it until we study small signal model and amplifier circuits.

Common Base Configuration:

When the base terminal is common to both, input and output circuit then the transistor is said to be in common base configuration.

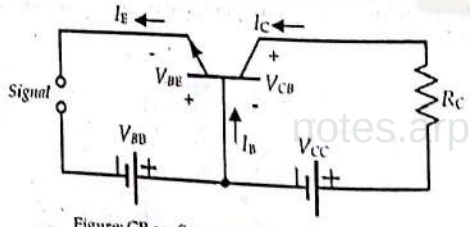


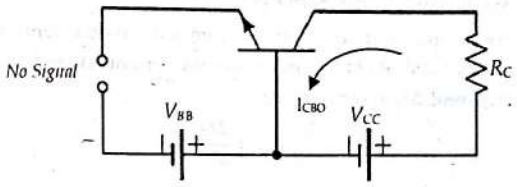
Figure: CB configuration for npn transistors

Here, input current is emitter current I_E and output current is collector current(I_C).

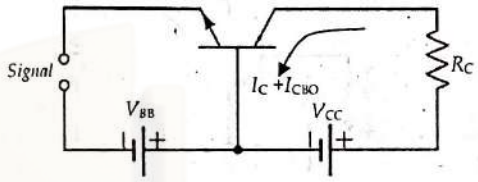
Here, input voltage is V_{BE} and output voltage is V_{CB}.

Current amplification factor (α): Current amplification factor (α) for common base configuration is defined as the ratio of change in output current (ΔI_C) to change in input current (ΔI_E) at constant collector base voltage V_{CB}. i.e. $\alpha = \frac{\Delta I_C}{\Delta I_E}$ at constant V_{CB}.

Leakage current I_{CBO}



So, practically when signal is applied,



Let us use open the emitter terminal. Then when we look at collector base junction it is reverse biased. In the reverse biased condition we assumed ideal diode does not conducts .But in reality small leakage current flows in the direction of the applied V_{CC}. Similarly, due to the practical reason there exist leakage current in base collector junction of the above common base configuration. It is denoted by I_{CBO} which means leakage current at collector base junction with emitter open.

$$\therefore \textit{Total collector current} = I_C + I_{CBO}$$

$$\therefore I_{C\,total} = \alpha I_E + I_{CBO}$$

Common Emitter Configuration

The transistor configuration in which the emitter terminal is common to both input and output circuit is known as common emitter configuration. Following circuit diagram shows the common emitter configuration for npn and pnp transistor:

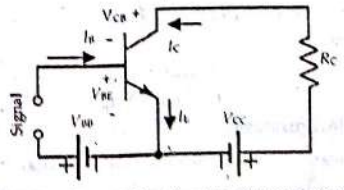


Figure: CE configuration for npn transistors

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Here the input current is Base current (I_B) and the output current is Collector current(I_C).

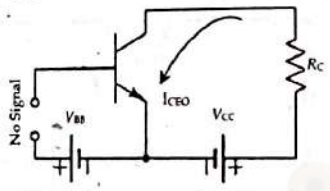
Similarly, input voltage is V_{BE} and output voltage is V_{CE}.

Base Current amplification factor (β):

Base current amplification factor for common emitter configuration is defined as the ratio of change in the output current ΔI_C to the change in the input current ΔI_B at constant V_{CE}.

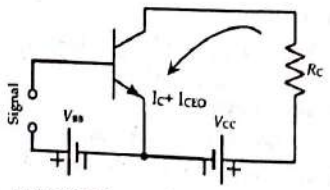
$$i.e\ \beta = \frac{\Delta I_C}{\Delta I_B}$$

Leakage current I_{CEO}



Due to the reverse bias leakage current I_{CEO} flow from collector to emitter when the base is made open in CE configuration .The current I_{CEO} is called as leakage current and it is read as collector to emitter current with base kept open.

So, in practical case, small amount of I_{CEO} appears in the circuit even when the signal is applied.



$$\therefore Total\ Collector\ current = \beta I_B + I_{CEO}$$
$$or, I_E = I_C + I_B + I_{CEO}$$
$$or, I_E = \beta I_B + I_B + I_{CEO}$$

For approximate study,

$$I_{CEO} = 0$$
$$\therefore I_E = (\beta + 1) I_B$$

Common Collector Configuration:

The transistor configuration in which collector terminal is common to both the input and output circuit is known as common collector configuration. Common collector configuration for npn transistor is shown below:

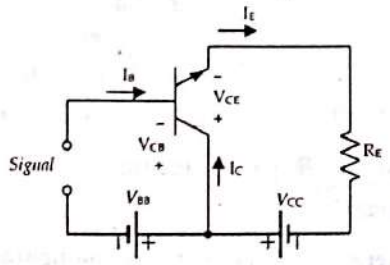


Figure: CC configuration for npn transistor

Here input current is base current I_B and output current is emitter current I_E. Similarly, input voltage is V_{CB} and output voltage is V_{CE}.

Current amplification factor (γ) for Common Collector configuration:

It is defined as the ratio of change in output current ΔI_E to the change in input current ΔI_B of common collector configuration.

$$\therefore Current\ amplification\ factor\ \gamma = \frac{\Delta I_E}{\Delta I_B}$$

Relationship between α, β and γ:

We know that,

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \qquad \beta = \frac{\Delta I_C}{\Delta I_B} \qquad \gamma = \frac{\Delta I_E}{\Delta I_B}$$

And,

$$\Delta I_E = \Delta I_B + \Delta I_C \ldots\ldots (i)$$

Dividing both sides of this equation by ΔI_C, we get,

$$\frac{\Delta I_E}{\Delta I_C} = \frac{\Delta I_B}{\Delta I_C} + \frac{\Delta I_C}{\Delta I_C}$$
$$or, \frac{1}{\alpha} = \frac{1}{\beta} + 1$$
$$or, \frac{1}{\alpha} = \frac{\beta + 1}{\beta}$$
$$\therefore \alpha = \frac{\beta}{\beta + 1} \ldots\ldots (ii)$$
$$or, \frac{1}{\beta} = \frac{1}{\alpha} - 1$$
$$or, \frac{1}{\beta} = \frac{1-\alpha}{\alpha}$$
$$\therefore \beta = \frac{\alpha}{1-\alpha} \ldots\ldots (iii)$$

Again,

$$\gamma = \frac{\Delta I_E}{\Delta I_B} = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

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$$or, \gamma = \frac{1}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}}$$

$$\therefore \gamma = \frac{1}{1-\alpha} \ldots\ldots (iv)$$

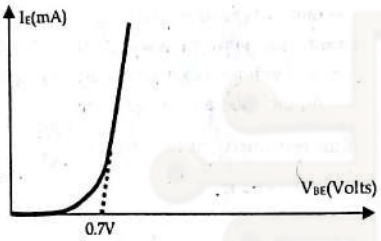
$$\therefore \gamma = \frac{1}{1-\frac{\beta}{\beta+1}} \ldots\ldots (iv) (\because Using\ equation\ ii)$$

$$\therefore \gamma = \beta + 1 \ldots\ldots (v)$$

Graphical Representation of Transistor Characteristics

A. Characteristics of Common Base configuration:

a. Input Characteristics



The plot of input current I_E and input voltage V_{BE} at constant output voltage V_{CB} is known as input characteristics in common base configuration.

$$Input\ resistance(r_i) = \frac{\Delta V_{BE}}{\Delta I_E}\ at\ constant\ V_{CB}$$

b. Output characteristics:

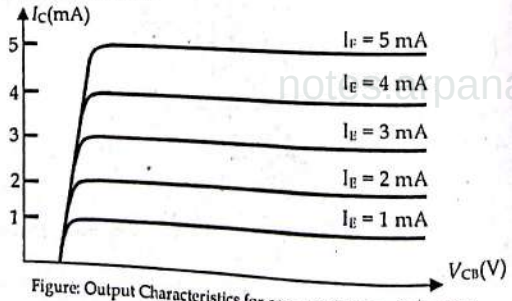


Figure: Output Characteristics for common base configuration

The plot of output current(I_C) versus output voltage (V_{CB}) at constant input current I_E is known as output characteristics in the common base configuration.

$$output\ resistance(r_o) = \frac{\Delta V_{CB}}{\Delta I_C}\ at\ constant\ I_E$$

B. Characteristics of the Common emitter configuration

a. Input Characteristics:

The graphical relationship between input current I_B and input Voltage V_{BE} at constant output voltage V_{CE} is called as input characteristics for CE configuration.

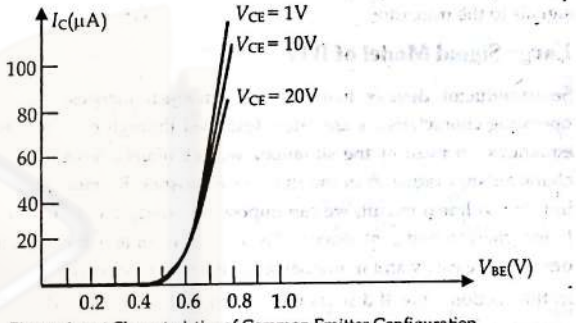


Figure: Input Characteristics of Common Emitter Configuration

$$\therefore Input\ resistance(r_i) = \frac{\Delta V_{BE}}{\Delta I_B}\ at\ constant\ V_{CE}$$

b. Output Characteristics:

The graphical relationship between output voltage V_{CE} and output current I_C at constant input current I_B is called as output characteristics in common emitter configuration.

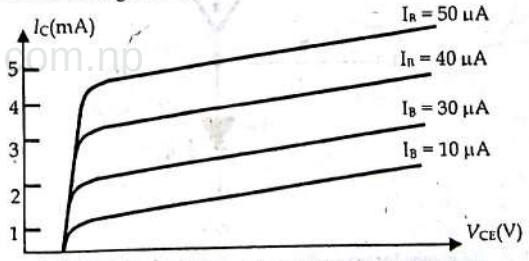


Figure: Output characteristics of Common Emitter Configuration

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$$\therefore Output\ resistance(r_o) = \frac{\Delta V_{CE}}{\Delta I_C} at\ constant\ I_B$$

BJT Transistor Modeling:

A model is a combination of circuit elements properly chosen that best approximates the actual behavior of the semiconductor devices under specified operating condition. Generally, for BJT there are two models:

i. Large Signal Models
ii. Small signal models

Large signal basically deals with the application of DC signal to the transistor whereas small signal model deals with the application of ac signals to the transistor.

Large Signal Model of BJT

Semiconductor devices like BJT are non-linear devices. Their internal operating characteristics are often described through complex non-linear equations. In most of the situations we are often interested in terminal characteristics rather than internal characteristics. By modeling transistor in terms of linear circuit, we can impose the condition of linearity to non-linear characteristics of device. Thus, we can obtain the parameters of device more easily and in mathematically convenient way.

In this section we will discuss the large signal model for BJT. As the name suggests large signal model of BJT is equivalent linear circuit representation of BJT when it is operating with large signals or DC voltages.

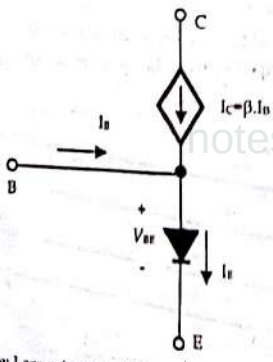

Figure: Large signal model for npn BJT in active mode.

Figure shows the large signal equivalent model of npn BJT in active mode. BJT can be replaced with this equivalent linear circuit while operating in active mode. In this model, the base emitter junction acts as a forward biased diode and has a drop of V_{BE}. And the collector current is dependent on base current. Thus, a dependent source is included in the collector circuit. The base current acts as control signal.

Note: We will frequently use this model in the analysis of DC biasing circuits for BJT. In the analysis we will not draw the circuit but keep this circuit in mind while applying KVL to obtain required currents and voltages in the circuit.

This model is valid as long as:

i. BJT is operating in active mode i.e. with EBJ forward biased and CBJ reversed biased.
ii. Signals applied are large signals.

Small Signal Equivalent circuit models

We discussed modeling of diode in the previous chapter. Similarly, transistor can be modeled to obtain the linearity. There are two kinds of models for transistor namely large signal model or DC model and small signal model or ac model. Let's move to small signal model.

The small signal modeling is concerned with the application of small signal to transistor. In this course there are two small signal models are studied namely Hybrid-π model and T-model.

Hybrid-π model:

This is a small signal model of BJT in which BJT is replaced by one of the following equivalent circuit:

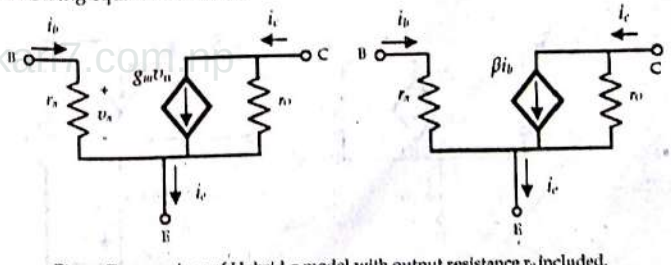

Figure: Two versions of Hybrid-π model with output resistance r_o included.

Following are the parameters of this model:

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Transconductance $g_m = \frac{I_C}{V_T}$,

I_C is obtained by dc analysis.

$r_\pi = \frac{V_T}{I_B} = \frac{\beta V_T}{I_C} = \frac{\beta}{g_m}$

$\therefore r_\pi = \frac{\beta}{g_m}$

T-model:

In this model a transistor is replaced by one of the following circuits

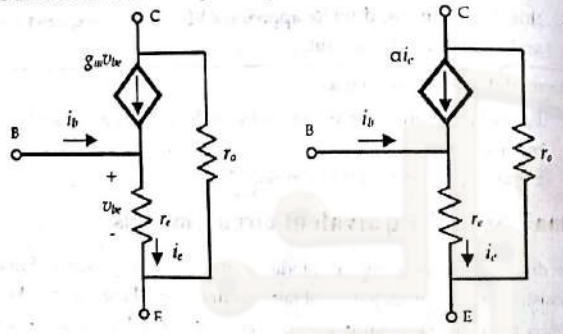

Figure: Two versions of T- model for npn BJT

Where,

$$r_e = \frac{\alpha}{g_m}$$

$$Transconductance\ g_m = \frac{I_C}{V_T},$$

I_C is obtained by dc analysis.

Relation between r_π and r_e:

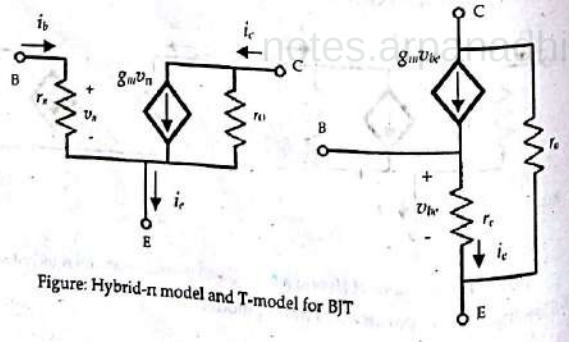

Figure: Hybrid-π model and T-model for BJT

Figure shows the hybrid-π model and T-model for BJT.

According to hybrid-π model the base emitter voltage drop is the drop across resistance r_π

$$\therefore v_{be} = v_\pi = i_b \times r_\pi$$
$$\therefore v_{be} = i_b \times r_\pi \ldots\ldots (i)$$

Similarly, according to T-model the base emitter voltage drop is the drop across the resistor r_e

$$\therefore v_{be} = i_e \times r_e \ldots\ldots (ii)$$

Since the base emitter voltage for a given transistor is constant. Therefore, equation (i) and (ii) must be identical.

$$i_b \times r_\pi = i_e \times r_e$$
$$or,\ i_b \times r_\pi = (\beta + 1) i_b \times r_e$$
$$\therefore\ r_\pi = (\beta + 1) r_e$$

This is the required relation.

DC and AC equivalent circuits:

Having learned large signal model and small signal model of BJT we now learn about DC and ac equivalent circuits. In DC analysis we utilize large signal model of the transistor. And in ac circuit analysis we utilize small signal model of the transistor.

In a transistor both ac and dc condition exists. Dc sources set up dc current and dc voltages whereas ac source set up ac currents and ac voltages. By adding the dc and ac current and voltage, we get the total current and voltage in the circuit.

DC equivalent circuit:

In this circuit only dc condition are considered. The steps to draw the dc equivalent circuits are:

i. Reduce all ac sources to zero.
ii. Open all capacitors.

AC equivalent circuits:

In this circuit, only ac condition is considered. The steps to draw the ac equivalent circuits are:

i. Reduce all the dc sources to 0V or Ground.
ii. Short all capacitors.

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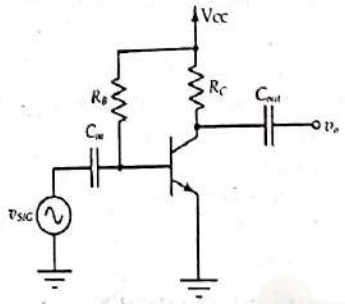
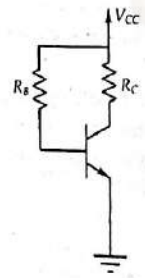
Note: For further analysis in DC equivalent circuit, we consider large signal model of BJT whereas for further analysis in AC equivalent circuit, we consider small signal model of BJT.

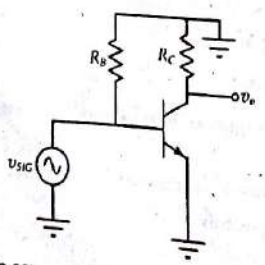
Example 1: Find the dc and ac equivalent circuits of the following transistor amplifier circuit.



Solution: The dc equivalent circuit after opening capacitors and shorting ac source is obtained as follows:



Also the ac equivalent circuit after shorting all capacitors and grounding dc voltage source is as follows:



It can be redrawn as:

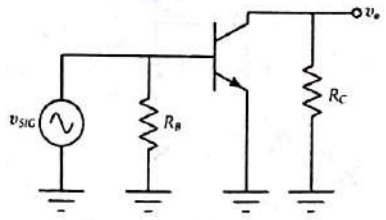
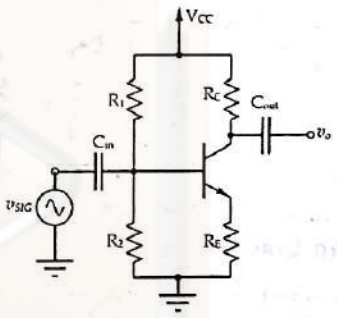


Figure: ac equivalent circuit

Example 2: Find the dc and ac equivalent circuits of the following transistor amplifier circuit. Assume source has some internal resistance in ac equivalent model.



Solution:

The dc equivalent circuit is:

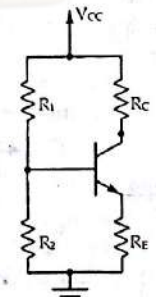


Figure: DC equivalent circuit

And the ac equivalent circuit is as follows:

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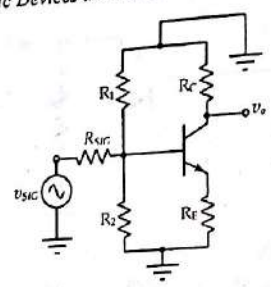


This circuit can be redrawn as:

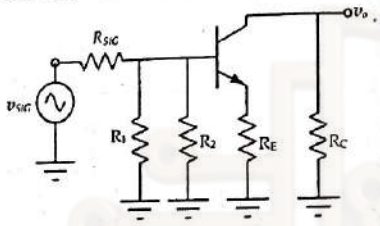


Figure: ac equivalent circuit

Graphical Load Line analysis

In the transistor circuit analysis, it is generally required to determine the collector current and collector emitter voltage. One of the methods used is to plot the output characteristics and find the collector current at any desired collector emitter voltage. However, a more convenient method can be used to solve such problems. This method is quite easy and is frequently used in the transistor circuit analysis. This method is called as load line analysis. By using this method we can easily determine the operating point and maximum current- voltage rating for transistor.

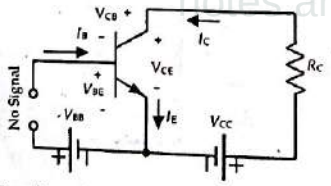
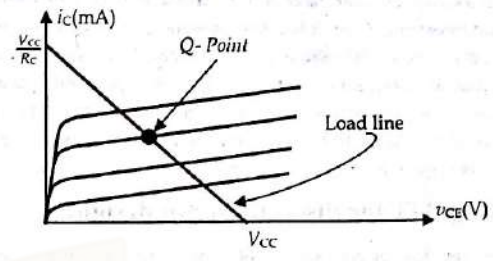


Figure: CE configuration for npn transistor when no signal applied.

Let us consider npn transistor in CE mode when no signal is applied as shown in the figure. So, DC conditions prevail in the circuit. It is therefore

the currents flowing through the circuit are called **zero signal currents** which are nothing but DC current.



Applying KVL in the output loop,
We get,

$$V_{CC} = I_C \times R_C + V_{CE}$$
$$\therefore V_{CE} = V_{CC} - I_C \times R_C$$
$$\therefore I_C = \frac{V_{CC}}{R_C} - \frac{1}{R_C} . V_{CE}$$

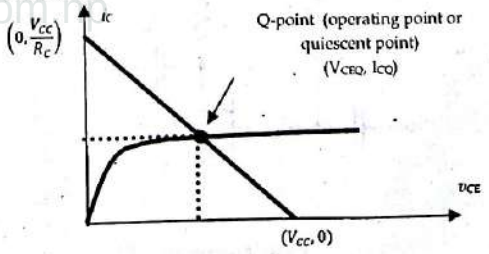
The above equation is a first degree equation of the form,

$$y = c - mx$$

Since V_{CC} and I_C are fixed. The different values of V_{CE} can be obtained for the different values of Load or resistance R_C. So, this line is called as load line.

When I_C=0 then, V_{CE}=V_{CC}
When V_{CE}=0 then, $I_C = \frac{V_{CC}}{R_C}$

Thus, the two points $(V_{CC}, 0)$ and $\left(0, \frac{V_{CC}}{R_C}\right)$ are the end points of the load line. By joining these two points load line is constructed. Also these points represent the maximum output voltage and maximum output current. The Q-point is the point of intersection of load line and output characteristics curve.



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Operating Point (V_{CEQ}, I_{CQ}):

The zero signal value of the collector current and collector emitter voltage is called operating point. It is called operating point because the variation of i_c and v_{ce} takes place about this point when the signal is applied. It is also called as quiescent point (or silent) point or Q-point because it is the point on i_C - v_{CE} characteristic when the transistor is silent i.e. there is the absence of the signal. It is beyond this point transistor starts to operate in the active region. It is represented as (V_{CEQ}, I_{CQ}).

Biasing BJT for discrete circuit design:

The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of the signal is known as transistor biasing. **The purpose of transistor biasing is to keep the emitter base junction forward biased and the collector base junction reversed biased during the passage of the signal.** This can be achieved by using the bias battery or by using biasing circuit. For the purpose of **simplicity and the economy we use biasing circuit** for the proper biasing of the transistor. The circuit which provides biasing is known as **biasing circuit.** There are four types of the biasing circuit. They are:

i. Base Bias (or fixed biasing)
ii. Emitter Bias
iii. Biasing with collector feedback resistor (collector biasing)
iv. Voltage divider biasing

Base Bias (or fixed biasing):

In this method a base resistor is connected between the power supply and base of the transistor.

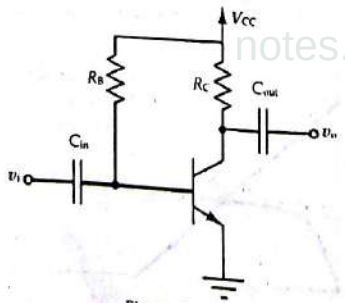
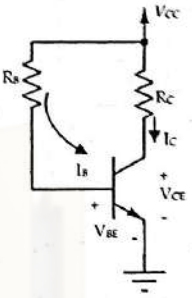


Figure: Base biasing

Here the function of capacitors is to isolate one stage of amplification from another stage of amplification. Though the terminals for signal input and output signal are shown our discussion is restricted to dc analysis only. For dc analysis capacitor acts as open circuit. And we are now going to determine zero signal collector current and collector emitter voltage. The dc equivalent circuit is:



Applying KVL at input loop,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\therefore I_C = \beta I_B$$

Applying KVL at output loop,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\therefore V_{CE} = V_{CC} - I_C R_C$$

Emitter Biasing:

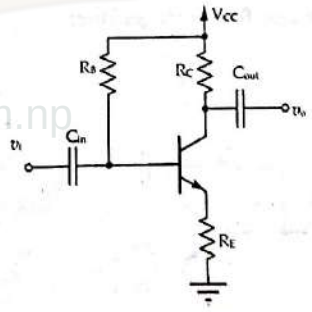


Fig: Emitter Biasing

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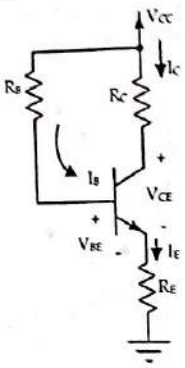
In this method a resistance is connected to the emitter terminal. Also present is a base resistor between base and power supply.

The dc equivalent circuit is:



Applying KVL at input loop,

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$
$$or, V_{CC} - V_{BE} = I_B R_B + I_E R_E$$
$$or, V_{CC} - V_{BE} = I_B R_B + (\beta + 1) I_B R_E$$
$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$
$$\therefore I_C = \beta I_B$$

Applying KVL at output loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$
$$\therefore V_{CE} = V_{CC} - I_C (R_C + R_E)$$
$$(\because I_C \approx I_E)$$

Biasing with collector feedback resistor:

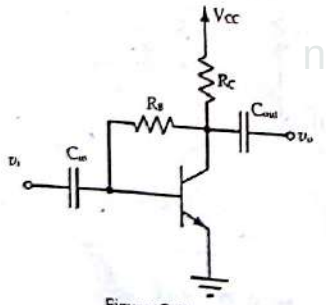


Figure: Collector feedback biasing

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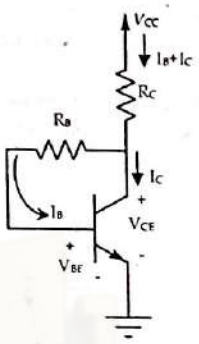
In this method a resistor forms the feedback path from collector to base. The emitter resistance is not used.

The dc equivalent circuit is:



Applying KVL in input loop,

$$V_{CC} - (I_B + I_C) R_C - I_B R_B - V_{BE} = 0$$
$$or, V_{CC} - I_E R_C - I_B R_B - V_{BE} = 0 (\because I_B + I_C = I_E)$$
$$or, V_{CC} - V_{BE} = I_E R_C + I_B R_B$$
$$or, V_{CC} - V_{BE} = (\beta + 1) I_B R_C + I_B R_B \quad (\because I_E = (\beta + 1) I_B)$$
$$\therefore I_B = \frac{V_{CC} - V_{BE}}{(\beta + 1) R_C + R_B}$$
$$\therefore I_C = \beta I_B$$

Applying KVL at output loop,

$$V_{CC} - (I_B + I_C) R_C - V_{CE} = 0$$
$$\therefore V_{CE} = V_{CC} - I_C R_C \ (\because (I_B + I_C) \approx I_C)$$

Voltage Divider Biasing

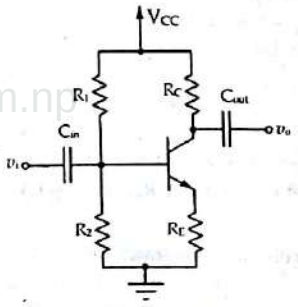


Figure: Voltage divider biasing

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Voltage divider biasing is most widely used biasing method. In this method four resistors are required. And this biasing enables the operation of transistor independent with parameter β. This means the operation of transistor does not get affected despite the variation in temperature. The circuit for the voltage divider biasing is shown in the figure.

For the voltage divider biasing we can find the zero signal collector current and collector emitter voltage using two methods namely approximate method and actual method(Thevenin's method). One can apply any of these two methods. Here we shall discuss both these methods.

The dc equivalent circuit is:

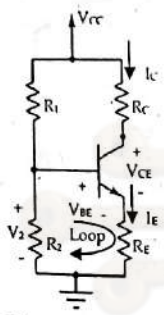


Figure: DC equivalent circuit

Approximate method

The voltage across resistor R_2 is given by:

$$V_2 = \frac{V_{CC} \times R_2}{(R_1 + R_2)}$$

Applying KVL in the loop I

$$V_2 - V_{BE} - I_E R_E = 0$$

$$\therefore I_E = \frac{V_2 - V_{BE}}{R_E}$$

$$\therefore I_C = \frac{V_2 - V_{BE}}{R_E} (\because I_C \approx I_E)$$

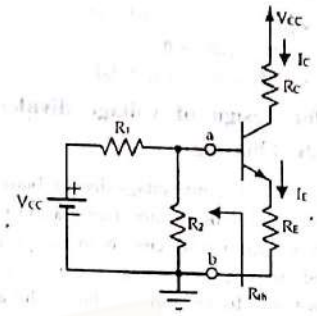
Applying KVL in output loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\therefore V_{CE} = V_{CC} - I_C(R_C + R_E) \quad (\because I_C \approx I_E)$$

Thevenin's method:

The dc equivalent circuit can be re-drawn as:



To Thevenize the left portion of circuit looking from terminals a and b, we have to calculate R_{th} and V_{th}.

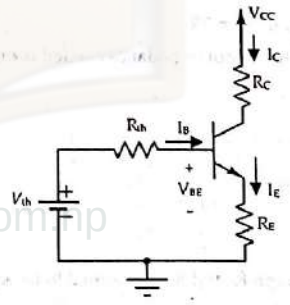
In order to find R_{th} we look from terminal a and b and short voltage source. Here resistors R_1 and R_2 appear parallel.

$$\therefore R_{th} = R_1 // R_2$$

In order to find Thevenin's voltage V_{th}, we have to find open circuit voltage. Here the voltage drop across R_2 is V_{th}.

$$\therefore V_{th} = \frac{V_{CC} \times R_2}{R_1 + R_2}$$

Replacing the left portion of circuit with Thevenin's equivalent circuit (i.e. V_{th} in series with R_{th}.), we get,



Applying KVL in base emitter loop,

We get,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$\therefore I_B = \frac{V_{th} - V_{BE}}{R_{th} + R_E(\beta + 1)}$$

$$\therefore I_C = \beta I_B$$

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Applying KVL in output loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\therefore \quad V_{CE} = V_{CC} - I_C(R_C + R_E) \quad (\because I_C \approx I_E)$$

General Guideline for design of voltage divider biasing circuit (or β- independent biasing)

In the previous section we studied about voltage divider biasing scheme. Since voltage divider biasing is the most important biasing method, we shall learn how to design voltage divider circuits in this section. In the earlier section we studied how to calculate zero signal currents and voltages. In this section we will discuss how to choose the appropriate values of resistance in the circuit. This is a crucial design step because if choice of resistances is incorrect, the voltage divider circuit may not establish the required bias to keep transistor in active mode. i.e. its emitter base junction forward biased and collector base junction reverse biased.

There are generally two guidelines for designing voltage divider circuits. They are:

Guideline 1 or 1/3rd rule ($V_{CC} \leq 10$ V)

This guideline is generally applicable when $V_{CC} \leq 10$ V.

In this guideline,

$$V_{th} = V_B = \frac{V_{CC}}{3}, \; V_{RC} = I_C R_C = \frac{V_{CC}}{3},$$

Above conditions will lead to: $R_1 = 2R_2$

For firm biasing: (Used when input impedance needed is high)

$$R_{th} = 0.1\beta R_E \approx \frac{2}{3} R_2$$

Where, $R_{th} = R_1 // R_2$

And R_1 is selected such that

$R_1 = 2R_2$

Or equivalently

In this scheme current through R_1 and R_2 is assumed to be equal to 0.1 I_E

i.e. $I_{R12} = 0.1\ I_E$

R_2 is found from

$V_B = I_{R12} R_2$

And using R_2 value of R_1 can be found from

$V_{CC} = I_{R12}(R_1 + R_2)$

For stiff biasing: : (Used when input impedance needed is less)

$$R_{th} = 0.01\beta R_E \approx \frac{2}{3} R_2$$

Where, $R_{th} = R_1 // R_2$

And R_1 is selected such that

$R_1 = 2R_2$

Or equivalently

In this scheme current through R_1 and R_2 is assumed to be equal to I_E

i.e. $I_{R12} = I_E$

R_2 is found from

$V_B = I_{R12} R_2$

And using R_2 value of R_1 can be found from

$V_{CC} = I_{R12}(R_1 + R_2)$

Guideline 2 or 1/10th rule ($V_{CC} \geq 15$ V)

This guideline is generally applicable when $V_{CC} \geq 15$ V.

In this guideline,

$$V_E = I_E R_E = \frac{1}{10} V_{CC}, \; V_{CE} = \frac{5}{10} V_{CC} \text{ and } V_{RC} = I_C R_C = \frac{4}{10} V_{CC}$$

Since $V_E = 0.4\ V_{CC}$ ($V_E << V_{CC}$)

Above conditions will lead to: $R_{th} = R_2$

For firm biasing: (Used when input impedance needed is high)

In this scheme current through R_1 and R_2 is assumed to be equal to 0.1 I_E

i.e. $I_{R12} = 0.1\ I_E$

And $R_{th} = R_2 = 0.1\beta\ R_E$

With R_2 known, the value of R_1 is determined from

$V_{CC} = I_{R12}(R_1 + R_2)$

For stiff biasing: (Used when input impedance needed is less)

In this scheme current through R_1 and R_2 is assumed to be equal to I_E

i.e. $I_{R12} = I_E$

And $R_{th} = R_2 = 0.01\beta R_E$

With R_2 known, the value of R_1 is determined from

$V_{CC} = I_{R12}(R_1 + R_2)$

Note:

When $10 < V_{CC} < 15$ then one may choose any one of the guidelines.

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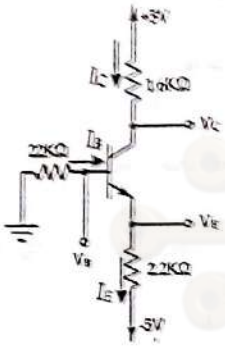
Analysis of the Transistor circuits at DC

Having studied large signal models of BJT and different schemes of biasing, analysis of transistor circuits at DC is straight forward. Lets us closely examine two examples to demonstrate the analysis of transistor circuits at DC.

Example 1:

Find the indicated voltages and currents in the following transistor circuit.



From the circuit, $V_B = 0$ and $I_B = 0$

Considering the transistor to be in active mode of operation, $\therefore V_{BE} = 0.7V$

The voltage V_E can be written as:

$V_E = -V_{BE} = -0.7\ V$

The current I_E is given by: $I_E = \frac{V_E - V_{EE}}{R_E} = \frac{-0.7 - (-5)}{2.2} = 1.95\ mA$

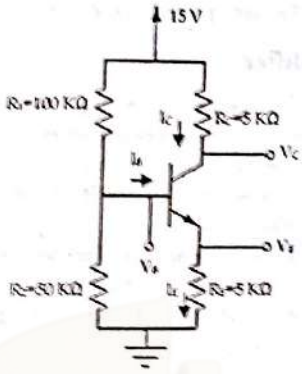
Assuming $I_C \cong I_E$, we get $I_C = 1.95$ mA

Therefore voltage V_C is given by:

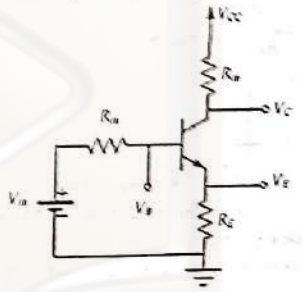
$$V_C = V_{CC} - I_C R_C = 5 - 1.95 \times 1.6 = 1.88\ V$$

Example 2:

Find the indicated voltages and currents in the following transistor circuit. Assume β=100.



Replacing above circuit with the Thevenin's equivalent circuit as described in biasing section. We get



$R_{th} = 100//50 = 33.33\ K\Omega$

$V_{th} = 15 \cdot \frac{50}{50+100} = 5\ V$

From the above equivalent circuit,

$$I_E = \frac{V_{th} - V_{BE}}{R_E + \frac{R_{th}}{\beta+1}} = \frac{5 - 0.7}{3 + \frac{33.33}{101}} = 1.29\ mA$$

$$\therefore I_B = \frac{I_E}{\beta+1} = \frac{1.29}{101} = 0.0128\ mA$$

$$\therefore I_C = \beta I_B = 100 \times 0.0128 = 1.28\ mA$$

$$\therefore V_E = I_E R_E = 1.29 \times 3 = 3.87\ V$$

$$\therefore V_B = V_{BE} + V_E = 3.87 + 0.7 = 4.57\ V$$

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$$\therefore V_C = V_{CC} - I_C R_C = 15 - 1.28 \times 5 = 8.6 \text{ V}$$

Transistor as an amplifier

An amplifier is a device which is capable of increasing the magnitude of incoming ac current or voltage signal. Generally amplifiers use dc power to raise the strength of the incoming ac signal. In any analog circuit the basic unit is an amplifier. So, analog circuits are built by using different combinations of amplifier circuits.

In this section we study the amplifying characteristics of the BJT transistor. For BJT to work as an amplifier we must operate it in active region of its characteristics. For this let us consider the following circuit.

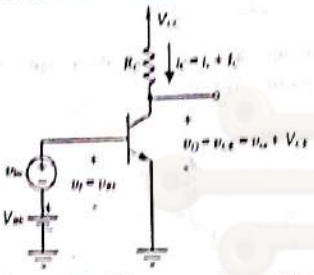

Figure a: Basic BJT common emitter amplifier circuit

Figure a shows the basic circuit diagram of BJT common emitter amplifier. In this circuit biasing arrangements are not shown so as to simplify the discussion however in a working circuit a good biasing arrangement must be done. The transitor input is formed by signal source v_{be} and dc voltage source V_{BE}. The main purpose of keeping dc source is to forward bias the emitter base junction throughout the operation. Thus the total input to the transistor is $v_I = v_{be} + V_{BE}$. This is also the total voltage across base emitter junction of the transistor. i.e. $v_I = v_{BE}$.

The collector current i_C is composed of both the ac component i_c and dc component I_C. Similarly the output voltage v_O or v_{CE} is composed of ac component v_{ce} and dc component V_{CE}.

Now the question that naturally arises is: How the circuit increases the magnitude of input signal ?The answer is:

The transistor transfers current from low resistance path to the high resistance path. The low resistance path of the transistor is the base emitter junnction(which is forward biased)while the high resistance path

of the transistor is the collector base junction(which is reversed biased). As a result output is naturally higher as compared to the input. This process is analogous to the increase in gravitational potential energy of an object when it is taken from low altitude to certain high altitude.

Applying KVL in the output loop, we get,

$$V_{CC} = i_C R_C + v_{CE}$$

$$v_{CE} = V_{CC} - i_C R_C$$

Differentiating above relation with respect to i_C, we get,

$$\frac{dv_{CE}}{di_C} = -R_C$$

The negative sign indicates that the output voltage is of opposite polarity as compared to the input signal. This is referred to as phase inversion. Also it signifies that the gain of the common emitter amplifier circuit is also negative.

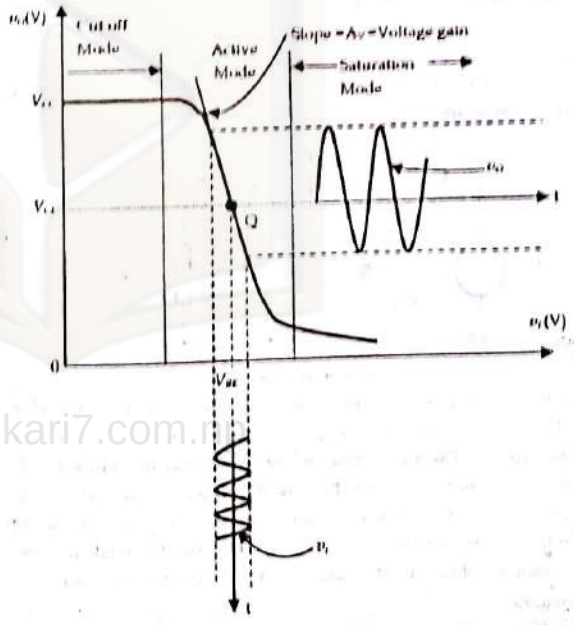

Figure b: Transfer Characteristics of the BJT common emitter amplifier circuit

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Figure b shows the transfer characteristics of the BJT common emitter amplifier circuit. The amplifier is biased at point Q. The small voltage signal v_{be} is superimposed on the dc bias voltage V_{BE}. The input signal can swing across V_{BE}. The resulting output signal v_O appears superimposed on the dc collector emitter voltage V_{CE}. The output signal can swing across V_{CE}. The amplitude of the output ac signal v_o is larger than the input signal v_{be} by the gain A_v. As shown in the figure the voltage gain of common emitter amplifier is negative.

Basic Single Stage BJT Amplifier Configurations

In the earlier section we studied amplifying nature of BJT. Having studied that, let us study some of the popular single stage transistor amplifier configurations. But before that we got to familiarize ourselves to different parameters of amplifier.

Characteristic parameters of Amplifier:

An amplifier may be unilateral or non- unilateral. An amplifier is called unilateral if its input impedance does not depend upon the load resistance. And an amplifier is called non- unilateral if its input impedance depends upon the load resistance.

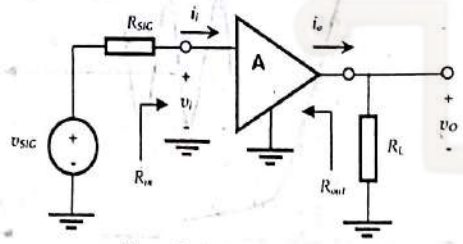

Figure: Typical amplifier circuit

Let us consider an amplifier circuit as shown in the figure. Here we have considered amplifier as a block in order to explain the basic parameters of the amplifier circuit. The signal source having internal resistance R_{SIG} is coupled to the amplifier. Thus, the effective input to amplifier is v_i (because of some drop in source internal resistance) as shown in the figure. The input current to the amplifier is i_i. The output v_o is taken across the load resistor R_L. Similarly, the output current of the amplifier is i_o.

Input Impedance:

The impedance of the amplifier circuit as seen from the input terminals is known as input impedance.

The input impedance of the above circuit is calculated as:

$$R_{in} = \frac{v_i}{i_i}$$

And the input impedance with no load resistor is given as:

$$R_{in} = \frac{v_i}{i_i} \text{ at } R_L = 0$$

Since the source internal resistance R_{SIG} and input impedance R_{in} form a voltage divider configuration. The input impedance of the amplifier should be as high as possible (ideally infinite) in order to effectively couple signal source to amplifier without loss in its internal resistance.

Output Impedance:

The impedance of the amplifier circuit as seen from the output terminals with the input terminals shorted is known as output impedance.

The output impedance of the above circuit is calculated as:

$$R_{out} = \frac{v_o}{i_o}$$

Since the load resistance R_L and output impedance of the amplifier R_{out} form a voltage divider configuration. The output impedance of the amplifier should be as low as possible (ideally zero) in order to effectively transfer output signal source to load resistance without loss in its output circuit.

Voltage gain:

The voltage gain of the amplifier A_v is defined as the ratio of the output voltage to the input voltage. It is given as:

$$A_v = \frac{v_o}{v_i}$$

The open circuit voltage gain of the amplifier A_v is defined as the voltage gain with load resistor opened.

i.e. $A_{vo} = A_v$ at $R_L \to \infty$

The overall voltage gain of the amplifier G_v is defined as the ratio of the output voltage v_o to the voltage of the signal source v_{SIG}.

$$G_v = \frac{v_o}{v_{SIG}}$$

Current gain:

It is the ratio of the output current to the input current. It is given as:

$$A_i = \frac{i_o}{i_i}$$

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Similarly the short circuit current gain A_{is} is the current gain of the amplifier with the load resistor shorted. i.e. at $R_L = 0$.

$$\therefore A_{is} = A_i \text{ at } R_L = 0$$

Power gain:

It is the ratio of output power to input power. It is denoted as A_P and is given by the product of current gain and voltage gain.

$$A_P = A_i \times A_V$$

Common Emitter Configurations:

In the common emitter configuration we study following different variations:

Common emitter amplifier using fixed biasing:

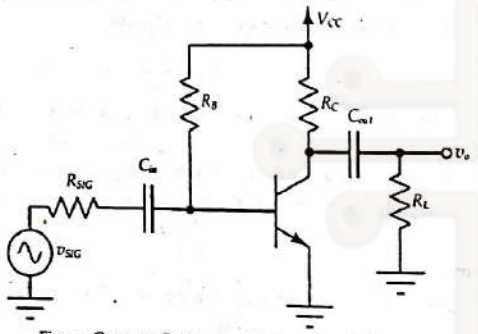


Figure: Common Emitter amplifier without emitter resistance

The ac equivalent circuit is:

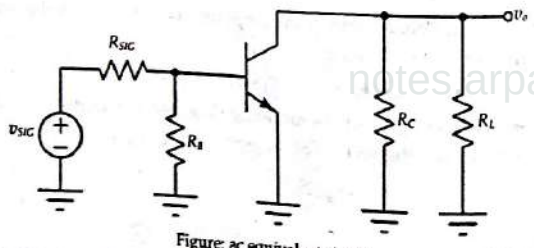
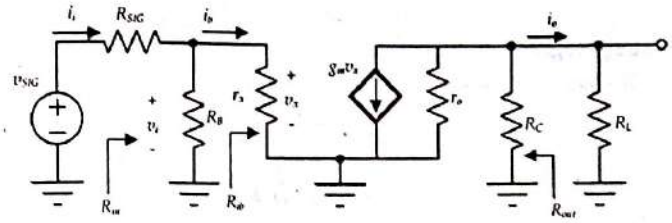


Figure: ac equivalent circuit

Replacing the transistor with its hybrid-π equivalent model, we get,



Here, $v_i = v_{SIG} \dfrac{R_{in}}{R_{in}+R_{SIG}}$

Input Impedance:

The input impedance of the amplifier can be found by looking into the amplifier input terminals.

$$R_{ib} = r_\pi$$

$$R_{in} = \frac{v_i}{i_i} = R_B // R_{ib}$$

$$\therefore R_{in} = R_B // r_\pi$$

Output Impedance:

The output impedance R_{out} is obtained by looking back into the output terminals of amplifier while shorting input i.e. $v_i = 0$. Since it will result in $v_\pi = 0$. So we obtain following expression for the output impedance:

$$R_{out} = R_C // r_o$$

Generally, $r_o >> R_C$

$$\therefore R_{out} \equiv R_C$$

Here we have used the fact that in parallel combination of resistance the total equivalent resistance of the combination is approximately equal to the value of lowest resistance in the combination.

Voltage gain:

The output voltage gain of the amplifier A_v is calculated as follows:

$$v_o = -g_m v_\pi (R_C // r_o // R_L)$$

Since $v_i = v_\pi$

$$\therefore A_v = \frac{v_o}{v_i} = -g_m (R_C // r_o // R_L)$$

Also the open circuit voltage gain A_{vo} can be obtained by setting $R_L \to \infty$ in the above expression. So we get,

$$A_{vo} = -g_m (R_C // r_o)$$

Also since $r_o >> R_C$

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$$A_{vo} = -g_m R_C$$

Current gain:

The output current i_o is given by:

$$i_o = -g_m v_\pi \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

$$i_o = -g_m v_i \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

$$i_o = -g_m i_i R_{in} \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

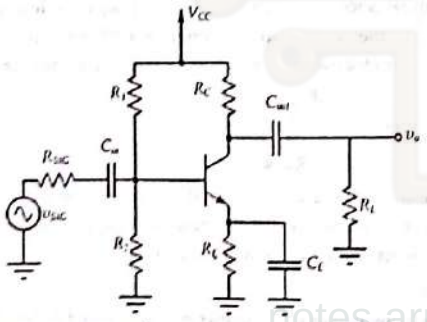
$$\frac{i_o}{i_i} = -g_m R_{in} \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

$$\therefore A_i = -g_m R_{in} \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

The short circuit current gain A_{is} is obtained by setting $R_L = 0$ in the above expression for A_i

$$\therefore A_{is} = -g_m R_{in}$$

β- Independent Common Emitter Amplifier with emitter bypassed capacitor:



Its ac equivalent circuit is:

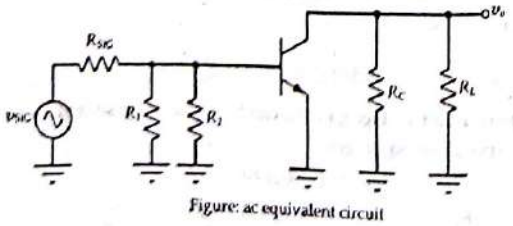
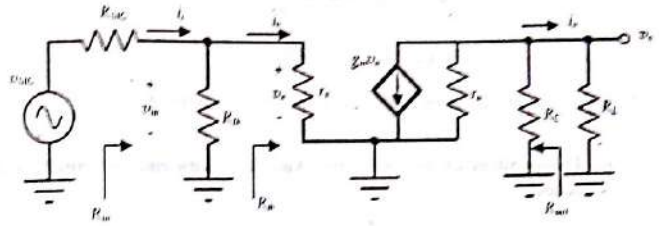


Figure: ac equivalent circuit

Replacing the transistor by hybrid-π model, we get



Where, $R_{in} = R_1 // R_2$

Input Impedance:

$R_{in} = r_\pi$

$R_{in} = R_{in} // R_{in}$

$\therefore R_{in} = R_1 // R_2 // r_\pi$

Output Impedance:

$R_{out} = R_C // r_o$

Since $r_o >> R_C$

$\therefore R_{out} \equiv R_C$

Voltage gain:

$v_o = -g_m v_\pi (r_o // R_C // R_L)$

Since $v_i = v_\pi$

or, $\frac{v_o}{v_i} = -g_m (r_o // R_C // R_L)$

$\therefore A_v = -g_m (r_o // R_C // R_L)$

The open circuit voltage gain A_{vo} can be found by setting $R_L \to \infty$

$A_{vo} = -g_m (r_o // R_C)$

Again, since $r_o >> R_C$

$A_{vo} = -g_m R_C$

Current gain:

The output current i_o is given by:

$$i_o = -g_m v_\pi \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

$$\text{or, } i_o = -g_m v_i \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

$$\text{or, } i_o = -g_m i_i R_{in} \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

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$$\text{or, } \frac{i_o}{i_i} = -g_m R_{in} \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

$$\therefore A_i = -g_m R_{in} \frac{(r_o // R_C)}{(r_o // R_C)+R_L}$$

The short circuit current gain A_{is} can be found by setting $R_L = 0$.

$$\therefore A_{is} = -g_m R_{in}$$

β - Independent Common Emitter Amplifier with emitter unbypassed capacitor:

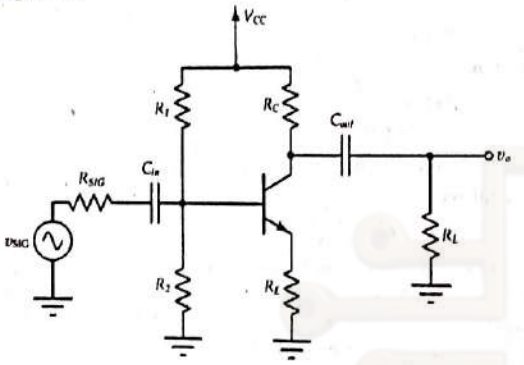


Figure: Common Emitter amplifier with voltage divider biasing scheme

The ac equivalent circuit is:

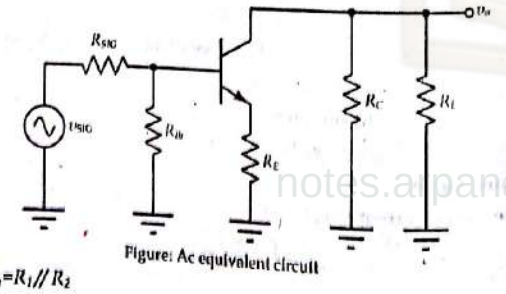
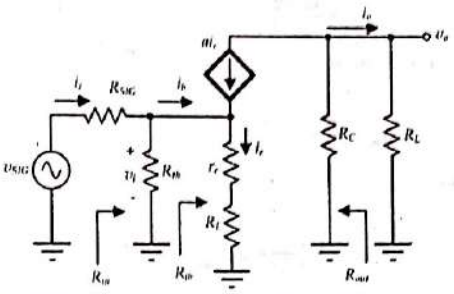


Figure: Ac equivalent circuit

Where, $R_{th} = R_1 // R_2$

Since the emitter resistance is present we shall use **T model** instead of hybrid-π model. Nevertheless hybrid-π model can also be used.



Note that we have not included the r_o of T- model so as to simplify the analysis. Also note that it does not have effect on the input and output impedance of the circuit. Thus its omission is justified.

Input Impedance:

$$R_{ib} = \frac{v_i}{i_b} = \frac{i_e(r_e+R_L)}{i_b}$$

$$\therefore R_{ib} = (\beta+1)(r_e+R_L) \quad \text{Since } i_e = (\beta+1)i_b$$

$$\therefore R_{in} = R_{th} // R_{ib}$$

$$\therefore R_{in} = R_1 // R_2 // (\beta+1)(r_e+R_L)$$

Output Impedance:

The output impedance is given as:

$$R_{out} = R_C$$

Voltage Gain:

The output voltage v_o is given by:

$$v_o = -\alpha i_e (R_C // R_L)$$

$$\text{or, } v_o = -\alpha(\beta+1) i_b (R_L // R_L)$$

$$\text{or, } v_o = \frac{-\beta i_i R_{ib}}{(R_{ib}+R_{ib})} (R_C // R_L) \quad \text{Because } \alpha = \frac{\beta}{\beta+1}$$

$$\text{or, } v_o = \frac{-\beta \frac{v_i}{R_{in}} R_{ib}}{(R_{ib}+R_{ib})} (R_C // R_L) \quad \text{Because } i_i = \frac{v_i}{R_{in}}$$

$$\text{or, } \frac{v_o}{v_i} = \frac{-\beta R_{ib}}{R_{in}(R_{ib}+R_{ib})} (R_C // R_L)$$

$$\text{or, } A_v = \frac{-\beta R_{ib}}{\frac{R_{ib}R_{ib}}{R_{ib}+R_{ib}}(R_{ib}+R_{ib})} (R_C // R_L) \quad \text{Because } R_{in} = R_{ib} // R_{ib}$$

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$$\therefore A_{io} = \frac{-\beta(R_C // R_L)}{R_{ib}}$$

If load resistor is not present i.e. if $R_L \to \infty$

Then,

$$\therefore A_{io} = \frac{-\beta R_C}{R_{ib}}$$

Current gain:

We know,

$$i_o = -\alpha i_e \frac{R_C}{(R_C+R_L)} = -\beta i_b \frac{R_C}{(R_C+R_L)}$$

$$\text{or, } i_o = -\beta i_b \frac{R_{th}}{(R_{th}+R_{ib})} \frac{R_C}{(R_C+R_L)}$$

$$\therefore A_i = \frac{-\beta R_{th}}{(R_{th}+R_{ib})} \frac{R_C}{(R_C+R_L)}$$

The short circuit current gain A_{is} is obtained by setting $R_L = 0$.

$$\therefore A_{is} = \frac{-\beta R_{th}}{(R_{th}+R_{ib})}$$

Common Base Configuration

β - Independent Common Base Amplifier

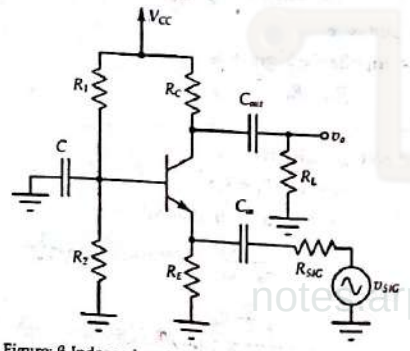
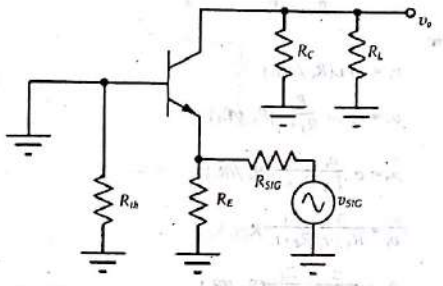


Figure: β-Independent Common base amplifier circuit

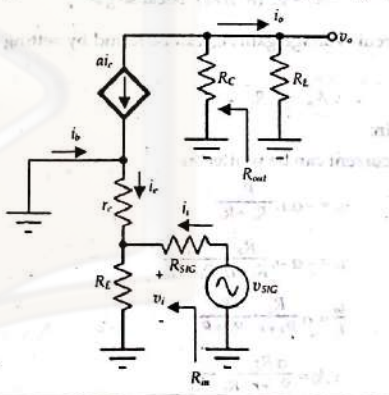
Figure shows the circuit diagram of common base amplifier using BJT. In this circuit input is applied via emitter terminal while output is taken from collector terminal. The name "common base" comes from the fact that the base terminal is common to both the input and output circuit.

The ac equivalent circuit is:



Where, $R_{th} = R_1 // R_2$

Since the resistor R_{th} is grounded at both of its terminals. So we can replace it by short and ignore it in our further calculations. Also since the emitter resistance is present we shall use T model for the small signal analysis. Replacing the transistor with T model and ignoring R_{th}, we get,



Also note that we have not included r_o of T model so as to simplify the analysis. Even if it is included in analysis it does not have any effect on the output impedance.

Input Impedance:

The input impedance of the circuit is given as:

$$R_{in} = R_E // r_e$$

Output Impedance:

The output impedance of the circuit is given as:

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$$R_{out} = R_C$$

Voltage Gain:

$$v_o = -\alpha\, i_c (R_C // R_L)$$

$$v_o = -\alpha \cdot -i_i \frac{R_E}{R_E + r_e}(R_C // R_L)$$

$$v_o = \alpha \cdot \frac{v_i}{R_{in}} \frac{R_E}{R_E + r_e}(R_C // R_L)$$

$$\frac{v_o}{v_i} = \frac{\alpha}{R_E // r_e} \frac{R_E}{R_E + r_e}(R_C // R_L)$$

$$A_v = \frac{\frac{\alpha}{R_E \cdot r_e}}{R_E + r_e} \frac{R_E}{R_E + r_e}(R_C // R_L)$$

$$A_v = \frac{\alpha}{r_e}(R_C // R_L)$$

$$\therefore A_v = g_m (R_C // R_L) \text{ Because } g_m = \frac{\alpha}{r_e}$$

The open circuit voltage gain A_{vo} can be found by setting the load resistor $R_L \to \infty$

$$\therefore A_{vo} = g_m R_C$$

Current Gain:

The output current can be written as:

$$i_o = -\alpha\, i_e \frac{R_C}{R_C + R_L}$$

$$i_o = -\alpha \cdot -i_i \frac{R_E}{R_E + r_e} \frac{R_C}{R_C + R_L}$$

$$\frac{i_o}{i_i} = \alpha \frac{R_E}{R_E + r_e} \frac{R_C}{R_C + R_L}$$

$$\therefore A_i = \frac{\alpha R_E}{R_E + r_e} \frac{R_C}{R_C + R_L}$$

The open circuit current gain A_{io} can be found by setting $R_L = 0$

$$\therefore A_{io} = \frac{\alpha R_E}{R_E + r_e}$$

Common Base amplifier with Emitter Biasing

This circuit circuit has same expressions for R_{in}, R_{out}, A_v, A_i as that for Common base amplifier using voltage divider biasing or β-independent biasing.

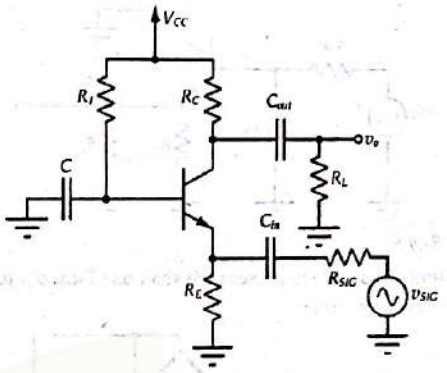

Figure: Common Base amplifier with Emitter Biasing

Common Collector Configuration: Emitter Follower

Common collector amplifier circuit is also called as **emitter follower** because output is the exact replica of the input. This circuit has the voltage gain of approximately unity which we shall see shortly. Following are the different variations of common collector amplifier:

β- Independent Common Collector amplifier: R_C Excluded

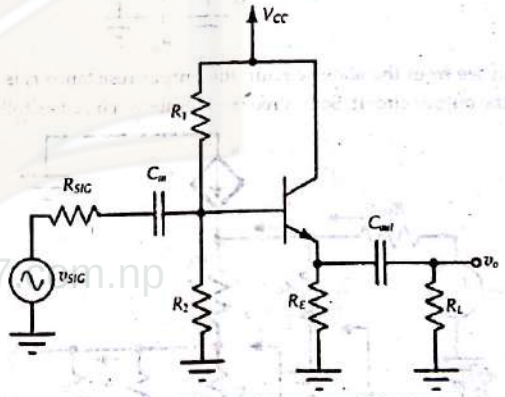

Figure: Common Collector amplifier circuit

The ac equivalent circuit is given below:

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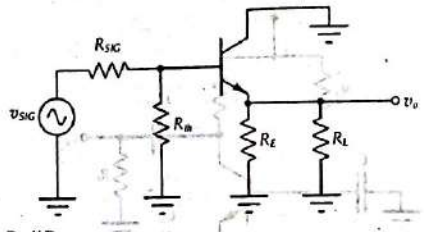
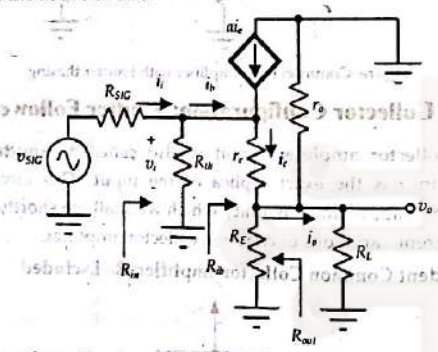
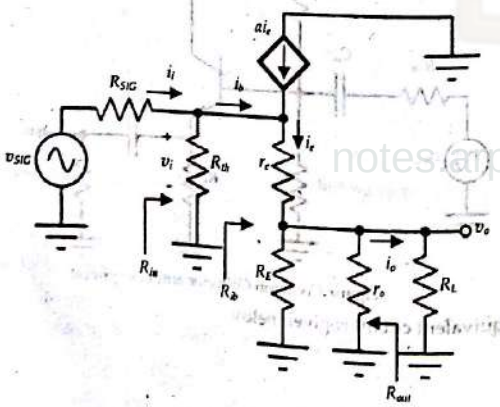


Where, $R_{th} = R_1 // R_2$

Since the emitter resistance is present we shall use T-model to replace the transistor in the above circuit.



As we can see from the above circuit, the output resistance r_o is reflected towards the output circuit. So lets redraw the above circuit as follows:



Input Impedance:

$$R_{in} = R_{th} // R_{ib}$$

$$R_{ib} = \frac{v_i}{i_b} = \frac{i_e(R_E // r_o // R_L + r_e)}{i_b} = (\beta+1)(R_E // r_o // R_L + r_e)$$

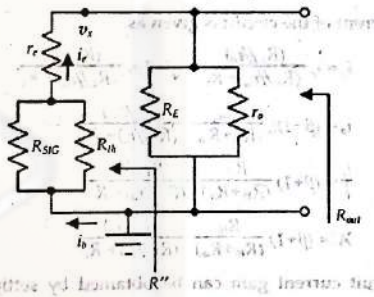
$$\therefore R_{in} = R_{th} // R_{ib}$$

$$\therefore R_{in} = R_1 // R_2 // (\beta+1)(R_E // r_o // R_L + r_e)$$

Here the input impedance depends upon the load resistance. Thus emitter follower is a non- unilateral amplifier.

Output Impedance:

The output impedance can be found by looking back at the output terminals with input source grounded. In order to make it simpler lets draw the equivalent circuit for obtaining output impedance.



Then the output resistance is given as:

$$R_{out} = R'' // (R_E // r_o)$$

$$R'' = \frac{v_x}{i_e} = \frac{i_e r_e + i_b R_{SIG} // R_{th}}{i_e}$$

$$R'' = r_e + \frac{R_{SIG} // R_{th}}{\beta+1}$$

$$\therefore R_{out} = \left(r_e + \frac{R_{SIG} // R_{th}}{\beta+1}\right) // (R_E // r_o)$$

Voltage Gain:

From the circuit the output voltage can be written as:

$$v_o = i_e (R_E // r_o // R_L)$$

$$\text{or, } v_o = (\beta+1) i_b (R_E // r_o // R_L)$$

$$\text{or, } v_o = \frac{(\beta+1) v_i}{R_{ib}} (R_E // r_o // R_L) \quad \text{Because } i_b = \frac{v_i}{R_{ib}}$$

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$$\text{or, } \frac{v_o}{v_i} = \frac{(\beta+1)(R_E//r_o//R_L)}{(\beta+1)(R_E//r_o//R_L+r_e)} \quad \text{Because } R_{ith} = (\beta+1)(R_L//r_o//R_L+r_e)$$

$$\text{or, } A_v = \frac{(R_E//r_o//R_L)}{(R_E//r_o//R_L+r_e)}$$

The value of the above expression is very close to unity.

So,

$$A_v \cong 1$$

Similarly for this configuration the open circuit voltage gain A_{vo} is also very close to unity.

So, $A_{vo} \cong 1$

Current Gain:

The output current of the circuit is given as:

$$i_o = i_e \frac{(R_E//r_o)}{(R_E//r_o)+R_L} = (\beta+1)i_b \frac{(R_E//r_o)}{(R_E//r_o)+R_L}$$

$$i_o = (\beta+1)i_i \frac{R_{th}}{(R_{th}+R_{ib})} \frac{(R_E//r_o)}{(R_E//r_o)+R_L}$$

$$\frac{i_o}{i_i} = (\beta+1) \frac{R_{th}}{(R_{th}+R_{ib})} \frac{(R_E//r_o)}{(R_E//r_o)+R_L}$$

$$A_i = (\beta+1) \frac{R_{th}}{(R_{th}+R_{ib})} \frac{(R_E//r_o)}{(R_E//r_o)+R_L}$$

The short circuit current gain can be obtained by setting $R_L=0$ in the above expression.

$$A_{is} = \frac{(\beta+1)R_{th}}{(R_{th}+R_{ib})}$$

β- Independent Common Collector amplifier: R_C included

The common collector amplifier circuit with R_C included is shown in the figure. This circuit has same expressions for $R_{in}, R_{out}, A_v, A_i$ as with the one without R_C. This is because the collector resistor R_C is not reflected in the output circuit.

To summarize, for this circuit:

$$R_{in} = R_1 // R_2 // (\beta+1)(R_E // r_o // R_L + r_e)$$

$$R_{out} = \left(r_e + \frac{R_{SIG} // R_1}{\beta+1}\right) // (R_C // r_o)$$

$$A_v = \frac{(R_E//r_o//R_L)}{(R_E//r_o//R_L+r_e)}$$

$$A_i = (\beta+1) \frac{R_{th}}{(R_{th}+R_{ib})} \frac{(R_L//r_o)}{(R_L//r_o)+R_L}$$

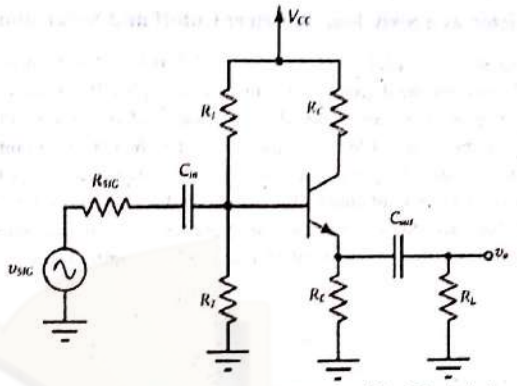
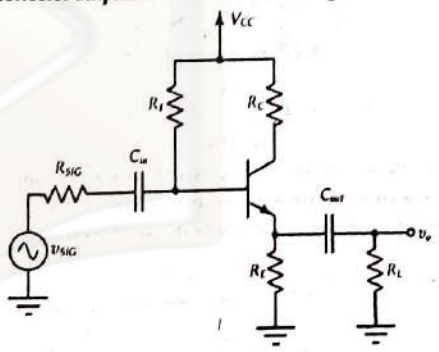


Figure: β- Independent Common Collector amplifier: R_C included

Common Collector amplifier with Emitter biasing:



This configuration has different expression for input impedance, output impedance and current gain. It can be easily obtained by replacing R_{th} by R_1.

$$R_{in} = R_1 // (\beta+1)(R_E // r_o // R_L + r_e)$$

$$R_{out} = \left(r_e + \frac{R_{SIG} // R_1}{\beta+1}\right) // (R_C // r_o)$$

$$A_v = \frac{(R_E//r_o//R_L)}{(R_E//r_o//R_L+r_e)}$$

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$$A_i = (\beta+1)\frac{R_1}{(R_1+R_{in})}\frac{(R_E//r_o)}{(R_E//r_o)+R_L}$$

Transistor as a Switch or inverter: Cutoff and Saturation

In this section we consider the operation of BJT as a switch. Switch is the basic element for the digital circuits. In every complex digital circuit like microprocessors there are hundreds of thousands of switch operating to implement the required logic. So the study of switch is important. The switch that we are going to discuss here is Resistor Transistor Logic (RTL) based switch. It is useful only in discrete circuit design. However in the IC design there are different families of digital circuits for implementing switch. You will study those in other course called "**Digital Logic**".

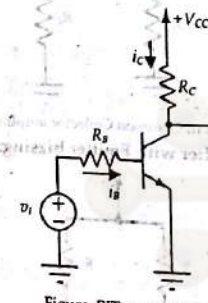

Figure: BJT as a switch

Figure shows the circuit for BJT as a switch. When the input v_I is below 0.5 V both the emitter base junction and collector base junction will be reversed biased. So the transistor will be in cut-off mode.

In cut-off mode,

$$i_B = 0$$
$$i_C = 0$$
$$v_C = V_{CC}$$

Thus, when input is low then output is high.

In order to turn on the transistor we should make the input v_I above 0.5 V. In fact for the appreciable current to flow input v_{BE} should be 0.7 V and v_I should be higher. The base current will be:

$$i_B = \frac{v_I - v_{BE}}{R_B}$$

And the collector current will be

$$i_C = \beta i_B$$

This relation applies only when the transistor is in **active mode**. Since the emitter base junction is forward biased, this will be applicable only when the collector base junction is reversed biased, that is as long as $v_{CB} < 0.4$ V. (This is because collector base junction can be forward biased with relatively low voltage > 0.4 V)

Then the output voltage v_C will be given by:

$$v_C = V_{CC} - i_C R_C$$

As v_I is increased, i_B will increase, as a result i_C will also increase. Then from the above equation output v_C will decrease. Eventually v_C decreases such that $v_{CB} > 0.4$ V. That is the collector base junction will be forward biased. Thus, both emitter base junction and collector base junction will be forward biased. So, the transistor will turn on or it will be in **saturation mode**. When v_I is high, i.e. equal to about V_{CC} then the output will be low. In terms of the output voltage v_C will be about 0.2 V. This is because for saturated BJT.

$$V_{CE\,SAT} = 0.2\ V$$

Thus, when input is high, the output will be low. In this way, the circuit in the figure acts as a digital switch or digital logic inverter.

These concepts will be even clearer if we utilize the **Voltage transfer characteristics** of the switch or inverter.

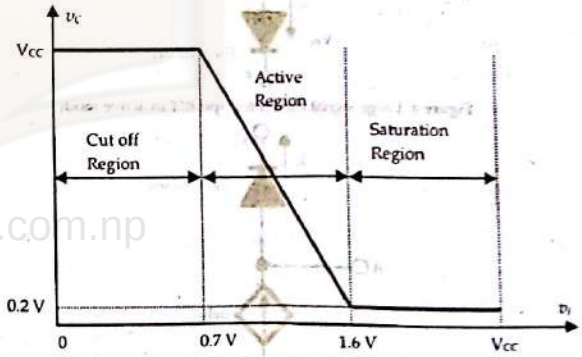

Figure: Voltage transfer characteristics of BJT switch or digital logic inverter.

Initially input is low, the transistor is in cut off region. Its emitter base junction and collector base junction are both reversed biased. In this case

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the output is high. Now if we keep on increasing input until 0.7 V, the output will be high because transistor is still in cut off mode.

If the input is made above 0.7 V, the base emitter junction will be forward biased. The collector base junction is still reversed biased. As a result transistor enter active region of operation. In this region transistor will have the negative gain given by the slope of the straight line. In this region output will decrease as we keep on increasing the input voltage.

If the input is made greater than 1.6 V, output will be 0.2 V i.e. output will be low. Now the collector base junction will be forward biased. Thus, now emitter base junction and collector base junction are both forward biased. The output will be low. At saturation the output of the transistor will be 0.2 V. It cannot fall below that. The output will be 0.2 V even if we increase the input all the way to V_{CC}.

A General Large Signal Model for the BJT: The Ebers-Moll Model

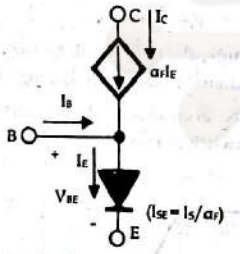

Figure a: Large signal model for npn BJT in active mode

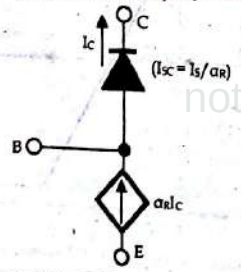

Figure b: Large signal model for npn BJT in reverse active mode.

When the npn transistor is operating in the active mode its large signal equivalent circuit model is as shown in figure a. Similarly, when the npn transistor is operating in reverse active mode then its large signal equivalent model is as shown in figure b.

Combining the above two models, we get Ebers- Moll Model. It is a generalized large signal model of BJT because it combines both the mode of operation namely active mode and reverse active mode.

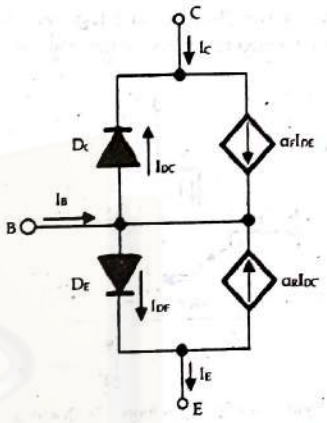

Figure: Ebers- Moll model for npn transistor.

From the above figure, we may write following equations:

$$I_E = I_{DE} - \alpha_R I_{DC} \quad \text{.... (i)}$$

$$I_C = -I_{DC} + \alpha_F I_{DE} \quad \text{....(ii)}$$

$$I_B = (1-\alpha_F) I_{DE} + (1-\alpha_R) I_{DC} \quad \text{....(iii)}$$

Diode current I_{DE} and I_{DC} can be expressed as:

$$I_{DE} = I_{SE}\left(e^{\frac{V_{BE}}{\eta V_T}} - 1\right) \quad \text{....(iv)}$$

$$I_{DC} = I_{SC}\left(e^{\frac{V_{BC}}{\eta V_T}} - 1\right) \quad \text{.....(v)}$$

Where,

I_{SE} and I_{SC} are the reverse saturation currents of base emitter diode D_E and collector base diode D_C respectively.

Substituting the values of I_{DE} and I_{DC} into the equation (i), (ii) and equation (iii), we get,

$$I_E = I_{SE}\left(e^{\frac{V_{BE}}{\eta V_T}} - 1\right) - \alpha_R I_{SC}\left(e^{\frac{V_{BC}}{\eta V_T}} - 1\right)$$

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$$I_C = -I_{SC}\left(e^{\frac{V_{BC}}{\eta V_T}}-1\right) + \alpha_F I_{SE}\left(e^{\frac{V_{BE}}{\eta V_T}}-1\right)$$

$$I_B = (1-\alpha_F) I_{SE}\left(e^{\frac{V_{BE}}{\eta V_T}}-1\right) + (1-\alpha_R) I_{SC}\left(e^{\frac{V_{BC}}{\eta V_T}}-1\right)$$

Darlington Pair (Super β transistor)

When the collector of two BJTs are tied together and emitter of one transistor is directly coupled to the base of the other we obtain Darlington pair transistor.

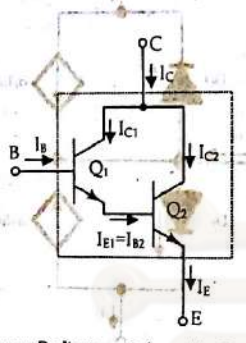

Figure: Darlington pair formed by Q_1 and Q_2

Figure shows the Darlington pair formed by two matched transistors Q_1 and Q_2. Let β_1 and β_2 be the current amplification factor for transistor Q_1 and Q_2 respectively.

$$I_{C1} = \beta_1 I_{B1}$$

$$\therefore I_{E1} = (\beta_1+1) I_{B1}$$

But $I_{E1} = I_{B2}$

$$I_{C2} = \beta_2 I_{B2}$$

$$\therefore I_{C2} = \beta_2(\beta_1+1) I_{B1}$$

Now,

$$I_C = I_{C1} + I_{C2}$$

$$= \beta_1 I_{B1} + \beta_2(\beta_1+1) I_{B1}$$

Since $I_{B1} = I_B$

$$\therefore I_C = \beta_1 I_B + \beta_2(\beta_1+1) I_B$$

$$\text{or, } \frac{I_C}{I_B} = \beta_1 + \beta_2 + \beta_1\beta_2$$

$$\text{or, } \beta_{DP} = \beta_1 + \beta_2 + \beta_1\beta_2$$

Where, β_{DP} is the current amplification factor of Darlington pair.

Since, $\beta_1\beta_2 >> \beta_1 + \beta_2$

$$\text{or, } \beta_{DP} = \beta_1\beta_2$$

Q_1 and Q_2 are generally matched transistors. Thus,

$$\beta_1 = \beta_2 = \beta$$

$$\therefore \beta_{DP} = \beta^2$$

Solved Numerical

1. Design a common base amplifier using β- independent dc biasing method. Use appropriate guideline to support your design. Given parameters are:

   V_{CC} = 24 V DC, I_C = 1mA and β = 200. Also find its voltage gain using its ac equivalent circuit. **2071 Shrawan**

Solution

Given: V_{CC} = 24 V DC, I_C = 1mA and β = 200

Since $V_{CC} \geq 15V$ we shall use guideline 2 (or 1/10th rule) for the design of bias circuit.

The dc biasing circuit is given below:

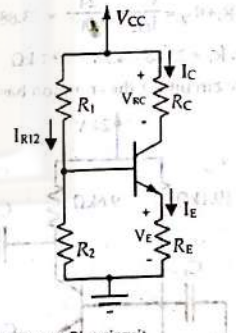

Figure: Bias circuit

$$I_B = \frac{I_C}{\beta} = \frac{1}{200} = 0.005 \text{ mA}$$

$$\therefore I_E = (\beta+1) I_B = 201 \times 0.005 = 1.005 \text{ mA}$$

Assuming V_T = 26 mV

$$g_m = \frac{I_C}{V_T} = \frac{1}{26} = 0.03846$$

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We know,

$$V_E = I_E R_E = 0.1\ V_{cc}$$

or, $1.005 \times R_E = 0.1 \times 24$

or, $R_E = \frac{2.4}{1.005}$

$\therefore R_E = 2.39\ k\Omega$

$$V_{RC} = I_C R_C = 0.4\ V_{cc}$$

or, $1 \times R_C = 0.4 \times 24$

$\therefore R_C = 9.6\ k\Omega$

Using Stiff voltage divider biasing,

$$R_2 = 0.01\ \beta\ R_E = 0.01 \times 200 \times 2.39 = 4.78\ k\Omega$$

For stiff biasing,

$$I_{R12} = I_E = 1.005\ mA$$

Where, I_{R12} is the series current flowing through R_1 and R_2.

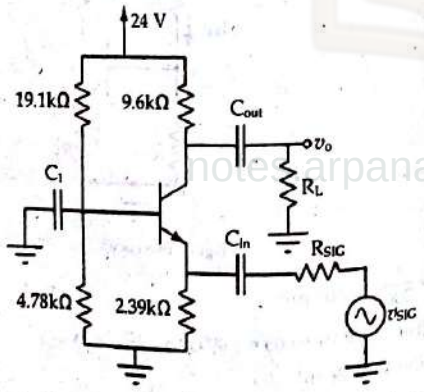
Therefore application of KVL results,

$$V_{CC} = I_{R12}(R_1 + R_2)$$

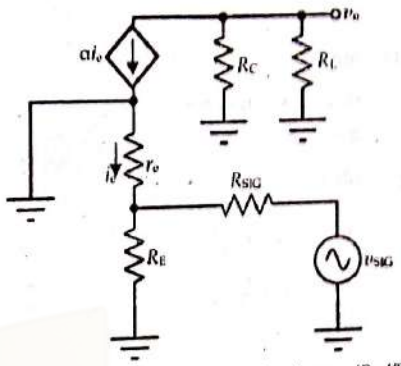
$$(R_1 + R_2) = \frac{V_{CC}}{I_{R12}} = \frac{24}{1.005} = 23.88\ k\Omega$$

$\therefore R_1 = 23.88 - 4.78 = 19.1\ k\Omega$

Thus, the complete circuit for the common base amplifier is:



The ac equivalent circuit of the above CB amplifier with transistor replaced by T- model is:



The voltage gain of the above circuit is given by: $A_v = g_m (R_C // R_L)$

Since the value of load resistance is unknown, we shall find the unloaded gain of the circuit. With load resistor opened,

Gain without load resistance is given by:

$$A_{vo} = g_m R_C$$

$$= 0.03846 \times 9.6 \times 1000 = 369.22$$

2. Design a voltage divider type dc biased common emitter amplifier to obtain β independent biasing. Use appropriate guidelines to support your design. Given parameters are $V_{CC} = 12$ V, $I_C = 2$ mA and $\beta = 150$. Also determine its voltage gain. **2070 Ashad**

Solution: Given parameters are $V_{CC} = 12$ V, $I_C = 2$ mA and $\beta = 150$

Since $V_{CC} = 12$ V, we shall use guideline 1 (1/3 rule) for the design of the bias circuit. The dc bias circuit is given below:

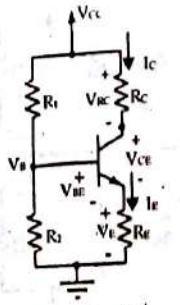


Figure: Bias Circuit

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$I_C = 2$ mA

$I_B = \frac{I_C}{\beta} = 0.01333$ mA

$I_E = (\beta+1)I_B = 151 \times 0.01333 = 2.0133$ mA

Assuming $V_T = 26$ mV,

$g_m = \frac{I_C}{V_T} = \frac{2}{26} = 0.07692$

We know, $V_{RC} = I_C R_C = \frac{V_{CC}}{3}$

or, $2 R_C = \frac{12}{3} = 4$

$R_C = 2\ k\Omega$

$V_B = \frac{V_{CC}}{3} = 4V$

or, $V_{BE} + V_E = 4$

or, $0.7 + 2.0133 \times R_E = 4$

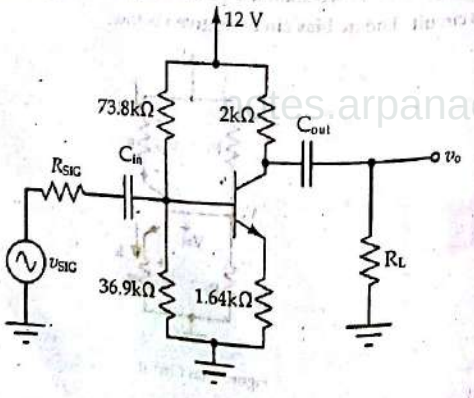
$R_E = \frac{4 - 0.7}{2.0133} = 1.64\ k\Omega$

Using firm biasing: $R_{th} = 0.1\beta R_E \simeq \frac{2}{3} R_2$

$R_{th} = 0.1\beta R_E = 0.1 \times 150 \times 1.64 = 24.6\ k\Omega$

$\therefore R_2 = \frac{3}{2} R_{th} = 36.9 k\Omega$

$\therefore R_1 = 2 \times R_2 = 73.8\ k\Omega$



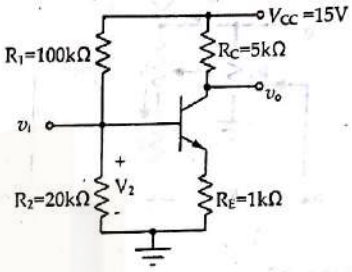
The complete circuit of common emitter amplifier is shown in the figure.

The unloaded voltage gain(i.e. Without R_L) of the above circuit is given as:

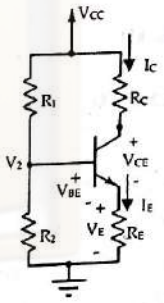
$A_{vo} = -g_m R_C$

$= -0.07692 \times 2 \times 1000 = -153.84$

3. Draw the ac equivalent for given circuit and find its input and output resistance. Assume $\beta = 100$ for the BJT. **2069 Chaitra**



Solution



From the approximate dc analysis of the circuit,

$V_2 = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{15 \times 20}{20 + 100} = 2.5V$

$I_E = \frac{V_2 - V_{BE}}{R_E} = \frac{2.5 - 0.7}{1} = 1.8$ mA

$I_C = \frac{\beta}{\beta+1} I_E = \frac{100}{101} \times 1.8 = 1.78$ mA

Assuming $V_T = 26$ mV

$g_m = \frac{I_C}{V_T} = \frac{1.78}{26} = 0.06846$

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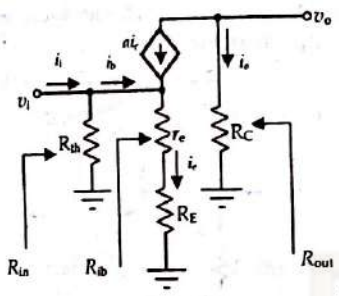
$$r_e = \frac{\alpha}{g_m} = \frac{\beta}{(\beta+1)g_m} = \frac{100}{101\times0.06846} = 14.46\ \Omega = 0.014462\ k\Omega$$

The ac equivalent circuit diagram with transistor replaced by T- model is as follows:



Where,

$R_{th} = R_1//R_2 = 16.67\ k\Omega$

Input Resistance:

$R_{ib} = (\beta+1)(r_e+R_E) = 102.460662\ k\Omega$

$R_{in} = R_{th}//R_{ib}$

$\therefore R_{in} = 16.67//102.460662 = 14.3373\ k\Omega$

Output Resistance:

$R_{out} = R_C = 5\ k\Omega$

Voltage Gain:

$$v_o = -i_o R_C$$

$$\text{or, } v_o = -\alpha i_e R_C$$

$$\text{or, } v_o = -\beta i_b R_C$$

$$\text{or, } v_o = -\beta i_i \frac{R_{th}}{R_{th}+R_{ib}} R_C$$

$$\text{or, } v_o = -\beta \frac{v_i}{R_{in}} \frac{R_{th}}{R_{th}+R_{ib}} R_C$$

$$\text{or, } \frac{v_o}{v_i} = -\beta \frac{1}{R_{in}} \frac{R_{th}}{R_{th}+R_{ib}} R_C$$

Substituting $R_{in} = R_{th}//R_{ib} = \frac{R_{th}R_{ib}}{R_{th}+R_{ib}}$

We get,

$$A_V = \frac{-\beta R_C}{R_{ib}}$$

$$\therefore A_V = \frac{-100\times5}{102.460662} = -4.8799$$

Current Gain:

$$i_o = -\alpha i_e$$

$$i_o = -\beta i_b$$

$$\text{or, } i_o = -\beta i_i \frac{R_{th}}{R_{th}+R_{ib}}$$

$$\text{or, } \frac{i_o}{i_i} = \frac{-\beta R_{th}}{R_{th}+R_{ib}}$$

$$\text{or, } A_i = \frac{-100\times16.67}{16.67+102.460662}$$

$$\therefore A_i = -13.993$$

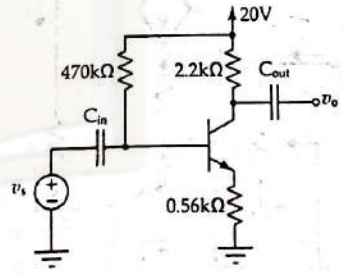
4. For the figure shown below with $\beta = 120$, find **2068 Chaitra**

a. Input impedance

b. Output Impedance

c. Voltage gain

d. Current Gain



Solution

The dc equivalent circuit is shown in the figure.

Applying KVL in input loop,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B+(\beta+1)R_E} = \frac{20-0.7}{470+121\times0.56} = 0.0358896$$

$I_C = \beta I_B = 120\times0.0358896 = 4.3067\ mA$

Assuming $V_T = 26\ mV$

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$$g_m = \frac{I_C}{V_T} = \frac{4.3076}{26} = 0.165644$$

$$r_e = \frac{\alpha}{g_m} = \frac{\beta}{(\beta+1)g_m} = \frac{120}{121\times0.165644} = 5.9871\ \Omega = 0.00598715\ k\Omega$$

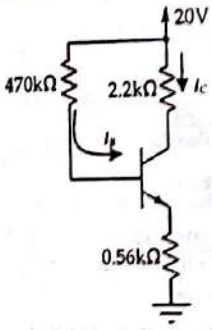
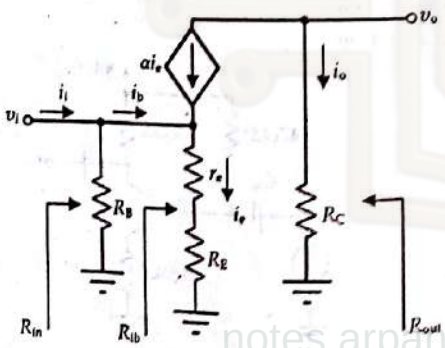


Figure: DC equivalent circuit

The ac equivalent circuit with the transistor replaced by T- model is given as:



$R_{ib} = (\beta+1)(r_e + R_E) = 121\times(0.00598596+0.56) = 68.48\ k\Omega$

Input Impedance:

$$R_{in} = R_B // R_{ib} = 470 // 68.48 = 59.77 k\Omega$$

Output Impedance:

$$R_{out} = R_C = 2.2\ k\Omega$$

Voltage Gain:

$$v_o = -i_o R_C$$

$$\text{or, } v_o = -\alpha i_e R_C$$

$$\text{or, } v_o = -\beta i_b R_C$$

$$\text{or, } v_o = -\beta i_i \frac{R_B}{R_B + R_{ib}} R_C$$

$$\text{or, } v_o = -\beta \frac{v_i}{R_{in}} \frac{R_B}{R_B + R_{ib}} R_C$$

$$\text{or, } \frac{v_o}{v_i} = -\beta \frac{1}{R_{in}} \frac{R_B}{R_B + R_{ib}} R_C$$

Substituting $R_{in} = R_B // R_{ib} = \frac{R_B R_{ib}}{R_B + R_{ib}}$

We get,

$$A_V = \frac{-\beta R_C}{R_{ib}}$$

$$\therefore A_V = \frac{-120\times2.2}{68.48} = -3.85$$

Current Gain:

$$i_o = -\alpha i_e$$

$$\text{or, } i_o = -\beta i_i \frac{R_B}{R_B + R_{ib}}$$

$$\text{or, } \frac{i_o}{i_i} = \frac{-\beta R_B}{R_B + R_{ib}}$$

$$\text{or, } A_i = \frac{-120\times470}{470+68.48}$$

$$\therefore A_i = -104.74$$

5. Design β- independent type dc biased common collector amplifier and find its current gain and input resistance. 2068 Baisakh

Given parameters are:

$V_{CC} = 20V$, $I_C = 2mA$, $\beta = 100$ and use firm biasing method.

Solution:

Given: $V_{CC} = 20$ V DC, $I_C = 2mA$ and $\beta = 100$

$$\therefore \alpha = \frac{\beta}{\beta+1} = 0.99$$

Since $V_{CC} \geq 15V$ we shall use guideline 2 (or 1/10th rule) for the design of bias circuit.

The dc biasing circuit is shown in the figure.

From the figure,

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$I_C = 2$ mA

$I_B = \frac{I_C}{\beta} = \frac{2}{100} = 0.02$ mA

$I_E = (\beta+1)I_B = 101 \times 0.02 = 2.02$ mA

Assuming $V_T = 26$ mV

$g_m = \frac{I_C}{V_T} = \frac{2}{26} = 0.07692$ and $r_e = \frac{\alpha}{g_m} = \frac{0.99}{0.07692} = 12.87\Omega = 0.01287$ kΩ

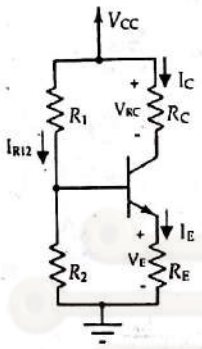


Figure: Bias circuit

We know,

$V_E = I_E R_E = 0.1\ V_{CC}$

or, $2.02 \times R_E = 0.1 \times 20$

or, $R_E = \frac{2.4}{2.02}$

$\therefore R_E = 0.99 \approx 1$kΩ

$V_{RC} = I_C R_C = 0.4\ V_{CC}$

or, $2 \times R_C = 0.4 \times 20$

$\therefore R_C = 4$ kΩ

Using firm voltage divider biasing,

$R_2 = 0.1\ \beta R_E = 0.1 \times 100 \times 1 = 10$ kΩ

For firm biasing, $I_{R12} = 0.1 I_E = 0.1 \times 2.02$ mA $= 0.202$ mA

Where, I_{R12} is the series current flowing through R_1 and R_2.

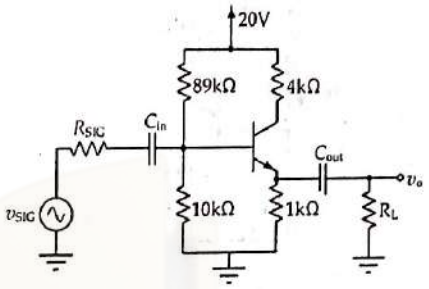
Therefore application of KVL results,

$V_{CC} = I_{R12}(R_1+R_2)$

$(R_1+R_2) = \frac{V_{CC}}{0.1 \times I_{R12}} = \frac{20}{0.202} = 99$kΩ

$\therefore R_1 = 99 - 10 = 89$kΩ

The complete circuit diagram for the common collector amplifier is given below:



Considering unloaded case, i.e. $R_L = \infty$

$R_{th} = R_1 // R_2 = 89//10 = 8.99$ KΩ

$R_{ib} = (\beta+1)(R_L + r_e) = 101 \times (1+0.01287) = 102.29$

$\therefore R_{in} = R_{ib}//R_{th} = 8.99//102.29 = 8.26$ kΩ

The short circuit current gain is given as:

$A_{is} = \frac{(\beta+1)R_{th}}{(R_{th}+R_{ib})} = \frac{101 \times 8.99}{8.99+102.29} = 8.15$

6. Design voltage divider bias(common emitter configuration) to get $I_{CQ} = 1.5$ mA. Assume power supply voltage $V_{CC} = 15$ V and beta of transistor is 110. 2071 Chaitra

Solution

Given parameters are $V_{CC} = 15$ V, $I_C = 1.5$ mA and $\beta = 110$

Since $V_{CC} = 15$ V, we shall use guideline 2 ($1/10^{th}$ rule) for the design of the bias circuit.

The dc bias circuit is shown in the figure.

From the dc equivalent circuit,

$I_C = 1.5$ mA

$I_B = \frac{I_C}{\beta} = 0.01364$

$I_E = (\beta+1)I_B = 111 \times 0.01364 = 1.51$ mA

Assuming $V_T = 26$ mV,

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$g_m = \frac{I_C}{V_T} = \frac{1.5}{26} = 0.05769$

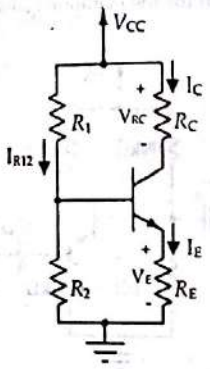


Figure: Bias circuit

From 1/10 rule:

$V_E = I_E R_E = \frac{V_{CC}}{10}$

or, $1.51 \times R_E = \frac{15}{10} = 1.5$

$R_E = 0.99\ k\Omega \approx 1k\Omega$

$V_{RC} = \frac{4}{10} V_{CC} = 0.4 \times 15$

or, $I_C R_C = 6$

or, $1.5 \times R_C = 6$

$R_C = 4\ k\Omega$

Using stiff voltage divider biasing,

$$R_2 = 0.01\ \beta\ R_E = 0.01 \times 110 \times 1 = 1.1\ k\Omega$$

For stiff biasing, $I_{R12} = I_E = 1.51$ mA

Where, I_{R12} is the series current flowing through R_1 and R_2.

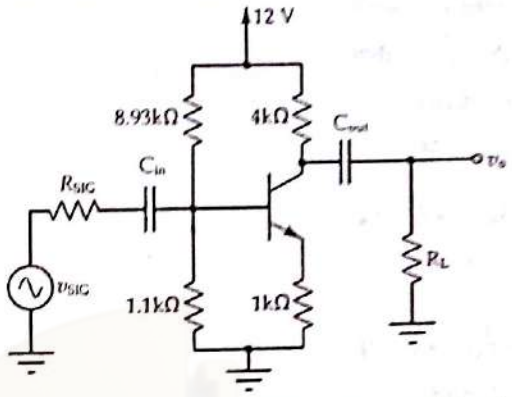
Therefore application of KVL results,

$$V_{CC} = I_{R12}(R_1 + R_2)$$

$$(R_1 + R_2) = \frac{V_{CC}}{I_{R12}} = \frac{15}{1.51} = 9.93\ k\Omega$$

$$R_1 = 9.93 - 1.1 = 8.93\ k\Omega$$

The complete circuit of common emitter amplifier is given below:



7. Design β independent type dc biased common emitter amplifier circuit. Given parameters are: V_{CC} = 24V DC, I_C = 1.5 mA, β= 150 and input impedance is comparatively large. Use appropriate guideline to support your design. Also determine its voltage gain. **2069 Ashad**

Solution

Given parameters are V_{CC} = 24 V, I_C = 1.5 mA and β = 150

Since V_{CC} = 24 V, we shall use guideline 2 (1/10 rule) for the design of the bias circuit.

The dc bias circuit is given below:

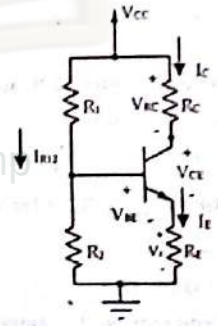


Figure: Bias Circuit

I_C = 1.5 mA

$I_B = \frac{I_C}{\beta}$ = 0.01 mA

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$\therefore I_E = (\beta+1)I_B = 151 \times 0.01 = 1.51$ mA

Assuming $V_T = 26$ mV,

$$g_m = \frac{I_C}{V_T} = \frac{1.5}{26} = 0.05769$$

From 1/10 rule:

$$V_E = I_E R_E = \frac{V_{CC}}{10}$$

$$\text{or, } 1.51 \times R_E = \frac{24}{10} = 2.4$$

$\therefore R_E = 1.59$ kΩ

$$V_{RC} = \frac{4}{10} V_{CC} = 0.4 \times 24$$

or, $I_C R_C = 9.6$

or, $1.5 \times R_C = 9.6$

$R_C = 6.4$ kΩ

Since the input impedance is relatively large, we shall use firm biasing.

Using firm voltage divider biasing,

$R_2 = R_{th} = 0.1\beta R_E$

$R_2 = 0.1\beta R_E = 0.1 \times 150 \times 1.59 = 23.85$ kΩ

For firm biasing,

$I_{R12} = 0.1 I_E$

$\therefore I_{R12} = 0.151$ mA

Where,

I_{R12} is the series current flowing through R_1 and R_2.

Therefore application of KVL results,

$$V_{CC} = I_{R12}(R_1+R_2)$$

$$(R_1+R_2) = \frac{V_{CC}}{I_{R12}} = \frac{24}{0.151} = 158.9 \text{ kΩ}$$

$$\text{or, } R_1 = 158.9 - 23.85$$

$\therefore R_1 = 135.05$ kΩ

The complete circuit of common emitter amplifier is shown in the figure.

The unloaded voltage gain(i.e. Without R_L) of the above circuit is given as:

$$A_{vo} = -g_m R_C$$

$$= -0.05769 \times 6.4 \times 1000 = -369.216$$

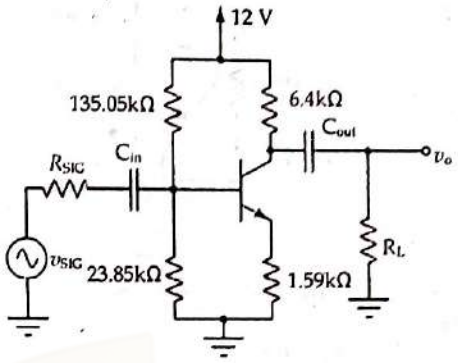
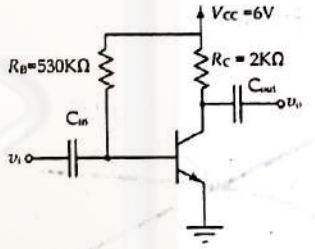


Figure: Circuit of designed common emitter amplifier

8. For the figure shown below β=100, determine operating point and draw the DC load line



Solution

The dc equivalent circuit is shown in the figure.

For operating point,

Applying KVL at input loop,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\therefore I_B = \frac{V_{CC}-V_{BE}}{R_B} = \frac{6-0.7}{530} = 0.01mA$$

$$\therefore I_C = \beta I_B = 100 \times 0.01 = 1mA$$

Applying KVL at output loop,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\therefore V_{CE} = V_{CC} - I_C R_C = 6 - 1 \times 2 = 4V$$

$\therefore$ Operating point (V_{CEQ}, I_{CQ})=(4V, 1mA)

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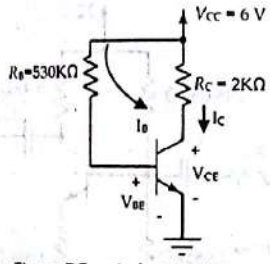


Figure: DC equivalent circuit

For load line,

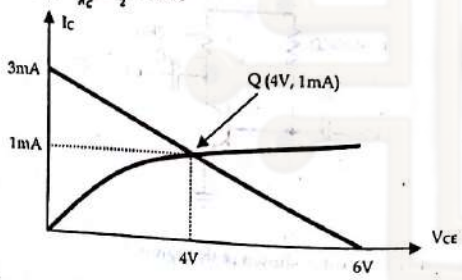
$$V_{CE} = V_{cc} - I_C R_C$$

When I_C=0,

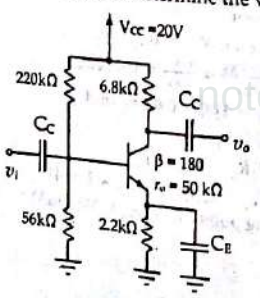
$$V_{CE}=V_{cc}=6V$$

When V_{CE}= 0,

$$I_C = \frac{V_{CC}}{R_C} = \frac{6}{2} = 3mA$$

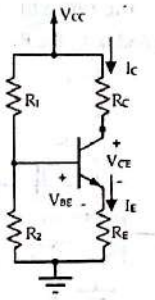


9. For the network shown below determine the value of R_{in}, R_{out} and A_V.



Solution:

The dc equivalent circuit is:



Using Thevenin's method,

$R_{th} = R_1 // R_2 = 220 // 56 = 44.64$ V

$$V_{th} = \frac{V_{CC}R_2}{R_1+R_2} = \frac{20\times56}{220+56} = 4.058 \text{ V}$$

$$I_E = \frac{V_{th} - V_{BE}}{\frac{R_{th}}{\beta+1}} = \frac{4.058 - 0.7}{2.2+\frac{44.64}{181}} = 1.37 \text{ mA}$$

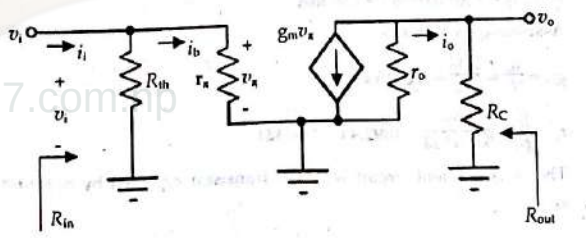
$$I_C = \alpha I_E = \frac{\beta}{\beta+1} I_E = \frac{180}{181} \times 1.37 = 1.36 \text{ mA}$$

Assuming, V_T = 26 mV

$$g_m = \frac{I_C}{V_T} = \frac{1.36}{26} = 0.052307$$

$$r_\pi = \frac{\beta}{g_m} = \frac{180}{0.052307} = 3.44 \text{ k}\Omega$$

The ac equivalent circuit with the transistor replaced by hybrid-π model is:



Input impedance: $R_{in} = R_{th} // r_\pi = 44.64 // 3.44 = 3.19 \text{ k}\Omega$

Output Impedance: $R_{out} = r_o // R_C = 50 // 6.8 = 5.98 \text{ k}\Omega$

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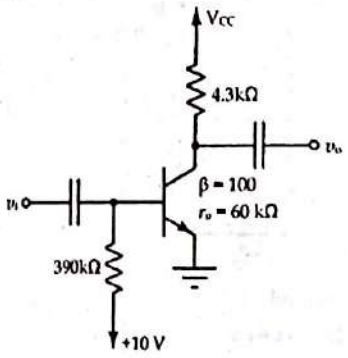
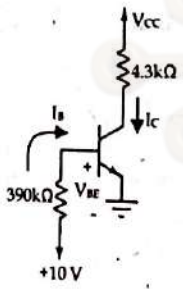
Voltage Gain: $A_V = -g_m(r_o // R_C) = -0.052307 \times 6.01 \times 1000 = -312.79$

10. For the circuit shown below find I_B, I_C, R_{in}, R_{out}, and A_V.



Solution

The dc equivalent circuit is:



$$I_B = \frac{10 - V_{BE}}{390k\Omega} = \frac{10 - 0.7}{390\,k\Omega} = 0.02385\ mA = 23.85\ \mu A$$

$I_C = \beta I_B = 100 \times 0.02385 = 2.38\ mA$

Assuming $V_T = 26\ mV$

$$g_m = \frac{I_C}{V_T} = \frac{2.38}{26} = 0.091538$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{0.091538} = 1092.44 = 1.09\ k\Omega$$

The ac equivalent circuit with the transistor replaced by hybrid-π model is:

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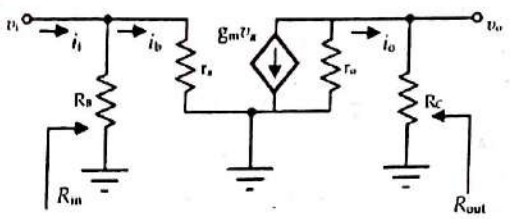
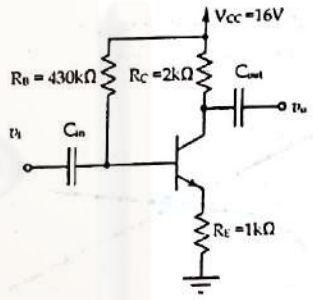


Input Impedance: $R_{in} = R_B // r_\pi = 390k\Omega // 1.09k\Omega = 1.08\ k\Omega$

Output Impedance: $R_{out} = r_o // R_C = 60k\Omega // 4.3k\Omega = 4.012\ k\Omega$

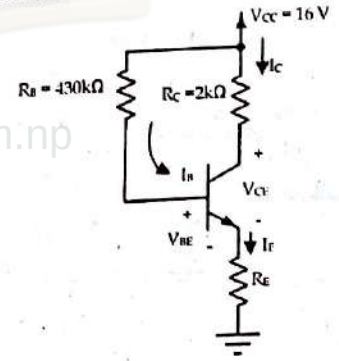
Voltage Gain: $A_V = -g_m(r_o // R_C) = -0.091538 \times (60 // 4.3) \times 1000 = -367.29$

11. For the circuit shown below β=75, determine Q-point and draw the load line.



Solution

The dc equivalent circuit is:



For load line,

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

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When I_C=0,

$$V_{CE}=V_{CC}=16V$$

When V_{CE}=0,

$$I_C = \frac{V_{CC}}{R_C+R_E} = \frac{16}{3} = 5.33mA$$

For operating point,

Applying KVL at input loop,

$$V_{CC} - I_BR_B - V_{BE} - I_ER_E = 0$$

$$or, V_{CC} - V_{BE} = I_BR_B + I_ER_E$$

$$or, V_{CC} - V_{BE} = I_BR_B + (\beta+1)I_BR_E$$

$$\therefore I_B = \frac{V_{CC}-V_{BE}}{R_B+(\beta+1)R_E} = \frac{16-0.7}{430+76\times1} = 0.030mA$$

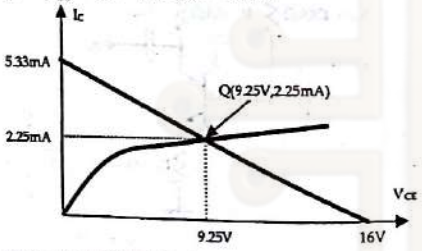
$$\therefore I_C = \beta I_B = 75 \times 0.030 = 2.25mA$$

Applying KVL at output loop,

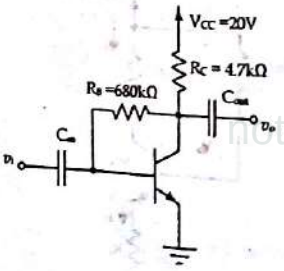
$$V_{CC} - I_CR_C - V_{CE} - I_ER_E = 0$$

$$or, V_{CE} = V_{CC} - I_C(R_C+R_E) \quad (\because I_C \approx I_E)$$

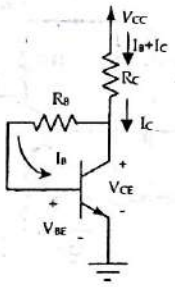
$$\therefore V_{CE} = 16 - 2.25 \times 3 = 9.25V$$



12. For the given circuit if β=120, determine Q-point and draw the load line.



Solution



Applying KVL in input loop,

$$V_{CC} - (I_B+I_C)R_C - I_BR_B - V_{BE} = 0$$

$$or, V_{CC} - I_ER_C - I_BR_B - V_{BE} = 0 (\because I_B + I_C = I_E)$$

$$or, V_{CC} - V_{BE} = I_ER_C + I_BR_B$$

$$or, V_{CC} - V_{BE} = (\beta+1)I_BR_C + I_BR_B \quad (\because I_E = (\beta+1)I_B)$$

$$\therefore I_B = \frac{V_{CC}-V_{BE}}{(\beta+1)R_C+R_B} = \frac{20-0.7}{121\times4.7+680} = 15.45\mu A$$

$$\therefore I_C = \beta I_B = 120 \times 15.45\mu A = 1.854mA$$

Applying KVL at output loop,

$$V_{CC} - (I_B+I_C)R_C - V_{CE} = 0$$

$$\therefore V_{CE} = V_{CC} - I_CR_C \quad (\because (I_B+I_C) \approx I_C$$

$$V_{CE} = 20 - 1.854 \times 4.7 = 11.29V$$

For load line,

$$V_{CE} = V_{CC} - (I_C+I_B)R_C$$

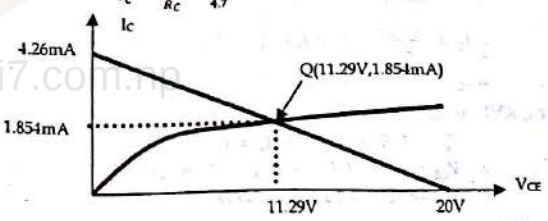
$$V_{CE} = V_{CC} - I_CR_C \ (\because I_C + I_B \approx I_C)$$

When I_C=0,

$$V_{CE}=V_{CC}=20V$$

When V_{CE}=0,

$$I_C = \frac{V_{CC}}{R_C} = \frac{20}{4.7} = 4.26mA$$



13. Determine the operating point and draw the DC load line and for the circuit given below. Given β=75. Use approximate method.

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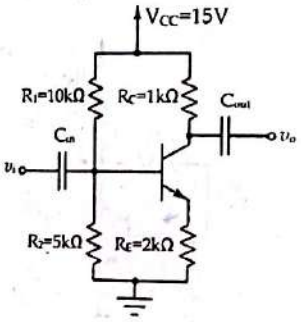
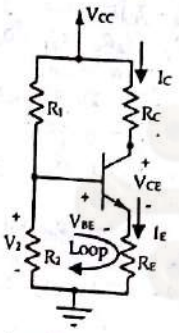


Solution

The dc equivalent circuit is:



The voltage across resistor R_2 is given by:

$$V_2 = \frac{V_{CC} \times R_2}{(R_1+R_2)} = \frac{15\times5}{10+5} = 5V$$

Applying KVL in the loop indicated,

$$V_2 - V_{BE} - I_E R_E = 0$$

$$\therefore I_E = \frac{V_2 - V_{BE}}{R_E}$$

$$\therefore I_C = \frac{V_2 - V_{BE}}{R_E} \ (\because I_C \approx I_E)$$

$$\therefore I_C = \frac{5-0.7}{2} = 2.15mA$$

Applying KVL in output loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\therefore V_{CE} = V_{CC} - I_C(R_C + R_E) \ (\because I_C \approx I_E)$$

$$V_{CE} = 15 - 2.15 \times 3 = 8.55V$$

For load line,

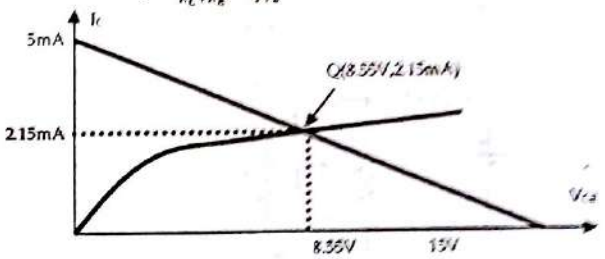
$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

When I_C=0,

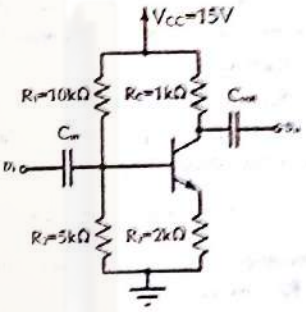
$$V_{CE} = V_{CC} = 15V$$

When V_{CE}=0,

$$I_C = \frac{V_{CC}}{R_C+R_E} = \frac{15}{1+2} = 5mA$$

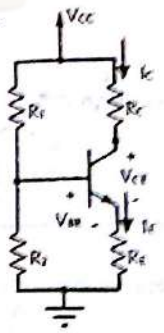


14. Determine the operating point draw the DC load line and for the circuit given below. Given that β=75. Use actual method (Thevenin's method)



Solution

The dc equivalent circuit is:



The dc equivalent circuit can be re-drawn as:

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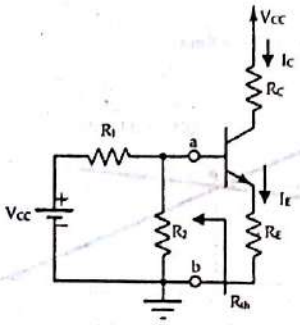


To Thevenize the left portion of circuit looking from terminals a and b, we have to calculate R_{th} and V_{th}.

In order to find R_{th} we look from terminal a and b and short voltage source. Here resistors R_1 and R_2 appear in parallel.

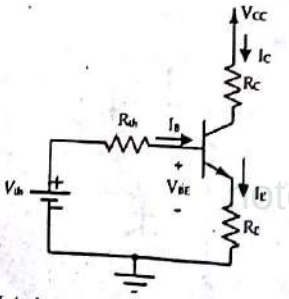
$$R_{th} = R_1 // R_2$$

$$\therefore R_{th} = 10 // 5 = 3.33\ k\Omega$$

In order to find Thevenin's voltage V_{th}, we have to find open circuit voltage. Here the voltage drop across R_2 is V_{th}.

$$\therefore V_{th} = \frac{V_{CC}R_2}{R_1+R_2} = \frac{15\times5}{10+5} = 5V$$

Replacing left portion of the circuit with Thevenin's equivalent circuit (i.e. V_{th} in series with R_{th}), we get,



Applying KVL in base emitter loop,
We get,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$\therefore I_B = \frac{V_{th}-V_{BE}}{R_{th}+R_E(\beta+1)} = \frac{5-0.7}{3.33+76\times2} = 0.02768mA$$

$$\therefore I_C = \beta I_B = 75 \times 0.02768 = 2.15mA$$

Applying KVL in output loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\therefore V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$\therefore V_{CE} = 15 - 2.1 \times 3 = 8.7V$$

Thus, operating point (V_{CEQ}, I_{CQ}) = (8.7V, 2.1 mA)

For load line,

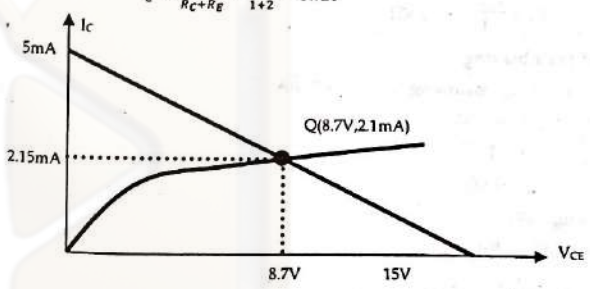
$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

When $I_C = 0$,

$$V_{CE} = V_{CC} = 15V$$

When $V_{CE} = 0$,

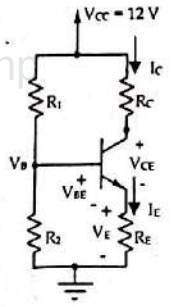
$$I_C = \frac{V_{CC}}{R_C+R_E} = \frac{15}{1+2} = 5mA$$



15. Design a voltage divider bias circuit.

Given: $V_{CC} = 12V$, $I_E = 1$ mA and $\beta = 100$

Solution



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Given: $V_{CC} = 12V$, $I_E = 1$ mA and $\beta = 100$

Assuming $I_C \approx I_E = 1$ mA

Since $V_{CC} = 12$ V, we shall use guideline 1.

$$V_{RC} = \frac{V_{CC}}{3} = \frac{12}{3} = 4$$

or, $I_C R_C = 4$

$\therefore R_C = 4\ k\Omega$

$$V_B = \frac{V_{CC}}{3} = \frac{12}{3} = 4$$

or, $V_{BE} + V_E = 4$

or, $0.7 + I_E R_E = 4$

$$\therefore R_E = \frac{4-0.7}{1} = 3.3\ k\Omega$$

Using Stiff biasing:

For stiff biasing assuming $I_{R12} = I_E = 1$ mA

$V_B = I_{R12} R_2$

or, $4 = 1 \times R_2$

$\therefore R_2 = 4\ k\Omega$

Applying KVL,

$V_{CC} = I_{R12}(R_1 + R_2)$

$$\therefore (R_1 + R_2) = \frac{V_{CC}}{I_{R12}} = \frac{12}{1} = 12$$

$\therefore R_1 = 12 - 4 = 8\ k\Omega$

Using Firm biasing:

For firm biasing assuming $I_{R12} = 0.1 I_E = 0.1$ mA

$V_B = I_{R12} R_2$

or, $4 = 0.1 \times R_2$

$\therefore R_2 = 40\ k\Omega$

Applying KVL,

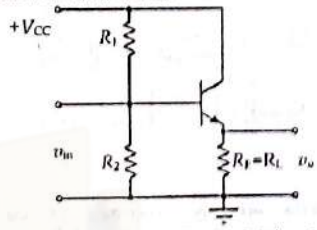
$V_{CC} = I_{R12}(R_1 + R_2)$

$$\therefore (R_1 + R_2) = \frac{V_{CC}}{I_{R12}} = \frac{12}{0.1} = 120$$

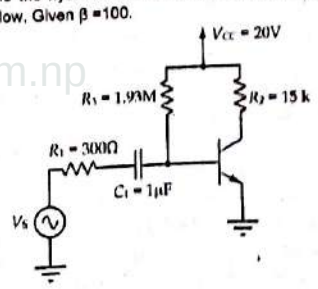
$\therefore R_1 = 120 - 40 = 80\ k\Omega$

EXERCISE

1. Design voltage divider bias (common emitter configuration) to get $I_{CQ} = 1.6$ mA. Assume power supply voltage $V_{CC} = 15$ V and beta of the transistor is 110. **2071 Chaitra**
2. Derive expression for A_v, R_{in} and R_{out}. **2071 Chaitra**



3. Describe in brief the operation of BJT as switch in cut off and saturation region. **2071 Shrawan**
4. Draw Ebers Moll model, low frequency π model and simple T- model for BJT. **2071 Shrawan**
5. Why common collector amplifier is called emitter follower? Draw the common collector transistor amplifier circuit and find its input impedance, output impedance and voltage gain. **2070 Chaitra**
6. Draw and describe Ebers Moll model for BJT. **2070 Chaitra**
7. Describe in brief the operation of BJT as switch in cut off and saturation region. **2070 Chaitra**
8. Draw the small signal model for common collector amplifier and find its input resistance and voltage gain. **2070 Ashad**
9. Define transconductance. Derive g_m for BJT. **2069 Chaitra**
10. Describe in brief the operation of BJT as a switch. **2069 Chaitra**
11. Describe the operation of BJT as a switch. **2069 Ashad**
12. Draw ac equivalent circuit of common collector amplifier. Find its input and output resistance. **2068 Chaitra**
13. Determine the hybrid π model for the low frequency analysis of the figure given below. Given $\beta = 100$. **2068 Shrawan**



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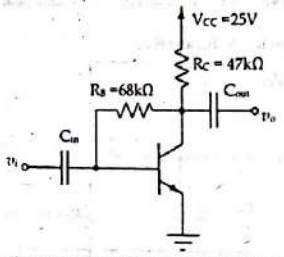
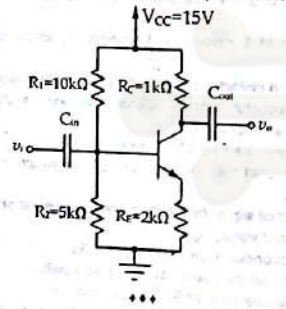
14. Derive an expression to find output resistance for emitter unbypassed common emitter amplifier circuit. 2068 Baisakh
15. Draw Ebers Moll model and ac equivalent T- Model for BJT. 2068 Baisakh
16. For the given circuit if β=100, determine Q-point and draw the load line.



17. Determine the operating point and draw the DC load line and for the circuit given below. Given β=75. Use exact method(Thevenins method).



Chapter 3

Field Effect Transistors

Introduction

Field effect transistors are other types of three terminal non-linear semiconductor devices which can perform the functions like amplification, switching etc as done by BJTs. Though the terminal characteristics are similar, working mechanism and construction of FET and BJT are completely different. Unlike BJTs, field effect transistors are **unipolar** devices which mean conduction is done by only one kind of charge carriers i.e. either by holes or by electrons. In field effect transistor, the output current is controlled by the field or voltage applied to input terminals. So, they are called field effect transistors as field affects the output. There are two types of FET namely JFET and MOSFET.

Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

Metal oxide semiconductor field effect transistor (MOSFET) is a type of field effect transistor. In this type of FET the oxide layer insulates the gate electrode from the device body and hence it is called **metal oxide semiconductor field effect transistor.**

MOSFET has very high input impedance than JFET and BJT. Due to the advent of MOSFET, JFET has become virtually obsolete. This is because MOSFET can perform all the functions performed by JFET with much more improvement. But the question is why BJTs are still used. The answer is because during the process of designing circuit there arises the conditions where the use of BJT along with MOSFET results best circuit for particular application. As compared to BJT, MOSFET can be made in small silicon area, their manufacturing process is relatively simple, cheap and they require low power to operate. MOSFET has specially dominated

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BJT in the design of analog and digital integrated circuits. This is because the scale of integration of MOSFET is more than BJT for same area of silicon chip. Semiconductor manufacturers believe that in the near future MOSFET will completely replace BJT for all kind of applications.

Types of MOSFET:

a. **On the basis of type of channel present:**

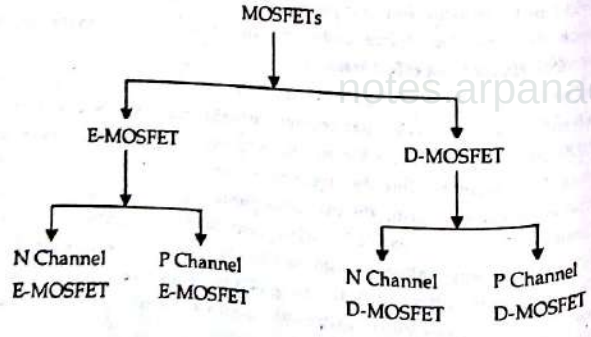
i. **n-channel MOSFET:** In this type of MOSFET the channel present is of n-type material which connects source and drain region for conduction. Channel is developed in substrate or body region of p type.

ii. **p-channel MOSFET:** In this type of MOSFET the channel present is of p-type material which connects source and drain region for conduction. Channel is developed in substrate or body region of n type.

b. **On the basis of presence or absence of channel**

i. **Enhancement MOSFET (E-MOSFET):** In this type of MOSFET the channel is initially absent. It has to be created by applying gate voltage. This kind of MOSFET operates in Enhancement mode only. Again E-MOSFET can be of two types namely n-channel E-MOSFET and p-channel E-MOSFET.

ii. **Depletion MOSFET (D-MOSFET):** In this type of MOSFET the channel is already present. The channel is already created by ion implantation. This type of MOSFET can operate in both enhancement mode and depletion mode. This is because channel can be enhanced and depleted by applying proper gate voltage. D-MOSFET is also of two types namely n-channel D-MOSFET and p-channel D-MOSFET.



Structure and Physical Operation of Enhancement Type MOSFET

Structure of N- channel E-MOSFET:

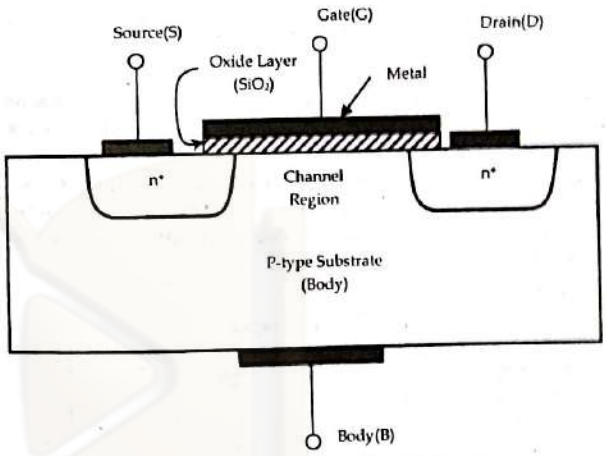


Fig: Physical structure of n-channel E-MOSFET

Figure shows cross-section of n-channel E-MOSFET. Whole transistor is fabricated on p-type substrate. It is on this substrate other structures are developed. Two heavily doped n^+ and n^+ regions are developed on the surface of substrate and are called **source** and **drain** respectively. MOSFET is also a symmetrical device. A thin layer of silicon dioxide is developed on the surface of the substrate. Here oxide layer acts as an excellent insulator and separates the gate electrode from substrate (or body).

Thus four terminals arise from source, drain, gate and body respectively. These terminals are marked as S, D, G and B. Generally the body terminal and gate terminal is shorted thus making MOSFET three terminal device with terminals S, G and D.

Note:

In IC MOSFET structure the terminal contacts are formed from n^+ source and n^+ drain regions. This is because if there were n source and n drain regions and if contacts were taken from them, the metal semiconductor

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junction will have rectifying property and will act like Schottky diode. And eventually it will disturb the normal operation of MOSFET. But with n⁺ region it will form simple ohmic contacts. Therefore, to eliminate the problem, n⁺ region is used in source and drain regions instead of n region in fabricating integrated circuit MOSFET.

Working of N channel E MOSFET

When v_{GS} is zero:

When v_{GS}=0 i.e. when no gate voltage is applied, two back to back pn junction diodes exist in series between drain and source. One diode is formed by the p-type substrate with n⁺ source region and other diode is formed by p-type substrate with n⁺ drain region. These two back to back diodes will prevent the flow of current between source and drain. In fact the region between source and drain has very high resistance which is of the order of 10^{12} kΩ.

When v_{GS} is positive:

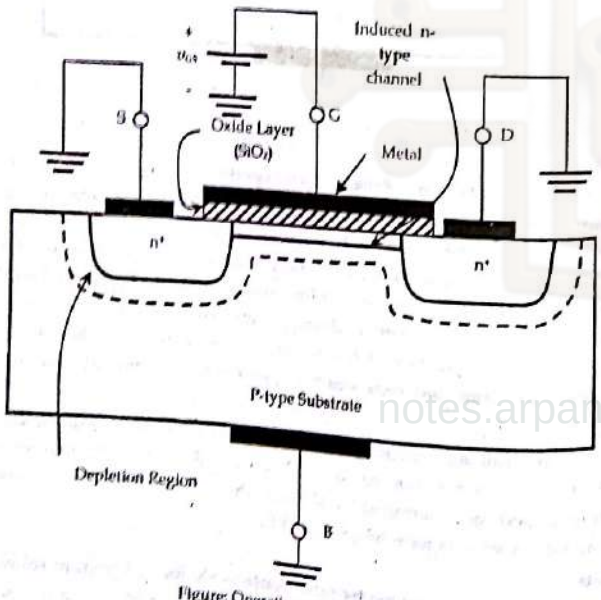
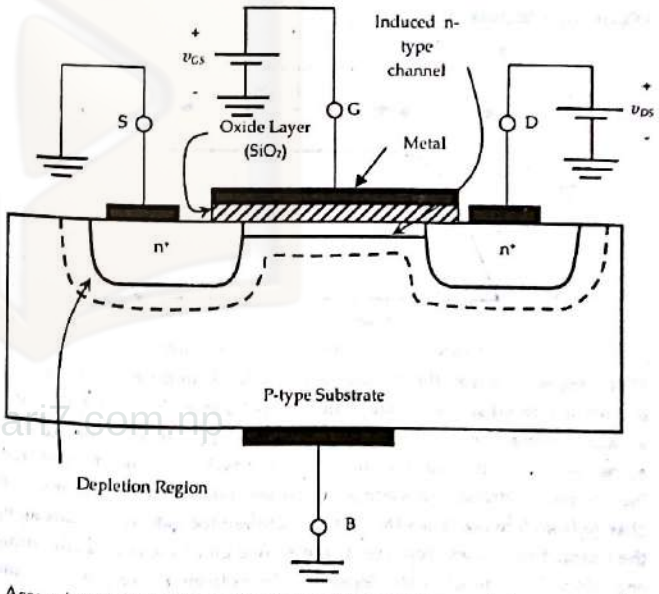


Figure: Operation when v_{GS} is positive

When v_{GS} is positive, at first instance, the holes in the substrate near the gate electrode are repelled and are pushed downward in the substrate leaving behind a carrier depletion region. The depletion region is populated by bound negative charge associated with acceptor atoms .These charges are uncovered because the neutralizing holes are pushed downwards in the substrate region. The positive gate voltage attracts the electrons from drain and source as well. Thus under gate electrode n-region is formed which connects the source and drain. This region is called channel. Due to positive voltage, a channel is enhanced and hence the name. The value of v_{GS} at which the channel is induced is called Threshold Voltage and it is designated as V_t. Once when the channel is created the current can flow through channel which is controlled by drain to source voltage v_{DS}. Thus by varying the voltage v_{DS} the drain current i_D flowing through the channel can be varied. Being symmetrical device, the source current of MOSFET i_S is equal to drain current i_D.

When v_{DS} is small and positive:



Assuming that the channel is induced by applying gate to source voltage v_{GS} greater than the threshold voltage V_t, let us consider the condition

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when a voltage v_{DS} is applied between the drain and source and the magnitude of v_{DS} is small, say 0.1 V or 0.2V, it causes current i_D to flow through the channel. The magnitude of current depends upon the excess gate voltage (v_{GS}-V_t).

When the v_{DS} is small enough, current i_D exhibits linear relationship with the drain to source voltage v_{DS} and with excess gate voltage (v_{GS}-V_t).

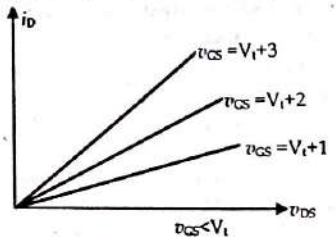

Figure: i_D Vs. v_{DS} curve for v_{DS} small

When v_{DS} is increased:

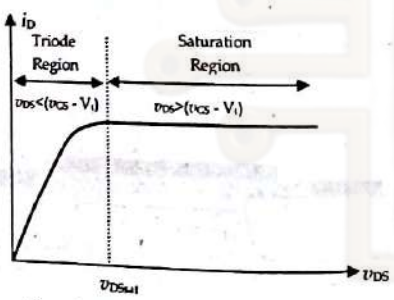

Figure: i_D Vs. v_{DS} curve for as v_{DS} is increased

When v_{DS} is increased the channel will not be of uniform depth and will acquire a tapered shape. To show this let us consider v_{GS} to be constant at a value greater than V_t. The voltage drop v_{DS} appears as a voltage drop across the length of channel. That is as we travel along the channel from the source to drain, the voltage measured relative to the source will change from 0 to v_{DS}. Thus the voltage between the gate and points along the channel decreases from v_{GS} at the source end to v_{GS}-v_{DS} at the drain end. Since the channel depth depends upon voltage (v_{GS}-v_{DS}), we can find that the channel is no longer of uniform depth. It will rather acquire a tapered shape being deepest at the source end and shallowest at the drain

end. As v_{DS} is increased further, the channel becomes more tapered and its resistance increases correspondingly. The i_D-v_{DS} curve no longer continue as a straight line but bend slightly as shown in the figure.

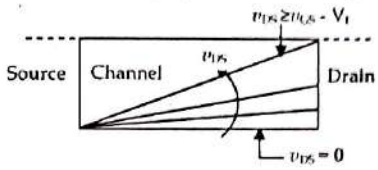

Figure: Increasing v_{DS} causes the channel to acquire a tapered shape. As v_{DS} reaches v_{GS}-V_t the channel is pinched off at the drain end. Increasing v_{DS} above v_{GS}-V_t has little effect on the channel shape.

Eventually, when v_{DS} is increased to the value that reduces the voltage between the gate and channel at the drain end to V_t i.e. v_{GS}-v_{DS} = V_t or v_{DS} = v_{GS} -V_t, the channel depth at the drain end reduces to almost zero value, and the channel is said to be pinched off. Increasing v_{DS} beyond this value has little effect on the channel shape, and the current through the channel remains constant at the value reached for v_{DS} = v_{GS} - V_t. The drain current saturates at this value and the MOSFET is said to have entered **saturation region** of operation. The voltage v_{DS} at which saturation occurs is denoted as v_{DSsat}

$$v_{DS} = v_{GS} - V_t$$

The device operates in the saturation region if $v_{DS} \geq v_{DSsat}$.

The region of i_D -v_{DS} characterjstics obtained for $v_{DS} \leq v_{DSsat}$ is called the **triode region.**

Symbols for n- channel E MOSFET:

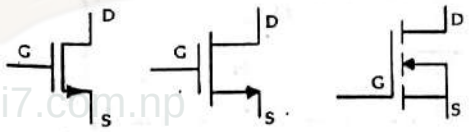

Figure: Different Symbols for n-channel E-MOSFET

Figure shows three different symbols for n- channel E-MOSFET. We can use any of them to represent n-channel E-MOSFET in the circuit. However for this book we shall use the first one.

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P-Channel E-MOSFET

The construction of p-channel E-MOSFET is similar to that of n-channel E-MOSFET except that the transistor is fabricated on n-type substrate with p^+ regions for source and drain. Thus, the channel induced is of p-type. The device operates in the same manner as the n-channel device except v_{GS} and v_{DS} are negative and threshold voltage V_t is also negative.

Symbols for P-channel E-MOSFET

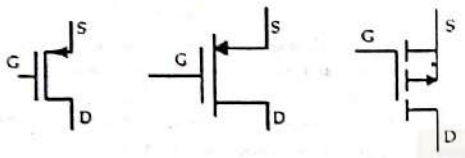

Figure: Different Symbols for p-channel E-MOSFET

Current Voltage Characteristics of Enhancement Type MOSFET

Characteristics of N-Channel E-MOSFET

For the proper operation of enhancement n channel MOSFET gate is always positive with respect to source and drain source voltage is such that the flow of charge carriers is from source to drain.

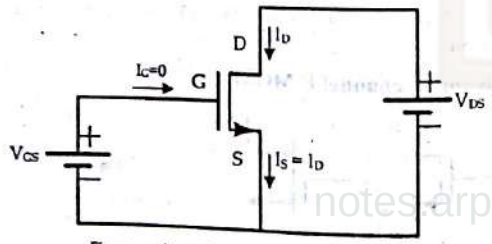

Figure: n-channel E-MOSFET biased for operation

Transfer characteristics of n-channel E-MOSFET:

The graphical relationship between gate to source voltage v_{GS} and drain current i_D is known as transfer characteristics.

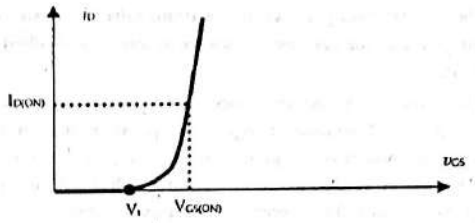

Figure: Transfer characteristics of n-channel E-MOSFET

The transfer characteristics curve is modeled by the equation,

$$i_D = K(v_{GS} - V_t)^2$$

Where,

$$K = \frac{I_{D(ON)}}{(V_{GS(ON)} - V_t)^2} \text{ in context of discrete MOSFET}$$

The value of $I_{D(ON)}$, $V_{GS(ON)}$, and V_t are provided in the data sheet of discrete E-MOSFET.

$$K = \frac{1}{2}\mu_n C_{OX}\left(\frac{W}{L}\right) \text{ in case of IC MOSFET}$$

μ_n = Electron Mobility

C_{OX} = Oxide capacitance

$\frac{W}{L}$ = Width to length ratio of MOSFET

Drain characteristics of n-channel E-MOSFET:

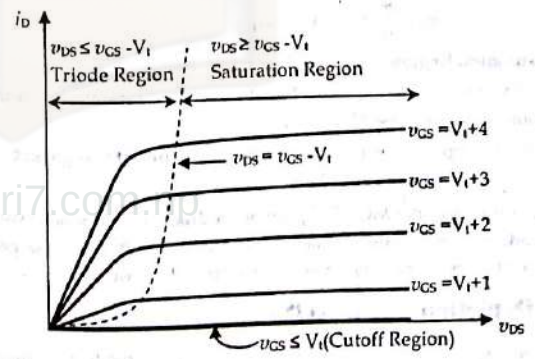

Figure: Drain characteristics of n-channel E-MOSFET

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The graphical relationship between the drain current i_D and drain to source voltage v_{DS} at constant gate to source voltage v_{GS} is called as drain characteristics.

Figure shows the drain characteristics of n- channel E-MOSFET. The characteristics have three distinct regions of operation: the cutoff region, saturation region and triode region. Saturation region is utilized when MOSFET is required to be used as an amplifier while the cutoff and the triode regions are utilized to operate MOSFET as a switch.

The Cutoff Region:

MOSFET enters the cutoff region of operation when $v_{GS} < V_t$.

The Triode Region:

After the channel is induced, MOSFET enters triode region of operation when $v_{DS} \leq v_{GS} - V_t$.

In triode region, MOSFET acts as a voltage controlled linear resistance. The i_D-v_{DS} relationship of MOSFET in the triode region is mathematically given by:

$$i_D = K\left[(v_{GS}-V_t)v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

If v_{DS} is sufficiently small we can neglect v_{DS}^2 term in the above expression.

$$i_D = K(v_{GS}-V_t)v_{DS}$$

The linear resistance of MOSFET r_{DS} in this region is given as:

$$r_{DS} = \frac{v_{DS}}{i_D} = \left[K(v_{GS} - V_t)\right]^{-1}$$

The Saturation Region:

After the channel is induced, MOSFET enters saturation region of operation when $v_{DS} \geq v_{GS} - V_t$.

In saturation region the expression for the drain current i_D is given by:

$$i_D = K(v_{GS} - V_t)^2$$

Thus in saturation the MOSFET provides a drain current whose value is independent of the drain voltage v_{DS} and is determined by the gate to source voltage v_{GS} according to the square law relationship.

The Depletion Type MOSFET

The structure of the depletion type MOSFET i.e. D-MOSFET is similar to that of E-MOSFET with one major difference and that is D-MOSFET has physically implanted channel which means the channel is already present. D-MOSFET can be of two types i.e. n-channel D-MOSFET and p-channel D-MOSFET. Thus, a n-channel depletion type MOSFET has the n-type silicon region connecting the n^+ source region and n^+ drain regions on the top of the p-type substrate. Thus if a voltage v_{DS} (drain to source voltage) is applied the current i_D will flow for v_{GS} (gate to source voltage) = 0.In other words, there is no need to induce a channel unlike n-channel E-MOSFET. Depletion type MOSFET can operate in two modes:

i. **Enhancement mode:** In this mode v_{GS} is positive for n channel D-MOSFET and negative for p channel D-MOSFET.As a result the channel width is increased and more current can flow through the channel. Thus, channel is said to be enhanced and this mode of operation is called as enhancement mode.

ii. **Depletion mode:** In this mode v_{GS} is negative (+ve for P channel device).As a result, the channel width is decreased because the electrons from channel are repelled to p-type substrate region and less current can flow through the channel. Here, the channel is said to be depleted and this mode of operation is called as depletion mode.

Symbols for n channel D-MOSFET

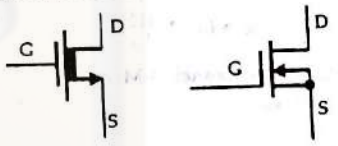

Figure: Symbols for n-channel D-MOSFET

Symbols for p channel D-MOSFET

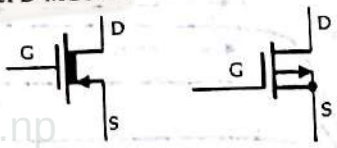

Figure: Symbols for p-channel D-MOSFET

Characteristics of n channel D-MOSFET

The characteristics of D-MOSFET are similar to that of E MOSFET except that D-MOSFET can operate in both the enhancement mode and depletion mode. When v_{GS} is positive, D-MOSFET operates in enhancement model and when v_{GS} is negative, D-MOSFET operates in

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depletion mode. The only difference is that the threshold voltage V_t of D MOSFET is negative.

Transfer characteristics of n-channel D-MOSFET:

It is the graph of i_D vs v_{GS}

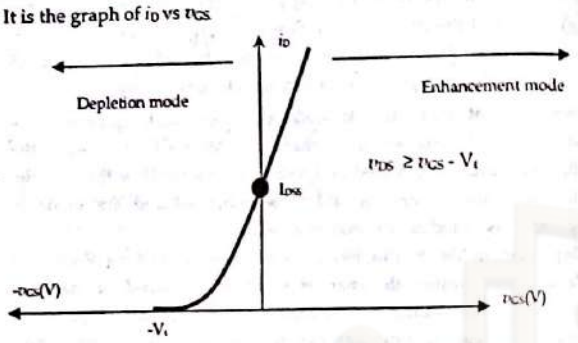


Figure: i_D vs v_{GS} Characteristics for n channel D MOSFET

In depletion mode, the current voltage relationship for D-MOSFET is identical to that for JFET, so for n channel D-MOSFET,

$$i_D = I_{DSS}\left(1 - \frac{v_{GS}}{V_p}\right)^2$$

Drain characteristics of n-channel D-MOSFET:

It is the graph of i_D Vs v_{DS}

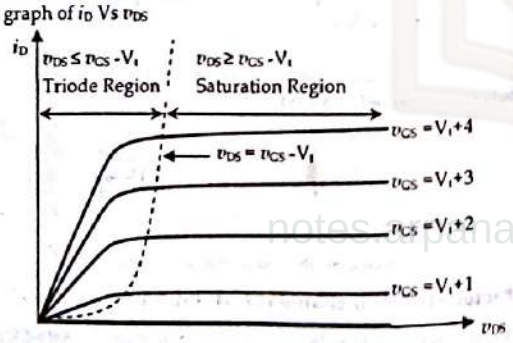


Figure: Drain characteristics for n-channel D-MOSFET

Here the threshold voltage V_t is negative.

Biasing in MOS Amplifier Circuits

Feedback Configuration:

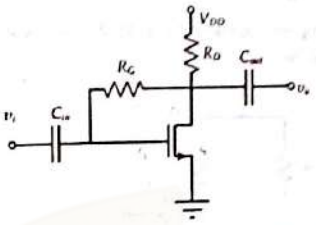
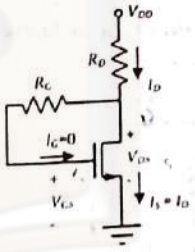


Figure shows the feedback biasing configuration for n- channel E MOSFET. Here a resistor R_G forms a feedback from the drain of the MOSFET to the gate. Function of resistor R_G is to provide sufficient gate voltage to turn MOSFET on.

The dc equivalent circuit is:



Applying KVL at the input side, we get,

$$V_{DD} = I_D R_D + V_{GS}$$

Since $I_G = 0$, $V_{GS} = V_{DS}$

$$\text{or, } V_{DD} = I_D R_D + V_{DS}$$

$$\text{or, } V_{DS} = V_{DD} - I_D R_D \quad \ldots\ldots\ldots \text{(i)}$$

Also the from the current-voltage relation of MOSFET,

$$I_D = K(V_{GS} - V_t)^2$$

$$\text{or, } I_D = K(V_{DS} - V_t)^2 \quad \ldots\ldots\ldots \text{(ii)}$$

Solving equation (i) and (ii) we get the operating point of the MOSFET. Generally operating point of MOSFET is written as (V_{DSQ}, I_{DQ}). Here V_{DSQ} and I_{DQ} are nothing but the value of V_{DS} and I_D obtained by solving the

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equation (i) and (ii). The 'Q' in the subscript is written to emphasize operating point or Q-point.

Voltage divider biasing:

A second popular biasing arrangement for E-MOSFET is voltage divider biasing. It is shown in the figure below:

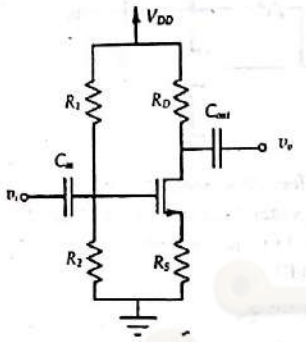
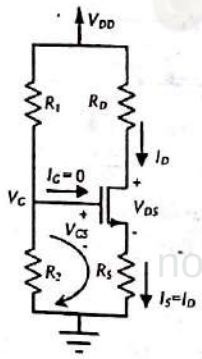


Figure: Voltage Divider biasing for E-MOSFET

The DC equivalent circuit is:



The voltage at gate can be found by calculating the drop across resistor R_2. Since R_1 and R_2 are in voltage divider configuration with supply V_{DD}.

$$V_G = V_2 = \frac{V_{DD} R_2}{R_1 + R_2}$$

Since $I_G = 0$,

Applying KVL in the loop indicated,

$$V_G = V_{GS} + I_D R_S$$

$$\therefore V_{GS} = V_G - I_D R_S \ldots\ldots\ldots(i)$$

Applying KVL in the output loop,

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D (R_D + R_S) \ldots\ldots\ldots(ii)$$

Also, the relationship between I_D and V_{GS} can be written as:

$$I_D = K (V_{GS} - V_t)^2 \ldots\ldots\ldots(iii)$$

Substituting the value of V_{GS} from equation *(i)* into the equation *(iii)*, we get,

$$I_D = K (V_G - I_D R_S - V_t)^2 \ldots\ldots\ldots(iv)$$

Solving the quadratic equation (iv), we get the value of I_D.

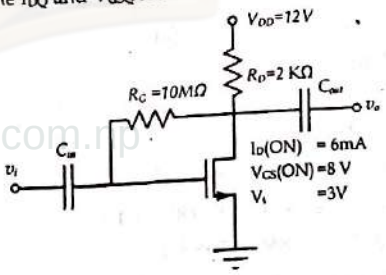
Once I_D is found we can calculate V_{GS} and V_{DS} from the equations (i) and (ii).

MOSFET Circuits at DC

Having learned MOSFET biasing, we are now ready to make simple DC analysis of the circuits containing MOSFET. The DC currents and voltages of the MOSFET determine its Q point. Specifically (I_D,V_{DS}) is called operating point of MOSFET. It is also written as (I_{DQ},V_{DSQ}). The operating point is a important parameter in designing amplifier circuits containing MOSFET. Let us closely examine few examples to learn how to analyze MOSFET circuits at DC.

Example 1:

Determine I_{DQ} and V_{GSQ} for the following figure.



Solution:

The DC equivalent circuit is shown below:

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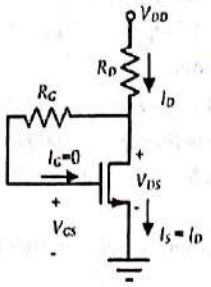


We know,

$$K = \frac{I_D(ON)}{(V_{GS}(ON) - V_t)^2} = \frac{6}{(8-3)^2} = 0.24$$

Applying KVL at the input side, we get,

$$V_{DD} = I_D R_D + V_{GS}$$

Since $I_G = 0$,

$$V_{GS} = V_{DS}$$

$$\therefore V_{DD} = I_D R_D + V_{DS}$$

$$\therefore V_{DS} = V_{DD} - I_D R_D$$

$$\therefore V_{DS} = 12 - 2I_D \ldots\ldots\ldots \text{(i)}$$

Also the from the current-voltage relation of MOSFET,

$$I_D = K(V_{DS} - V_t)^2$$

$$\therefore I_D = 0.24\,(V_{DS} - 3)^2 \ldots\ldots\ldots \text{(ii)}$$

From equation (i) and (ii),

We get,

$$V_{DS} = 12 - 2\times 0.24\,(V_{DS} - 3)^2$$

$$\text{or, } V_{DS} = 12 - 0.48\,(V_{DS}^2 - 6V_{DS} + 9)$$

$$\text{or, } V_{DS} = 12 - 0.48\,V_{DS}^2 + 2.88V_{DS} - 4.32$$

$$\therefore 0.48\,V_{DS}^2 - 1.88V_{DS} - 7.68 = 0$$

Solving using calculator, we get, $V_{DS} = 6.41$ and $V_{DS} = -2.49$

Since V_{DS} must be positive so we ignore negative value.

$$\therefore V_{DS} = 6.41 \text{ V}$$

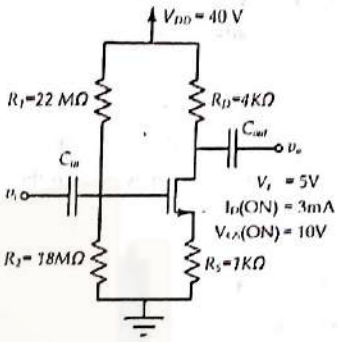
From equation (ii), we get,

$$I_D = 0.24\,(6.41 - 3)^2 = 2.79 \text{ mA}$$

Hence, $I_{DQ} = 2.79$ mA and $V_{DSQ} = 6.41$ V

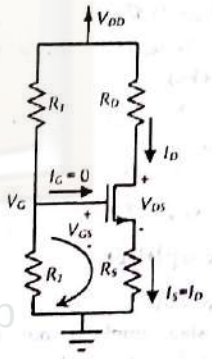
Example 2:

Find the I_{DQ} and V_{DSQ} for the following circuit.



Solution:

The DC equivalent circuit is:



We know,

$$K = \frac{I_D(ON)}{(V_{GS}(ON) - V_t)^2} = \frac{3}{(10-5)^2} = 0.12$$

$$V_G = V_2 = \frac{V_{DD} R_2}{R_1 + R_2} = \frac{40*18}{18+22} = \frac{720}{40} = 18\text{V}$$

Since $I_G = 0$,

Applying KVL in the loop indicated,

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$$V_G = V_{GS} + I_D R_S$$

$$\text{or, } 18 = V_{GS} + I_D$$

$$\text{or, } V_{GS} = 18 - I_D \quad \ldots\ldots(i)$$

Applying KVL in the output loop, we get,

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

$$\text{or, } V_{DS} = 40 - 5\,I_D \ldots\ldots(ii)$$

Also we know,

$$I_D = K(V_{DS} - V_t)^2$$

$$\therefore I_D = 0.12 * (V_{GS} - 5)^2 \ldots\ldots(iii)$$

Substituting the value of V_{GS} from equation (i) into the equation (iii), we get,

$$I_D = 0.12 \times (18 - I_D - 5)^2$$

$$\text{or, } I_D = 0.12 \times (13 - I_D)^2$$

$$\text{or, } I_D = 0.12\,I_D^2 - 3.12\,I_D + 20.28$$

$$\text{or, } 0.12\,I_D^2 - 4.12\,I_D + 20.28 = 0$$

Solving the above quadratic equation using calculator,

$I_D = 5.96$ mA and $I_D = 28.37$ mA

Since I_D =28.37 mA results in negative value of V_{GS} and I_D =5.96 results in positive value of V_{GS} we select,

I_D =5.96 mA

From equation (i),

V_{GS} =18 - 5.96 =12.04 V

Hence, I_{DQ} =5.96 mA and V_{GSQ} =12.04 V

MOSFET as an Amplifier

In the earlier chapter we studied about the amplifying characteristics of BJT and different single stage amplifier configurations for BJT. Similarly MOSFET can also be used as an amplifier. In this section we discuss amplifying nature of MOSFET. In order to operate MOSFET as an amplifier it should be operated in saturation mode. In order to operate MOSFET in saturation, channel must be enhanced by applying v_{GS} greater than the threshold voltage V_t and the drain voltage must be above the gate voltage by at least the threshold voltage. i.e. $v_{DS} \geq v_{GS} - V_t$

Figure shows the circuit diagram of common source MOSFET amplifier.

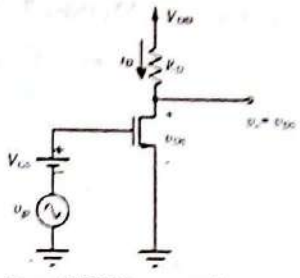

Figure MOSFET as an amplifier

The MOSFET input is formed by signal source v_g and dc voltage source V_{GS}. The main purpose of keeping dc source is to provide positive gate voltage so as to keep MOSFET in saturation mode throughout the operation. Thus the total input to the gate of MOSFET is $v_i = v_g + V_{GS}$. This is also the total voltage between gate and source of the MOSFET i.e. $v_i = v_{GS}$.

The drain current i_D is composed of both the ac component i_d and dc component I_D. Similarly the output voltage v_O or v_{DS} is composed of ac component v_o, and dc component V_{DS}.

Applying KVL in the output loop, we get,

$$v_O = V_{DD} - i_D R_D$$

Differentiating above relation with respect to i_D, we get,

$$\frac{dv_O}{di_D} = -R_D$$

The negative sign indicates that the output is 180° phase inverted as compared to the input.

The voltage gain of the above circuit is given by:

$$A_v = -g_m R_D$$

This can be readily proved by the interested reader by using one of the small signal models of MOSFET in the fashion that we did for BJT.

Now again the question that naturally arises is: how does the MOSFET amplify input signal? The answer is: The MOSFET uses relatively small voltage signal to cause appreciable current to flow through the channel and through drain resistance R_D. When the input signal varies around the dc voltage, large drain current will vary accordingly. When this large current is dropped in drain resistance or load resistance, an amplified version of input is obtained.

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Transconductance of E- MOSFET:

Transconductance of MOSFET is defined as the ratio of change in drain current, Δi_d to change in gate to source voltage Δv_{GS}.

Mathematically, $g_m = \frac{di_d}{dv_{GS}}$

We know for E - MOSFET, $i_D = k (v_{GS} - V_t)^2$

Differentiating the above relation with respect to v_{GS}, we get,

$$\frac{di_D}{dv_{GS}} = \frac{d}{dv_{GS}} [K (v_{GS} - V_t)^2]$$

$$= \frac{d}{d(v_{GS} - V_t)} [K (v_{GS} - V_t)^2] \frac{d}{dv_{GS}} [(v_{GS} - V_t)]$$

$$= 2K (v_{GS} - V_t) \times 1$$

$$= 2K (v_{GS} - V_t)$$

$$\therefore g_m = 2K (v_{GS} - V_t)$$

Small Signal models for MOSFET

Hybrid-π Model:

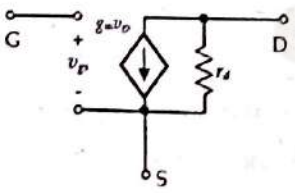

Figure: Hybrid- π model for MOSFET

T- model:

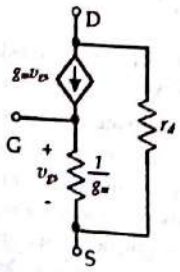

Figure: T-model for MOSFET

Junction Field Effect Transistor (JFET)

Junction Field Effect transistor is the simplest type of field effect transistor. In this type of FET a PN junction is formed between the p-type and n-type material so it is called Junction field effect transistor. There are two types of JFET. They are p-channel JFET and n-channel JFET. The construction of p-channel and n-channel JFET are shown in the figure.

Construction of JFET

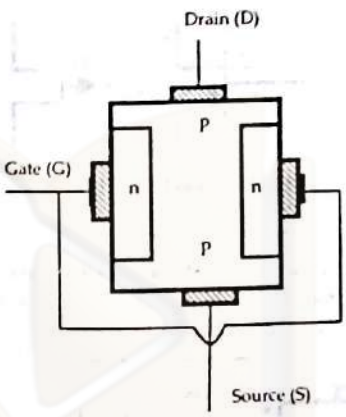

Fig: p-channel JFET

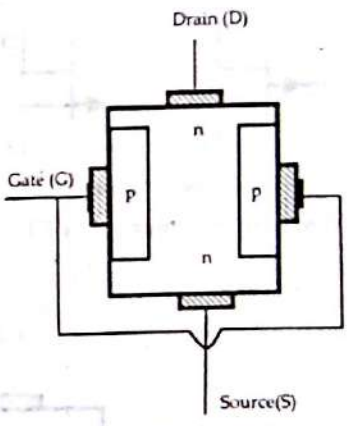

Fig: n-channel JFET

Figure above shows the construction of p-channel JFET and n-channel JFET. In p-channel JFET the p-type silicon bar is doped with n-type materials in same way as belt surrounds our waist. Here, the p-type material forms the channel for conduction and hence the name p-channel JFET.

Similarly, in n-channel JFET the n-type silicon bar is doped with p-type materials in same way as belt surrounds our waist. Here, n-type material forms the channel for conduction and hence the name n-channel JFET.

As shown in figure there are three terminals namely gate, drain and source. It is to be noted that JFET is symmetrical device. When connected to external circuit, the drain and source connection can be interchanged without affecting the overall performance of the circuit.

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Symbols for n-channel JFET:

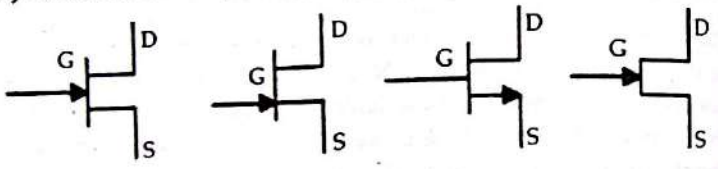

Figure: Different varieties of symbols for n channel JFET

Symbols for p-channel JFET:

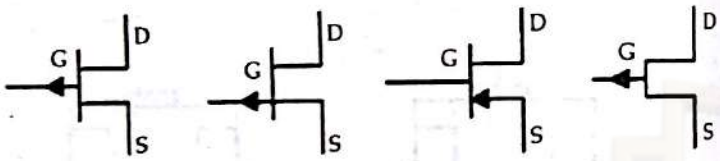

Figure: Different varieties of symbols for p channel JFET

Working of n channel JFET

For proper operation of JFET, gate is always reversed biased with respect to source (i.e. v_{GS} is negative) and drain source terminals are so biased that the flow of charge carriers is from source to drain.

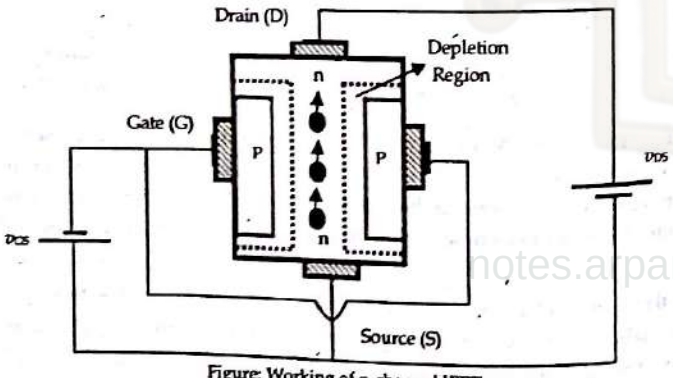

Figure: Working of n-channel JFET

Figure shows the circuit to demonstrate working of n channel JFET. Here gate is reversed biased with respect to source by gate to source voltage v_{GS}. Also drain source terminals are connected to v_{DS} so as to enable flow of electrons from source to drain. Here as we can see, n type material forms the channel for conduction hence the name n channel JFET. Also electron is the charge carrier responsible for conduction of current.

The p type material and n type material forms pn junction. So there exists a depletion layer as shown in the figure. The width of the depletion layer depends upon the gate to source voltage v_{GS}.

When v_{GS}= 0 normal drain current flows through the circuit. Electrons in the n type channel are repelled by negative terminal of v_{DS} and attracted by the positive terminal of v_{DS} .The conventional direction of current flow is from drain to source while flow of charge carriers (electrons) is from source to drain. The drain current is designated as i_D and source current is designated as i_S. Being symmetrical device, JFET has $i_S = i_D$.

If v_{GS} is increased from certain negative value (Note that v_{GS} cannot exceed 0V because gate should be reversed biased with respect to the source), reverse bias on the pn junction increases and it causes width of depletion layer to increase. But as the width of depletion layer is increased, channel width is decreased and hence flow of charge carriers is decreased. Thus when v_{GS} is increased drain current i_D (=i_S, source current) is decreased.

If v_{GS} is decreased, reverse bias on the pn junction decreases and it causes width of depletion layer to decrease. But as the width of depletion layer is decreased, channel width is increased and hence flow of charge carriers is increased. Thus when v_{GS} is decreased drain current i_D (=i_S, source current) is increased.

Thus in conclusion, current in the channel flows due to v_{DS} and gate to source voltage v_{GS} controls the drain current i_D.

At constant v_{GS}, when v_{DS} is small, similar to MOSFET, i_D of JFET exhibits linear relationship with v_{DS}. Also the channel is of uniform shape.

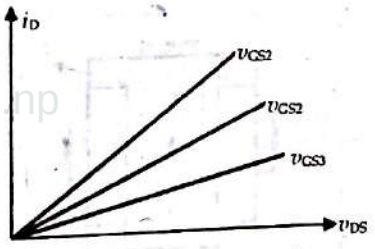

Figure: i_D - v_{DS} relationship for small v_{DS}

When v_{DS} is increased at constant v_{GS}, the channel acquires the tapered shape as in MOSFET. The resistance of the channel increases and the relationship between i_D and v_{DS} will no longer be linear. It will slightly bend as v_{DS} is increased.

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When v_{DS} is sufficiently increased, a point is reached at which the channel is pinched off at the drain end. The values of v_{GS} for which channel is pinched off is called as **pinch off voltage** and it is designated as V_p or sometimes as $V_{GS(off)}$. The drain current saturates once the channel is pinched off. The value of saturated drain current at pinch off voltage is designated as I_{DSS}.

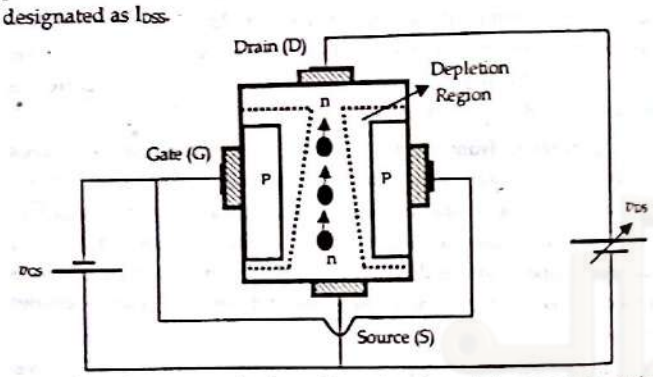

Figure: Increasing v_{DS} causes the channel to acquire a tapered shape and ultimately for $v_{GS} = V_p$ the channel is pinched off

Working of p-channel JFET

Working of p channel JFET is similar to n channel JFET. In p channel JFET channel is of p type. So charge carriers responsible for conduction are holes. As in n channel JFET, gate is reverse biased with respect to source by a voltage v_{SG} and drain source terminals are biased with voltage v_{SD} so that the flow of charge carriers (holes) is from source to drain.

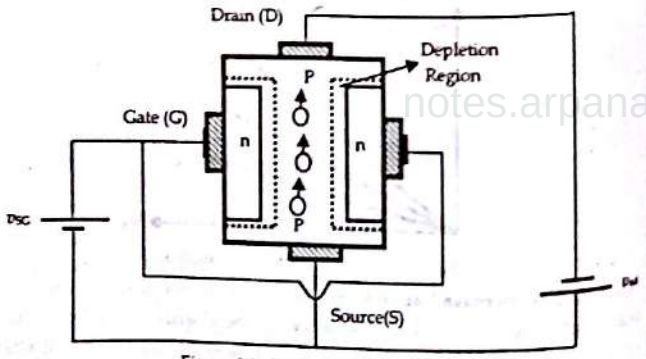

Figure: Working of p-channel JFET

Static Characteristics of JFET

a. **Drain Characteristics:** The graphical relationship between drain current i_D and drain to source voltage v_{DS} at constant gate to source voltage v_{GS} is called as drain characteristics.

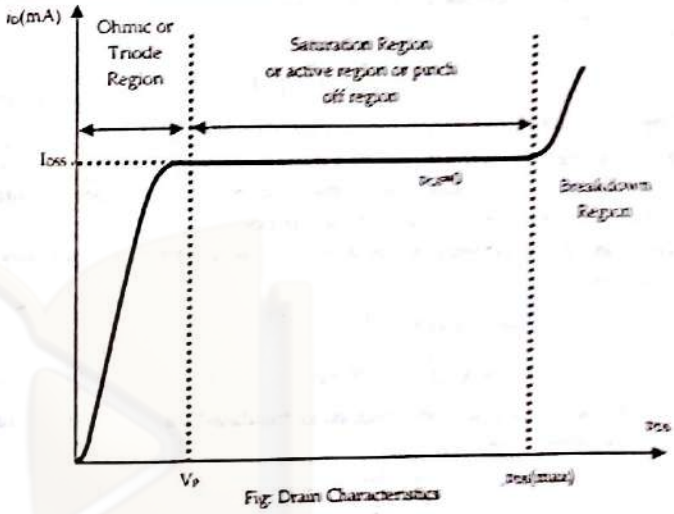

Fig: Drain Characteristics

Figure shows the drain characteristics of n channel JFET. The characteristics consist of three distinct regions of operation: the **cutoff region**, the **triode region or ohmic region**, the **saturation region or active region or pinch off region**.

The cutoff region:

The JFET enters this region of operation if:

$$v_{GS} \leq V_p$$

In this region of operation, the drain current i_D becomes zero.

$$i_D = 0$$

The triode region or ohmic region:

The JFET enters this region of operation if:

$$V_p \leq v_{GS} \leq 0 \text{ and } v_{DS} \leq v_{GS} - V_p$$

In this region of operation, the expression for the drain current is given by:

$$i_D = I_{DSS}\left[2\left(1-\frac{v_{GS}}{V_p}\right)\left(\frac{v_{DS}}{-V_p}\right)-\left(\frac{v_{DS}}{V_p}\right)^2\right]$$

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The saturation region or active region or pinch off region:

The JFET enters this region of operation if:

$$V_P \le v_{GS} \le 0 \text{ and } v_{DS} \ge v_{GS} - V_P$$

In this region of operation, the expression for the drain current is given by:

$$i_D = I_{DSS}\left(1 - \frac{v_{GS}}{V_P}\right)^2$$

This relationship is also known as transfer characteristics.

b. Transfer Characteristics:

The graphical relationship between the drain current i_D and gate to source voltage v_{GS} is known as transfer characteristics.

The transfer characteristics curve of JFET is governed by Shockley's equation,

$$i_D = I_{DSS}\left(1 - \frac{v_{GS}}{V_P}\right)^2$$

$$\text{or, } i_D = I_{DSS}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)^2$$

V_P or $V_{GS(off)}$ is usually specified in the datasheet of the JFET. It has negative value.

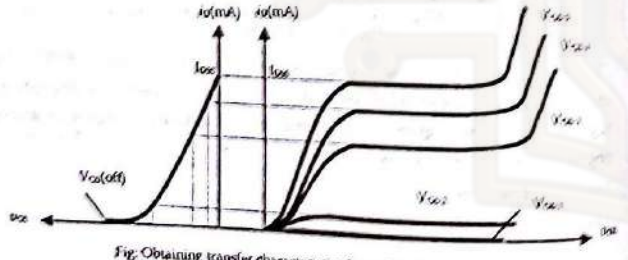

Fig: Obtaining transfer characteristics from drain characteristics

JFET small signal parameters:

AC drain resistance

It is the ratio of change in drain to source voltage Δv_{DS} to the change in drain current Δi_D at constant v_{GS}.

Mathematically,

$$r_d = \frac{\Delta v_{DS}}{\Delta i_D} \text{ at constant } v_{GS}$$

Transconductance

It is given by the ratio of change in drain current Δi_D to the corresponding change in gate to source voltage Δv_{GS}.

Mathematically,

$$g_m = \frac{\Delta i_D}{\Delta v_{GS}} \text{ Siemens(S)}$$

Amplification factor (μ)

It is given by the ratio of change in drain to source voltage, Δv_{DS} to the corresponding change in the gate to source voltage, Δv_{GS}.

Mathematically,

$$\mu = \frac{\Delta v_{DS}}{\Delta v_{GS}}$$

μ is related to g_m as follows:

$$\mu = -g_m r_d$$

Expression for the Transconductance of JFET:

As we know already, transconductance of JFET is defined as the ratio of change in drain current Δi_D to the corresponding change in gate to source voltage Δv_{GS}.

i.e. $$g_m = \frac{\Delta i_D}{\Delta v_{GS}}$$

From transfer characteristics of JFET,

$$i_D = I_{DSS}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)^2$$

Differentiating the above expression w.r.t v_{GS}, we get,

$$\frac{di_D}{dv_{GS}} = \frac{d}{dv_{GS}}\left\{I_{DSS}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)^2\right\}$$

$$\text{or, } g_m = \frac{d}{d\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)}\left\{I_{DSS}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)^2\right\} \cdot \frac{d}{dv_{GS}}\left\{\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)\right\}$$

$$= 2I_{DSS}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right) \cdot \frac{-1}{V_{GS(off)}}$$

$$\therefore g_m = \frac{-2I_{DSS}}{V_{GS(off)}}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)$$

$$\text{At } v_{GS} = 0,\ g_{m0} = \frac{-2\,I_{DSS}}{V_{GS(off)}}$$

$$\therefore g_m = g_{m0}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)$$

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JFET Biasing:

The biasing of JFET deals with the setting a fixed level of current which should flow through the FET with a desired fixed voltage across FET junction. For proper operation of JFET, gate must be negative with respect to source. And drain source terminals should be biased to cause flow of carriers form source to drain. This can be achieved by using bias batteries in gate and drain terminals or with a circuit known as biasing circuit. Obviously biasing JFET with biasing circuit is more elegant and economical compared to biasing with bias batteries.

Fixed Biasing

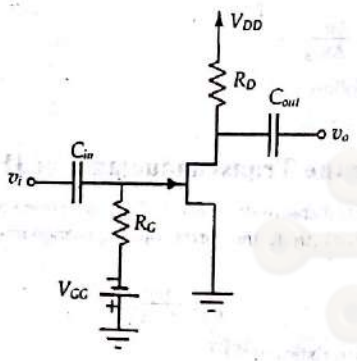

Figure shows the fixed bias configuration for n channel JFET. The battery V_G ensures gate is always reverse biased w.r.t source and no current flows through the resistor R_G and gate terminal.

i.e. $I_G = 0$.

DC equivalent circuit is shown in the figure.

Applying KVL to the input side, we get,

$$V_{GG} + V_{GS} = 0$$

$$V_{GG} = -V_{GS}$$

$$\therefore V_{GS} = -V_{GG}$$

The drain current can be calculated from the following equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

Applying KVL in the output circuit, we get

$$V_{DD} - I_D R_D - V_{DS} = 0$$

$$\therefore V_{DS} = V_{DD} - I_D R_D$$

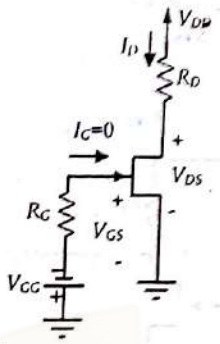

Figure: DC equivalent circuit

Self Bias:

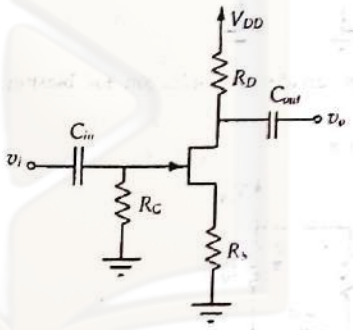

Figure shows the circuit for self bias configuration of JFET. This biasing scheme eliminates the extra power supply. Also a resistor R_S is included in the source terminal.

Applying KVL to the input side, we get,

$$V_G = V_{GS} + I_D R_S$$

Since $I_G = 0 \therefore V_G = 0$

$$\therefore V_{GS} = -I_D R_S$$

This voltage reverse biases the gate terminal w.r.t source.

The drain current can be obtained as:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

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Applying KVL to the output side, we get,

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D(R_D + R_S)$$

Voltage Divider biasing:

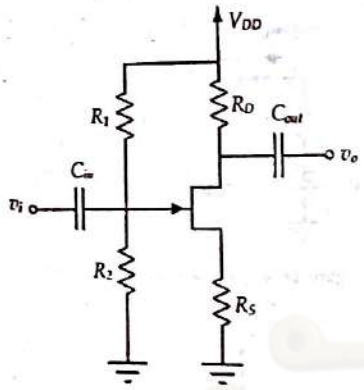
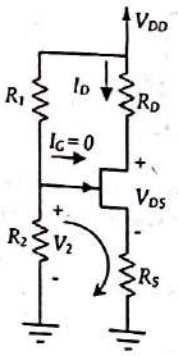


Figure shows the voltage divider configuration for biasing n channel JFET.

The DC equivalent circuit is:



Since resistors R_1 and R_2 forms a voltage divider configuration, the drop across R_2 is given as:

$$V_2 = V_{DD} \frac{R_2}{R_1 + R_2}$$

Applying KVL in the loop indicated, we get

$$V_2 - V_{GS} - I_D R_S = 0$$

$$\therefore V_{GS} = V_2 - I_D R_S$$

The circuit is so designed that $I_D R_S$ is greater than V_2. Therefore V_{GS} will be negative and thus will reverse bias gate w.r.t. source which is essential for the operation of JFET.

The drain current can be obtained as:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

Applying KVL to the output side, we get,

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D(R_D + R_S)$$

Small signal models of JFET:

The small signal models (i.e. hybrid-π model and T-model) for JFET are identical to that for MOSFET. The only difference is that for JFET expression for g_m is given by:

$$g_m = \frac{-2I_{DSS}}{V_{GS(off)}}\left(1 - \frac{v_{GS}}{V_{GS(off)}}\right)$$

Common Source Voltage divider JFET amplifier with source bypassed capacitor:

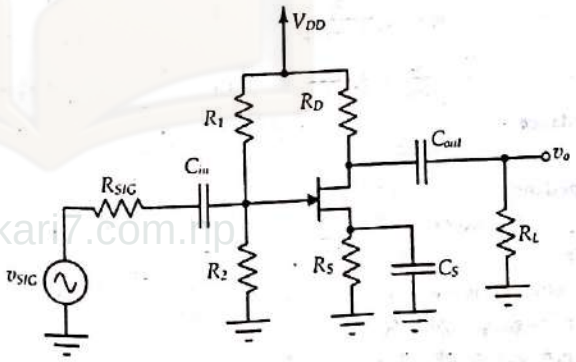


Figure shows the circuit for common source JFET amplifier with source bypass capacitor C_S.

The ac equivalent circuit is as follows:

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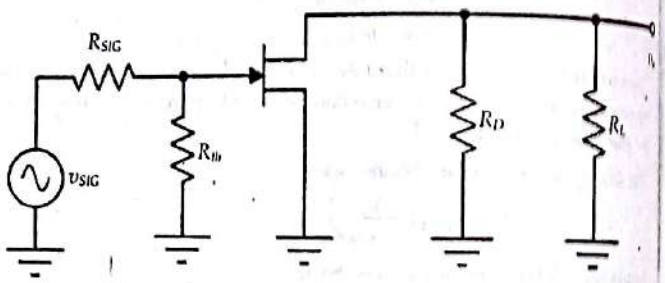
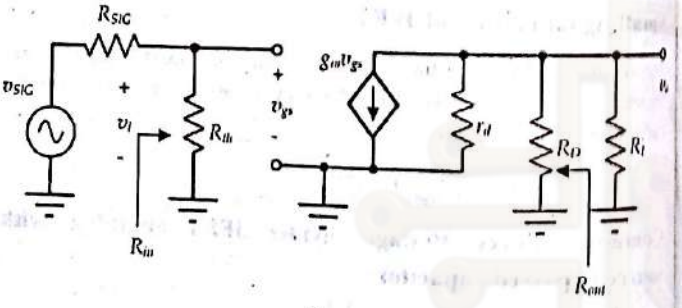


Where, $R_{th} = R_1//R_2$

Replacing JFET with its hybrid -π equivalent circuit, we get,



$$v_i = v_{SIG}\frac{R_{in}}{R_{in}+R_{SIG}}$$

$$\therefore \frac{v_i}{v_{SIG}} = \frac{R_{in}}{R_{in}+R_{SIG}} \quad \ldots(i)$$

Input Impedance

$$R_{in} = R_{th} = R_1//R_2$$

Output Impedance:

$$R_{out} = R_D//r_d$$

Voltage gain:

The output voltage v_o is given as:

$$v_o = -g_m v_{gs}(r_d//R_D//R_L)$$

$$\text{or, } v_o = -g_m v_i (r_d//R_D//R_L)$$

$$\frac{v_o}{v_i} = -g_m(r_d//R_D//R_L)$$

$$\therefore A_v = -g_m(r_d//R_D//R_L)$$

The open circuit voltage gain can be found by setting $R_L \to \infty$

$$\therefore A_{vo} = -g_m(r_d//R_D)$$

The overall voltage gain G_v is given by:

$$G_v = \frac{v_o}{v_{sig}}$$

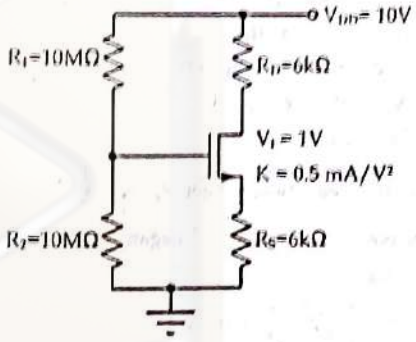
$$= \frac{v_o}{v_i}\frac{v_i}{v_{sig}}$$

$$\therefore G_v = -g_m(r_d//R_D//R_L)\frac{R_{in}}{R_{in}+R_{sig}} \text{ (Using eqn(i))}$$

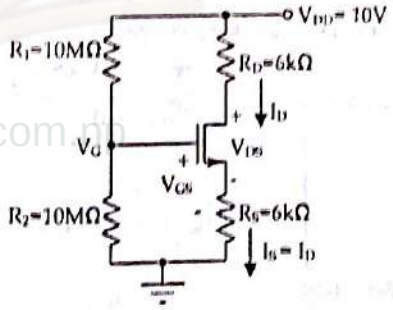
Current gain is meaningless since input current is zero.

Solved Numerical:

1. Find the value of I_D and V_{DS} in the given circuit. **2068 Chaitra**



Solution:



From the circuit,

$$V_G = \frac{V_{DD}R_2}{R_1+R_2} = \frac{10\times10}{10+10} = 5V$$

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Applying KVL around gate source loop, we get,

$$V_G = V_{GS} + I_D R_S$$

or, $5 = V_{GS} + 6I_D$

$6I_D = 5 - V_{GS}$ .....(i)

Applying KVL in the output loop, we get,

$$V_{DD} = I_D R_D + V_{DS} + I_D R_S$$

$V_{DS} = V_{DD} - (R_D + R_S)I_D$

$V_{DS} = 10 - 12I_D$ .....(ii)

From the transfer characteristics,

$$I_D = k(V_{GS} - V_t)^2$$

or, $I_D = 0.5(V_{GS} - 1)^2$

or, $6I_D = 6 \times 0.5(V_{GS} - 1)^2$

Using equation (i)

$5 - V_{GS} = 3(V_{GS}^2 - 2V_{GS} + 1)$

or, $5 - V_{GS} = 3V_{GS}^2 - 6V_{GS} + 3$

$3V_{GS}^2 - 5V_{GS} - 2 = 0$

Solving above quadratic equation, we get, $V_{GS} = 2, -\frac{1}{3}$

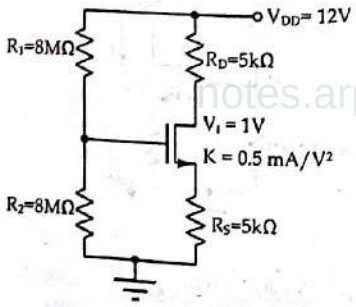
Since V_{GS} must be positive we can ignore negative value.

$\therefore V_{GS} = 2$ V

From equation (i), $I_D = \frac{5 - V_{GS}}{6} = \frac{5-2}{6} = 0.5$ mA

From equation (ii), $V_{DS} = 10 - 12I_D = 10 - 12 \times 0.5 = 4$ V

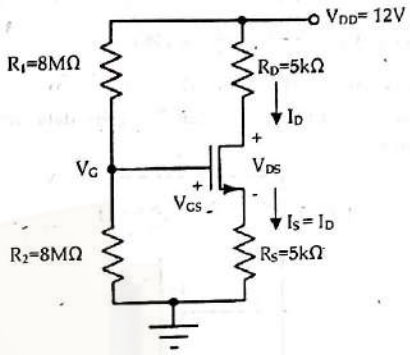
2. Find the value of I_D and V_{DS} in the given circuit. 2071 Chaitra



Solution:

$V_t = 1$V

$K = 0.5$ mA/V²



For the circuit,

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2} = \frac{12 \times 8}{8+8} = 6\text{V}$$

Applying KVL around gate source loop, we get,

$$V_G = V_{GS} + I_D R_S$$

or, $6 = V_{GS} + 5I_D$

$5I_D = 6 - V_{GS}$ .....(i)

Applying KVL in the output loop, we get,

$$V_{DD} = I_D R_D + V_{DS} + I_D R_S$$

$V_{DS} = V_{DD} - (R_D + R_S)I_D$

$V_{DS} = 12 - 10I_D$ .....(ii)

From the transfer characteristics,

$$I_D = K(V_{GS} - V_t)^2$$

or, $I_D = 0.5(V_{GS} - 1)^2$

or, $5I_D = 5 \times 0.5(V_{GS} - 1)^2$

Using equation (i)

$6 - V_{GS} = 2.5(V_{GS}^2 - 2V_{GS} + 1)$

or, $6 - V_{GS} = 2.5V_{GS}^2 - 5V_{GS} + 2.5$

$2.5V_{GS}^2 - 4V_{GS} - 3.5 = 0$

Solving above quadratic equation, we get,

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$V_{GS} = 2.23, -0.63$

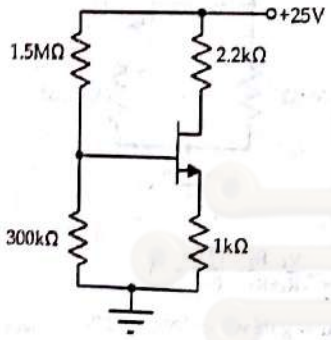
Since V_{GS} must be positive we can ignore negative value.

$\therefore V_{GS} = 2.23$

From equation (i), $I_D = \frac{6-V_{GS}}{5} = \frac{6-2.23}{5} = 0.75$ mA

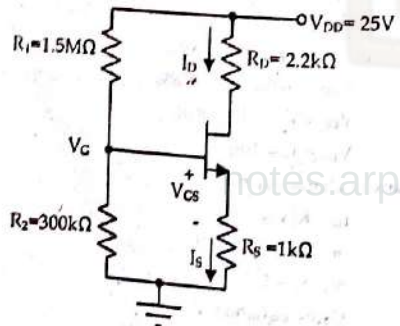
From equation (ii), $V_{DS} = 12 - 10I_D = 12 - 10\times0.75 = 4.5$V

3. Find I_D and V_{DS} for the given circuit. The given data are: $V_P = -4$V and $I_{DSS} = 10$ mA 2070 Chaitra



Solution

Given data are: $V_P = -4$V and $I_{DSS} = 10$ mA



From the circuit,

$$V_G = \frac{V_{DD}R_2}{R_1+R_2} = \frac{25\times300}{1500+300} = 4.17 \text{ V}$$

Applying KVL in gate source loop,

$$V_G = V_{GS} + I_DR_S$$

$$\text{or, } 4.17 = V_{GS} + I_D$$

$$I_D = 4.17 - V_{GS} \quad \text{.....(i)}$$

From the transfer characteristics,

$$I_D = I_{DSS}(1-\frac{V_{GS}}{V_P})^2$$

$$\text{or, } 4.17 - V_{GS} = 10(1+\frac{V_{GS}}{4})^2$$

$$\text{or, } \frac{10}{16}V_{GS}^2 + 6V_{GS} + 5.83 = 0$$

Solving above equation, we get,

$V_{GS} = -1.097, -8.5$

The second solution is impossible since V_G cannot maintain this magnitude of reverse bias between gate and source terminals.

$\therefore V_{GS} = -1.097$

From equation (i),

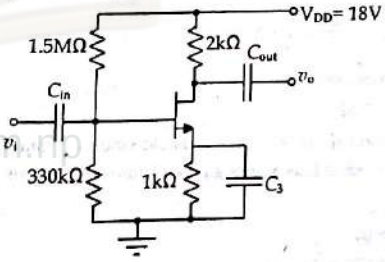
I_D = 4.17 + 1.2995 = 5.267 mA

Applying KVL in the output loop, we get,

$V_{DS} = V_{DD} - (R_D+R_S)I_D$

$\therefore V_{DS} = 25 - 3.2\times5.267 = 8.15$ V

4. Find I_D and V_{DS} for the given circuit. Given data are: $V_P = -5.5$V and $I_{DSS} = 12$ mA and assume that all the capacitors are ideal and check whether it is operating in pinch off region or not? 2070 Ashad



Solution

Given: $V_P = -5.5$V and $I_{DSS} = 12$ mA

The dc equivalent circuit is:

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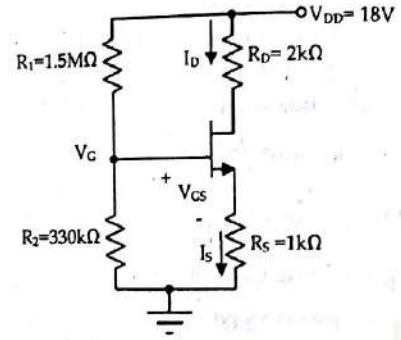

From the circuit,

$$V_G = \frac{V_{DD}R_2}{R_1+R_2} = \frac{18\times330}{1500+330} = 3.25\ V$$

Applying KVL in gate source loop,

$$V_G = V_{GS} + I_DR_S$$

$$\text{or, } 3.25 = V_{GS} + I_D$$

$$I_D = 3.25 - V_{GS} \quad \text{.....(i)}$$

From the transfer characteristics,

$$I_D = I_{DSS}(1-\frac{V_{GS}}{V_P})^2$$

$$\text{or, } 3.25 - V_{GS} = 12(1+\frac{V_{GS}}{5.5})^2$$

$$\text{or, } \frac{48}{121}V_{GS}^2 + \frac{59}{11}V_{GS} + 8.75 = 0$$

Solving above equation, we get,

V_{GS} = -1.8977, -11.62

The second solution for V_{GS} is impossible since V_G cannot maintain this magnitude of reverse bias across gate and source terminals.

V_{GS} = -1.8977

From equation (i),

I_D = 3.25 –(-1.8977) = 5.15 mA

Applying KVL in the output loop, we get,

$V_{DS} = V_{DD} - (R_D+R_S)I_D$

$\therefore V_{DS} = 18 - 3\times5.15 = 2.55$ V

Test for pinch off condition

$V_{DS} \geq V_{GS} - V_P$

or, $2.55 \geq -1.8977 + 5.5 = 3.6$

or, 2.55 ≥ 3.6(False)

Thus JFET is not operating in pinch off condition.

5. An n-channel JFET has a pinch off voltage of -4.5V and I_{DSS} = 9mA. At what values of V_{GS} will I_D be equal to 3 mA? What is its g_m at this I_D?

Solution

Given: V_P = -4.5 V and I_{DSS} = 9 mA

We know,

$$I_D = I_{DSS}(1-\frac{V_{GS}}{V_P})^2$$

For I_D = 3 mA

$$3 = 9(1+\frac{V_{GS}}{4.5})^2$$

$$\text{or, } \frac{1}{3} = (1+\frac{V_{GS}}{4.5})^2$$

$$\text{or, } \frac{1}{\sqrt{3}} = (1+\frac{V_{GS}}{4.5})$$

$$\text{or, } \frac{V_{GS}}{4.5} = (\frac{1}{\sqrt{3}} - 1)$$

$$\therefore V_{GS} = 4.5(\frac{1}{\sqrt{3}} - 1) = -1.902\ V$$

Also, $g_{mo} = \frac{-2I_{DSS}}{V_P} = \frac{-2\times9}{-4.5} = 4$ mA/V

$$\therefore g_m = g_{mo}(1-\frac{V_{GS}}{V_P}) = 4(1-\frac{1.902}{4.5}) = 2.31\ \text{mA/V}$$

6. Figure shows the circuit of a common source FET amplifier. Determine the value of output voltage, input impedance, output impedance. Take g_m = 5000μS. Neglect drain resistance(r_d).

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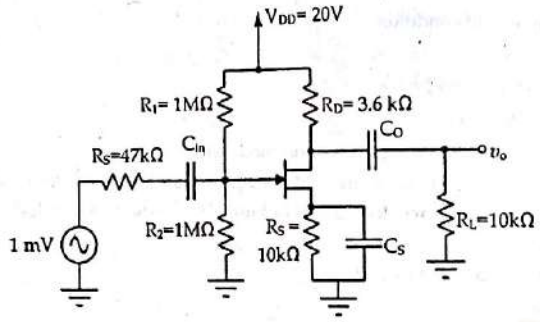
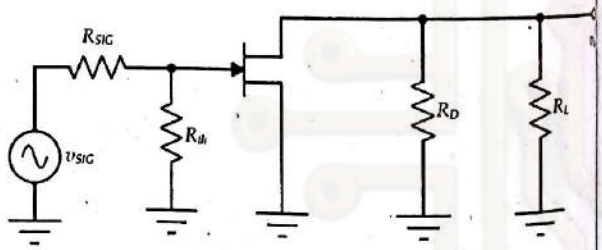


Solution:

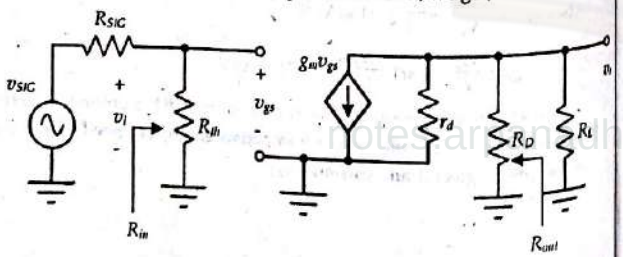
$g_m = 5000\mu S = 5000\times10^{-6} = 0.005$ S

The ac equivalent circuit is as follows:



Where, $R_{th} = R_1 // R_2$

Replacing JFET with its hybrid –π equivalent circuit, we get,



$$v_i = v_{SIG}\frac{R_{in}}{R_{in}+R_{SIG}}$$

$$\therefore \frac{v_i}{v_{SIG}} = \frac{R_{in}}{R_{in}+R_{SIG}} \quad \text{....(i)}$$

Input Impedance

$$R_{in} = R_{th} = R_1 // R_2 = 1 // 1 = 1M\Omega$$

Output Impedance:

$$R_{out} = R_D = 3.6k\Omega$$

Voltage gain:

The output voltage v_o is given as:

$$v_o = -g_m v_{gs}(R_D // R_L)$$

$$\text{or, } v_o = -g_m v_i (R_D // R_L)$$

$$\frac{v_o}{v_i} = -g_m (R_D // R_L)$$

$$\therefore A_v = -g_m (R_D // R_L)$$

The overall voltage gain G_v is given by:

$$G_v = \frac{v_o}{v_{SIG}}$$

$$= \frac{v_o}{v_i}\frac{v_i}{v_{SIG}}$$

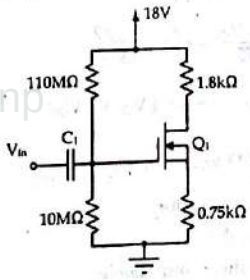
$$\therefore G_v = -g_m (R_D // R_L)\frac{R_{in}}{R_{in}+R_{SIG}} \text{ (Using eqn(i))}$$

$$= -0.005\times(3.6//10)\times1000\times\frac{1000k\Omega}{1000k\Omega+47k\Omega}$$

$$= -12.64$$

$\therefore v_o = G_v v_{SIG} = -12.64\times1$ mV $= -12.64$ mV

7. For n-channel D MOSFET as shown below. Find I_{DQ} and V_{GSQ}. Given that $I_{DSS} = 6$ mA and $V_P = -3$ V



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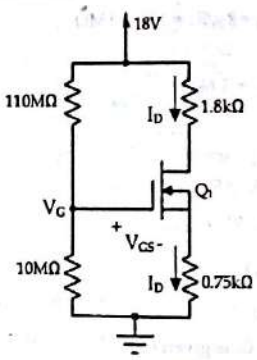
Solution

Given: I_{DSS} = 6 mA and V_P = -3 V

The dc equivalent circuit is:



$$\therefore V_G = \frac{18\times10}{10+110} = 1.64 \text{ V}$$

Applying KVL around gate source loop,

$$V_G = V_{GS} + I_D R_S$$

$$\text{or, } 0.75\, I_D + V_{GS} = 1.64$$

$$\therefore V_{GS} = (1.64 - 0.75\, I_D)$$

We know, for depletion type MOSFET, current voltage relation is identical to that of JFET,

$$I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_P}\right)^2$$

$$\text{or, } I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_P}\right)^2$$

$$\text{or, } \frac{1.64 - V_{GS}}{0.75} = 6\left(1+\frac{V_{GS}}{3}\right)^2$$

$$\text{or, } 2.19 - 1.33\, V_{GS} = 6 + 4\, V_{GS} + \frac{6}{9}V_{GS}^2$$

$$\text{or, } \frac{6}{9}V_{GS}^2 + 5.33\, V_{GS} + 3.81 = 0$$

Solving using calculator,

V_{GS} = -0.7935 , -7.2

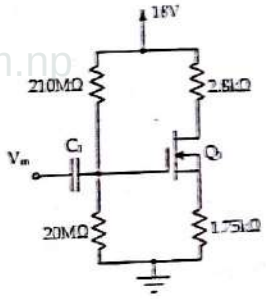
The second value is impossible.

$$\therefore V_{GSQ} = -0.7935 \approx -0.8 \text{ V}$$

$$\therefore I_{DQ} = \frac{1.64 - V_{GS}}{0.75} = \frac{1.64 - (-0.8)}{0.75} = 3.25 \text{ mA}$$

EXERCISE

1. Describe the operation of JFET in ohmic and active regions of operations with the help of graph and expressions. 2071 Chaitra
2. Describe the principle of operation of N-channel JFET with the help of drain and transfer characteristics graphs and mathematical expressions. 2071 Shrawan
3. Derive an expression to obtain transconductance of EMOSFET. 2071 Shrawan
4. Describe the construction and working principle of EMOSFET with help of drain characteristics curve and mathematical expressions. 2071 Chaitra
5. Derive an expression to obtain the transconductance of JFET. 2070 Chaitra
6. Explain why self DC biasing MOSFET circuit is better than fixed DC biasing MOSFET circuit. 2070 Ashad
7. Derive an expression to find the transconductance for JFET. 2070 Ashad
8. Describe with necessary graphs and expressions the principle of operation of N-channel JFET. 2069 Chaitra
9. Describe the working principle of n-channel JFET with the help of I_D-V_{GS} characteristics curve. Find the condition and expression for it to operate in active mode of operation and write the expression for drain current. 2069 Ashad
10. Describe the physical structure of N-channel JFET and explain its working principle and characteristics clearly marking the various regions of operation. 2068 Chaitra
11. Derive the expression to obtain the transconductance of E-MOSFET. 2068 Chaitra
12. Derive an expression to find the transconductance for JFET. 2068 Baisakh
13. Describe the principle of operation of E-MOSFET with the help of IV characteristics curves and algebraic expressions. Also show its ac equivalent circuit model. 2068 Baisakh
14. Describe the principle of operation of EMOSFET with the help of I_D-V_{GS} characteristics .Find the expressions when it operates in ohmic region and also in pinch off region of operations. 2067 Ashad
15. For n-channel D MOSFET as shown below. Find I_{DQ} and V_{GSQ}. Given that I_{DSS} = 9 mA and V_P = -4 V



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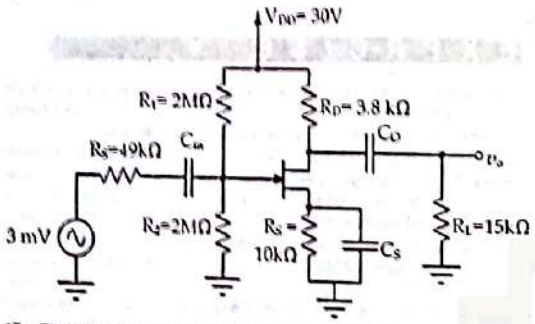
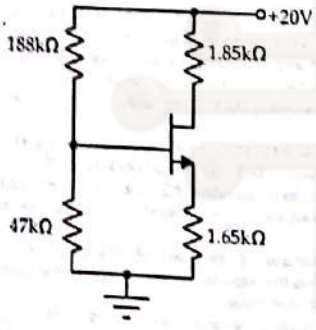
16. Figure shows the circuit of a common source FET amplifier. Determine the value of output voltage, input impedance, output impedance. Take g_m = 4000μS. Neglect drain resistance (r_d).



17. Find I_D and V_{DS} for the given circuit. The given data are: V_P = - 5V and I_{DSS} = 18 mA



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Chapter 4

Output Stages and Power Amplifiers

Introduction

In the earlier chapters we discussed about amplifying nature of BJT and FET. We also discussed about simple single stage amplifier configurations utilizing both BJT and FETs. In fact the circuits that we studied are just a unit of practical amplifier circuit. In practical amplifier circuits there are a lot of stages. Generally a complex amplifier circuit consists of three main stages. The first stage is input stage which is concerned with coupling the signal from the source to the amplifier circuit. Second stage is concerned with providing sufficient gain to the input signal and the third or final stage is concerned with providing output to the load. The output stage should have as low resistance as possible so as to effectively transfer the entire output signal to the load without any drop in its output circuit. That is output stage should deliver required amount of power to the load in an efficient manner. Thus the power dissipation in transistor of output stage should be as low as possible. It is one of the most challenging requirements in the design of amplifier circuits.

Since the output stages have to deal with relatively large signals and voltages so the small signal approximations are either not applicable or must be used with care. In this chapter we shall discuss output stages of high power amplifiers. In our context high power means power greater than 1W. Power amplifiers are those amplifiers which dissipate high power in the output circuit.

Classification of output stages

The output stage of an amplifier circuit is formed by different combination and configuration of transistor(s). The basic principle is to

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make output stage to have low impedance and reduce distortion that may be introduced when providing output signal to the load.

Output stages are classified based on the waveform of collector current that results when an input signal is applied. Following are the different output stages used in power amplifiers:

i. Class A output stage
ii. Class B output stage
iii. Class AB output stage
iv. Class C output stage
v. Class D output stage

Class A output stage

Class A output stages are those output stage in which the output current flows for the full cycle of the applied input signal for each transistor forming the output stage. The amplifiers utilizing class A at the output stage are known as class A amplifiers. Thus conduction angle for class A output stage is 360°

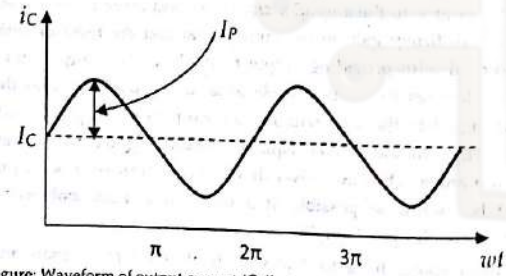

Figure: Waveform of output current (Collector current) of transistor in class A amplifier

Figure shows the waveform of collector current of the transistor in class A amplifier. The collector current has Q-point current I_C and the ac collector current i_c is superimposed upon the dc collector current or Q- point collector current. The output ac current swings above and below the Q-point current with peak value of I_P. From the figure it is seen that output current appears with conduction angle 360° which means it appears for complete time period.

In fact the amplifiers circuits that we discussed till now are of class A type. The main advantage of class A amplifier is that it provides relatively

distortion free output. But the major disadvantage is that class A amplifiers have relatively low operating efficiency.

Series Fed Class A amplifier:

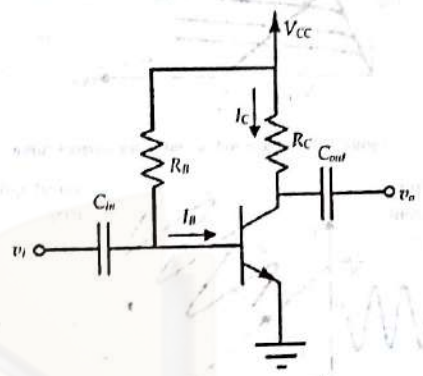

Figure: Series Fed class A amplifier

Figure shows the circuit diagram of series fed class A amplifier. The name "series fed" derives from the fact that resistor R_C forms a series load of this amplifier.

DC operating condition:

Applying KVL in the input loop:

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B$$

Applying KVL in the output loop, we get,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\therefore V_{CE} = V_{CC} - I_C R_C$$

When $V_{CE} = 0$, $I_C = \frac{V_{CC}}{R_C}$

When $I_C = 0$, $V_{CE} = V_{CC}$

Thus dc load line can be constructed by joining $(0, V_{CC}/R_C)$ and $(V_{CC}, 0)$ on the i_C Vs v_{CE} characteristics of the transistor.

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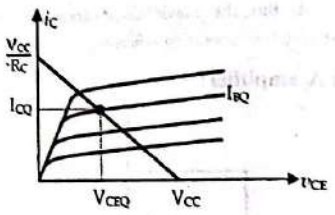

Figure: DC load line and output characteristics curve

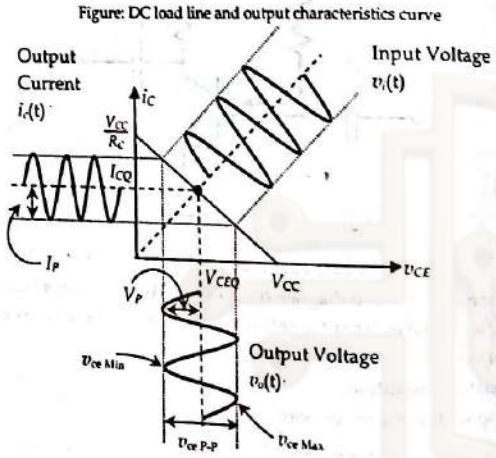

Figure: Input and output signals superimposed upon bias current and voltage

Input Power:

The dc input power to the amplifier is given by:

$$P_{in\ DC} = I_{CQ}\, V_{CC}$$

From the figure $I_{CQ} = \frac{V_{CC}}{2R_C}$

$$P_{in\ DC} = \frac{V_{CC}}{2R_C} V_{CC}$$

$$\therefore P_{in\ DC} = \frac{V_{CC}^2}{2R_C} \quad \text{.....(i)}$$

Output Power:

The ac output power from the amplifier is given as:

$$P_{O\ AC} = V_{CE\ RMS}\, I_{C\ RMS} = \frac{v_{ce\ P\text{-}P}}{2\sqrt{2}} \frac{i_{c\ P\text{-}P}}{2\sqrt{2}}$$

$$P_{O\ AC} = \frac{v_{ce\ P\text{-}P}^2}{8R_C} \quad \text{..... (ii)} \qquad \left(i_{c\ P\text{-}P} = \frac{v_{ce\ P\text{-}P}}{R_C}\right)$$

General Efficiency:

The efficiency is given by the ratio of output ac power to input dc power.

$$\eta = \frac{P_{O\ AC}}{P_{in\ DC}}$$

$$\therefore \eta = \frac{\frac{v_{ce\ P\text{-}P}^2}{8R_C}}{\frac{V_{CC}^2}{2R_C}}$$

$$\therefore \eta = \frac{v_{ce\ P\text{-}P}^2}{4V_{CC}^2} \quad \text{..... (iii)}$$

From figure we may write,

$v_{ce\ P\text{-}P} = v_{ce\ Max} - v_{ce\ Min}$

$V_{CC} = v_{ce\ Max} + v_{ce\ Min}$

Substituting these values in equation (iii), we will get general efficiency:

$$\eta = \frac{1}{4} \frac{(v_{ce\ Max} - v_{ce\ Min})^2}{(v_{ce\ Max} + v_{ce\ Min})^2}$$

$$\therefore \eta = 0.25 \frac{(v_{ce\ Max} - v_{ce\ Min})^2}{(v_{ce\ Max} + v_{ce\ Min})^2} \quad \text{.....(iv)}$$

Maximum Efficiency:

By looking at the figure, we can say that the maximum peak- peak value of output voltage, $v_{ce\ P\text{-}P\ Max}$ is V_{CC}

Using equation (iii), the maximum efficiency, η_{max} is given by:

$$\eta_{max} = \frac{V_{CC}^2}{4V_{CC}^2}$$

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$$= \frac{1}{4} = 0.25 = 25\%$$

$$\therefore \eta_{max} = 25\%$$

Hence the maximum efficiency of class A amplifier is only 25%. This relatively low efficiency makes class A amplifier unattractive in designing audio and power amplifiers.

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Transformer Coupled Class A amplifier:

The efficiency of class A amplifier can be increased by using transformer to couple the load. The efficiency is increased because of two reasons. First, transformer at the output provides impedance matching thereby causing maximum power transfer. And second, the dc resistance of the primary coil of transformer is very small (ideally zero).

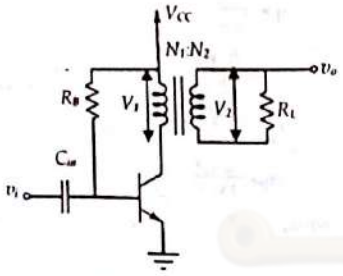

Figure: Transformer coupled Class A amplifier

In this amplifier a transformer is used to couple the output to the load resistor R_L. The output resistance as seen by the transistor at collector is R_L'.

Let $N_1(N_2)$, $I_1(I_2)$,$V_1(V_2)$ be the number of turns, current and voltage at primary(secondary) coil of the transformer.

Then, we know,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

Similarly,

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}$$

Combining above relations, we get,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}\frac{N_1^2}{N_2^2}$$

$$R_L' = a^2 R_L \text{ .....(i)}$$

Where,

$a = \frac{N_1}{N_2}$ is called turn ratio of transformer

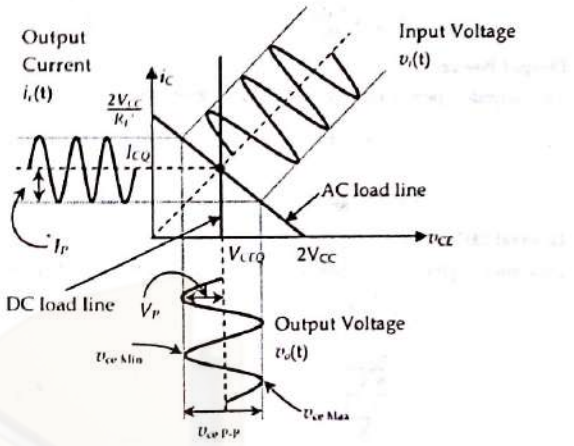

Figure above shows input and output waveforms of transformer coupled class A amplifier. Because primary winding of transformer have ideally zero resistance, the DC load line is a straight vertical line given by:

$$v_{CE} = V_{CC}$$

Similarly transformer's primary coil in the collector of transistor provides ac load of resistance R_L'. Therefore AC load line corresponding to R_L' will be as shown in the figure. Instead of extending to V_{CC} in v_{CE} axis and to $\frac{V_{CC}}{R_L'}$ in i_C axis, the ac load line extends up to $2V_{CC}$ and upto $\frac{2V_{CC}}{R_L'}$ in v_{CE} and i_C axis respectively. This is due to counter emf effect of the transformer.

Q-point of the amplifier is obtained at the intersection of dc and ac load line. The input and output current/voltage waveforms swing across the Q-point as shown in the figure.

Input Power:

The input power to the amplifier is given by:

$$P_{in\,DC} = I_{CQ} V_{CC}$$

From the figure $I_{CQ} = \frac{V_{CC}}{R_L'}$

$$P_{in\,DC} = \frac{V_{CC}}{R_L'} V_{CC}$$

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$$\therefore P_{in\,DC} = \frac{V_{CC}^2}{R_L'} \quad \text{.....(i)}$$

Output Power:

The output ac power from the amplifier is given as:

$$P_{O\,AC} = \frac{V_{CE\,RMS}^2}{R_L'}$$

$$P_{O\,AC} = \frac{v_{ce\,P\text{-}P}^2}{8R_L'} \quad \text{.....(ii)}$$

General Efficiency:

Efficiency is given by the ratio of output ac power to input dc power.

$$\eta = \frac{P_{O\,AC}}{P_{in\,DC}}$$

$$= \frac{\frac{v_{ce\,P\text{-}P}^2}{8\,R_L'}}{\frac{V_{CC}^2}{R_L'}}$$

$$\therefore \eta = \frac{v_{ce\,P\text{-}P}^2}{8V_{CC}^2} \quad \text{.....(iii)}$$

From the figure we can clearly see,

$v_{ce\,P\text{-}P} = v_{ce\,Max} - v_{ce\,Min}$

$2V_{CC} = v_{ce\,Max} + v_{ce\,Min}$

Substituting these values in equation (iii), we will get general efficiency:

$$\eta = \frac{1}{8} 4\frac{(v_{ce\,Max} - v_{ce\,Min})^2}{(v_{ce\,Max} + v_{ce\,Min})^2}$$

$$\therefore \eta = 0.5\frac{(v_{ce\,Max} - v_{ce\,Min})^2}{(v_{ce\,Max} + v_{ce\,Min})^2} \quad \text{.....(iv)}$$

Maximum Efficiency:

By looking at the figure, we can say that the maximum peak- peak value of output voltage, $v_{ce\,P\text{-}P\,Max}$ is $2V_{CC}$

Using equation (iii), the maximum efficiency, η_{max} is given by:

$$\eta_{max} = \frac{4V_{CC}^2}{8V_{CC}^2}$$

$$= \frac{1}{2} = 0.5 = 50\%$$

$$\therefore \eta_{max} = 50\%$$

Thus, the maximum efficiency of transformer coupled class A amplifier is 50%. However in practice the efficiency is lower than 50%. This is because in our analysis we have assumed ideal conditions ignoring losses in transformer, resistance of primary coil etc.

Class B output stage

Class B output stages are those output stage in which for each transistor the collector current flows for the half cycle of the applied input signal. The amplifiers utilizing class B at the output stage are known as class B amplifiers. Thus conduction angle for class B output stage is 180°.

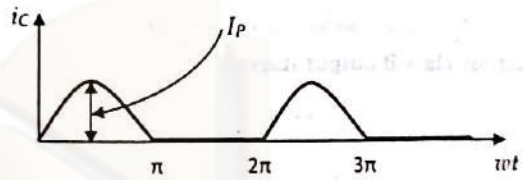

Figure: Collector current waveform of a transistor operating in class B output stage

Class B output stage generally use two transistors to conduct for positive and negative half cycles of the applied input signal. The currents from both the transistors are then combined to form the load current. The class B amplifiers are generally used in RF amplifiers in which the output of the amplifier circuit is fed to the transmitting antenna.

Push Pull Circuits:

Two signals of opposite polarity are known as push pull signals. Push pull signals are widely used in the input of class B amplifiers. The circuits which produce push pull signals are also known as phase splitters. Different methods of generating push pull signals are as follows:

i. Using Transformer:

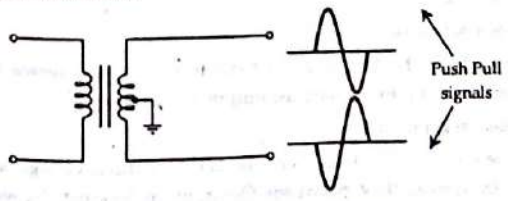

Figure: Transformer based phase splitter

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ii. Using Transistor:

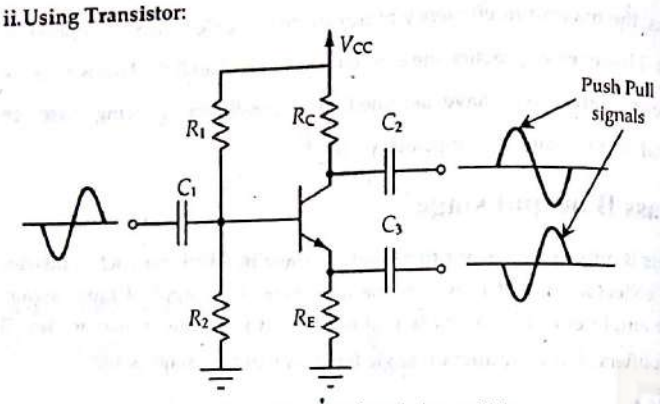

Figure: Transistor based phase splitter

Basic circuit for class B output stage

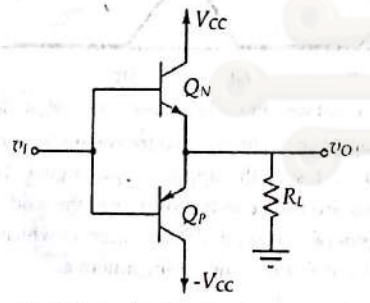

Figure: Basic circuit for class B output stage

Figure shows the basic circuit diagram for class B output stage. It consists of two complementary types of transistor (i.e. pnp and npn transistors) connected in **push pull fashion**. It is called so because transistor Q_N pushes (or sources) current to the load and transistor Q_P pulls (or sinks current) from the load. This circuit is so designed that only one transistor can be "on" at a time.

When input v_I is zero:

In this case both the transistors will be in cut-off and hence will not conduct any current. Eventually the output will be zero.

When input v_I is positive:

In this case transistor Q_N turns "on". In fact if the input voltage is greater than cut in voltage 0.5V transistor Q_N starts to conduct. However for

significant current to flow the input must exceed 0.7 V. At the same time when input is increased, transistor Q_P will be reversed biased and eventually turned off by the base emitter drop of Q_N, i.e. v_{BEN}.

The output voltage when transistor Q_N is fully on is given as:

$$v_O = V_{CC} - V_{CENsat}$$

Where, V_{CENsat} is the output voltage v_{CE} of transistor Q_N at saturation. It is a small leakage voltage that appears when transistor Q_N is "on".

This positive output voltage v_O will push current into the load.

The input voltage corresponding to maximum positive output is given as:

$$v_I = V_{CC} - V_{CENsat} + V_{BEN}$$

When input v_I is negative:

In this case transistor Q_P turns "on". In fact, if the input voltage is greater (in reverse sense) than cut in voltage - 0.5 V, transistor Q_P starts to conduct. However for significant current to flow the input must exceed -0.7 V. At the same time transistor Q_N will be reversed biased and eventually turned off by the base emitter drop of Q_P, i.e. v_{EBP}.

utput voltage when transistor Q_P is fully on is given as:

$$v_O = - V_{CC} + V_{ECPsat}$$

Where, V_{ECPsat} is the output voltage v_{EC} of transistor Q_P at saturation. It is a small leakage voltage that appears when transistor Q_P is "on"

This negative output voltage will pull current from the load.

The input voltage corresponding to maximum negative output is given as:

$$v_I = - V_{CC} + V_{ECPsat} - V_{EBP}$$

Transfer characteristics of class B output stage:

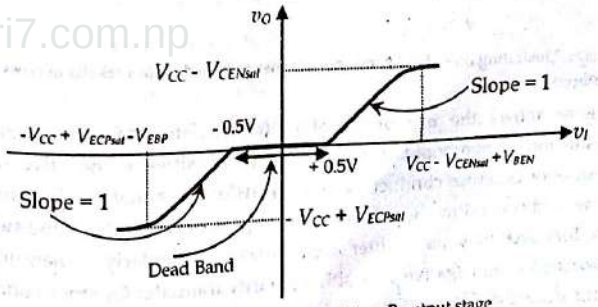

Figure: Transfer Characteristics of class B output stage

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Figure shows the transfer characteristics of class B output stage. From the figure we can see that within 0.5V on the either side, none of the two transistors is on. This region is known as **dead band**. This region will give rise to the cross over distortion as discussed next.

Cross Over Distortion in class B output stage:

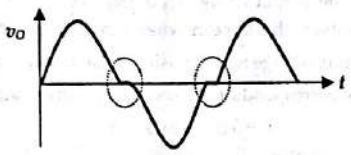

Figure: Cross over distortion in class B amplifiers

Cross over distortion occurs in class B amplifiers as a result of one transistor cutting off before the other transistor starts conducting. The region shown by dotted circles denotes cross over distortion in output waveform.

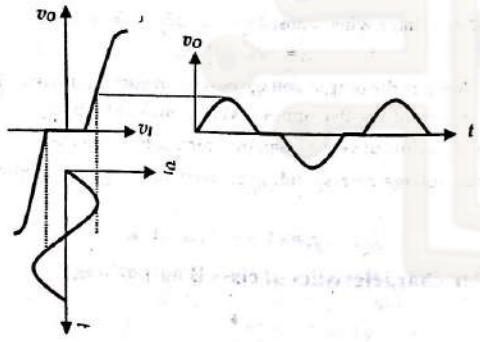

Figure: Illustrating how dead band in the transfer characteristics results in cross over distortion.

Figure shows the use of transfer characteristics to show cross over distortion. When input is switched from positive to negative polarity transistor Q_N stops conducting and transistor Q_P cannot conduct instantly. It will start conducting only after input goes down to the value sufficient to forward bias its emitter base junction. Similarly, when input is switched from negative to positive polarity transistor Q_P stops conducting and transistor Q_N cannot conduct instantly. It will start conducting only after input goes up to the value sufficient to forward bias its emitter base

junction. Thus, in both cases there will be delay in getting one of the transistors forward biased. This delay will give rise to corresponding delay in output that appears across the load. Since the positive and negative portions of waveform cannot meet perfectly at zero crossings, this type of distortion is called as **cross over distortion.**

Cross over distortion can be eliminated by removing dead band in the transfer characteristics. Dead band can be removed by applying a slight forward bias to base emitter junction of both transistors forming the output stage. This is exactly what will be done in class AB amplifiers, which is to be discussed next. But before leaving this topic, there is another way to reduce cross over distortion in class B amplifiers. It is by using high gain op- amp in the negative feedback configuration as shown in the circuit below:

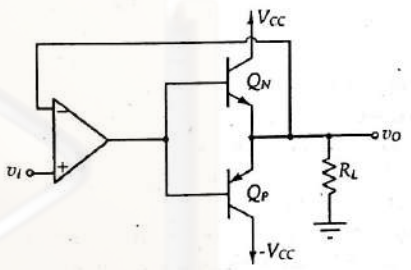

Figure: Class B output stage with op-amp connected in negative feedback connection to reduce cross over distortion.

The use of op amp will reduce the dead band from ±0.5 V to $\pm\frac{v_O}{A}$ V. Thus there is considerable decrease in the cross over distortion.

Transformer coupled class B push pull amplifier:

Figure shows the circuit diagram of transformer coupled class B push pull amplifier. The name push pull derives from the fact that the input transformer supplies push pull signals to the bases of two npn transistors Q_1 and Q_2. In this context transformer is also known as phase splitter. Since the push pull signals are 180° out of phase, two transistors cannot be "on" at the same time. Also the transistors are biased such that in the absence of input signal both transistor remains in cut off mode and conduct no current. Thus under no signal condition transistor dissipate

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negligible power. In this way, efficiency is increased. Let V_P and I_P be the peak voltage and peak current flowing through the load.

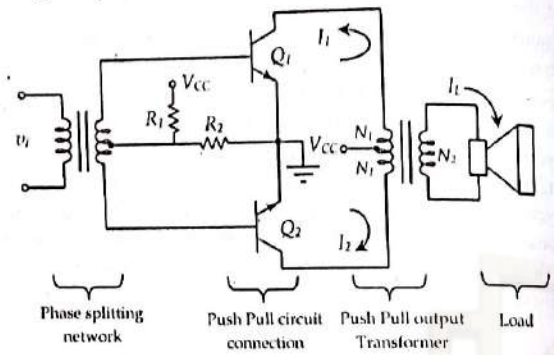

Figure: Transformer coupled class B push pull amplifier

When the input v_i is positive:

In this case, transistor Q_1 is "on" while the transistor Q_2 is "off". Current I_1 flows through the circuit in the direction indicated. Load current I_L flows through the load.

When the input v_i is negative:

In this case, transistor Q_2 is "on" while the transistor Q_1 is "off". Current I_2 flows through the circuit in the direction indicated. Load current I_L flows through the load.

In both case the direction of the load current is same.

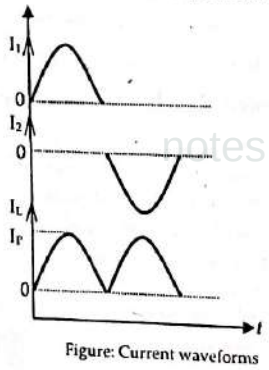

Figure: Current waveforms

Figure shows the current waveforms in class B transformer coupled amplifier. The direction of I_1 and I_2 are different but the direction of load current I_L is same. Thus load current I_L is a full wave rectified sine wave as shown in the figure.

Since the current waveform at load is in the form of full wave rectified sine wave, the rms value, I_{RMS}, and average value or dc value, I_{av}, of the load current I_L are given as:

$$I_{RMS} = \frac{I_P}{\sqrt{2}} \text{ and } I_{av} = \frac{2I_P}{\pi}$$

Output power:

The ac output power delivered to the load is given as:

$$P_{O\,AC} = V_{RMS}\, I_{RMS} = \frac{V_P}{\sqrt{2}} \frac{I_P}{\sqrt{2}}$$

$$P_{O\,AC} = V_{RMS}\, I_{RMS} = \frac{V_P}{\sqrt{2}} \frac{V_P}{\sqrt{2}\, R_L}$$

$$\therefore P_{O\,AC} = V_{RMS}\, I_{RMS} = \frac{V_P^2}{2R_L} \quad \text{.....(i)}$$

Input Power:

The input dc power of the amplifier is given as:

$$P_{in\,DC} = I_{av}\, V_{CC} = \frac{2I_P V_{CC}}{\pi} \quad \text{.....(ii)}$$

General Efficiency:

It is given by the ratio of output ac power to input dc power.

$$\eta = \frac{P_{O\,AC}}{P_{in\,DC}} = \frac{\frac{V_P^2}{2R_L}}{\frac{2I_P V_{CC}}{\pi}}$$

$$\eta = \frac{\pi}{4} \frac{V_P^2}{I_P R_L V_{CC}}$$

$$\eta = \frac{\pi}{4} \frac{V_P^2}{\frac{V_P}{R_L} R_L V_{CC}}$$

$$\therefore \eta = \frac{\pi}{4} \frac{V_P}{V_{CC}} \quad \text{.....(iii)}$$

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Maximum Efficiency:

The maximum value of peak output voltage V_P is V_{CC}. Therefore, substituting V_{CC} for V_P in equation (iii) we get maximum efficiency of the amplifier. If we do so, we get,

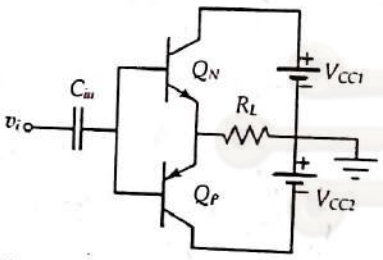
$$\eta_{max} = \frac{\pi}{4}\frac{V_{CC}}{V_{CC}} = \frac{\pi}{4} = 0.785 = 78.5\%$$

$$\therefore \eta_{max} = 78.5\%$$

Thus the maximum efficiency of class B push pull amplifier is 78.5% which is very good as compared to that of class A amplifier.

Complementary symmetry class B push pull amplifier:

The main disadvantage of transformer coupled class B push pull amplifier is the costly and bulky transformer. Complementary class B push pull amplifier eliminates the need of transformer.



When V_{in} is positive:

In this case transistor Q_N is on and transistor Q_P is off. Output current flows through the load resistor.

When V_{in} is negative:

In this case transistor Q_P is on and transistor Q_N is off. Output current flows through the load resistor.

In both cases, the direction of load current is same.

Complementary symmetry pushpull circuit using Darlington pair

A more practical version of complementary push pull amplifier is shown in the figure. Note that the load is driven as the output of an emitter follower so that the resistance of the load is matched by the low output resistance of the driving source. The circuit uses complementary Darlington connected transistors to provide higher current and lower output resistance.

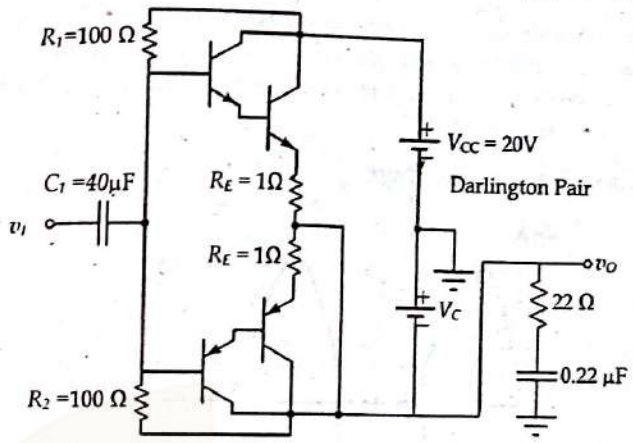


Figure: Complementary class B push pull amplifier

Quasi Complementary class B amplifier:

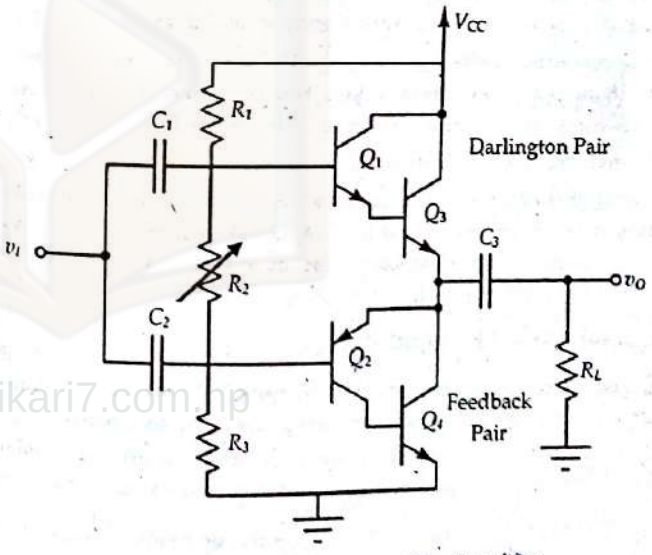


Figure: Quasi complementary Class B amplifier

Figure shows the circuit diagram of quasi complementary class B amplifier. It is one of the most popular configurations for power amplifier. In this circuit push pull action is achieved by the npn and pnp transistors Q_1 and Q_2 while the output is provided by the two matched

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npn transistors Q_3 and Q_4. The connection of Q_2 and Q_4 is called feedback pair. Its main purpose is to reduce the output impedance of the circuit so as to effectively drive the load. Variable resistor R_2 can be adjusted to reduce cross over distortion.

Class AB output stage:

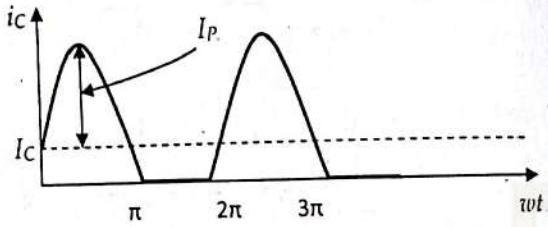

Figure: Output collector current waveform of a transistor operating in Class AB amplifier.

Class AB output stage is a type of output stage in which transistor is so biased that it conducts for an interval slightly greater than half cycle. The resulting conduction angle is greater than 180° but much less than 360° .Class AB output stage generally uses two transistors in which one transistor conducts for positive half cycle with conduction angle slightly greater than 180° and another transistor conducts for negative half cycle with conduction angle slightly greater than 180° . The currents from both the transistors are then combined to form the load current. In this way class AB output stage eliminates cross over distortion which was introduced in class B amplifier.

Circuit for Class AB output stage:

Cross over distortion can be eliminated by using class AB output stage. As shown in the figure, a small bias voltage V_{BB} is applied between the bases of Q_N and Q_P. Since transistors Q_N and Q_P are of similar characteristics, any voltage V_{BB} applied between the bases will divide equally in $\frac{V_{BB}}{2}$ and $\frac{V_{BB}}{2}$ as shown. This voltage sets up required amount of quiescent current required by the amplifier. Thus, even when input signal is zero, both transistors will conduct and cross over distortion is completely eliminated.

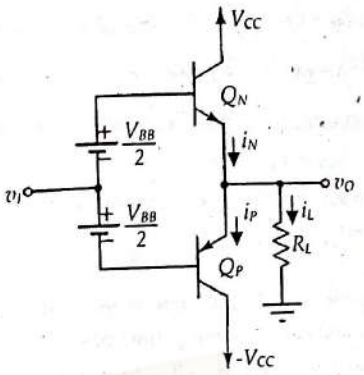

Figure: Circuit for class AB output stage

When v_I is very small or zero:

Considering base emitter junction of transistors as forward biased diode; forward biased by $\frac{V_{BB}}{2}$ volts, we can write:

$$i_N = i_P = I_Q = I_S e^{\frac{V_{BB}}{2\eta V_T}}$$

The value of V_{BB} is selected so as to yield the required quiescent current I_Q.

When v_I is positive:

In this case transistor Q_N acts as an emitter follower and supplies current to pnp transistor and load. Therefore we may write,

$$i_N = i_P + i_L$$

Due to positive input voltage v_I, the base emitter voltage drop of transistor Q_N, v_{BEN}, also increases. Thus output voltage is positive and given by following expression after applying KVL:

$$v_O = v_I + \frac{V_{BB}}{2} - v_{BEN}$$

As the output voltage v_O is increased, the load current i_L is also increased. The current i_N has to increase. As a result, base emitter voltage of npn transistor v_{BEN} also increases. But since the total voltage between bases of two transistors is almost fixed at V_{BB}, this will lead to corresponding decrease in base emitter voltage drop of pnp transistor i.e v_{EBP} and hence current i_P is decreased.

The relationship between the current i_N and i_P is obtained as follows:

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$$v_{BEN} + v_{EBP} = V_{BB}$$

$$\text{or, } \eta V_T \ln\left(\frac{i_N}{I_S}\right) + \eta V_T \ln\left(\frac{i_P}{I_S}\right) = V_{BB}$$

Which, may be simplified using properties of logarithms to obtain,

$$i_N i_P = I_Q^2$$

Since the product is constant, the increase in current i_N causes decrease in current i_P and vice versa.

When v_I is negative:

In this case reverse happens. Transistor Q_P acts as an emitter follower and supplies currents to load and current through the transistor Q_N decrease.

Thus in conclusion, we can say that class AB output stage operates in the manner similar to class B output stage but with one major exception. That is for small input voltage both transistors conduct and as input is increased one of the two transistors takes over the operation. Because of smooth transition the cross over distortion is almost completely eliminated.

Transfer characteristics of class AB output stage

The transfer characteristics for class AB output stage is given in the figure. As we can see from figure, the transition is smooth and thus there will be no distortion.

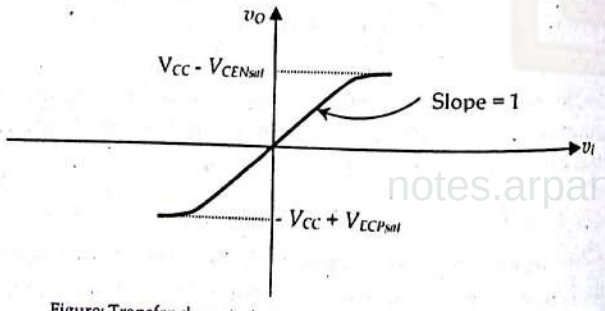
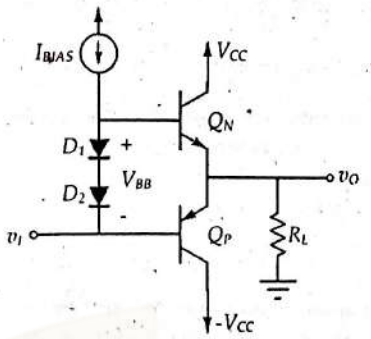


Figure: Transfer characteristics of class AB output stage

Biasing in Class AB output stage:

The voltage V_{BB} required for class AB output stage as discussed can be done in two ways. They are as follows:

Biasing Using Diodes:



In this scheme, the bias voltage V_{BB} is generated by passing current I_{BIAS} through two diodes D_1 and D_2 as shown in the figure. Thus V_{BB} in this case will be equal to two diode drops.

Biasing Using V_{BE} multiplier:

This is another scheme for generating voltage V_{BB} in class AB output stage.

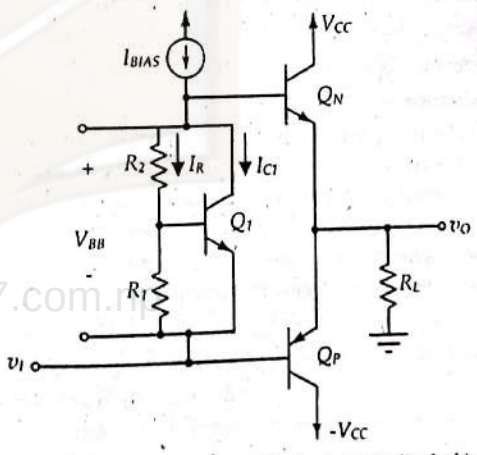


Figure: Class AB output stage utilizing V_{BE} multiplier for biasing

Neglecting the base current of transistor Q_1 we can write,

$$I_R = \frac{V_{BE1}}{R_1}$$

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$$V_{BB} = I_R(R_1+R_2)$$
$$= \frac{V_{BE1}}{R_1}(R_1+R_2)$$
$$\therefore V_{BB} = V_{BE1}\left(1+\frac{R_2}{R_1}\right)$$

Since the voltage V_{BB} so produced is obtained by multiplying base emitter voltage of transistor Q_1 with the factor $(1+R_2/R_1)$, so this circuit has got its name as V_{BE} multiplier.

I_{BIAS} supplies current I_R and I_{C1}.

Power BJT

Power BJTs are the transistors which are designed to handle relatively large currents in the range of amperes and to withstand power dissipation in the watts and tens of watts ranges. Their physical structure, packaging and specification are different from that of small signal transistors.

Junction Temperature:

Power transistors dissipate large amount of power in their collector base junction. The dissipated power raises the temperature of junction. The temperature at the junction is denoted as T_j. The junction temperature should not exceed the maximum limit $T_{j\,max}$, otherwise the junction may be permanently damaged. For silicon based transistors maximum junction temperature is in the range of 150° C to 200° C.

Thermal Resistance:

Let us consider a BJT operating in free air condition. The heat dissipated by the junction is conducted away to the transistor case and from the case into the surrounding at ambient condition. Let us assume BJT is dissipating power P_D watts when its junction temperature is T_j .Let T_A denote ambient temperature. Then the temperature rise of the junction relative to the ambient temperature is expressed as:

$$T_j - T_A = \theta_{JA} P_D$$

Where, θ_{JA} is the **thermal resistance** between the junction and ambience. It has a unit of degree Celsius per watt. Thermal resistance gives the increase in the temperature of the junction for each watt of power dissipated. Since the junction temperature should not exceed the maximum value $T_{j\,max}$, thermal resistance of the transistor should be as low as possible so as to prevent the increase in junction temperature when large amount of power is dissipated in the transistor. The value of thermal resistance is usually specified in the manufacturer's data sheets.

The thermal conduction process can be thought of as electrical circuit as shown below in which power dissipated is analogous to current, thermal resistance is analogous to resistor and the node voltages are analogous to temperature of junction and ambient condition respectively.

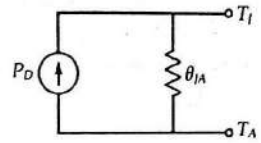

Figure: Electrical analogy of thermal conduction process

Power Derating Curve:

The graphical relationship between the maximum allowable power dissipation and the ambient temperature of a transistor is called as **power derating curve**. Power derating curve is generally given in the datasheets by the manufacturers. This curves shows that transistor can dissipate power P_{D0} safely within the ambient temperature T_{A0}. However if the temperature is increased beyond T_{A0} the power dissipation in the transistor must be derated accroding to the straignt line as shown in the figure.

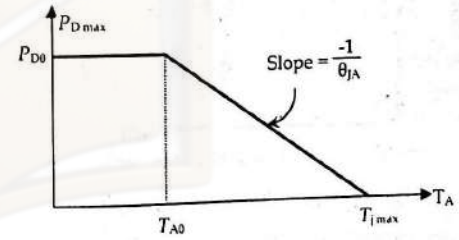

Figure: Power Derating curve of BJT

Apart from derating curve, if $P_D(T_0)$ is the power dissipation at temperature T_0 and $P_D(T_1)$ is the power dissipation at temperature T_1. Then we can use following relation to find $P_D(T_1)$ when $P_D(T_0)$ is known.

$$P_D(T_1) = P_D(T_0) - (T_1 - T_0)\,\theta_{JA}$$

Transistor Case and heat sinks

Heat is conducted from the junction of the semiconductor material towards the case and into the surrounding air. It is radiated from the surface of the case. To improve conduction good physical contact is made between the junction and the device's metal case. In many power

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transistor collector is in direct contact with the case. To improve the conduction and radiation of heat from the case to the surrounding, power devices are often equipped **heat sink**. These are attached to the device to be cooled and conduct the heat outwards. To enhance the cooling operation one can use cooling fans at the surface of the heat sink of the case.

The total junction to ambient thermal resistance can be expressed as

$$\theta_{JA} = \theta_{SA} + \theta_{JC} + \theta_{CS}$$

Where,

θ_{SA} = heat sink to ambient thermal resistance

θ_{JC} = junction to case thermal resistance

θ_{CS} = case to heat sink thermal resistance

When heat sink is used the electrical equivalent circuit for the thermal conduction process is given as follows:

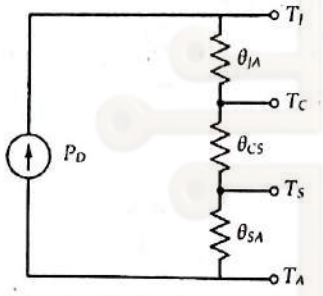

Figure: Equivalent analogy for thermal conduction process when heat sink is used.

Class C output stage

Class C output stage is the output stage in which a transistor is biased such that the collector current flows for less than 180 of the input signal.

The collector current waveform of a transistor operating in class C is given below:

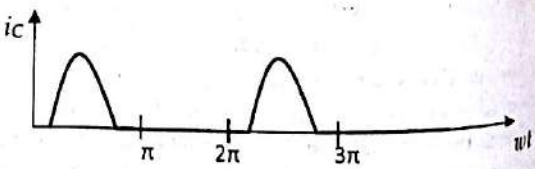

Figure: Collector current waveform of transistor operating in class C

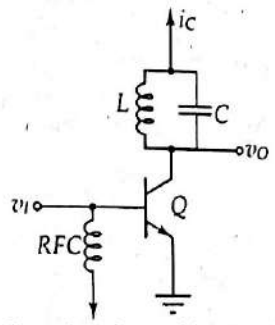

Figure: circuit diagram of class C amplifier

Figure shows the circuit of class C amplifier. The tuned circuit in the collector will provide a full cycle of output signal only for the fundamental frequency of the tuned circuit. So class C amplifiers are used in single frequency circuits such as communication circuit.

T d Amplifiers:

Tuned amplifiers are those amplifiers which amplify specific band of frequency or narrow band of frequency. Tuned amplifier makes use tuned circuit.

Because of the resonance phenomenon in tuned circuit, tuned amplifier is capable of selecting a particular band of frequency for amplification.

Class A tuned amplifier

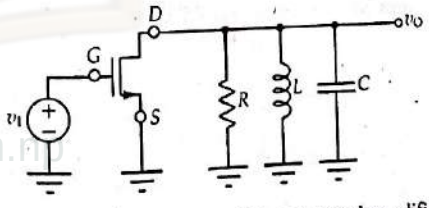

Figure shows the circuit diagram of class A tuned amplifier utilizing N channel Enhancement MOSFET. It consists of tuned circuit at the drain of the MOS transistor. The output of the circuit is taken across the parallel RLC circuit. Input v_I is applied to the gate of MOS transistor. The above circuit can only amplify inputs signals with frequencies in the range lying within the bandwidth of the amplifier. So this amplifier is called tuned

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amplifier. It is amplifier of class A type. This means output current flows for complete cycle of input.

The output current i_o can be written as:

$$i_O = Y v_O$$

Where, Y is the admittance of parallel RLC circuit.

$$Y = \frac{1}{R} + j(wC - \frac{1}{wL})$$

$$\therefore i_O = [\frac{1}{R} + j(wC - \frac{1}{wL})] v_O$$

$$\therefore i_O = (G + jB)v_O \quad \text{.....(i)}$$

Where, $G = \frac{1}{R}$ .....(ii)

and

$$B = (wC - \frac{1}{wL}) \quad \text{.....(iii)}$$

At lower frequency B is large due to large value of inductive susceptance. As a result Y is large and v_O is small.

At resonant frequency w_O, $B = 0$. As a result Y is least and v_O is largest. Thus at resonance frequency, we have,

$$B = 0$$

$$\text{or, } w_O C - \frac{1}{w_O L} = 0$$

$$\therefore w_O = \frac{1}{\sqrt{LC}} \quad \text{.....(iv)}$$

Similarly, at cutoff frequency of w_C (3-dB points)

$$|B| = |G|$$

$$\text{or, } B^2 = G^2$$

$$\text{or, } B = \pm G$$

$$\therefore B = \pm \frac{1}{R} \quad \text{..... (v)}$$

From equation (v) and (iii) at w_C

$$(w_C C - \frac{1}{w_C L}) = \pm \frac{1}{R}$$

At lower cutoff frequency w_{C1},

$$(w_{C1} C - \frac{1}{w_{C1} L}) = \frac{-1}{R} \quad \text{.....(vi)}$$

At upper cutoff frequency w_{C2},

$$(w_{C2} C - \frac{1}{w_{C2} L}) = \frac{1}{R} \quad \text{.....(vii)}$$

Solving equations (vi) and (vii), we get,

$$w_{C1} = \frac{-1}{2RC} + \sqrt{(\frac{1}{2RC})^2 + \frac{1}{LC}}$$

$$w_{C2} = \frac{1}{2RC} + \sqrt{(\frac{1}{2RC})^2 + \frac{1}{LC}}$$

Therefore bandwidth is given by:

$$BW = w_{C2} - w_{C1}$$

$$BW = \frac{1}{RC}$$

The quality factor (Q) is given by:

$$Q = \frac{w_O}{BW}$$

$$\therefore Q = w_O RC$$

Lastly, the frequency response curve (Gain Vs w) of the circuit is given below:

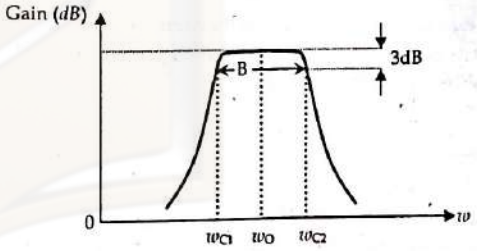

Figure: Frequency response curve of class A tuned amplifier

Note:

The frequency response curve is similar to that of band pass filter. This gives the reason for why tuned amplifier amplifies only over the narrow range of input signal. Also if the reader is interested s/he can find the transfer function of the circuit and take its magnitude which is nothing but the gain of the amplifier. The expression will verify why the shape of curve is as shown.

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Advantages of Tuned Amplifiers

Followings are the advantages of the tuned amplifiers:

i. Small power losses
ii. High selectivity
iii. Smaller collector supply voltage

Solved Numerical

1. A class B audio amplifier is providing 20V peak sine wave signal to 8Ω speaker with power supply of 25V (=V_{CC}). At what efficiency is it operating? **2068 Chaitra**

Solution:

We have,

Supply voltage, V_{CC} =25 V

Load resistance, $R_L = 8\ \Omega$

$$\eta = \frac{\pi}{4}\frac{V_P}{V_{CC}}$$

$$\text{or, } \eta = \frac{\pi}{4}\frac{20}{25}$$

$$\therefore \eta = 0.628 = 62.8\%$$

Thus the amplifier is working at 62.8% efficiency.

2. Calculate the efficiency of transformer coupled push pull power amplifier for a supply voltage of 20V and output of: **2069 Chaitra**

V_P =20V

V_P =16V

Solution:

We have,

Supply voltage, $V_{CC} = 20$ V

We know for transformer coupled push pull amplifier efficiency is given as:

$$\eta = \frac{v_{ce\ P\text{-}P}^2}{8V_{CC}^2}$$

Since $v_{ce\ P\text{-}P} = 2\,v_P$

$$\eta = \frac{v_P^2}{2V_{CC}^2}$$

When $v_P = 20$V

$$\eta = \frac{20^2}{2 \cdot 20^2} = 0.5 = 50\%$$

When $v_P = 16$ V

$$\eta = \frac{16^2}{2 \cdot 20^2} = 0.32 = 32\%$$

3. Calculate the efficiency of class B amplifier for a supply voltage of V_{CC} = 24V with peak output voltage of

i. $V_P = 22$V ii. $V_P = 6$ V

Solution

We know the efficiency of class B amplifier is given as:

$$\eta = \frac{\pi}{4}\frac{V_P}{V_{CC}}$$

i. $$\eta = \frac{\pi}{4}\frac{V_P}{V_{CC}} = \frac{\pi}{4}\frac{22}{24} = 0.72 = 72\%$$

ii. $$\eta = \frac{\pi}{4}\frac{V_P}{V_{CC}} = \frac{\pi}{4}\frac{6}{24} = 0.196 = 19.6\%$$

4. A transformer coupled class A large signal amplifier has maximum and minimum values of collector- emitter voltages of 25V and 2.5V. Determine its collector efficiency.

Solution:

$v_{ce\ Max} = 25$ V

$v_{ce\ min} = 2.5$ V

We know the collector efficiency is given as:

$$\eta = 0.5\frac{(v_{ce\ Max} - v_{ce\ Min})^2}{(v_{ce\ Max} + v_{ce\ Min})^2}$$

$$\text{or, } \eta = 0.5\frac{(25 - 2.5)^2}{(25 + 2.5)^2} = 0.335$$

$$\therefore \eta = 33.5\%$$

5. A class A transformer coupled amplifier uses 25:1 transformer to drive 4Ω load. Calculate the effective load as seen by the collector of the transistor.

Solution:

Load resistance, $R_L = 4\ \Omega$

Turn ratio, $a = 25$

Therefore the load resistance as seen by the collector side of transistor is given as:

$$R_L' = a^2 R_L = 25^2 \cdot 4 = 2500\ \Omega = 2.5\ k\Omega$$

6. For a class B amplifier using a supply of V_{CC} = 30V and driving a load of 16Ω determine the maximum input power, output power and transistor dissipation.

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Solution:

We have,

Load resistance, $R_L = 16\Omega$

Supply voltage $V_{CC} = 30V$

We know, for class B amplifier,

Input dc power, $P_{in\,DC} = \frac{2}{\pi}\frac{V_P V_{CC}}{R_L}$

Output AC power, $P_{O\,AC} = \frac{V_P^2}{2R_L}$

For maximum input and output power, $V_P = V_{CC}$

Therefore,

$P_{in\,DC\,Max} = \frac{2}{\pi}\frac{V_{CC}^2}{R_L} = \frac{2}{\pi}\frac{30^2}{16} = 35.81$ Watts

$P_{O\,AC\,Max} = \frac{V_{CC}^2}{2R_L} = \frac{30^2}{2.16} = 28.125$ Watts

Power dissipation in two transistors,

$P_{2Q} = P_{in\,DC\,Max} - P_{O\,AC\,Max} = 35.81 - 28.125 = 10.68$ Watts

Power dissipation in single transistor, $P_Q = \frac{P_{2Q}}{2} = \frac{10.68}{2} = 5.34$ Watts

7. For class B amplifier providing 20V peak signal to 16 Ω load (speaker) and a power supply of V_{CC} =30 V, determine the input power, output power and circuit efficiency.

Solution:

Supply voltage, $V_{CC} = 30V$

Peak output, $V_P = 20$ V

Load resistance $R_L = 16\ \Omega$

Now,

Peak current, $I_P = \frac{V_P}{R_L} = \frac{20}{16} = 1.25$ A

∴ Input power, $P_{in\,DC} = \frac{2I_P V_{CC}}{\pi} = \frac{2 \times 1.25 \times 30}{\pi} = 23.87$ Watts

∴ Output power, $P_{O\,AC} = \frac{V_P}{\sqrt{2}}\frac{I_P}{\sqrt{2}} = \frac{20 \times 1.25}{2} = 12.5$ Watts

∴ Circuit Efficiency = $\frac{P_{O\,AC}}{P_{in\,DC}} \times 100 = \frac{12.5}{23.87} = 52.37\%$

8. Determine what maximum dissipation will be allowed for an 80-W silicon transistor (rated at 25° C) if derating is required above 25°C by a derating factor of 0.5 W/°C at a case temperature of 125°C.

Solution

$T_0 = 25°C$

$T_1 = 125$ °C

$\theta_{10} = 0.5$ W/°C

We know,

$P_D(T_1) = P_D(T_0) - (T_1 - T_0)\theta_{10}$

$= 80 - (125 - 25)0.5$

$\therefore P_D\,(125°C) = 30$ Watts

9. A silicon power transistor is operated with a heat sink (θ_{SA} =1.5 °C/W). The transistor rated at 150W (25°C) has θ_{JC} = 0.5 °C/W and the mounting insulation has θ_{CS} = 0.6 °C/W. What maximum power can be dissipated if the ambient temperature is 40°C and $T_{J\,Max}$ = 200°C ?

Solution:

We know,

$\theta_{JA} = \theta_{SA} + \theta_{JC} + \theta_{CS}$

$\therefore \theta_{JA} = 1.5 + 0.5 + 0.6 = 2.6$ °C/W

Also we know,

$T_J - T_A = \theta_{JA} P_D$

or, $T_{J\,Max} - T_A = \theta_{JA} P_{D\,Max}$

or, $200 - 40 = 2.6\ P_{D\,Max}$

$\therefore P_{D\,Max} = 61.5$ Watts

10. A BJT is specified to have maximum power dissipation P_{D0} of 2 watts at an ambient temperature of 25°C, and a maximum temperature $T_{J\,max}$ of 150°C. Find the following:

i. The thermal resistance θ_{JA}

ii. The maximum power that can be safely dissipated at an ambient temperature of 50°C

iii. The junction temperature if the device is operating at T_A = 25°C and is dissipating 1 W.

Solution:

$T_J - T_A = \theta_{JA} P_D$

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i. $\theta_{JA} = \frac{T_{J\max} - T_{A0}}{P_{D0}} = \frac{150 - 25}{2} = 62.5\ ^\circ C/W$

ii. $P_{D\,Max} = \frac{T_{J\max} - T_A}{\theta_{JA}} = \frac{150 - 50}{62.5} = 1.6\ W$

iii. $T_J = T_A + \theta_{JA} P_D = 87.5\ ^\circ C$

EXERCISE

1. What is cross over distortion and how can it be eliminated? 2071 Shrawan
2. Draw a circuit diagram of tuned amplifier. Determine the range of frequency in which it gives the maximum gain within 3-db range. 2071 Shrawan
3. Why the efficiency of class A amplifier is low? Obtain the expression of the general efficiency of series fed class – A amplifier circuit. 2071 Shrawan
4. What is the maximum efficiency of class B amplifier? State the condition when it occurs. 2070 Chaitra
5. When are tuned amplifiers used? Draw class A tuned amplifier circuit and find its 3db bandwidth. 2070 Chaitra
6. Draw the circuit diagram of complementary symmetry class-AB amplifier using Darlington pair transistor. 2070 Chaitra
7. Draw the circuit of class A series fed amplifier and its corresponding characteristic graph. And find its general efficiency. 2070 Ashad
8. Draw the circuit diagram of transformer coupled class B push pull amplifier. And show that the maximum efficiency is 25π %. 2070 Ashad
9. Determine the general efficiency of transformer coupled class A power amplifier. 2069 Chaitra
10. Draw the circuit diagram of complementary symmetry class AB amplifier. 2069 Chaitra
11. Calculate the efficiency of transformer coupled push pull power amplifier for a supply voltage of 20V and output of: 2069 Chaitra
    $V_P = 20V$
    $V_P = 16V$
12. Draw the circuit diagram of class B push pull amplifier with output transformer and explain how push pull action is achieved. Determine the general efficiency of class B push pull amplifier. 2068 Chaitra
13. Draw class A tuned amplifier circuit and derive the expression for 3db bandwidth of the amplifier. 2068 Chaitra
14. A class B audio amplifier is providing 20V peak sine wave signal to 8Ω speaker with power supply of 25V (=V_{CC}). At what efficiency is it operating? 2068 Chaitra
15. Define cross over distortion in class B amplifiers. Draw quasi complementary symmetry class AB amplifier. And explain how cross over distortion is eliminated in class AB amplifier. 2068 Balsakh
16. What is the maximum efficiency of class B amplifier and state the condition when it occurs? 2068 Balsakh
17. Why heat sink is necessary in power transistor? Explain it with the help of thermal ohm's law or thermal resistance method. 2068 Balsakh

♦♦♦

Chapter 5

Signal Generators and Waveform Shaping Circuits

Introduction

In this chapter we will discuss signal generators and waveform shaping circuits. Signal generators are the oscillators which produce ac signals of desired frequency and shape. Signal generators may be of different types depending upon the waveform of signal generated. Among different oscillators that exist, **sinusoidal oscillators** are the most popular ones. This is primarily because of the fact that **sinusoids** (sine and cosine function) have got the **linearity property**. That means under the linear operations like integration, differentiation etc on sinusoidal signal the resulting signal is again a sinusoid of same frequency but with different phase of course. This is a very important and peculiar property because if we apply sinusoidal signal at the input, all the signals within the circuit will have same shape weather it is passed to capacitor, inductor, resistor or any active components.

Feedback

Feedback is one of the important terms associated with the amplifier circuits. Feedback may be of two types either positive feedback or negative feedback. Almost all the amplifier circuits use the feedback mechanism for performing various functions. Positive feedback is used when we have to design an oscillator circuit where as negative feedback is used in most of the practical amplifier circuits so as to improve circuit performance.

The process of injecting a fraction of output signal from the output terminal to input terminal either in phase/anti-phase with the input signal is known as feedback.

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Types of Feedback

1. **Positive Feedback (or Regenerative feedback):** When the output signal is injected from output terminal to input terminal in phase with the input signal, it is called as positive feedback. Positive feedback is the basis for the design of oscillator circuits. For other purpose it is seldom used. This is because the gain of the amplifier is already so high that positive feedback leads to further increase in the noise level and ultimately cause distortion of signal.

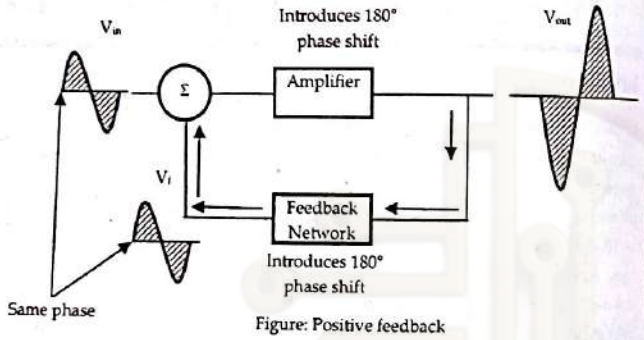

Figure: Positive feedback

2. **Negative Feedback (or Degenerative feedback):** When the output signal is injected from the output terminal to the input 180° out of phase i.e. anti phase with the input signal, it is called as negative feedback. Most of the practical amplifier circuit includes the negative feedback network to reduce the noise level in the output due to the introduction of undesired signal at input.

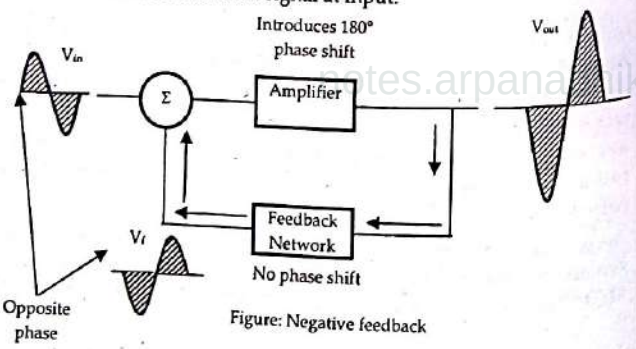

Figure: Negative feedback

Notes:

- A practical amplifier generally makes 180° phase shift.
- Feedback network either makes no phase shift or introduces 180° phase shift depending upon whether amplifiers output is phase inverted or not.

Oscillators:

Those circuits which are used to produce a continuously varying waveform (like sinusoidal, square, triangular etc) without any input signals (not to be confused with biasing dc supply) are known as oscillators. Generally an amplifier having positive feedback at its input works as an oscillator. These circuits require only a slight trigger to start the operation.

Following are the criteria for the amplifiers circuit to work as an oscillator

i. It should employ positive feedback mechanism.
ii. It should follow Barkhausen criteria.ie loop gain must be equal to unity ($A\beta=1$).

Principle of Sinusoidal Oscillators:

Let us consider a sinusoidal oscillator as shown in the figure. It consists of an amplifier with open loop gain A. The feedback network with feedback factor β forms positive feedback to the input of amplifier. With certain conditions fulfilled this circuit will work as an oscillator to produce sinusoidal signal at its output.

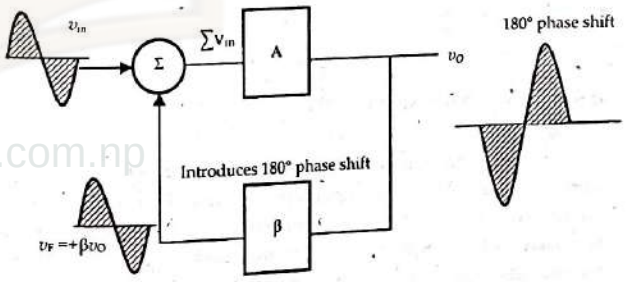

Figure: Amplifier using positive feedback

Let v_{in} be input signal and v_F be feedback voltage then,

$$v_F = \pm \beta v_O$$

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Since we are using positive feedback, therefore we may write,

$$v_F = +\beta v_O$$

Therefore, total input to the amplifier is

$$\sum v_{in} = v_{in} + v_f$$

$$\therefore \sum v_{in} = v_{in} + \beta v_O$$

Since A is the open loop gain of the amplifier then,

$$v_O = A \sum v_{in}$$

$$\text{or, } v_O = A (v_{in} + \beta v_O)$$

$$\text{or, } v_O (1 - A\beta) = A\, v_{in}$$

$$\text{or, } \frac{v_O}{v_{in}} = \frac{A}{1-A\beta}$$

$$\therefore A_F = \frac{A}{1-A\beta}$$

Where,

A_F = amplifier gain with feedback

This is the required expression for the amplifier gain with feedback for positive feedback amplifier.

This expression shows that feedback amplifier gain is independent of the input and output voltage. Thus, we can change the amplifier gain by changing the feedback factor β.

Barkhausen Criteria

The amplifier gain with feedback for the amplifier with the positive feedback is given by:

$$\therefore A_F = \frac{A}{1-A\beta}$$

If Aβ=1 in the above expression, then

$$A_F = \infty$$

i.e. the gain of the amplifier with feedback becomes infinite. This means there is output without any input signal applied. This criterion (Aβ = 1) for amplifier with positive feedback is known as **Barkhausen Criteria**. It is a basis for the design of oscillators and function generators. Here, the infinite gain means that even the vanishingly small input will give rise to the finite output. When the circuit receives a slight trigger or noise at its input, it will start to produce output.

Thus to summarize, the basic condition for a positive feedback amplifier to work as an oscillator are as follows:

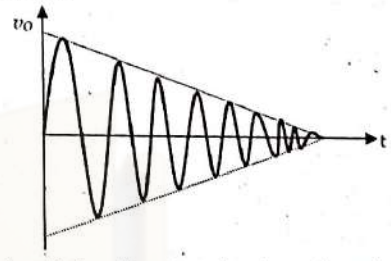
i. It should employ positive feedback mechanism.

ii. It should satisfy Barkhausen criteria.

The above points may also be known as **basic principles of sinusoidal oscillators.**

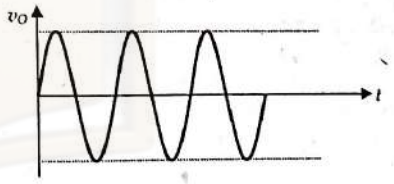
Nature of oscillations

Damped Oscillations:



In this type of oscillation, the output signal gets damped after certain cycles of oscillations. Damped oscillations occur if the criterion of sinusoidal oscillator is not fulfilled.

Undamped Oscillations:



In this type of oscillation, the output signal oscillates for indefinite time period. This type of oscillations occurs if the criterion of sinusoidal oscillator is fulfilled.

Op - amp RC oscillator Circuits

As the name suggests op- amp RC oscillator circuits are the sinusoidal oscillators which produces sinusoidal waveform using op- amp, resistors and capacitors. Followings are the different Op- amp RC oscillators that we will study:

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Wien - Bridge Oscillator:

This is one of the simplest types of Op- amp RC oscillator available. The circuit diagram of Wien bridge oscillator is as shown below:

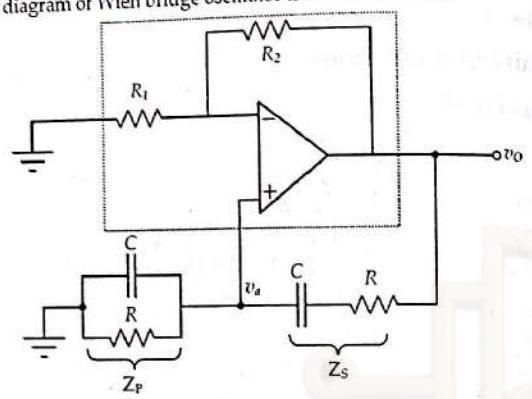

Figure: Wien- Bridge oscillator

Figure shows the circuit diagram of Wien bridge oscillator using op- amp. The circuit portion enclosed by the dotted rectangle is an amplifier with gain given by $(1+R_2/R_1)$ (Since input is applied to non-inverting input from feedback). The RC network forms a feedback network. It consists of two impedances in series forming a voltage divider with the output terminal. The first one is series impedance Z_S formed by series combination of R and C. And the second one is the parallel impedance Z_P formed by the parallel combination of R and C. The voltage drop across Z_P is fed to the non inverting input terminal of the op amp.

The feedback voltage to the non inverting input is v_a. It is the voltage drop across impedance Z_P.

$$v_a = v_O \frac{Z_P}{Z_S + Z_P}$$

$$v_a = \frac{Z_P}{Z_S + Z_P} v_O$$

Comparing it with $v_a = \beta v_O$

We get, Feedback factor, $\beta = \frac{Z_P}{Z_P + Z_S}$

Since the gain of the amplifier for input through non- inverting terminal is

$$A = (1+\frac{R_2}{R_1})$$

Therefore, the loop gain of the amplifier is given by:

$$L(s) = A\beta$$

$$L(s) = (1+\frac{R_2}{R_1}) \frac{Z_P}{Z_P + Z_S}$$

$$L(s) = (1+\frac{R_2}{R_1}) \frac{R.\frac{1}{Cs}}{R+\frac{1}{Cs}} \Big/ \frac{R.\frac{1}{Cs}}{R+\frac{1}{Cs}} + R + \frac{1}{Cs}$$

$$L(s) = (1+\frac{R_2}{R_1}) \frac{R.\frac{1}{Cs}}{R.\frac{1}{Cs} + R^2 + 2R\frac{1}{Cs} + \frac{1}{(Cs)^2}}$$

$$L(s) = (1+\frac{R_2}{R_1}) \frac{1}{1 + RCs + 2 + \frac{1}{RCs}}$$

$$L(s) = \frac{(1+\frac{R_2}{R_1})}{3 + RCs + \frac{1}{RCs}}$$

Substituting $s = jw$, we get,

$$L(jw) = \frac{(1+\frac{R_2}{R_1})}{3 + RCjw + \frac{1}{RCjw}}$$

$$L(jw) = \frac{(1+\frac{R_2}{R_1})}{3 + j(RCw - \frac{1}{RCw})}$$

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For zero phase shift, term contributing to phase shift must be zero,

$$\text{i.e. } RCw_O - \frac{1}{RCw_O} = 0$$

$$\text{or, } RCw_O = \frac{1}{RCw_O}$$

$$\therefore w_O = \frac{1}{\sqrt{RC}}$$

Thus frequency of oscillation in Hertz is given as:

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$$f_o = \frac{1}{2\pi\sqrt{RC}}$$

To satisfy Barkhausen criteria, magnitude of loop gain must be unity.

i.e. $|L(jw)| = 1$

$$\text{or, } \frac{1+\frac{R_2}{R_1}}{3} = 1$$

$$\text{or, } \frac{R_2}{R_1} = 2$$

$$\therefore R_2 = 2R_1$$

RC Phase Shift Oscillator:

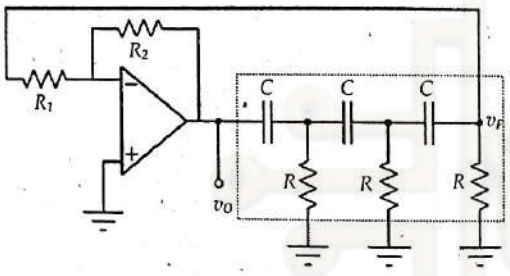


Figure: RC phase shift oscillator

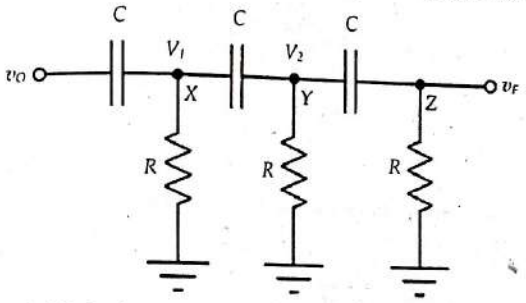
Figure shows the circuit diagram of RC phase shift oscillator. It consists of an op - amp connected in inverting configuration. The RC network forms feedback between the output and inverting input terminal.

Since the op- amp is connected in inverting configuration the output of the op- amp will be phase inverted. In order to feedback this output in same phase at input terminal (i.e. positive feedback), the feedback network must provide 180° phase shift. The RC network of above circuit provides the 180° phase shift and confirms positive feedback to the amplifier. This is the first requirement of sinusoidal oscillator.

Secondly the passive components are so chosen that magnitude of loop gain of this circuit is greater than or equal to unity.

In this way phase shift oscillator generates sinusoidal output.

Let us consider the feedback network in order to derive the expression for loop gain and frequency of oscillation



Let V_1 and V_2 be the node voltages at the nodes X and Y respectively.

Applying KCL at nodes X, Y and Z, we get,

$$(V_1 - v_O)\, jwC + (V_1 - V_2)\, jwC + \frac{V_1}{R} = 0 \quad \text{.....(i)}$$

$$(V_2 - V_1)\, jwC + (V_2 - v_F)\, jwC + \frac{V_2}{R} = 0 \quad \text{.....(ii)}$$

$$(v_F - V_2)\, jwC + \frac{v_F}{R} = 0 \quad \text{.....(iii)}$$

Solving equations (i),(ii) and (iii), we get,

$$= \frac{v_F}{v_O} = \frac{(jwRC)^3}{(jwRC)^3 + 6\,(jwRC)^2 + 5jwRC + 1}$$

Therefore loop gain is given as:

$$L(jw) = A\,\beta(jw) = \frac{A(jwRC)^3}{(jwRC)^3 + 6\,(jwRC)^2 + 5jwRC + 1}$$

$$L(jw) = \frac{-Aj(wRC)^3}{-j(wRC)^3 - 6\,(wRC)^2 + 5jwRC + 1}$$

To satisfy Barkhausen criteria:

i. Phase shift must be zero

For this the real part of denominator in the expression for loop gain must be zero (Because this cancels out "j" term)

Thus,

$$-6(wRC)^2 + 1 = 0$$

The corresponding frequency w_O is then obtained as:

$$-6(w_ORC)^2 + 1 = 0$$

$$6(w_ORC)^2 = 1$$

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$$w_o^2 = \frac{1}{6(RC)^2}$$

$$w_o = \frac{1}{\sqrt{6}RC} \quad \text{.....(iv)}$$

$$\therefore f_o = \frac{1}{2\pi\sqrt{6}\ RC} \quad \text{.....(v)}$$

ii. And magnitude of loop gain must be unity at w_o

i.e. $|L(jw_o)| = 1$

$$\text{or, } \frac{-Aj(w_oRC)^3}{-j(w_oRC)^3 + 5jw_oRC} = 1$$

$$\text{or, } \frac{-A(w_oRC)^2}{-(w_oRC)^2 + 5} = 1$$

$$\text{or, } -A(w_oRC)^2 = -(w_oRC)^2 + 5$$

From equation (iv), we can write, $(w_oRC)^2 = \frac{1}{6}$

So,

$$-A.\frac{1}{6} = -\frac{1}{6} + 5$$

$$\therefore A = -29$$

Also since $|L(jw_o)| = A\beta = 1$

$$\therefore \beta = -\frac{1}{29}$$

Quadrature Oscillator

Quadrature oscillator is another type of op- amp RC oscillator.

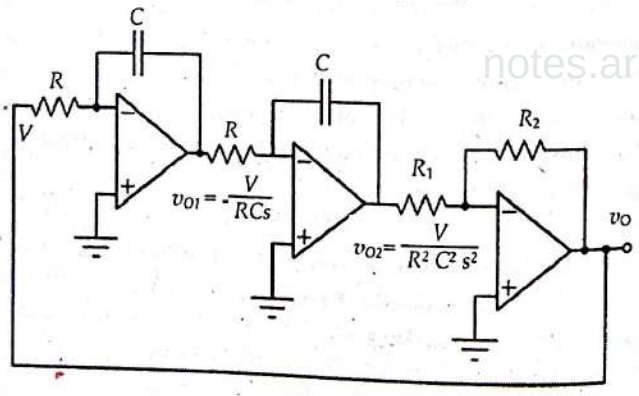

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Figure: Quadrature Oscillator

Figure shows the circuit diagram of Quadrature oscillator using op-amp. It consists of two inverting integrator connected in cascade followed by an inverting amplifier. The name quadrature oscillator comes from the fact that the signal at the output of first op- amp and the second op- amp are at 90 degree phase difference.

Let us consider input V is available at first. Let s be a frequency variable. Then the output of first integrator is given by:

$$v_{O1} = -\frac{V}{RCs}$$

(Integration in time domain with respect to t corresponds to division by s in the frequency domain.)

This voltage becomes input to the second integrator. The output of the second integrator is given as:

$$v_{O2} = \frac{V}{R^2C^2s^2}$$

Thus the output of the whole circuit is given as:

$$v_O = \frac{-R_2}{R_1}v_{O2}$$

$$v_O = \frac{-R_2}{R_1}\frac{V}{R^2C^2s^2}$$

Loop transfer function is given as:

$$L(s) = \frac{v_O}{V} = \frac{\frac{-R_2}{R_1}}{R^2C^2s^2}$$

Substituting $s = jw$, we get,

$$L(jw) = \frac{\frac{R_2}{R_1}}{R^2C^2w^2}$$

Here the positive loop gain expression shows that output is in phase with the input.

In order to satisfy Barkhausen criteria, magnitude of loop gain must be equal to unity.

This happens when,

$R^2C^2w^2 = 1$ and $R_1 = R_2$ in above expression

The corresponding frequency of oscillation is given as:

$$w_O = 1/CR$$

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The trigger input V need not be applied further as output of the circuit maintains the oscillation.

LC and Crystal Oscillators:

Op- amp RC oscillators are useful for generating sinusoidal signal in the frequency range 10kHz to 100kHz (Sometimes up to 1 MHz as well). For generating high frequency it is either limited by the size of passive components or by the slew rate limitation of the op- amp. In order to generate high frequency sinusoids, we use LC and crystal oscillators in conjunction with transistor. LC oscillators are those oscillator circuits which contain tuned components or tank circuit along with transistors. Generally tuned components are formed by combination of inductors and capacitors. Tuned components set frequency of oscillation by means of resonance phenomenon.

Tank Circuit:

A circuit which produces electrical oscillation of any desired frequency is known as an oscillator circuit or tank circuit.

Generally, inductor and capacitor connected in parallel acts as simple tank circuit.

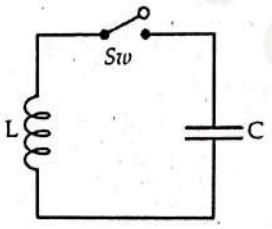

Figure: LC based tank circuit

An LC circuit, oscillating at its natural resonant frequency, can store electrical energy. A capacitor stores energy in the electric field (E) between its plates, depending on the voltage across it, and an inductor stores energy in its magnetic field (B), depending on the current through it.

In above LC circuit, initially switch *sw* is open. Capacitor is charged up to certain volts by means of external source. This creates certain electric field across the capacitor.

Now the switch *sw* is closed. The capacitor discharges through inductor. Current flows through the inductor and keeps on increasing magnetic field around the inductor. At the same time voltage across capacitor goes

on decreasing. Finally voltage across capacitor drops to zero and current through the inductor become maximum and so is the magnetic field.

After the capacitor is fully discharged, and since inductor opposes change of current through it, the magnetic field around the inductor causes current to flow through the circuit in the opposite direction charging capacitor in opposite polarity as compared to previous polarity. As the charge on capacitor is increased, the magnetic field around the inductor is decreased. The capacitor is charged till the magnetic field around the inductor drops to zero.

This process repeats itself again and again. The charge flows back and forth between the plates of the capacitor, and through the inductor creating oscillation of current/voltages.

Resonance occurs when an LC circuit is driven from an external source at a frequency w_O at which the inductive and capacitive reactances are equal in magnitude. The frequency at which this equality holds for the particular circuit is called the **resonant frequency**. The resonant frequency of the LC circuit is

$$w_O = \frac{1}{\sqrt{LC}}$$

The equivalent resonant frequency in units of hertz is:

$$\therefore f_O = \frac{w_O}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

Note:

In the design of LC transistor oscillators, one may use different combinations of inductors and capacitors to form a tank circuit. The configuration and value of inductors and capacitors determine the frequency of oscillation of the tank circuit.

LC Oscillators:

Colpitts Oscillator

Figure shows the circuit diagram of Colpitts oscillator using transistor. In this circuit transistor works as an amplifier which is biased by means of voltage divider formed by R_1 and R_2. RFC stands for Radio Frequency Choke and its function is to provide high reactance at w_O but low dc resistance. Capacitors C_{C1} and C_{C2} are used for coupling purpose. Capacitor C_E bypasses emitter resistance R_E. The tank circuit which forms the feedback network consists of a series combination of two capacitors connected in parallel with an inductor. However the common node

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between the capacitors is grounded. The output of the circuit is taken from the inductor by coupling with another inductor.

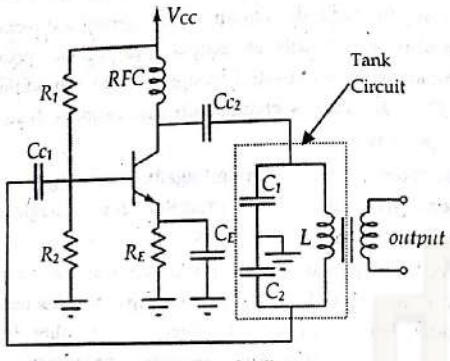


Figure: Colpitts Oscillator

The output of the transistor is 180° phase inverted. But since the feedback must be positive, the tank circuit provides an additional phase shift of 180° so that the total phase shift around the loop is 360°. Also the passive components are chosen so as to maintain the value of loop gain equal to or slightly greater than unity. Thus under certain conditions fulfilled this circuit will produce sinusoidal oscillation.

The frequency of oscillation is given as:

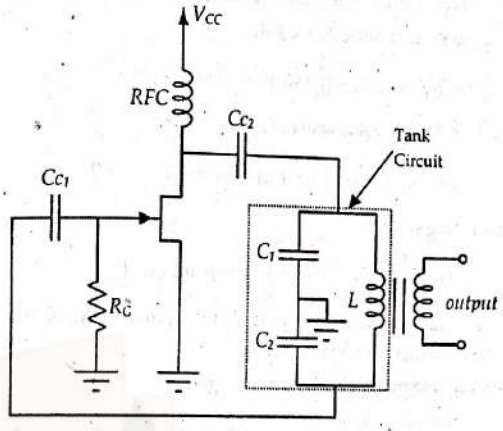
$$w_O = \frac{1}{\sqrt{LC_{eq}}}$$

$$w_O = \frac{1}{\sqrt{L\frac{C_1C_2}{C_1+C_2}}}$$

$$\therefore f_O = \frac{1}{2\pi\sqrt{L\frac{C_1C_2}{C_1+C_2}}}$$

JFET implementation of Colpitts oscillator:

The figure shows the JFET implementation of Colpitts oscillator. This circuit is similar to the BJT circuit. The expression for frequency of oscillation is same as that given for BJT version.



Derivation of frequency of oscillation of Colpitts oscillator:

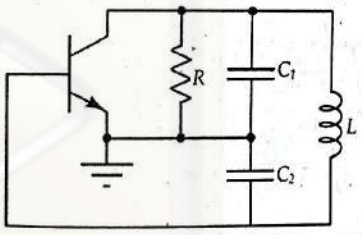


Figure a: Simplified Colpitts oscillator circuit

Figure shows the simplified circuit diagram of the Colpitts oscillator. In this circuit biasing arrangement is omitted and only tank circuit is included so as to explore the basics and reduce complications.

Replacing transistor by the hybrid - π equivalent model, we get,

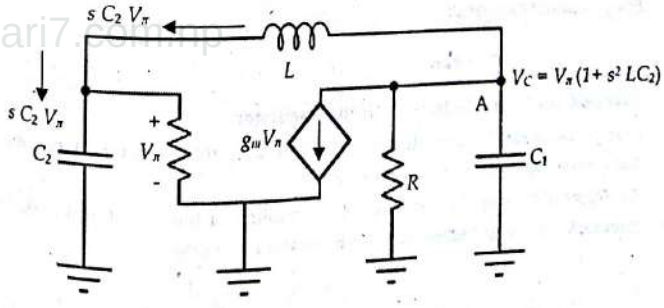


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Figure b: Equivalent circuit with transistor replaced by hybrid – π model

Applying KCL at node A, we get,

$$sC_2V_\pi + g_mV_\pi + (\frac{1}{R}+sC_1)(1 + s^2 LC_2)\ V_\pi = 0$$

Since $V_\pi \neq 0$, rearranging above expression to get,

$$s^3LC_1C_2 + s^2 (L\frac{C_2}{R}) + s(C_1 + C_2) + (g_m+\frac{1}{R}) = 0$$

Substituting $s = j\omega$

$$(g_m + \frac{1}{R} - \frac{\omega^2LC_2}{R}) + j[\omega(C_1+ C_2) - \omega^3LC_1C_2] = 0$$

For oscillation to start, real part and imaginary part of the of the above expression must be equal to zero.

Equating imaginary part to zero, we get,

$$s\omega_O(C_1+ C_2) - \omega_O^3LC_1C_2 = 0$$

Solving we get,

$$\omega_O = \frac{1}{\sqrt{L.\frac{C_1C_2}{C_1+ C_2}}}$$

$$f_O = \frac{1}{2\pi\sqrt{L.\frac{C_1C_2}{C_1+ C_2}}}$$

Equating real part to zero,

$$g_m + \frac{1}{R} - \frac{\omega_O^2LC_2}{R} = 0$$

Substituting value of ω_O will result into the condition,

$$\frac{C_2}{C_1} = g_mR$$

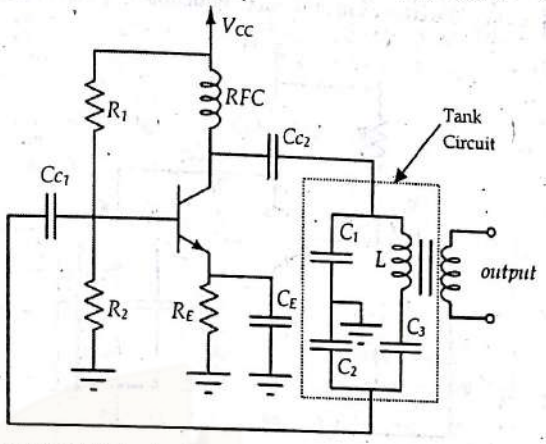
For sustained oscillation,

$$\frac{C_2}{C_1} > g_mR$$

Clapps Oscillator: Modified Colpitts Oscillator

Clapps oscillator is a modified version of Colpitts oscillator. It has an additional capacitor C_3 in the tank circuit.

Addition of C_3 improves the frequency stability of the circuit and makes operaiton of circuit independent of transistor parameters.



The frequency of oscillation is given as:

$$\omega_O = \frac{1}{\sqrt{LC_{eq}}}$$

$$\text{or, } \omega_O = \frac{1}{\sqrt{L\frac{C_1C_2C_3}{C_1C_2+C_2C_3+C_3C_1}}}$$

$$\text{or, } \omega_O = \sqrt{\frac{C_1C_2+C_2C_3+C_3C_1}{L(C_1C_2C_3)}}$$

$$\text{or, } f_O = \frac{1}{2\pi}\sqrt{\frac{1}{LC_1}+\frac{1}{LC_2}+\frac{1}{LC_3}}$$

Hartley Oscillator

In hartley oscillator, tank circuit consists of series combination of inductors of inductances L_1 and L_2 (With their common node being grounded) in parallel with the capacitor of capacitance C.

The frequency of oscillation of the circuit is determined by the oscillation frequency of the tank circuit.

It is given as:

$$\omega_O = \frac{1}{\sqrt{L_{eq}C}}$$

$$\text{or, } \omega_O = \frac{1}{\sqrt{(L_1+L_2+2M)C}}$$

$$f_O = \frac{1}{2\pi\sqrt{(L_1+L_2+2M)C}}$$

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Where M is the coefficient of mutual induction between L_1 and L_2.

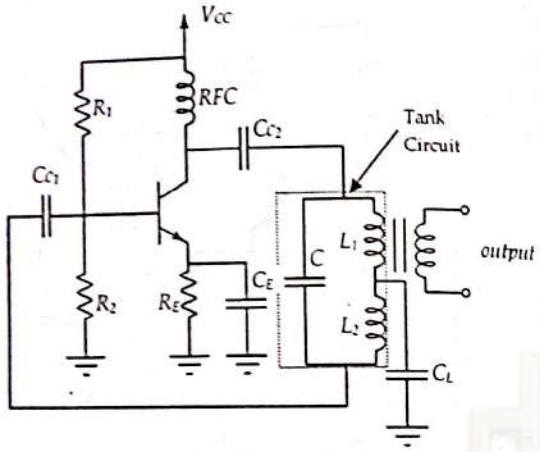
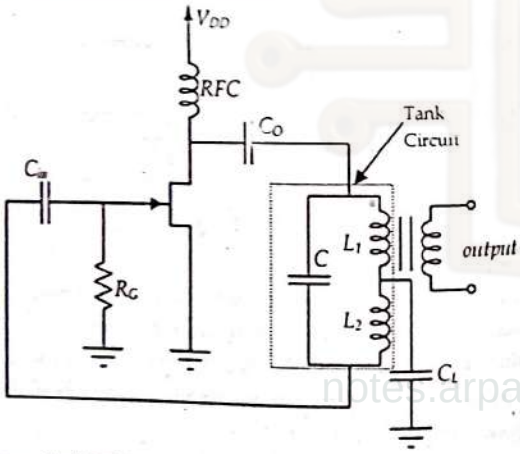


Figure: Hartley Oscillator

JFET Implementation of Hartley Oscillator



The figure shows the JFET implementation of Colpitts oscillator.

Crystal Oscillators

In LC crystal oscillators, electrical resonant circuit or a tank circuit comprising of inductances and capacitances is used. The frequency of oscillation in those circuits is determined by the configuration and values of inductances and capacitances employed.

Instead of using tank circuit, we may use a crystal oscillator. Crystal oscillator is capable of producing required frequency of oscillation by means of mechanical vibration when alternating voltage is applied across it.

The symbol of crystal oscillator is given below:

Figure: Crystal Oscillator

The oscillating property of crystal oscillator can be modeled by an equivalent LC electrical resonant circuit as shown below:

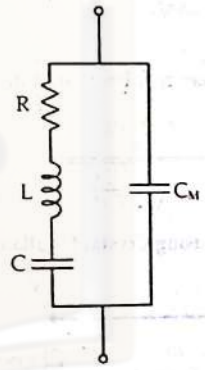


Figure: Electrical equivalent circuit of crystal oscillator

Figure shows the electrical equivalent circuit of crystal oscillator. It consists of a series RLC circuit and a mounting capacitance C_M in parallel with series RLC circuit.

The impedance of the combination can be expressed as:

$$Z = \frac{1}{sC_M} \frac{s^2 + (1/LC)}{s^2 + [(C_M + C)/LCC_M]}$$

Where, s is a frequency variable.

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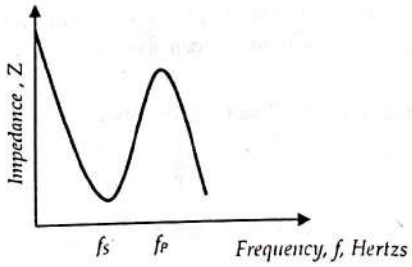


Figure shows the variation of the impedance of the crystal oscillator with frequency. The resonance occurs at two frequencies.

The resonance frequency f_S because of series RLC resonance is given as:

$$f_S = \frac{1}{2\pi\sqrt{LC}}$$

Similarly, the resonance frequency f_P because of the parallel resonance of the combination is given as:

$$f_P = \frac{1}{2\pi\sqrt{L\frac{C.C_M}{C+C_M}}}$$

Sinusoidal Oscillator circuit using Crystal Oscillator:

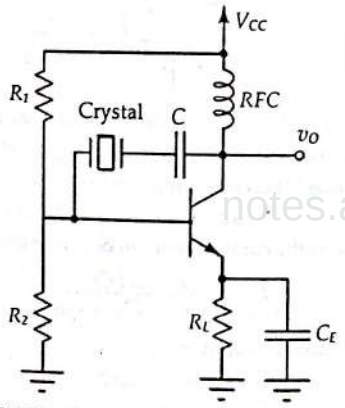


Figure: NPN BJT based implementation of Crystal Oscillator

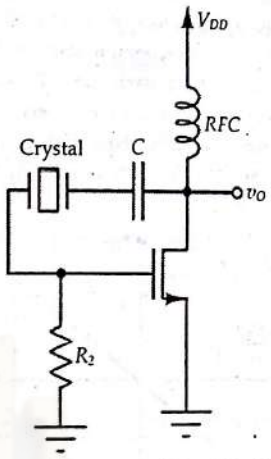


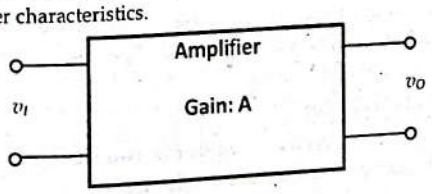
Figure: NMOS based implementation of Crystal Oscillator

In LC oscillator circuits we used tuned components to set the frequency of oscillation. Now in crystal oscillator based sinusoidal oscillator, crystal oscillator is used to set up the frequency of oscillation.

Note than in following oscillator circuit, crystal oscillator is placed in the feedback path of the amplifier. The biasing circuit is so designed that Barkhausen criteria is fulfilled.

Amplifier Saturation:

Until now we studied that amplifiers are linear device formed by utilizing almost linear portion of non- linear characteristics curve of transistors. We also studied that Q- point should be located within the middle of linear portion of transfer characteristics curve so that input signal and output signals do not get clipped off when they swing across the bias point. In previous sections we did not make any comments about the amplifier saturation and the range of input signal for which amplifier provides linear transfer characteristics.



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Let us consider an amplifier circuit with gain A. The output of the amplifier can go up to certain positive maximum value and it can go down to certain negative manimum based on the polarity and magnitude of the input applied. These maximum values of output signal level of an amplifier are known as **saturation levels**. The positive one is called positive saturation level. It is designated as $+V_{Sat}$ and the negative one is called negative saturation level. It is designated as $-V_{Sat}$.

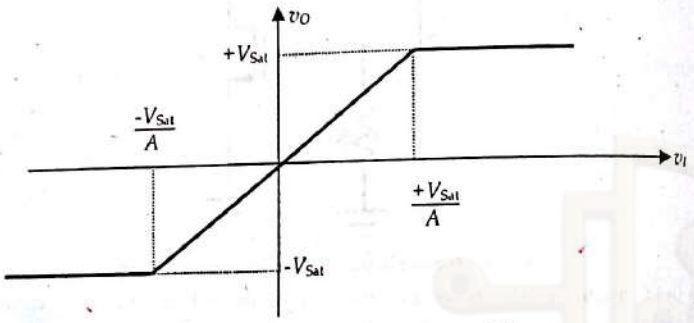

Figure: Transfer Characteristics of an amplifier

Figure shows the transfer characteristics of an amplifier with positive saturation level $+V_{Sat}$ and negative saturation level $-V_{Sat}$. Beyond saturation levels the output cannot increase or decrease any further for any additional increase in input. The input corresponding to the positive and negative saturation level are $\frac{+V_{Sat}}{A}$ and $\frac{-V_{Sat}}{A}$. Thus any signal v_I that falls within the range

$$\frac{-V_{Sat}}{A} \leq v_I \leq \frac{+V_{Sat}}{A}$$

will be amplified without any distortion because this input range falls within linear portion of transfer characteristics curve.

Multivibrators:

A multivibrator is a circuit which is used to implement simple two state systems such as oscillators, timers and flip-flops. Following are the different types of multivibrators:

i. **Astable Multivibrator**

In this type of multivibrator neither of two states is stable. Astable multivibrators are used to make oscillators.

ii. **Monostable Multivibrator**

In this type of multivibrator one of the two states is stable while the other one is not. Monostable multivibrators are used to make timers.

iii. **Bistable Multivibrator**

In this type of multivibrator both of the two states are stable. It can remain in either state indefinitely. Bistable multivibrators are used to make flip-flops.

Bistable multivibrator using op- amp:

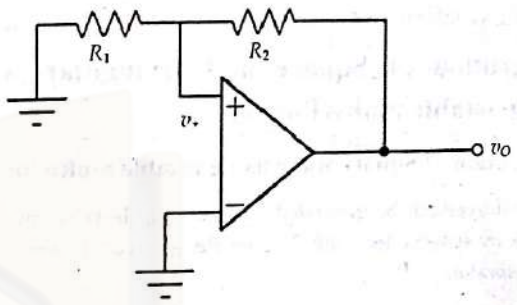

Figure: Bistable multivibrator using op-amp

Figure s. vs the circuit of a bistable multivibrator using op- amp having open loop gain A. The inverting input terminal is grounded. The input at non- inverting input is designated as v_+. Resistors R_1 and R_2 are configured to control the gain of the amplifier.

In any electronic circuits noise in inevitably present. So it is reasonable to assume that v_+ has a slight positive value above the ground because of noise. This input v_+ is amplified by the large gain of the amplifier resulting into amplified output at the op-amp terminal. At the same time this output is fed back at the non-inverting input with feedback factor β given by:

$$\beta = R_1/(R_1+R_2).$$

Thus $v_+ = \beta\, v_O$ is fed to the non- inverting input and even large output appears at the output terminal. This regeneration process will continue till the output of the op-amp saturates at the positive saturation level $+V_{Sat}$. At this output the input to the non- inverting terminal is $v_+ = +\beta V_{Sat}$. This input will keep the output at positive saturation level. This is one stable state of the circuit.

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Similarly if we had assumed v_+ is slightly negative due to noise, the output would be saturated at the negative saturation level $-V_{Sat}$. At this output the input to the non- inverting input is $v_+ = -\beta V_{Sat}$ which will keep the output at negative saturation level. This is another stable state of the circuit.

Thus the output of this circuit can be in two stable state indefinitely based upon the nature of input at the non- inverting input terminal. So this circuit is called as **bistable multivibrator.**

The bistable circuit can be arranged to change its state periodically so as to realize astable multivibrator. This is discussed in the next topic.

Generation of Square and triangular waveform using astable multivibrator

Generation of Square wave using astable multivibrator:

Square waves can be generated by arranging bistable multivibrator to change its state periodically. The resulting circuit is known as **astable multivibrator.**

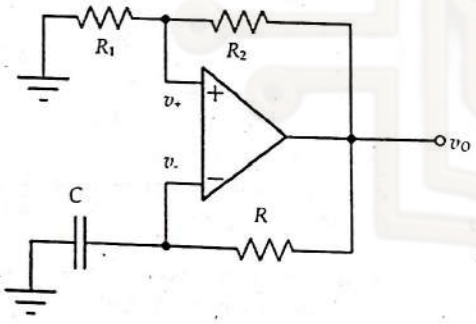

Figure: Astable Multivibrator using op amp

Figure shows the circuit diagram of astable multivibrator using op amp. Two resistors R_1 and R_2 forms a voltage divider network with feedback factor $\beta = R_1/(R_1+R_2)$. The voltage at the non inverting input is denoted as v_+ and that at inverting input is designated as v_-. The addition of RC network to the bistable circuit has resulted in the circuit of astable multivibrator. Inclusion of the RC network will cause the output of the amplifier to switch its state periodically so as to generate square waves at the output terminal.

Circuit Operation

Suppose at a certain instant output of the op-amp is at positive saturation level $+V_{Sat}$. The feedback voltage $v_+ = +\beta V_{Sat}$ is fed to the non inverting input of the op amp. At the same time capacitor gets charging current through the resistor R. As a result voltage across the capacitor will increase. When the voltage drop across the capacitor slightly exceeds $+\beta V_{Sat}$, a net negative voltage appears across the op amp which drives the output of op amp to the negative saturation level $-V_{Sat}$.

Now the output of op amp is $-V_{Sat}$. The feedback voltage $v_+ = -\beta V_{Sat}$ is fed to the non inverting input of the op amp. At the same time capacitor gets charging current through resistor R but in the reverse direction as compared to previous case. As a result voltage across the capacitor will drop. As the voltage across the capacitor drops slightly below $-\beta V_{Sat}$ a net positive voltage appears across the op amp which drives the output of op amp to positive saturation level V_{Sat}.

In this way, output of the astable multivibrator will periodically switch between V_{Sat} and $-V_{Sat}$ to generate a train of square wave as shown in the figure.

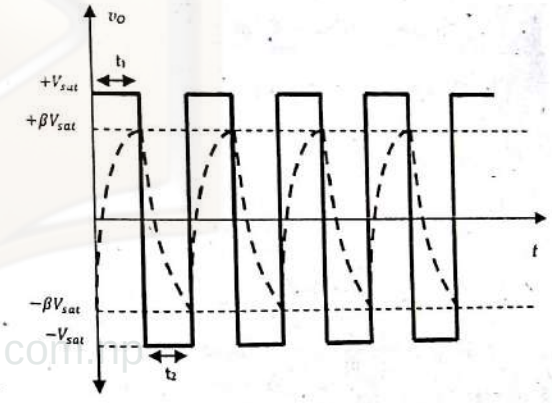

Figure shows the output waveform of astable multivibrator. The output waveform has square shape. So astable multivibrator is also known as square wave generator. The dotted waveform denotes the waveform of voltage across the capacitor during the operation.

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Expression for time period and frequency of oscillation

The time period of square wave generated is the sum of the time for which output is $+V_{sat}$ (t_1) and the time for which output is $-V_{sat}$ (t_2). If T is the time period. Then,

$$T = t_1 + t_2$$

For symmetric square wave, $t_1 = t_2$

$$T = 2t_1 \quad \text{....(i)}$$

From the transient analysis of RC circuit, the voltage across the capacitor at any instant is given by:

$$v_C(t) = V_{Sat} - [V_{Sat} - V_{C\,initial}]\, e^{\frac{-t}{RC}} \quad \text{.....(ii)}$$

Initially the output of op amp is at V_{Sat}. The voltage across the capacitor at this condition is $-\beta V_{Sat}$. The voltage across the capacitor becomes $+\beta V_{Sat}$ after time t_1. Thus,

$$v_{C\,initial} = -\beta V_{Sat} \text{ and } v_C(t_1) = +\beta V_{Sat}$$

So from equation (ii),

$$+\beta V_{Sat} = V_{Sat} - [V_{Sat} - (-\beta V_{Sat})]\, e^{\frac{-t_1}{RC}}$$

$$\text{or, } +\beta V_{Sat} = V_{Sat} - [V_{Sat} + \beta V_{Sat}]\, e^{\frac{-t_1}{RC}}$$

$$\text{or, } +\beta V_{Sat} = V_{Sat} - [V_{Sat} + \beta V_{Sat}]\, e^{\frac{-t_1}{RC}}$$

$$\text{or, } V_{Sat}(1+\beta)\, e^{\frac{-t_1}{RC}} = V_{Sat}(1-\beta)$$

$$\text{or, } (1+\beta)\, e^{\frac{-t_1}{RC}} = (1-\beta)$$

$$\text{or, } e^{\frac{-t_1}{RC}} = \frac{(1-\beta)}{(1+\beta)}$$

Taking ln on both sides, we get,

$$\text{or, } t_1 = -RC \ln \frac{(1-\beta)}{(1+\beta)}$$

$$\therefore t_1 = RC \ln \frac{(1+\beta)}{(1-\beta)}$$

From equation (i),

$$T = 2t_1 = 2RC \ln \frac{(1+\beta)}{(1-\beta)}$$

If $R_1 = R_2$, then, $\frac{(1+\beta)}{(1-\beta)} = 3$

$$\therefore T = 2\,RC \ln(3) \cong 2RC$$

The frequency of oscillation is given by:

$$f = \frac{1}{T}$$

$$\therefore f = \frac{1}{2RC \ln \frac{(1+\beta)}{(1-\beta)}}$$

If $R_1 = R_2$, then

$$f = \frac{1}{2RC}$$

Note: RC can be replaced by tau τ in all of above expressions. The constant τ in this case is called **RC time constant**.

Generation of triangular waveform using astable multivibrator

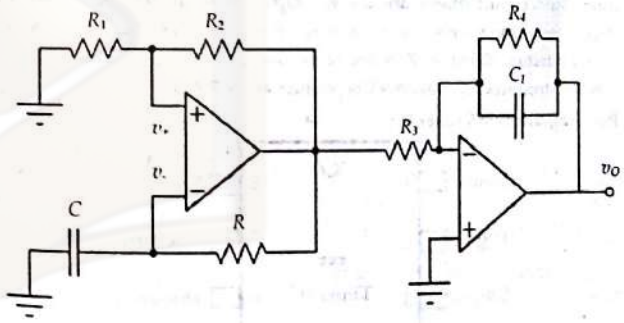

Figure: Triangular Wave generator using op-amp

Triangular wave can be generated by integrating square waves generated by square wave generator.

The circuit for triangular wave generator using op-amp is shown in the above figure. Note that square wave generator circuit is followed by integrator circuit. In addition a resistance R_f is connected in parallel to the capacitor C_1 of the integrator circuit. This is because the placement of this resistor ensures the integration operation at low frequency. Amplitude of generated triangular waveform is less than that of square wave.

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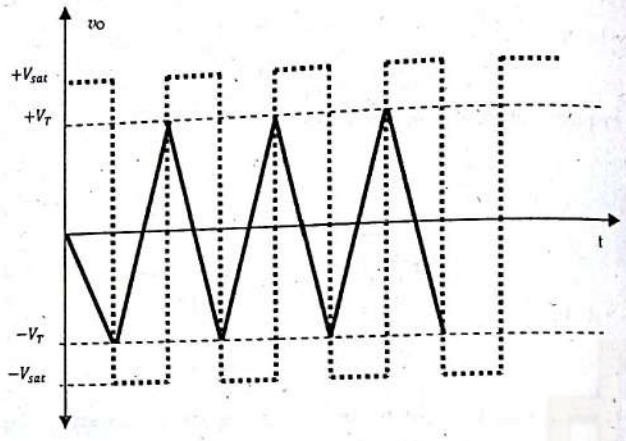

Figure: Waveform generated by triangular wave generator

Integrated Circuit Timers

Integrated circuit timers are the IC chips which are basically used to implement monostable and astable multivibrators having precise characteristics. 555 timer IC is one of the popular integrated circuit timers. It was introduced by Signetics Corporation in 1972 AD.

Pin diagram of 555 timer IC:

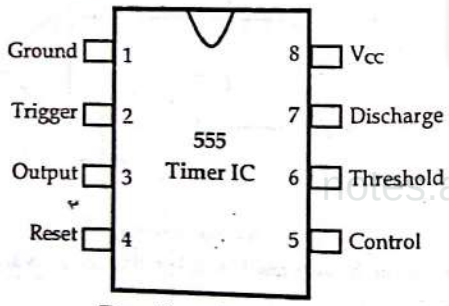

Figure: Pin out diagram of 555 timer IC

The figure shows the pin out diagram of 555 timer IC. This IC is made up of several components which is further illustrated in the block diagram given below:

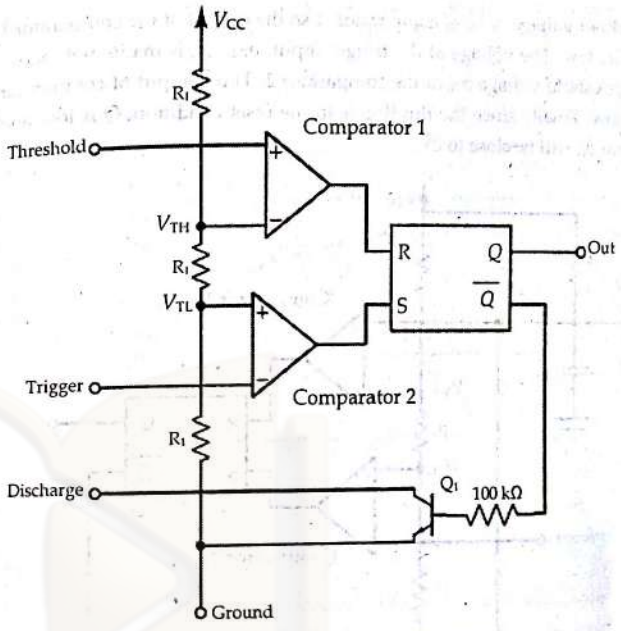

Figure: Block diagram of 555 timer IC

Figure shows the block diagram of 555 timer IC. It consists of two comparators namely comparator 1 and comparator 2, SR flip-flop and transistor Q_1 which operates as a switch. Three resistors having same magnitude R_1 forms voltage divider configuration with supply voltage V_{CC} and establish reference voltages for two comparators. For first comparator reference voltage is $\frac{2}{3}V_{CC}$ and for second comparator reference voltage is $\frac{1}{3}V_{CC}$.

555 timer IC as monostable multivibrator:

Figure shows the monostable multivibrator implemented using 555 timer IC along with external resistor R and external capacitor C.

In the stable state, SR flip flop is in reset condition. This means its Q output will be low and $\bar{Q}$ output will be high. Because $\bar{Q}$ is high transistor Q_1 turns on. Thus voltage across capacitor, v_C, drops to zero because of the short provided by Q_1 to ground. Since v_C is low as compared to

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threshold voltage, V_{TH}, of comparator 1 so the output of the comparator 1 will be low. The voltage at the trigger input, $v_{TRIGGER}$ is made high above the threshold voltage v_{TL} of the comparator 2. Thus output of comparator 2 is low. Finally since the flip flop is in the reset condition, Q is low and output v_O will be close to 0V.

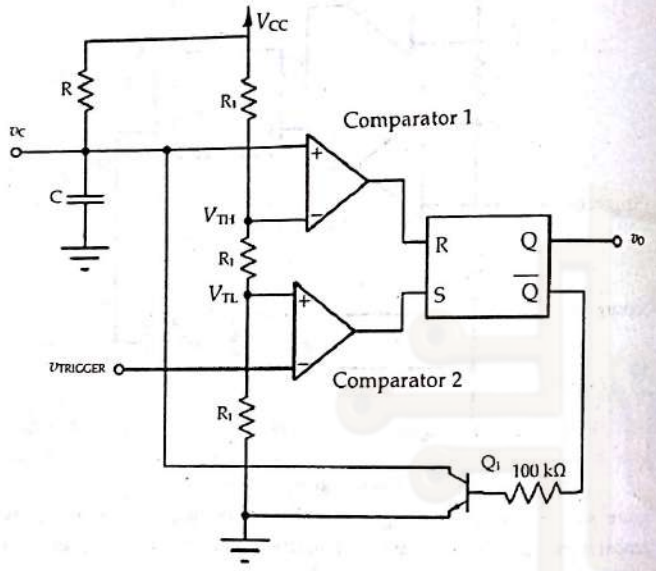

Figure: 555 timer IC as monostable multivibrator

Now consider a negative pulse of short duration applied to the trigger input terminal. Since $v_{TRIGGER}$ is less as compared threshold voltage V_{TL} of comparator 2, the output of comparatror 2 will be high. Thus SR flip flop will set and its Q output will become high. At the same time $\bar{Q}$ becomes low. Because $\bar{Q}$ is low transistor Q_1 turns off. During this time capacitor C gets charged through the resistor R. The voltage v_C will rise exponentially towards V_{CC}. The monostable multivibrator is now in the quasi stable state. This state prevails until v_C reaches and begins to exceed the threshold voltage V_{TH} of comparator 1. Thus output of the comparator 1 will be low. Thus SR flip flop will be in reset condition. The output at $\bar{Q}$ will be high and turns on transistor Q_1. In turn transistor Q_1 will rapidly discharge capacitor C causing v_C to go to 0V. Also at reset Q output of the flip flop goes low and v_O goes back to 0V.

The monostable multivibrator is now in stable state and is ready to receive new trigger pulse.

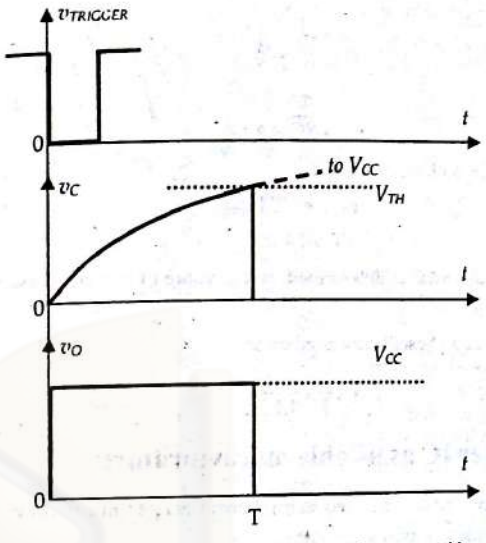

Figure: Various waveforms in operation of 555 timer IC as monostable multivibrator

Figure shows the waveform of $v_{TRIGGER}$, v_C and output v_O. As seen in the figure output has only one stable state and that is v_O being low (close to 0V). The output is high only for fixed interval T after the trigger input is applied. So this circuit is also known as one shot.

The voltage across the capacitor can be written as:

$$v_C = V_{CC}(1 - e^{\frac{-t}{RC}}) \quad \ldots\ldots (i)$$

At $t = T$, $v_C = V_{TH} = \frac{2}{3}V_{CC}$

From equation (i),

$$\text{or, } \frac{2}{3}V_{CC} = V_{CC}(1 - e^{\frac{-T}{RC}})$$

$$\text{or, } \frac{2}{3} = 1 - e^{\frac{-T}{RC}}$$

$$\text{or, } \frac{2}{3} = 1 - e^{\frac{-T}{RC}}$$

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$$\text{or, } e^{\frac{-T}{RC}} = 1 - \frac{2}{3}$$

$$\text{or, } e^{\frac{-T}{RC}} = \frac{1}{3}$$

$$\text{or, } e^{\frac{T}{RC}} = 3$$

Taking ln on both sides, we get,

$$\text{or, } T = RC \ln(3)$$

$$\therefore T \cong 1.1\ RC$$

Hence pulse width is determined by the value of external resistance and capacitance.

The frequency of oscillation is given as:

$$f = \frac{1}{T} = \frac{1}{1.1\ RC}$$

555 timer IC as astable multivibrator:

555 IC timer can be arranged to implement astable multivibrator so as to generate square waves at its output.

Figure shows the astable multivibrator implemented using 555 timer IC along with external resistors R_A and R_B and capacitor C. In the stable state lets' assume capacitor is fully discharged and SR flip flop is in the set state. Thus v_O is high and transistor Q_1 is off. At this time capacitor C will charge through the series combination of R_A and R_B. As a result voltage across the capacitor v_C will rise exponentially towards V_{CC}. As v_C becomes slightly greater than or equal to the threshold voltage V_{TL} of the comparator 2, the output of the comparator 2 goes low. This has no effect on the output of the circuit and flip flop remains in set state(Because R=0 and S=0 will preserve state in SR flip flop).This will continue until the voltage v_C becomes slightly greater than or equal to the threshold voltage V_{TH} of the comparator 1.Thus output of comparator 1 becomes low and SR flip flop will be in reset condition. The Q output of flip flop is low and v_O will be close to 0V. Also because $\bar{Q}$ is high transistor Q_1 turns on. The saturated transistor Q_1 causes voltage at the node between R_A and R_B to be approximately 0V. Thus capacitor will discharge with the time constant CR_B towards the 0V. When v_C reaches threshold voltage V_{TL} of comparator 2, output of comparator 2 becomes high and sets the flip flop. The output voltage v_O goes high, $\bar{Q}$ goes low turning off transistor Q_1.

Capacitor C begin to charge through the series combination of R_A and R_B and v_C will rise exponentially towards V_{CC} with time constant $C(R_A+R_B)$.This rise continues until v_C reaches V_{TH} at which output of comparator 1 goes high resetting the flip flop.

And this cycle continues.

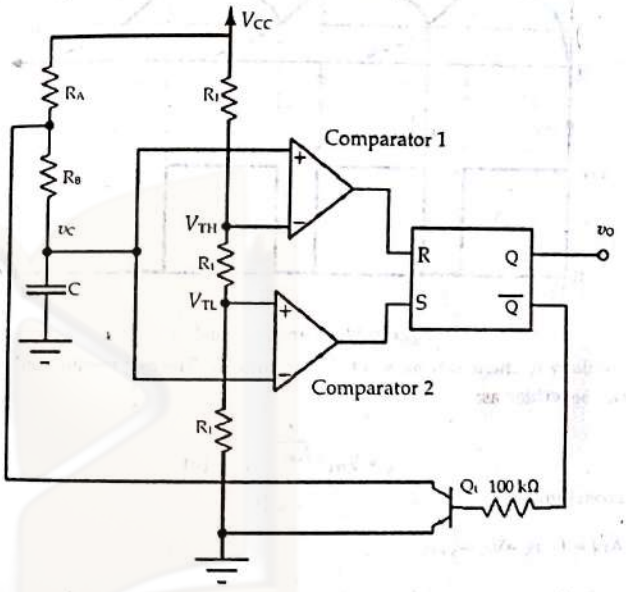

Figure a: 555 timer IC as astable multivibrator

Figure b shows the waveforms for v_C and v_O. The output is a square wave with $T_H > T_L$.

v_C changes from V_{TL} to V_{TH} in time T_H. The exponential rise of v_C can be written as:

$$v_C = V_{CC} - (V_{CC} - V_{TL})\, e^{\frac{-t}{(R_A+R_B)C}} \quad \text{.....(i)}$$

From figure at $t = T_H$, $v_C = V_{TH} = \frac{2}{3}V_{CC}$

Also since $V_{TL} = \frac{1}{3}V_{CC}$

Substituting these values in equation (i) we get,

$$T_H = C(R_A+R_B)\ln 2$$

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$$\therefore T_H = 0.69\,C(R_A+R_B)$$

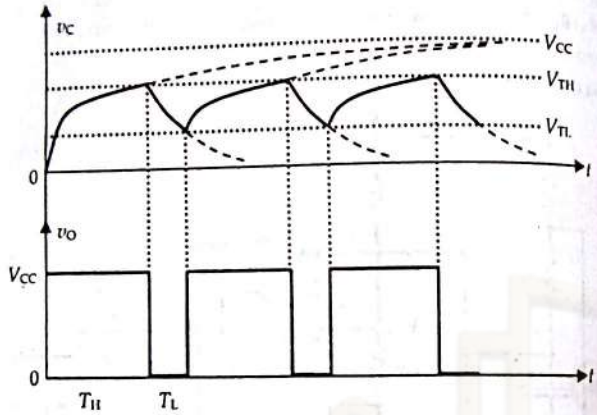

Figure b: Waveforms for v_C and v_O

Similarly v_C changes from V_{TH} to V_{TL} in time T_L. The exponential fall of v_C can be written as:

$$v_C = V_{TH}\,e^{\frac{-t}{R_BC}} \quad \text{.....(ii)}$$

From figure,

At $t = T_L$, $v_C = V_{TL} = \frac{1}{3}V_{CC}$

Also we know, $V_{TH} = \frac{2}{3}V_{CC}$

Substituting these values in equation (ii), we get,

$$T_L = CR_B \ln 2$$

$$\therefore T_L = 0.69\,CR_B$$

The total time period T for one complete cycle is given as:

$$T = T_L+T_H = 0.69\,C(R_A+2R_B)$$

The frequency of oscillation is given as:

$$f = \frac{1}{T} = \frac{1}{0.69C(R_A+2R_B)}$$

$$\therefore f = \frac{1.45}{C(R_A+2R_B)}$$

Duty cycle is defined as the ratio of time for which output is high (T_H) to the total time period (T).

$$\text{Duty Cycle} = \frac{T_H}{T} = \frac{R_A+R_B}{R_A+2R_B}$$

555 timer IC as a voltage controlled oscillator:

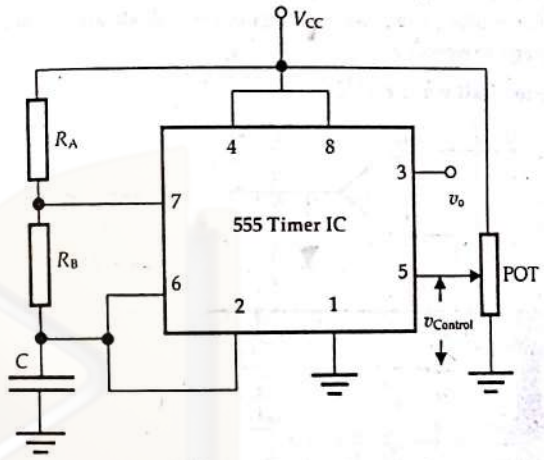

Figure: 555 timer IC as voltage controlled oscillator

Figure shows the circuit connection for the operation of 555 timer IC as a voltage controlled oscillator. This circuit is also sometimes called as voltage to frequency converter because the output frequency can be changed by changing the control voltage. Control voltage is applied at pin 5 by means of a potentiometer POT. Normally control voltage is $+\frac{2}{3}V_{CC}$

The frequency of output voltage can be changed by changing the control voltage.

Precision Rectifier Circuits:

Rectifier circuits are used to convert ac current/voltage into DC current/voltage utilizing unidirectional current conduction property of diode. In these circuits, the input voltage being considered has very high value as compared to the diode forward voltage drop which is normally

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0.7V for Si based diodes. We normally ignore diode drop in the operation of the rectifier circuits.

However, if the input signal to be rectified has peak amplitude of very small value, say 0.1V, it is impossible to use conventional rectifier circuits. This is because 0.1V is very much small as compared to diode forward voltage drop. In these conditions we use precision rectifier circuits. Precision rectifier circuits use the combination of diode and op amp for the purpose of rectification.

Precision half wave rectifier:

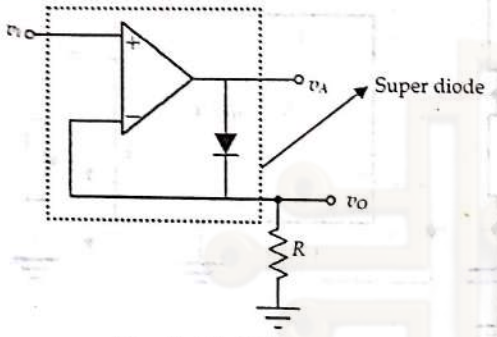

Figure: Precision Half wave rectifier

Figure shows the circuit diagram of precision half wave rectifier. It consists of a special combination of diode and op amp. This combination is popularly referred to as "**Super diode**". The diode is connected in the negative feedback path between the op amp output terminal and op amp inverting input terminal. A resistance R is connected as load of the rectifier circuit.

When input v_I is positive:

Output voltage v_A of op amp will become positive and diode will conduct. Thus a negative feedback path is established between the op amp output terminal and op amp inverting input terminal. Thus a virtual ground will appear between the op amps two input terminals. Thus voltage at the non inverting input terminal which is also the output voltage v_O is equal to the input v_I at the non inverting input terminal.

Note that the offset voltage of 0.5V to 0.7V presented in the simple half wave rectifier is no longer present. The input v_I has to exceed only a negligibly small voltage equal to the diode drop divided by the op amps

open loop gain. Thus the transfer characteristic of the circuit is a straight line passing through the origin. It makes the circuit very useful for rectification involving very small signals.

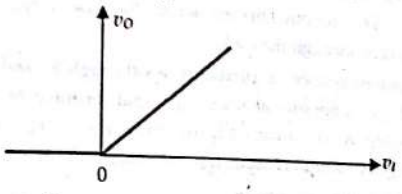

Figure: Transfer characteristics of precision half wave rectifier circuit.

When input v_I is negative:

The op amp output voltage v_A will also be negative. In this case diode will not conduct. Thus output v_O will zero.

Precision Full wave rectifier:

Figure shows the circuit diagram of precision full wave rectifier. It consists of two op amps A_1 an A_2. Resistor R forms the rectifier load.

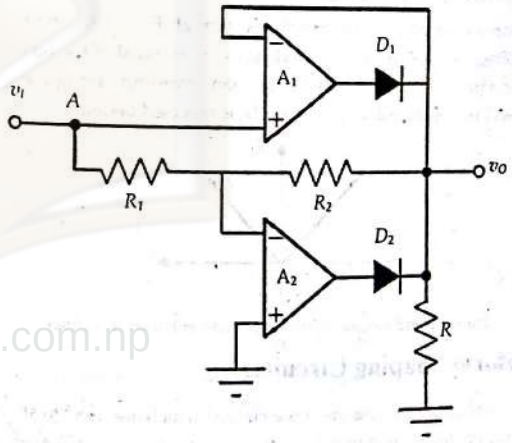

Figure: Precision full wave rectifier

When the input v_I is positive:

Since the op amp A_1 is connected in non inverting configuration, in this case its output is positive thus forward biasing the diode D_1. This closes the feedback loop around the op amp A_1 forcing the virtual short between

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the inverting and non inverting input terminals of op amp A_1. The input voltage v_I applied to the non inverting terminal of op amp A_1 will become equal to the voltage at inverting input terminal, which is also the output voltage v_O. This output appears across the load resistor R and causes current to flow through the load.

At the same time since no current flows through R_1 and R_2, the positive input voltage v_I appears at inverting input terminal of the op amp A_2. Since op amp A_2 is connected in the inverting configuration, its output will be negative and thus diode D_2 is reversed biased.

When the input v_I is negative:

Since the op amp A_2 is connected in inverting configuration, its output is positive thus forward biasing the diode D_2. This closes the feedback path around the op amp A_2 forcing the virtual short between the inverting and non inverting input terminals of op amp A_2. The input voltage v_I applied to the inverting input terminal of op amp A_2 will become equal to the voltage at the non inverting input terminal, which is also the output voltage v_O. This output appears across the load resistor R and causes current to flow through the load.

At the same time since no current flows through R_1 and R_2, the negative input voltage v_I appears at non inverting input terminal of the op amp A_1. Since op amp A_1 is connected in the non inverting configuration, its output will be negative and thus diode D_1 is reversed biased.

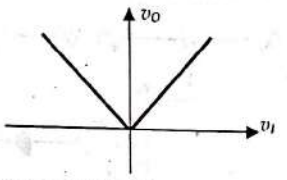

Figure: Transfer characteristics of a precision full wave rectifier

Waveform Shaping Circuits:

Waveform Shaping Circuits are those circuits which are used to shape the waveform of the input signal so as to generate the output waveform of the desired shape. In our earlier course "Basic Electronics Engineering" we learnt about diode based waveform shaping circuits like clippers, clampers, rectifiers etc. However there are other classes of wave shaping circuits.

One of the most widely used wave shaping requirement in the area of electronics engineering is sine wave shaping. There are two distinct methods for the sine wave generation by means of waveform shaping. They are:

i. Break Point Method:

In the breakpoint method the desired non linear transfer characteristics (in our case the sine function) is implemented as a piecewise linear curve. Diodes are utilized as switches that turn on at various breakpoints of transfer characteristics thus switching into the circuit additional resistors that cause the transfer characteristics to change slope.

ii. The Non- linear amplification method:

In non- linear amplification method input wave is fed to the amplifier circuit having non linear transfer function.

As an example sine wave can be generated by feeding triangular wave at the input of the differential amplifier in which a degeneration resistor is connected between the emitters.

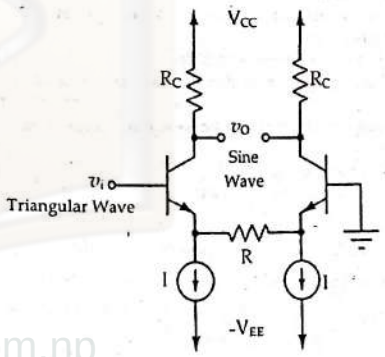

Figure: Use of differential amplifier with degeneration resistor between emitters for shaping triangular wave into sine wave

EXERCISE

1. Draw voltage controlled oscillator circuit using IC 555 and derive expression for frequency of oscillation. **2071 Chaitra**
2. Explain working principle of Wien bridge oscillator with necessary expressions. **2071 Chaitra**

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3. Define Barkhausen criteria for sinusoidal oscillation. Draw the circuit diagram of RC phase shift oscillator and derive its frequency of oscillation. **2071 Shrawan**
4. Define the term multivibrator. Explain the operation of op amp based astable multivibrator with the help of circuit diagram and waveform. **2071 Shrawan**
5. Write the applications of tuned LC oscillators. Draw the Colpitts oscillator circuit and derive the expression for frequency of oscillation. **2070 Chaitra**
6. Draw AMV circuit using IC 555 or BJT. **2070 Chaitra**
7. State barkhausen criteria for sine wave oscillator. **2070 Chaitra**
8. Define Barkhausen criteria for sinusoidal oscillation. Draw the circuit diagram of RC phase shift oscillator and derive its frequency of oscillation. **2070 Ashad**
9. Draw the square wave generator circuit using operational amplifier. **2070 Ashad**
10. Draw Wien bridge oscillator circuit and derive the expression for the frequency of oscillation and the gain of amplifier circuit. **2069 Chaitra**
11. Draw the circuit diagram of square wave generator. **2069 Chaitra**
12. Draw RC sine wave oscillator circuit and explain its operation with the help of Barkhausen criteria for oscillation. Determine the expression for oscillating frequency and gain of the amplifier. **2069 Ashad**
13. Describe the operation of IC 555 as square wave oscillator and find its frequency of oscillation. **2068 Chaitra**
14. Define multivirator. Explain how 555 timers can be used in astable mode to generate square waves. **2068 Shrawan**
15. Write short notes on Wien bridge oscillator. **2068 Shrawan**
16. State Barkhausen criteria and explain the principle of oscillation. **2068 Baisakh**
17. Draw wien bridge oscillator circuit and write the expression for the frequency of oscillation. **2068 Baisakh**
18. Draw crystal oscillator circuit. **2068 Baisakh**

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Chapter 6

Power Supplies, Breakdown diodes and Voltage Regulators

Introduction

In many applications of electronics, it is required to use DC power instead of AC power. Electronic components like diode, transistor and other components along with various integrated circuits require DC power to operate. For this reason the AC power is to be converted into DC power which means AC current/voltage needs to be converted to DC current/voltage. The process of conversion of AC current/voltage in to DC current/voltage is known as rectification. The output of rectifier is in the form of pulsating DC which can be smoothened by using a filter circuit. However filtered output consists of ripples. In order to get a stable DC output, the output of filter circuit must be regulated by means of a voltage regulator. Voltage regulators are those devices which provide a stable output DC voltage to the load regardless of the input variations.

Power supply is a complete system consisting of transformer (to step down ac voltage), rectifier (to convert ac into dc), filter (to smoothen pulsating dc) and finally a voltage regulator (to get stable dc output).

Unregulated DC power Supply

DC Power supply without the voltage regulator circuit at its output is known as unregulated dc power supply.

The block diagram of unregulated power supply is:

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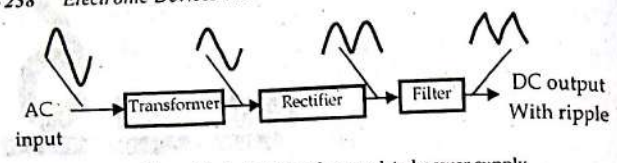

Figure: Block diagram of unregulated power supply

The ac signal from the supply is stepped down using step down transformer. The output obtained from the transformer is fed to the rectifier circuit. The output of rectifier is a pulsating dc which requires smoothing further using the filter circuit. The output of the filter circuit is the final dc output voltage. The output of this unregulated power supply consists of ripples which might not be suitable for all DC devices. In order to get a stable DC output we must use a voltage regulator at the output of the filter circuit to form a regulated dc power supply which is to be discussed next.

Before leaving to next section, one of the possible implementation of unregulated power supply is as shown below.

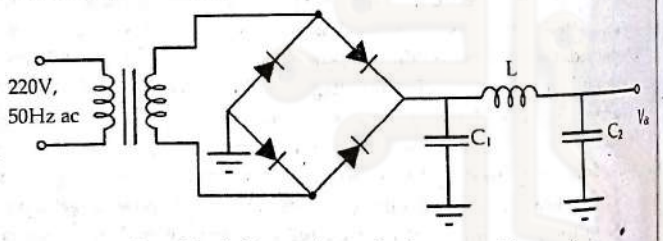

Figure: Circuit diagram of unregulated power supply

Figure shows the circuit diagram of unregulated dc power supply. The step down transformer steps down ac voltage from the 220V ac supply to some small value appropriate to form input to the bridge rectifier. Generally, it is stepped down to 12V ac. The bridge rectifier performs the job of rectification. The 12V ac is converted into the pulsating dc voltage by the bridge rectifier. The LC network formed by C_2, L and C_1 implement a filter. This network converts the pulsating dc into dc voltage with ripples. Thus the final output of the circuit is dc voltage with ripple.

Regulated DC power Supply

The DC power supply system with the voltage regulator at the output is known as regulated dc power supply.

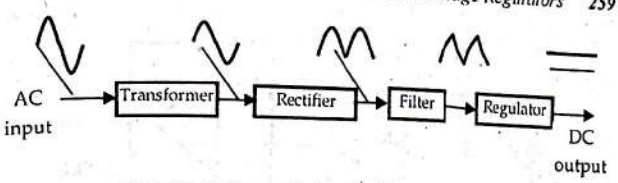

Figure: Block diagram for Regulated DC power supply

In this power supply system voltage regulator converts the dc voltage with ripples at the output of filter into a stable DC output.

One of the possible implementation of regulated dc power supply is as shown below:

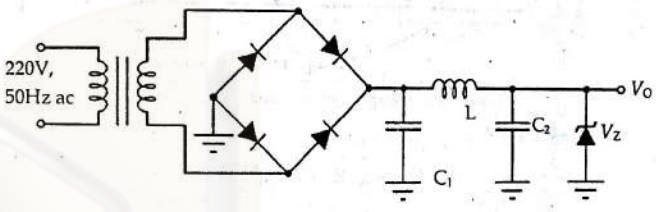

Figure: Circuit diagram of regulated dc power supply

In the above circuit zener diode is used as a voltage regulator. This power supply is also sometimes referred to as **"Zener regulated power supply."**

Due to the advent of IC voltage regulators having precise characteristics and protection mechanisms, zener diodes have lost a great deal of popularity for voltage regulation in the last decade.

Band gap Reference Voltage

Band gap reference voltage source is a circuit which functions like a precise zener diode. It generates a reference voltage that has positive temperature coefficient characteristics which compensates and corrects the negative temperature coefficient of V_{BE}. When they are added together they produce a constant dc voltage which has zero temperature coefficient characteristics. The zero temperature coefficient characteristics imply that the dc voltage output will be independent of variations in temperature.

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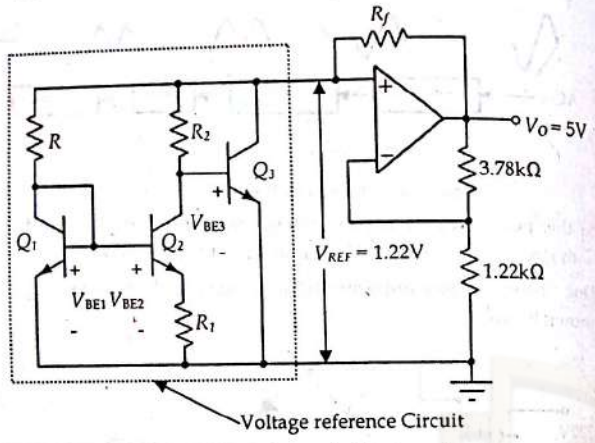

Figure: 5V dc voltage regulator using bandgap reference

From the figure,

$$V_{BE1} - V_{BE2} = I_{C2} R_1 = V_T \ln \frac{I_{C1}}{I_{C2}}$$

$$\therefore I_{C2} \propto V_T$$

Since current I_{C2} is proportional to thermal voltage, the voltage drop across resistor R_2 has positive temperature coefficient characteristics.

Figure shows the circuit of 5V dc voltage regulator implemented using bandgap reference voltage source. Transistors Q_1 and Q_2 form a bandgap voltage reference source. Combination of transistor Q_1 and Q_2 will generate voltage across R_2 that has positive temperature coefficient characteristics. The voltage across R_2 is added with the base emitter voltage of transistor Q_3 so as to generate the reference voltage V_{REF}. The positive temperature coefficient characteristics of voltage drop across the resistor R_2 cancels the negative temperature coefficient of base emitter voltage drop V_{BE3} of transistor Q_3. In this way the reference voltage V_{REF} has zero temperature coefficient characteristics. Typically value of reference voltage is designed to be approximately around 1.22 V. This voltage is the band gap voltage for Si based devices. It is therefore the reference source is called bandgap reference voltage source.

The reference voltage is amplified by op amp so as to generate 5V dc output. In order to confirm this, appropriate values of resistors must be chosen to set the gain of op amp. If it is required to design voltage

regulator to produce output voltage other than 5V then we can do so by changing the gain of the op-amp with appropriate resistance values.

Series Voltage Regulators

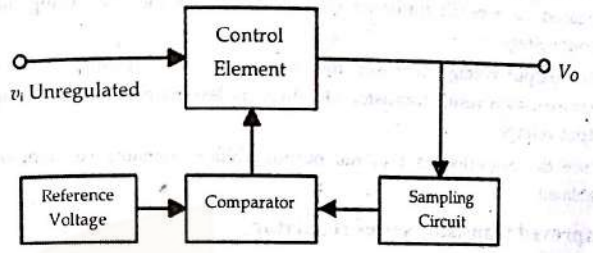

Figure: Series Voltage Regulator

Figure shows the block diagram of series voltage regulator. The series control element controls the amount of input that gets to the output. The output voltage is sampled by a circuit that provides a feedback voltage to be compared to a reference voltage.

If the output voltage increases, the comparator circuit provides control signal to the series control element to decrease the amount of output voltage thereby maintaining the output voltage.

If the output voltage decreases, the comparator circuit provides control signal to the series control element to increase the amount of output voltage thereby maintaining the output voltage.

Transistor Series Regulator

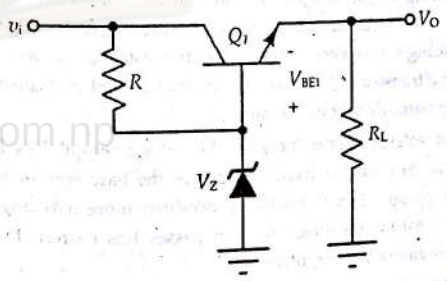

Figure: Transistor series regulator

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Figure shows the circuit diagram of simple transistor series regulator. In this circuit zener diode provides reference voltage. Where as transistor Q_1 forms the series control element.

When output voltage decreases, the base emitter voltage drop of Q_1 is increased, as a result transistor Q_1 conducts more thereby raising the output voltage.

When output voltage increases, the base emitter voltage drop of Q_1 is decreased, as a result transistor Q_1 conducts less thereby reducing the output voltage.

Hence the net effect is that the output voltage remains constant or regulated.

Improved transistor series regulator

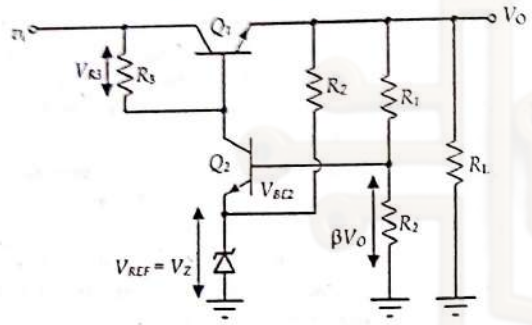

Figure: Improved transistor series regulator

Figure shows the circuit diagram of improved transistor series regulator. In this circuit resistors R_1 and R_2 act as a sampling network with zener diode providing a reference voltage. The transistor Q_2 controls the base current to the transistor Q_1 to vary the current passed by transistor Q_1 to maintain the output voltage constant.

If the output voltage tries to increase, the voltage sampled by R_1 and R_2 increases. The increased voltage βV_O causes the base emitter voltage of transistor to go up. Thus transistor Q_2 conducts more reducing the base current of transistor Q_1 which in turn passes less current to the load thereby maintaining the output voltage.

If the output voltage tries to decrease, the reverse happens. The voltage sampled by R_1 and R_2 decreases. Thus βV_O is decreased which decreases the base emitter voltage of transistor Q_2. As a result transistor Q_2 conducts

less increasing the base current to the transistor Q_1 which in turn passes more current to the load thereby maintaining the output voltage.

From the circuit diagram,

$$V_{BE2} + V_Z = \beta V_O = \frac{R_2}{R_1 + R_2} V_O$$

Rearranging the above equation, we get,

$$V_O = \frac{R_1 + R_2}{R_2} (V_Z + V_{BE2})$$

$$\therefore V_O = \left(1 + \frac{R_1}{R_2}\right) (V_Z + V_{BE2})$$

Op amp series regulator

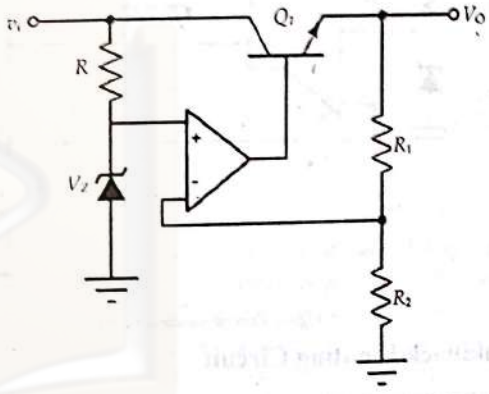

Figure: Op amp series regulator

Figure shows the circuit diagram of series regulator implemented using op amp. The op amp compares the reference voltage provided by the zener diode with the feedback voltage provided by the resistors R_1 and R_2. If the output varies, the conduction of the transistor Q_1 is controlled to maintain the output voltage constant.

The output voltage is maintained at the value of:

$$V_O = \left(1 + \frac{R_1}{R_2}\right) V_Z$$

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Current Limiting Circuit

Current limiting circuit is used for the short circuit protection. Figure shows the circuit diagram of current limiting circuit. As the load current I_L increases, the voltage drop across the short circuit sensing resistor R_{SC} is also increased. As a result transistor Q_2 turns on diverting the base current to transistor Q_1. Thus Q_1 conduct less and load current through Q_1 is reduced preventing any additional current to the load R_L. Hence the action of R_{SC} and transistor Q_2 provides limiting of the maximum load current.

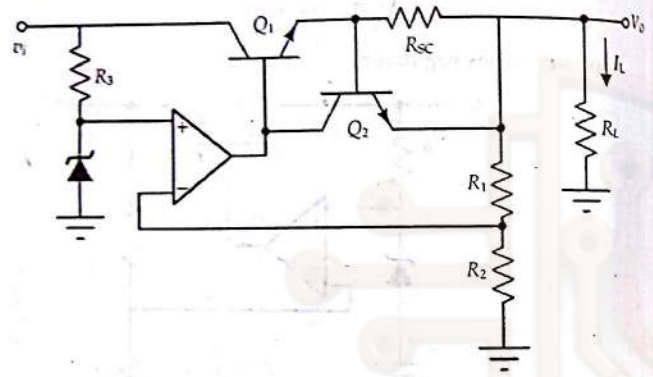

Figure: Current Limiting Circuit

Foldback Limiting Circuit

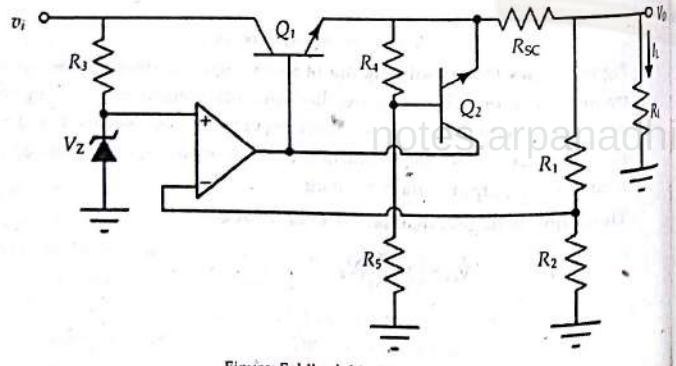

Figure: Foldback Limiting

Current limiting circuit provides protection when current through the load increases beyond the limiting value. The circuit of the figure called foldback limiting circuit reduces both the output voltage and the output current protecting the load from over current as well as protecting the regulator.

Shunt Voltage Regulators

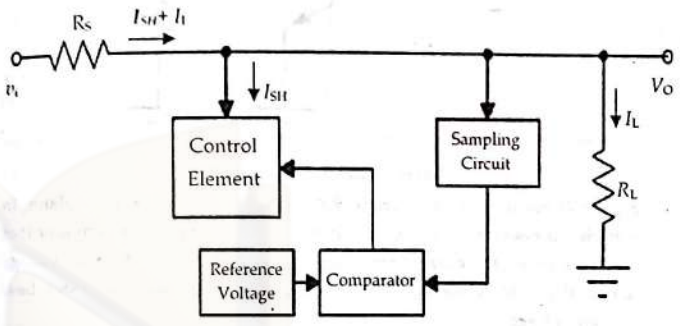

Figure: Shunt voltage Regulator

Figure shows the block diagram of shunt voltage regulator. In this circuit normally the input voltage supplies current to the load. When output voltage tries to change, some of the current is pulled away by the control element to maintain the regulated output voltage across the load.

If the load voltage tries to change due to the change in the load, the sampling circuit provides a feedback voltage to the comparator. The comparator compares the feedback signal with the reference voltage to generate control signal for the control element. Based on control signal control element varies the amount of current to be shunted away from the load in order to keep output voltage constant.

When the output voltage increases, sampling circuit provides feedback signal to the comparator. The output of comparator causes the control element to shunt away more current providing less current to the load thereby keeping the regulated voltage from rising.

When the output voltage decreases, reverse happens. In this case control element shunts away less current thereby providing the sufficient current to the load maintaining the load voltage.

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Transistor Shunt Regulator

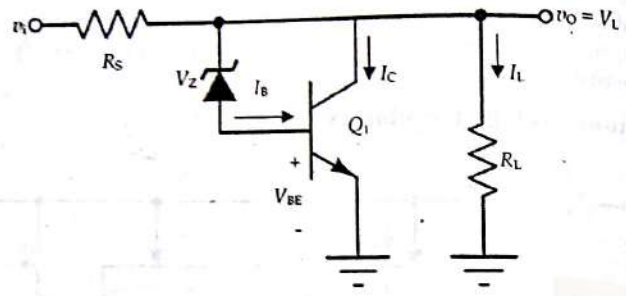

Figure: Transistor Shunt regulator

Figure shows the circuit diagram of a simple transistor shunt regulator. In this circuit resistor R_S drops the unregulated voltage by an amount that depends upon the current supplied to the load resistor R_L. The voltage across the load is set by the zener diode voltage and transistor base emitter voltage.

If the load resistance decreases, a reduced drive current to the base of Q_1 results, shunting less collector current. The load current is thus larger, thereby maintaining the regulated voltage across the load.

The output voltage V_O or V_L is given as:

$$V_O = V_L = V_Z + V_{BE}$$

Improved Shunt Regulator

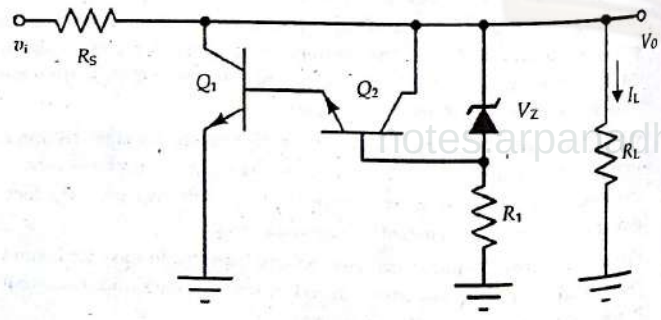

Figure: Improved shunt regulator

Figure shows the circuit diagram of improved shunt regulator using transistor. The zener diode provides a reference voltage so that the voltage across the R_1 senses the output voltage. As the output voltage tries to change, the current shunted by the Q_1 is varied to maintain the output voltage constant. Transistor Q_2 provides larger base current to the transistor Q_1 than in the normal shunt regulator. Thus regulator can handle relatively larger load current.

The output voltage is given as:

$$V_O = V_L = V_Z + V_{BE2} + V_{BE1}$$

Op amp Shunt Voltage Regulator

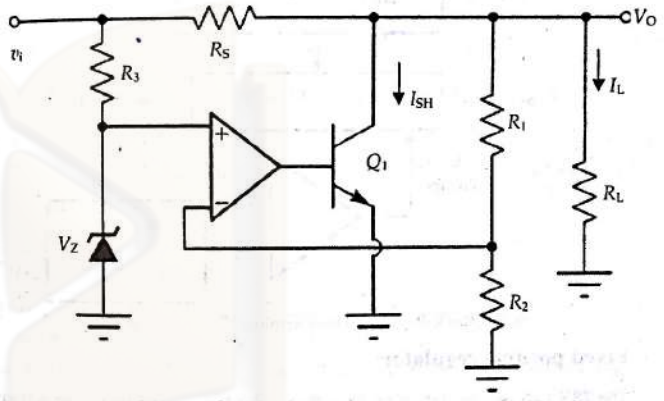

Figure: Op amp shunt voltage regulator

Figure shows the circuit diagram of op amp shunt regulator. In this circuit op amp is used as a voltage comparator. The zener voltage is compared with the feedback voltage obtained from the voltage divider formed by R_1 and R_2 to provide the control drive current to shunt element Q_1. The current through R_S is thus controlled to drop a voltage across R_S so that the output voltage is maintained.

IC voltage Regulators

The voltage regulator circuits which are available in the form of single IC are known as IC voltage regulators. The IC voltage regulators contain the circuitry for reference source, comparator amplifier, control device and overload protection all in a single IC. Though the internal construction of

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IC voltage regulators is somewhat different from the discrete voltage regulators but the external operations is much more similar. IC voltage regulators provide regulation of fixed positive voltage, a fixed negative voltage or an adjustably set voltage.

IC voltage regulators can be connected to the ordinary unregulated power supply comprising of transformer, rectifier, and filter so as to form the regulated power supply.

Generally IC voltage regulators are available as the three terminal IC voltage regulators.

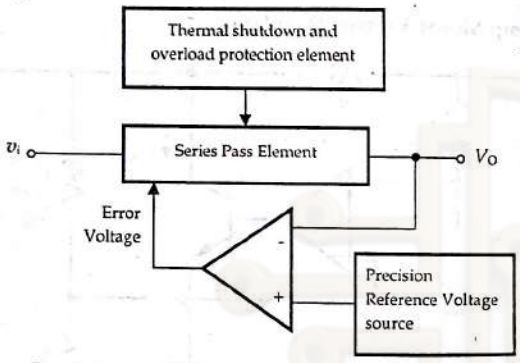

Figure: Block diagram of three terminal IC voltage regulator

Fixed positive regulator:

The 78XX series regulators provide fixed regulated positive voltages from the 5V to 24V. Let us consider an IC 7805 from 78XX series. This IC provides regulated output of 5V. IC 7805 can be connected as follows so as to get a regulated positive output voltage of 5V.

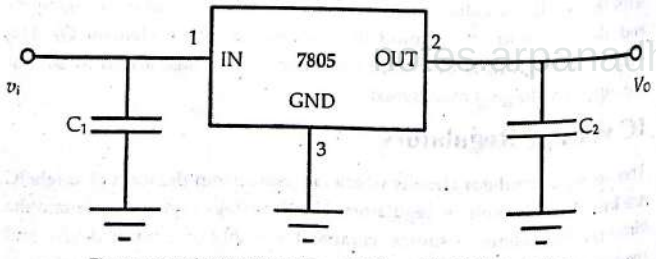

Figure : Use of 7805 IC to get a regulated positive 5V DC outptut

Figure shows the connection of 7805 IC to give a regulated 5V DC output. Pin 1 of IC is connected to unregulated input voltage. Pin 3 is grounded. And pin 2 provides regulated output voltage. Function of capacitors C_1 and C_2 is to reduce the effects of noise.

Table below shows some of the regulators from 78XX series, their output voltage and corresponding minimum input voltages.

Table : Positive voltage regulators in the 78XX series

IC	Output Voltage(V_O)	Minimum input(v_i)
7805	+5	7.3
7806	+6	8.3
7808	+8	10.5
7810	+10	12.5
7812	+12	14.6
7815	+15	17.7
7818	+18	21.0
7824	+24	27.1

Fixed negative regulator:

The 79XX series regulators provide fixed regulated negative voltages from the -5V to -24V.

Let us consider an IC 7905 from 79XX series. This IC provides regulated output of - 5V. IC 7905 can be connected as follows so as to get a regulated negative output voltage of -5V.

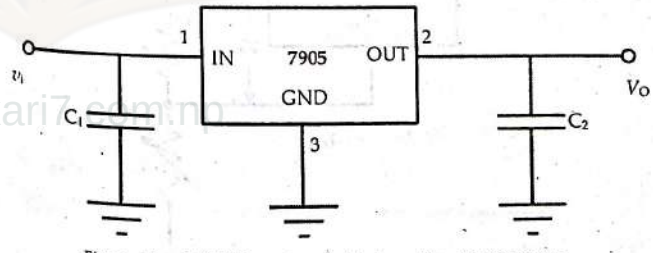

Figure : Use of 7905 IC to get a regulated negative -5V DC outptut

Figure shows the connection of 7905 IC to give a regulated -5V DC output. Pin 1 of IC is connected to unregulated input voltage. Pin 3 is

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grounded. And pin 2 provides regulated output voltage. Function of capacitors C_1 and C_2 is to reduce the effects of noise.

Table below shows some of the regulators from 79XX series, their output voltage and corresponding minimum input voltages.

Table : Negative voltage regulators in the 79XX series

IC	Output Voltage(V_O)	Minimum input(v_i)
7905	-5	-7.3
7906	-6	-8.3
7908	-8	-10.5
7910	-10	-12.5
7912	-12	-14.6
7915	-15	-17.7
7918	-18	-21.0
7924	-24	-27.1

Adjustable Voltage regulator:

The voltage regulators that allow the user to set the output voltage to a desired regulated value are known as adjustable voltage regulator. LM 317, for example can be operated with the output voltage regulated at any setting over the range of voltage from 1.2 V to 37V. Figure below shows how the regulated output voltage of LM 317 can be set.

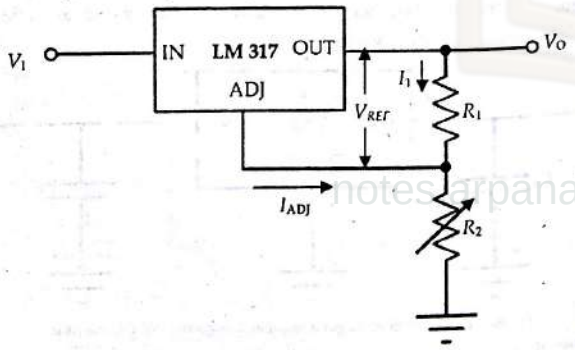

Figure: Adjustable voltage regulator using IC LM317

In the above circuit resistors R_1 and R_2 sets the output to any desired voltage over the adjustment range (1.2V to 37 V). The output voltage desired can be calculated using

$$V_O = V_{REF}\left(1+\frac{R_2}{R_1}\right) + I_{ADJ}\,R_2$$

The typical values of V_{REF} and I_{ADJ} are

$$V_{REF} = 1.25\text{ V and } I_{ADJ} = 100\ \mu A$$

Specification of a Regulator

Load Regulation:

It indicates the change in the output voltage when the load current changes from no load to full load condition. It is denoted by LR.

Load Regulation (LR) = $V_{NL} - V_{FL}$

Where,

V_{NL} = output voltage at no load.

V_{FL} = output voltage at full load

$$\%\text{ Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

Line Regulation:

It indicates the change in load voltage for a specified range of input voltage. It is denoted by SR.

$SR = V_{LH} - V_{LL}$

Where,

V_{LH} = load voltage with high line voltage

V_{LL} = load voltage with low line voltage

$$\%SR = \frac{V_{LH} - V_{LL}}{V_{NOM}}$$

Where, V_{NOM} is the nominal load voltage under the typical condition.

Line regulation is also sometimes called as source regulation.

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Performance parameters of voltage regulators

Voltage Stability Factor (S_V):

It is defined as the ratio of change in the output voltage to the change in input voltage for a constant load current and temperature. It is denoted as S_V.

Mathematically,

$$S_V = \frac{\Delta V_O}{\Delta V_I} \text{ at constant } I_L \text{ and } T$$

Temperature Stability Factor (S_T)

It is defined as the ratio of change in output voltage to the change in temperature at constant load current and input voltage. It is denoted as S_T.

Mathematically,

$$S_T = \frac{\Delta V_O}{\Delta T} \text{ at constant } I_L \text{ and } V_{in}$$

Output Resistance:

It is defined as the ratio of the change in the output voltage to the change in the load current for constant input voltage and temperature. It is denoted as R_O.

Mathematically,

$$R_O = \frac{\Delta V_O}{\Delta I_L} \text{ at constant } V_{in} \text{ and } T$$

This resistance is not a real resistance. It is included to see the effect of the load current on load voltage.

Practically, all the parameters S_T, S_V, R_O should be as low as possible.

Thus the total change in the output voltage is given as:

$$\Delta V_O = S_V \Delta V_{in} + R_O \Delta I_L + S_T \Delta T$$

Expression of Voltage Stability factor (S_V) for series regulator

The figure shows the circuit diagram of series regulator.

From the circuit, application of KVL results,

$v_i = V_{R3} + V_{BE1} + V_O$

Since the base emitter voltage drop of transistor Q_1 is a constant quantity and output is desired to be constant, we may assume the change in input voltage is approximately equal to the change in voltage across the resistor R_3 when current through it varies.

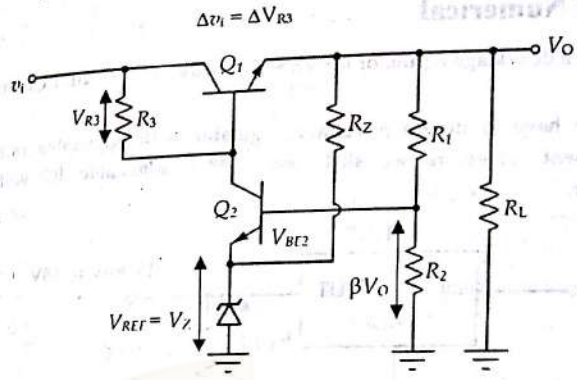

Figure: Series voltage regulator circuit redrawn to derive the expression for voltage stability factor S_V

The gain A of the transistor Q_2 is expressed as:

$$A = \frac{\text{Total resistance in the collector of } Q_2}{\text{Total resistance in the emitter of } Q_2}$$

$$\text{or, } A = \frac{R_3}{r_{e2} + r_Z}$$

Since the feedback voltage βV_O is input to the base of Q_2 and V_{R3} is the output of Q_2. Therefore we may write,

$$V_{R3} = A \times \beta V_O$$

Let ΔV_O be the small change in the output voltage and ΔV_{R3} be the small change in the voltage across the resistor R_3.

Then,

$$\Delta V_{R3} = A \times \beta \Delta V_O$$

$$\text{or, } \frac{\Delta V_O}{\Delta V_{R3}} = \frac{1}{A\beta}$$

$$\text{or, } \frac{\Delta V_O}{\Delta v_i} = \frac{1}{\left(\frac{R_3}{r_{e2} + r_Z}\right)\left(\frac{R_2}{R_1 + R_2}\right)}$$

$$\therefore S_V = \left[\left(\frac{r_{e2} + r_Z}{R_3}\right)\left(1 + \frac{R_1}{R_2}\right)\right]$$

This is the required expression for the voltage stability factor of transistor series regulator.

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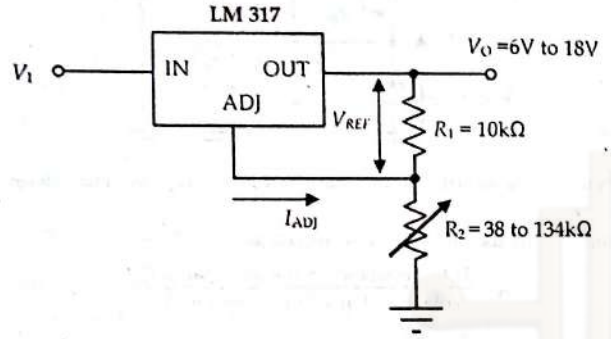
Solved Numerical

1. Design a dc voltage regulator for V_O = 6V to 18V. **2071 Chaitra**

Solution

Here we have to design a voltage regulator with adjustable output requirement. Therefore we shall use LM317 adjustable IC voltage regulator.



For the circuit,

$$V_O = V_{REF}\left(1+\frac{R_2}{R_1}\right) + I_{ADJ} R_2$$

Typically, V_{REF} = 1.25V and I_{ADJ} = 100 μA

Since the value of I_{ADJ} is negligibly small we shall ignore its value in our further calculations.

$$V_O = V_{REF}\left(1+\frac{R_2}{R_1}\right) = 1.25\left(1+\frac{R_2}{R_1}\right)$$

Rearranging the equation,

$$R_2 = R_1\left(\frac{V_O}{1.25}-1\right)$$

When V_O = 6 V,

$$R_2 = R_1\left(\frac{6}{1.25}-1\right) = 3.8R_1$$

$$\therefore R_2 = 3.8\ R_1$$

When V_O = 18 V,

$$R_2 = R_1\left(\frac{18}{1.25}-1\right) = 13.4\ R_1$$

$$\therefore R_2 = 13.4\ R_1$$

Taking R_1 = 10 kΩ,

For V_O = 6 V,

$$R_2 = 3.8\ R_1 = 3.8\times10 = 38\ k\Omega$$

For V_O = 18 V,

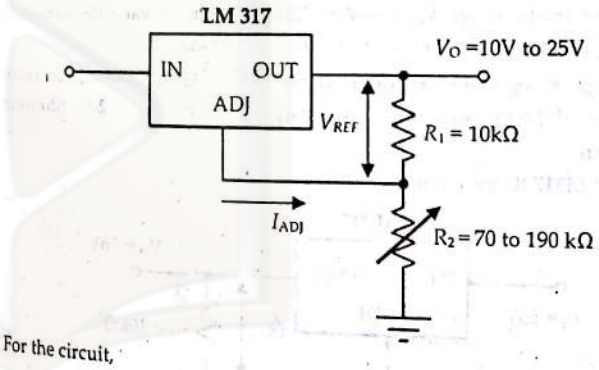
$$R_2 = 13.4\ R_1 = 13.4\times10 = 134\ k\Omega$$

Thus, if we take R_1 = 10 kΩ, then the output voltage can be obtained in the range V_O = 6V to 18V by taking a variable resistor R_2 having resistance in the range R_2 = 38kΩ to 134kΩ.

2. Design a (10-25) V variable DC series voltage regulator using LM317 IC. **2070 Chaitra**

Solution

Here we have to design a voltage regulator with adjustable output requirement. Therefore we shall use LM317 adjustable IC voltage regulator.



For the circuit,

$$V_O = V_{REF}\left(1+\frac{R_2}{R_1}\right) + I_{ADJ} R_2$$

Typically, V_{REF} = 1.25V and I_{ADJ} = 100 μA

Since the value of I_{ADJ} is negligibly small we shall ignore its value in our further calculations.

$$V_O = V_{REF}\left(1+\frac{R_2}{R_1}\right) = 1.25\left(1+\frac{R_2}{R_1}\right)$$

Rearranging the equation,

$$R_2 = R_1\left(\frac{V_O}{1.25}-1\right)$$

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When $V_O = 10$ V,

$$R_2 = R_1\left(\frac{10}{1.25} - 1\right) = 7R_1$$

$$\therefore R_2 = 7\,R_1$$

When $V_O = 25$ V,

$$R_2 = R_1\left(\frac{25}{1.25} - 1\right) = 19R_1$$

$$\therefore R_2 = 19\,R_1$$

Taking $R_1 = 10\ k\Omega$,

For $V_O = 10$ V,

$$R_2 = 7R_1 = 7 \times 10 = 70\ k\Omega$$

For $V_O = 25$ V,

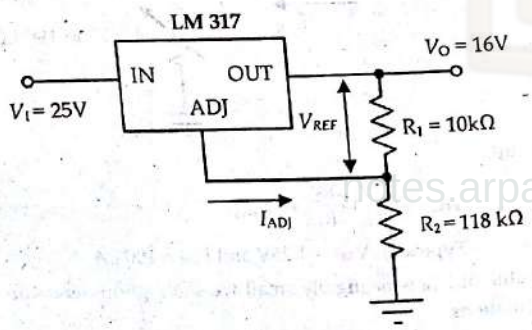
$$R_2 = 19\,R_1 = 19 \times 10 = 190\ k\Omega$$

Thus, if we take $R_1 = 10\ k\Omega$ resistor, than the output voltage can be obtained in the range $V_O = 10V$ to 25V by taking a variable resistor R_2 having resistance in the range $R_2 = 70\ k\Omega$ to $190\ k\Omega$.

3. Design a regulator circuit to obtain 16 VDC. Choose approximate values of the parameters. Input voltage is 25 VDC. **2071 Shrawan**

Solution

Using LM317 IC the circuit is as follows:



We know,

$$R_2 = R_1\left(\frac{V_O}{1.25} - 1\right)$$

For $V_O = 16$ V

$$R_2 = R_1\left(\frac{V_O}{1.25} - 1\right)$$

$$R_2 = R_1\left(\frac{16}{1.25} - 1\right)$$

$$R_2 = 11.8\,R_1$$

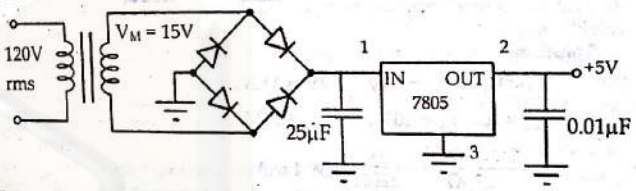
Taking $R_1 = 10\ k\Omega$

$$R_2 = 11.8\,R_1 = 11.8 \times 10$$

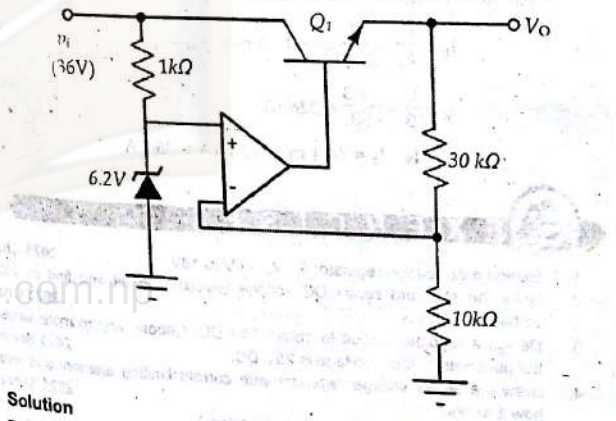
$$\therefore R_2 = 118\ k\Omega$$

4. Draw a voltage regulator supply using a full wave bridge rectifier, capacitor filter and IC regulator to provide output of 5V.

Solution



5. Calculate the regulated output voltage in the circuit shown below



Solution

The output of the circuit is given as:

$$V_O = \left(1 + \frac{R_1}{R_2}\right)V_Z$$

$$\text{or, } V_O = \left(1 + \frac{30}{10}\right)6.2 = 4 \times 6.2$$

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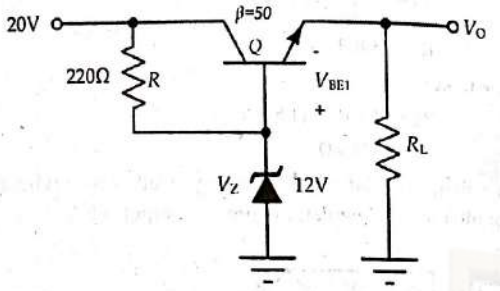
$$V_O = 24.8 \text{ V}$$

6. Calculate the output voltage and the zener current in the regulator circuit as given below for $R_L = 1k\Omega$.



Solution

$V_O = V_Z - V_{BE} = 12V - 0.7V = 11.3V$

$V_{CE} = v_i - V_O = 20V - 11.3V = 8.7 \text{ V}$

$$I_R = \frac{20V - 12V}{220\Omega} = \frac{8V}{220\Omega} = 36.4 \text{ mA}$$

For $R_L = 1 \text{ k}\Omega$,

$$I_L = \frac{V_O}{R_L} = \frac{11.3V}{1k\Omega} = 11.3 \text{ mA} \approx I_E \approx I_C$$

$$I_B = \frac{I_C}{\beta} = \frac{11.3}{50} = 226\mu A$$

$I_Z = I_R - I_B = 36.4 \text{ mA} - 226 \mu A = 36 \text{mA}$

EXERCISE

1. Design a dc voltage regulator for $V_O = 6V$ to 18V **2071 Chaitra**
2. Draw the standard series DC voltage regulator circuit and find its voltage stability factor(S_v). **2071 Chaitra**
3. Design a regulator circuit to obtain 16V DC. Choose approximate values of the parameters. Input voltage is 25V DC. **2071 Shrawan**
4. Draw the series voltage regulator with current limiting element and explain how it works. **2071 Shrawan**
5. Draw block diagram for IC voltage regulator **2071 Shrawan**
6. Design a (10-25)V variable dc series voltage regulator using LM 317 IC. **2070 Chaitra**
7. Draw the circuit of current limiting circuit in dc voltage regulator. **2070 Chaitra**
8. Find voltage stability factor of series dc voltage regulator. **2070 Chaitra**

9. Draw the circuit diagram of variable series voltage regulator with transistor as error amplifier. And find its voltage stability factor and explain how change in output voltage can be improved. **2070 Ashad**
10. Design a voltage circuit using LM 317 to obtain 16V DC with the input 24V DC. **2070 Ashad**
11. Draw standard series DC voltage regulator and find its voltage stability factor(S_v) **2069 Chaitra**
12. Design a 4.2 V to 12 V variable dc voltage regulator using IC LM 317 **2069 Chaitra**
13. Define the regulated power supply. Determine the voltage stability factor(line regulation) factor of variable series voltage regulator circuit with transistor as an error amplifier. **2069 Ashad**
14. Design a 10V- 15V variable dc series voltage regulator using LM 317 IC **2069 Ashad**
15. Estimate voltage stability factor (S_v) for standard dc voltage regulator using BJT. Also explain the operation of overload protection circuit that could be used in series voltage regulator circuit. **2068 Chaitra**
16. Draw the transistorized series voltage regulator circuit and explain its operation. **2068 Shrawan**
17. Write short notes on IC voltage regulator. **2068 Shrawan**
18. Draw standard series dc voltage regulator circuit and find its voltage stability factor(S_v) **2068 Baisakh**
19. Draw a voltage regulator circuit using LM 317. **2068 Baisakh**
20. Draw a circuit diagram for bandgap reference voltage source. **2068 Baisakh**

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